AWI–ESM3 (v3.4, CORE3 / TCO95) PI–control tuning campaign

`Tuning_test_06*` — progress report

Evaluation via `release_evaluation_tool2` (part2 / part8 / part4)

2026–06–30 (rev. **2026–10–15**)

Abstract

AWI–ESM3 v3.4 (OpenIFS 48r1 / FESOM2 / LPJ-GUESS, TCO95L91) is being tuned for a pre-industrial spin-up against two faults: a positive TOA and surface radiative imbalance, and a NH high-latitude boreal-summer cold bias severe enough to lose the boreal forest. An AMIP testbed (§11) runs the atmosphere alone so the vegetation–albedo feedback is open and the root cause can be hunted without the coupled runaway closing over it.

Where we are (2026–10–15). The CMIP7 spin-up runs on the tagged carbon-cycle build from PHC3 with icebergs from the model's own mass budget; after 160 years it is 0.4–0.5 K colder than the tuning line at the same age and its carbon drift is 0.41 $\mathrm{GtC\,yr^{-1}}$ against a target of 0.1 (§75). Five freshwater and sea-ice levers left the Southern Ocean cap where it was, and a second control showed that single-run decadal differences are twice as noisy as our test assumed (§70–§72). The deep drift was located and judged unstoppable by run length, which decided the restart (§73, §74).

As of 2026–10–06. AWI-ESM3 v3.5.0 is tagged in every component (§65) and the line has two stable candidates on hemispheric GM: `cvmix_TKE` alone, with a nearly still deep ocean and a slowly eroding Antarctic summer pack, and TKE + IDEMIX, which holds the Antarctic and buries heat at 700–1500 m (§64). An attempt to return to KPP with a uniform GM of 1000 quieted the abyss and opened the Southern Ocean (§67). The diagnosis that followed is the present basis of the work: the Southern Ocean surface cap is too weak in every run, 78 % of observed and 31 % in the Weddell region, because the applied GM coefficient is half the CORE2 value, the wind stress is about 10 % too strong and the iceberg freshwater enters too far north (§68). The plan is §69. The carbon-cycle configuration had been exchanging gases against zero pressure and is fixed (§66).

As of 2026–10–03. The campaign runs on albedo, on a line that carries a boreal forest: **PICAL_crunveg** restarts LPJ-GUESS from the CRUNCEP state and keeps 81 % of its larch after 19 years, at a cost of about $2 \times 10^3\,\mathrm{km^3}$ of NH ice (§58). The Labrador Sea ice-capped regime was traced across 61 runs to Round 36's GM 2500 (§59), whose heat-uptake benefit turned out to be Southern Ocean ventilation at 34–55 S (§60); of four 20-year GM arms at the production step, **hemispheric GM** (2500 south, 1000 north) restores Labrador convection and changes nothing else significantly, and is adopted, with an overshoot to correct. Against a pre-industrial reference the model is **+0.46 K too warm**, in summer mid-latitudes, while its Arctic winter is cold (§61.3): too warm and a positive imbalance now point the same way. The Antarctic winter warm ring sits over a too-diffuse pack whose drift over-responds to correct winds (§62). Put on one axis for the first time, the 290-year line shows a mid-depth warm layer that has built throughout and that IDEMIX, not TKE, feeds (§63); a TKE-only continuation is running, and the carbon-cycle configuration is paused for performance work.

As of 2026–09–25. Screening is over and the campaign is producing: **PICAL_ccnice** has run 1940–2100 on the v3.5 defaults with `cvmix_TKE+cvmix_IDEMIX`. It is not converged. It cools 0.21 K/decade at the surface while net TOA is positive, because total ocean heat content is flat and the upper 700 m is losing heat to below 700 m at 0.32 $\mathrm{W\,m^{-2}}$ equivalent (§57.2). The Antarctic sits at 62 % of observed winter volume, the ACC at 98.9 Sv against 134 ± 11.2 observed, and the boreal forest is gone to grass competition (§57). Five Southern Ocean levers were tried and excluded (§54); the GM setting that improved the CDW cost 16.7 Sv of Drake Passage (§57.4). Work is moving to albedo for queue time; see `HANDOFF_albedo_spinup_continuation.md` and §77.

What is left. Three routes are closed by measurement rather than opinion: transpiration→cloud (§76.2), snow-melt albedo (§27.2), and land surface albedo, which buys at most one sixth of the universal bias (§30.3). Skin conductivity is falsified for both the winter route (§28.1) and the boreal organic layer (§30.1). **Round 22 relocated the two remaining targets** (§38): the Southern Ocean is confirmed as the largest regional term and grows period-cleanly to +7.46 $\mathrm{W\,m^{-2}}$, but two thirds of it is missing cloud *area* that no lever — including the adopted INP branch — has ever moved; and the **global tropospheric cold bias** is now located, at the *extratropical tropopause* rather than in tropical convection, where the atmosphere is both 1.6 K too cold and 10 % too dry, two large errors that cancel in OLR and so cannot be fixed one at a time.

Read §76 for the causal web and what is still dark; §25 for what the observations can and cannot settle; §78 for every mechanism this campaign has killed, with the measurement that killed it. Parts I–III are the coupled work that preceded the AMIP testbed; Part IV is the atmosphere-only campaign in the order it happened.

Front-matter note — new-sea-ice branch `Tuning_test_09A/09B/09C`

The dated account of this branch is §12 in the timeline; what follows is the run-identity reference only.

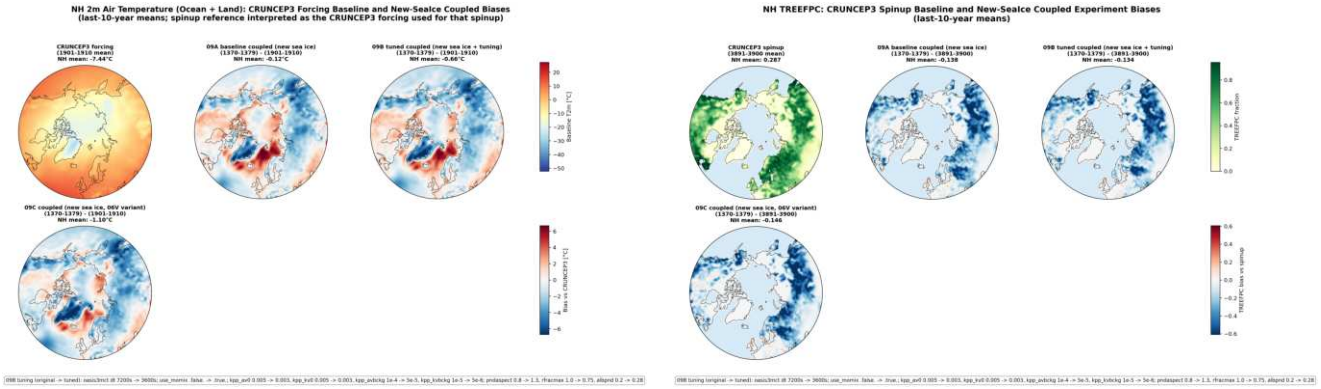
This report now includes the new three-way coupled comparison branch: `09A` (baseline + new sea ice), `09B` (new sea ice + tuned stack), and `09C` (new sea ice, 06V variant). All three completed the same 30-year segment and are compared on the same last-10-year window (model years 1370–1379).

Run identities

- `09A`: `Tuning_test_09A_lpjguess_Baseline_coupled_fromCRUNCEP_newSeaIce`
- `09B`: `Tuning_test_09B_06T_1hCPL_MOSPP_KPPLOW_CRUNCEPinit_newSeaIce`
- `09C`: `Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`

09B tuning block documented on the figures (original → tuned)

- `oasis3mct.time_step`: 7200 s → 3600 s
- `use_momix`: `.false.` → `.true.`
- `kpp_av0`: 0.005 → 0.003, `kpp_kv0`: 0.005 → 0.003
- `kpp_avbckg`: 1e-4 → 5e-5, `kpp_kvbckg`: 1e-5 → 5e-6
- `pndaspect`: 0.8 → 1.3, `rfracmax`: 1.0 → 0.75, `albpnd`: 0.2 → 0.28



(a) NH T2m bias vs CRUNCEP3 spinup reference (ocean + land). **(b)** NH TREEFPC bias vs CRUNCEP3 spinup reference.

Figure 1: New-sea-ice branch update: common last-10-year means (1370–1379) for `09A/09B/09C`.

Part I

Tuning away the high TOA imbalance

*We set out to cut the AWI–ESM3 PI-control's +1.5 W/m^2 TOA imbalance. Every effective lever does it by **cooling** the model, and the aggressive coupling/ocean-mixing levers cut the imbalance hardest — but push the planet too cold against ERA5. This part documents that trade-off across 22 runs and traces the aggressive lever to a Southern-Ocean ocean-convection mechanism.*

1 Goal & setup

Reduce the net top-of-atmosphere (TOA) radiative imbalance of the AWI–ESM3 pre-industrial control so it equilibrates faster, *without* degrading the CMIP6-normalised performance index (CMPI, `part4`) or inflating the 2 m-temperature RMSE vs ERA5 (`part8`). 22 experiments, each a 30-year segment (model years 1350–1379) of the CORE3/TCO95 configuration (FESOM2 CORE3 mesh, OpenIFS TCO95/A096, LPJ-GUESS), all branched from one **Baseline**. The baseline PI-control still drifts with a positive TOA imbalance of $\approx +1.5\,\mathrm{W/m^2}$ (energy *into* the

system — the signature of unequilibrated spin-up and/or ongoing ocean heat uptake). **A caution on the target.** The tuning levers change the model's *equilibrium*, not its rate of approach, so "lower imbalance" is a useful *ranking* objective but not automatically "more equilibrated": a low-imbalance-but-cold state is not obviously better than a high-imbalance-but-unbiased one. This is why the campaign's real aim (§9) is a *warm-floor* state where the imbalance is cut by a *moderate* lever, not by cooling into a cold bias — i.e. we treat imbalance and 2 m-T bias as joint targets, not imbalance alone.

Structural note. Almost every non-baseline run also carries a *common melt-pond shift* (`albpnd` 0.2 → 0.28, `rfracmax` 1.0→0.75, `pndaspect` 0.8→1.3). Differences below are each run's *distinguishing* change on top of that.

Four parts. (I) evaluates the 22 imbalance-tuning runs and exposes the imbalance↔cold-bias trade-off, tracing the aggressive lever to a Southern-Ocean mechanism. **(II)** shows that the cold bias which makes that lever costly is largely *not* the tuning but a **structural** LPJ-GUESS boreal-forest collapse (§6) — a *counter-lever*: fixing it warms the NH high latitudes, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. Feedback-free spin-ups then quantify the sensitivity to forcing and isolate the comparatively small contribution from replacing net radiation alone. **(III)** records the original warm-floor work plan. **(IV)** supersedes its ordering where newer evidence requires it: first fix the atmospheric forcing on AMIP, then re-spin LPJG and tune only the residual (§15, §16).

2 Headline: the imbalance ↔ cold-bias trade-off

Every lever that meaningfully lowers the TOA imbalance does so by *cooling* the model, which worsens the cold bias against ERA5. Two families separate cleanly: the sea-ice/albedo/melt-pond changes move gently down-left (small imbalance gain, small bias cost); the coupling + ocean-mixing changes cut the imbalance hard (30–57 %) but push the planet 1.7–2.1 K too cold.

The cold-bias axis is two things at once. It mixes (i) a *shared structural floor* — every run, including the baseline and the "clean" melt-pond runs, carries a strong Siberia/N. Canada cold bias that is *not* a tuning artefact but an LPJ-GUESS boreal-forest-cover bug (§6); and (ii) the *differential*, tuning-induced global cooling that the aggressive runs add on top. The structural floor is the same for all points and is independently fixable; the tuning only moves runs vertically relative to it. This is why the "warming lever" in §9 is the LPJG fix, not another atmospheric retune.

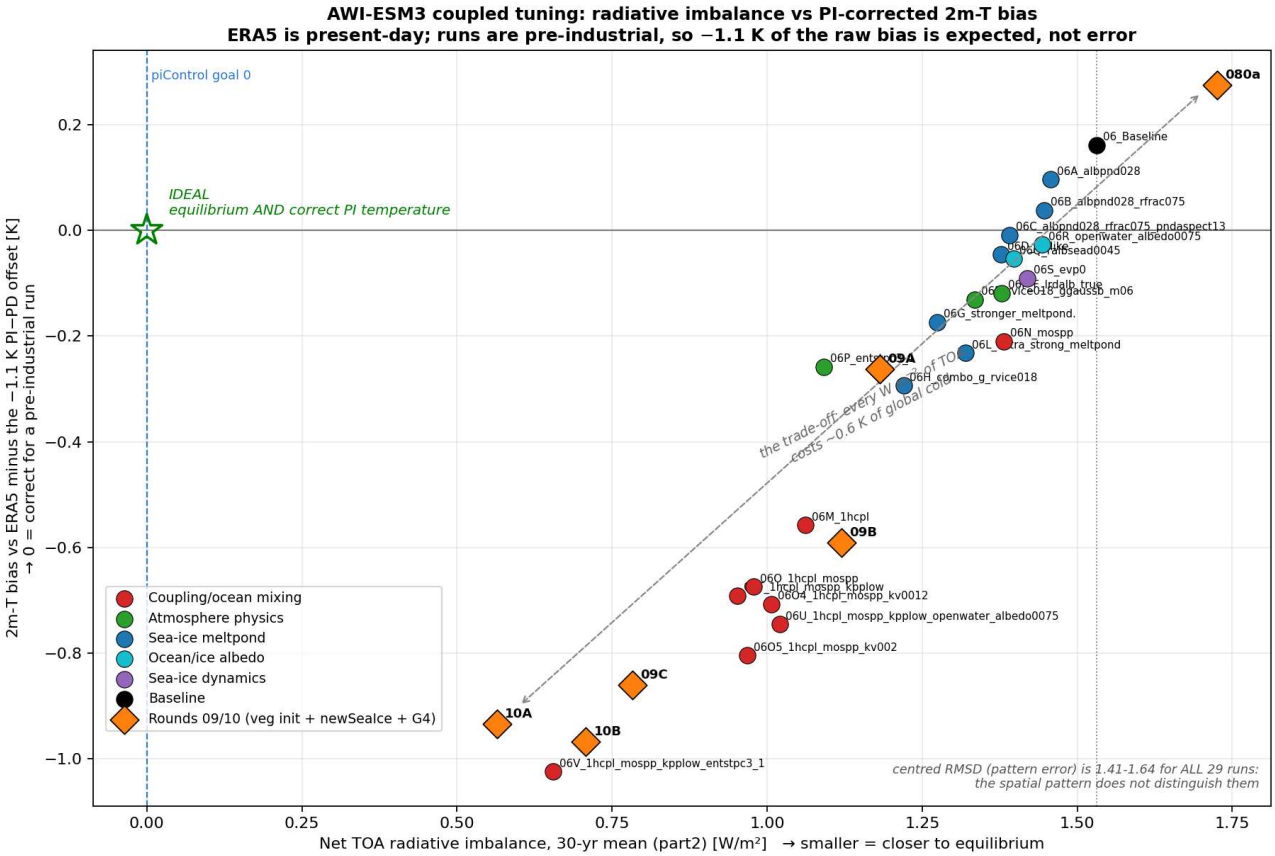


Figure 2: Net TOA imbalance (30-yr mean, `part2`) against the 2 m-T bias vs ERA5 *corrected for the reference period*, now including rounds 09 and 10 (orange diamonds, common window 1354–1379). **ERA5 is present-day and the runs are pre-industrial, about 1.1 K apart, so that much of the raw cold bias is expected rather than error**; the y axis subtracts it, and 0 is correct for a PI run. Ranking by raw RMSD penalises exactly the runs that are correctly cold, which is why the earlier version of this figure was misleading. Dashed blue = the corrected piControl goal of 0 (§37); the star marks the ideal, which nothing occupies. **The runs lie on a line, not in a cloud**: every $\mathrm{W\,m^{-2}}$ of TOA improvement costs $\approx$0.6 K of global cold, from 080a at $(1.73, +0.28)$ to 10A at $(0.57, -0.93)$. And the *pattern* error is not what distinguishes them — centred RMSD, $\sqrt{\mathrm{RMSD}^2 - \mathrm{bias}^2}$, which removes any uniform offset, is 1.41–1.64 K across all 29 runs, a $\pm 8\,\%$ spread against a raw-RMSD spread of 64 %. Colour = parameter family.

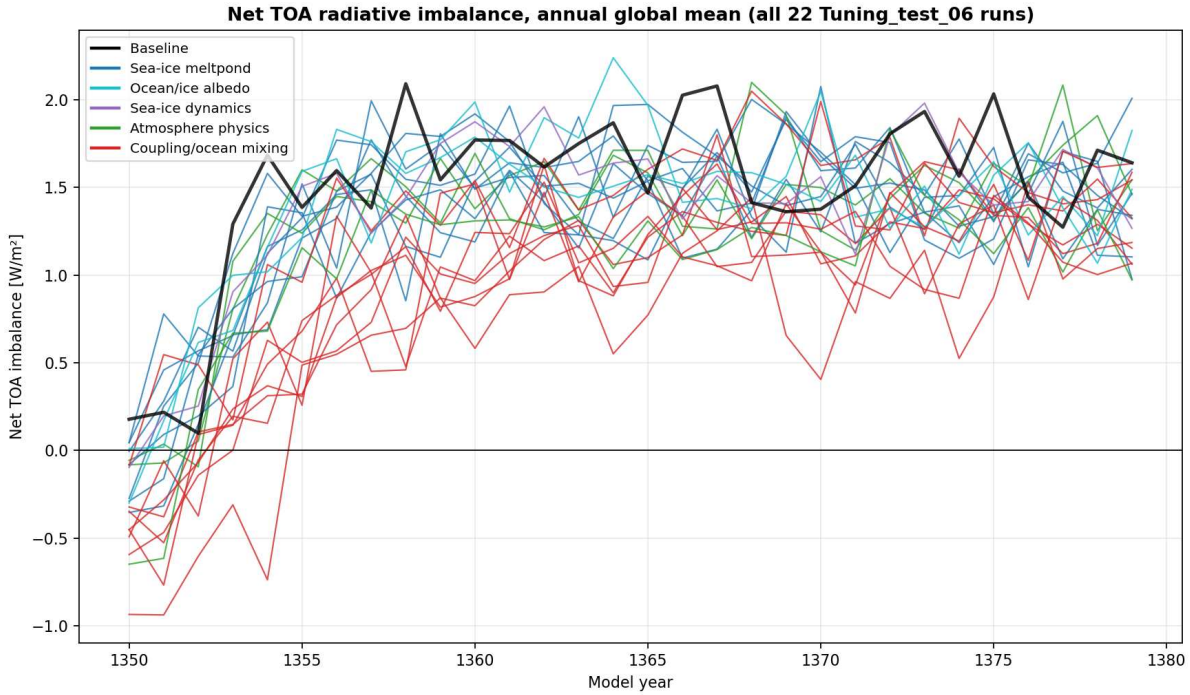


Figure 3: Annual global-mean net TOA imbalance, all 22 runs (independent CDO calculation; matches `part2`). None has flattened within 30 yr, so treat the mean level as a *ranking*, not a final equilibrium value.

3 Master metric table (coupled runs, sorted by TOA imbalance)

Run	TOA imb. [W/m^2]	vs base	2m RMSD [K]	2m bias [K]	CMPI
06V_1hcpl_mospp_kpplow_entstpc3_1	0.655	−57%	2.671	−2.124	0.803
06T_1hcpl_mospp_kpplow	0.952	−38%	2.385	−1.791	0.773
06O5_1hcpl_mospp_kv002	0.968	−37%	2.511	−1.904	0.788
06O_1hcpl_mospp	0.978	−36%	2.389	−1.774	0.772
06O4_1hcpl_mospp_kv0012	1.007	−34%	2.387	−1.807	0.781
06U_1hcpl_mospp_kpplow_openw0075	1.020	−33%	2.422	−1.845	0.776
06M_1hcpl	1.062	−31%	2.315	−1.658	0.773
06P_entstpc3_1	1.091	−29%	2.076	−1.358	0.759
06H_combo_g_rvice018	1.220	−20%	2.082	−1.394	0.757
06G_stronger_meltpond	1.274	−17%	1.990	−1.274	0.753
06L_extra_strong_meltpond	1.320	−14%	2.050	−1.332	0.751
06E_rvice018_ggaussb_m06	1.335	−13%	1.969	−1.231	0.754
06D_HRlike	1.377	−10%	1.892	−1.146	0.750
06F_lrdalb_true	1.378	−10%	1.962	−1.219	0.752
06N_mospp	1.382	−10%	2.011	−1.310	0.751
06C_albpnd028_rfrac075_pndaspect13	1.391	−9%	1.884	−1.109	0.751
06Q_ralbsead0045	1.397	−9%	1.891	−1.154	0.748
06S_evp0	1.419	−7%	1.935	−1.191	0.745
06R_openwater_albedo0075	1.443	−6%	1.860	−1.127	0.744
06B_albpnd028_rfrac075	1.447	−5%	1.847	−1.062	0.741
06A_albpnd028	1.457	−5%	1.811	−1.004	0.748
06_Baseline	**1.531**	–	**1.717**	−0.939	**0.745**
Rounds 09/10 (added 2026-08-07), same window 1354–1379.					
Bias column also given PI-corrected (raw +1.1 K); cRMSD = pattern error.					
10A (06V + G4)	**0.565**	−63%	2.522	−2.034 (−0.93)	cR 1.49
10B (10A + `tanh` depletion)	0.708	−54%	2.547	−2.067 (−0.97)	cR 1.49
09C (06V + newSeaIce)	0.783	−49%	2.475	−1.960 (−0.86)	cR 1.51
09B (06T + newSeaIce)	1.120	−27%	2.238	−1.691 (−0.59)	cR 1.47
09A (baseline + newSeaIce)	1.182	−23%	2.042	−1.363 (−0.26)	cR 1.52
080a (CRUNCEP veg init)	1.726	+13%	**1.629**	−0.825 (+0.28)	cR 1.41

The 09/10 block is the campaign's central trade-off in one place, but read the bias column with the reference period in mind: ERA5 is present-day and these runs are pre-industrial, ≈1.1 K apart, so the PI-corrected value in brackets is the one that means

anything. On that basis 10A is 0.93 K too cold where 06_Baseline is 0.16 K too warm — a real gap, but a third of what the raw numbers suggest. **And the spatial pattern is not what changes**: centred RMSD is 1.41–1.64 across all 29 runs, so the entire raw-RMSD spread is the global mean, counted twice. Nothing yet occupies the low-TOA, correct-temperature corner. Over their own longer window (1370–1399) 10A reads 0.802/2.565 and 10B 0.916/2.383, so the ranking is window-dependent and only the common window is tabulated. CMPI is left blank for these rows rather than guessed — it needs `part4_cmpi.py` and its own reference set. TOA = 30-yr global-mean net TOA from `part2_rad_balance.py`. RMSD/bias from `part8` methodology (cos-lat weighted, model−ERA5, yrs 1354–1379); cross-checked vs the tool's on-figure values (baseline 1.717 vs 1.721; 06V 2.671 vs 2.678). CMPI from `part4_cmpi.py`: < 1 = better than CMIP6 ensemble-mean error; lower is better.

The per-run distinguishing parameter change (which knob each `06*` run alters on top of the common melt-pond shift) is catalogued in App. A.

4 Core diagnostics: radiation, 2 m temperature, CMPI

4.1 Radiation budget (`part2`)

Surface net flux tracks TOA to within $\approx 0.1\,\mathrm{W/m^2}$ in every run — i.e. the atmospheric column stores essentially no energy on these multi-year means, as expected. Note this does *not* by itself localise the *cause*: a TOA $\approx$ surface match is equally consistent with a genuine TOA radiative error and with ocean heat uptake / unequilibrated spin-up drift passing cleanly through the surface (§5, §6.3). What it rules out is an atmosphere-only bookkeeping leak, so the imbalance and the surface energy budget can be read interchangeably. Baseline $\approx +1.53\,\mathrm{W/m^2}$; the coupling+mixing stack drives it down to $+0.66\,\mathrm{W/m^2}$ (06V).

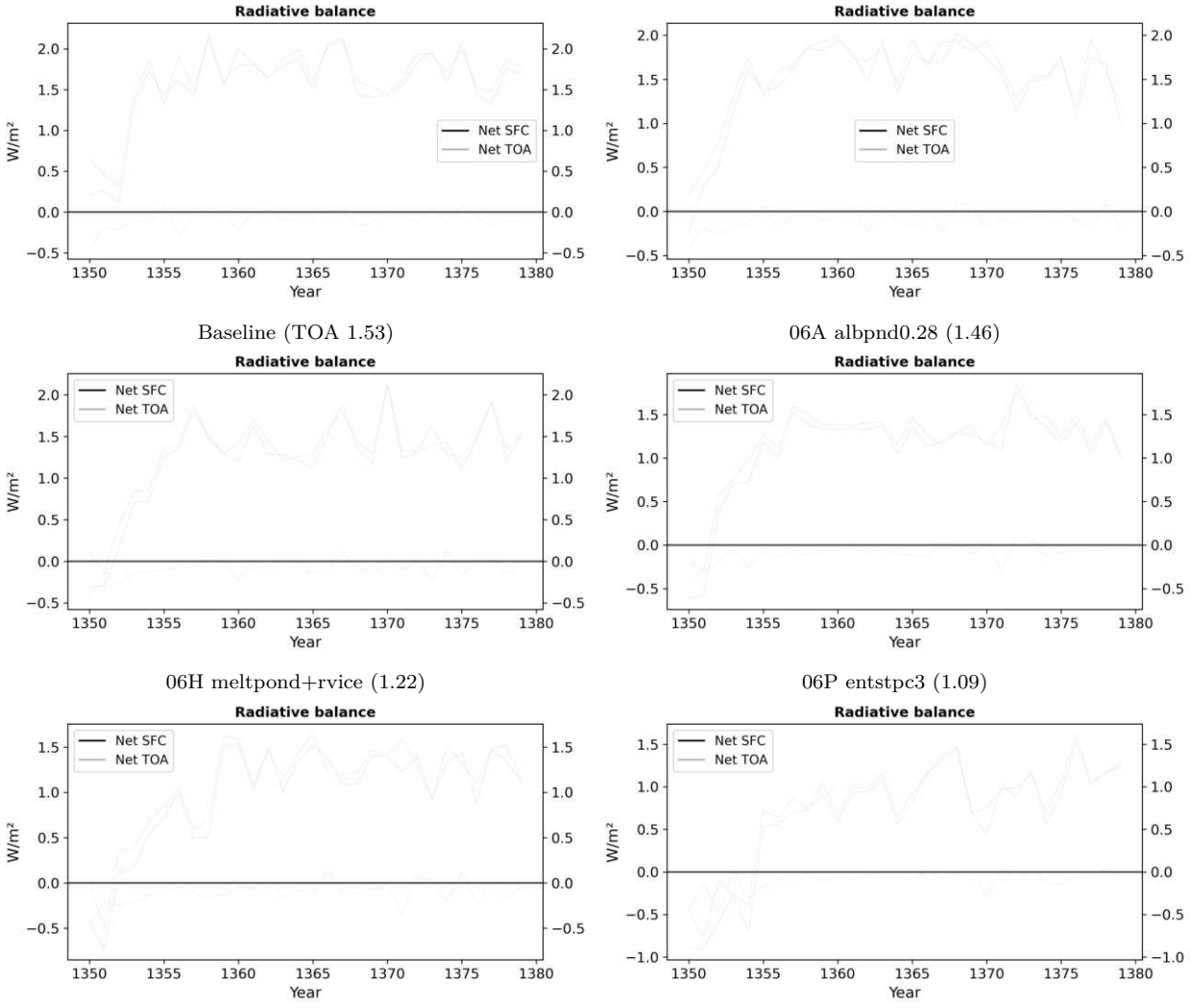
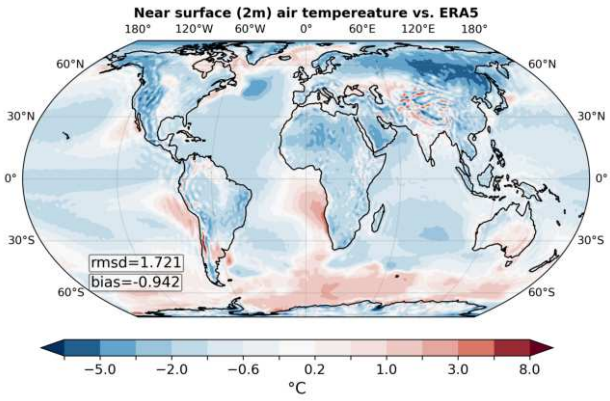


Figure 4: `part2` radiation budget: grey = net TOA, blue = net surface, dark = their difference ($\mathrm{W/m^2}$).

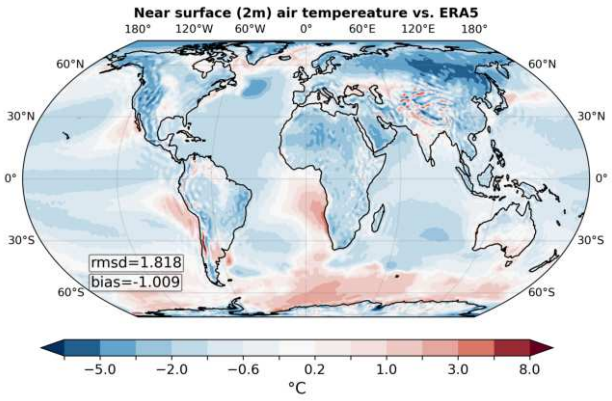
4.2 2 m temperature bias vs ERA5 (`part8`)

Baseline: cold over NH continents, warm over the Southern Ocean and tropical upwelling. The aggressive coupling runs remove the Southern-Ocean warm bias but turn the whole planet cold (06V is cold almost everywhere). *Caveat:* this is PI-control vs present-day ERA5, so ≈ 0.5–0.7 K of global cold offset is physically expected; the coupling runs' −1.7 to −2.1 K is clearly too much, the melt-pond family (−1.0 to −1.4 K) is more defensible.

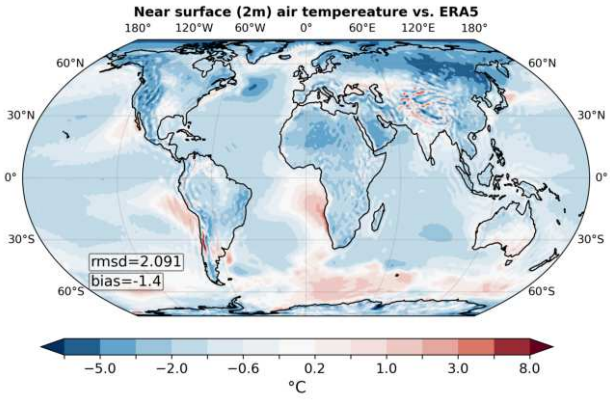
The deep **Siberia / N. Canada cold lobe is present in *every* panel** — baseline and the clean melt-pond runs included — and barely changes between them: it is the structural LPJ-GUESS forest-cover signal (§6), not a product of the imbalance tuning. The aggressive coupling runs add a broad, near-global cooling *on top* of that fixed lobe.



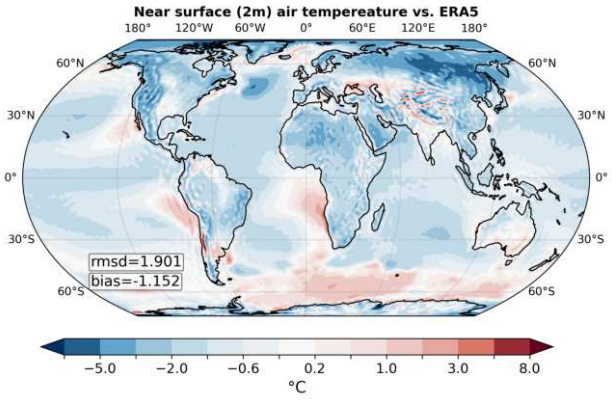
Baseline (RMSD 1.72, bias −0.94)



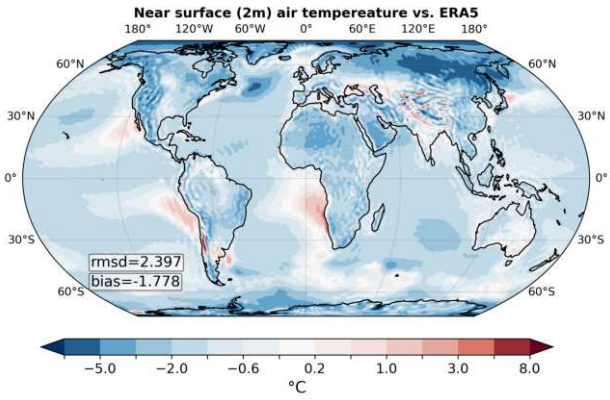
06A (1.81, −1.00)



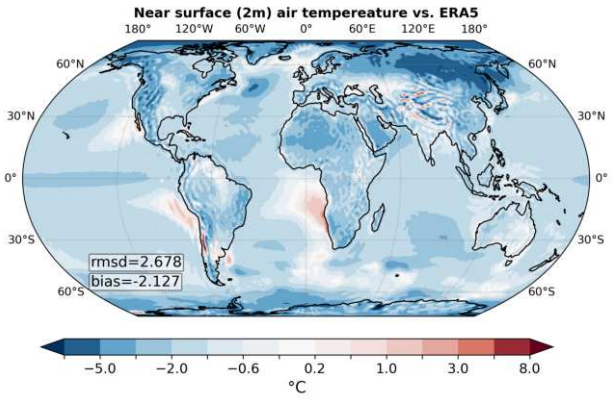
06H (2.08, −1.39)



06D HRlike (1.89, −1.15)



06O (2.39, −1.77)



06V (2.67, −2.12)

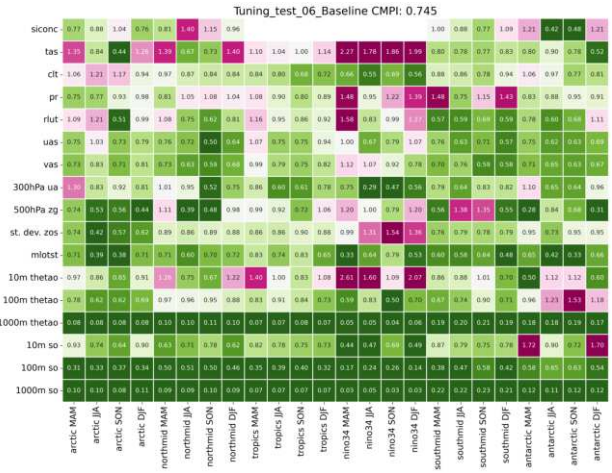
Figure 5: `part8` 2 m air-temperature bias (model − ERA5), EqualEarth, ±8 K scale.

4.3 CMPI (`part4`)

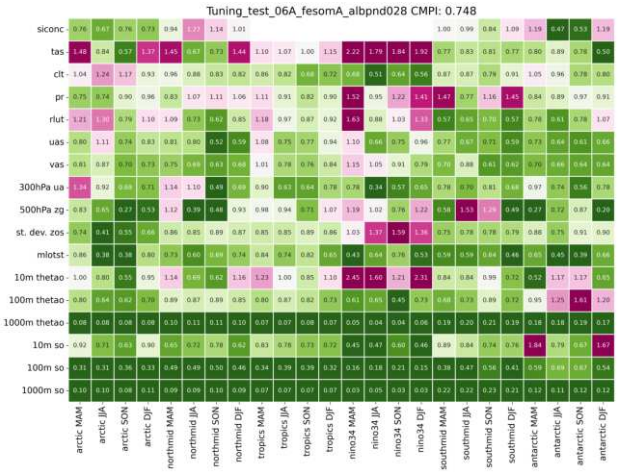
CMPI integrates 13 fields (siconc, tas, clt, pr, rlut, uas, vas, ua@300, zg@500, zos, mlotst, thetao, so) over regions/seasons, normalised against 30 CMIP6 models; < 1 **= better than the CMIP6 ensemble-mean error**. Run on a subset spanning the trade-off front (now complete for all 22 runs).

How to read this (important). CMPI here compares a *PI-control* run to *present-day* observations, so a moderate global cold offset — and hence some CMPI rise — is *expected* and is not by itself a disqualifier. The informative question is not "did CMPI go up?" but "*where* did the error change, and is that consistent with the parameter that was altered?" A sea-ice-albedo change should move `siconc` in the polar regions and little else; if it instead degrades the tropics, that is a red flag. The per-variable / per-region breakdown below is the tool for that judgement; the global number is a summary, not a verdict.

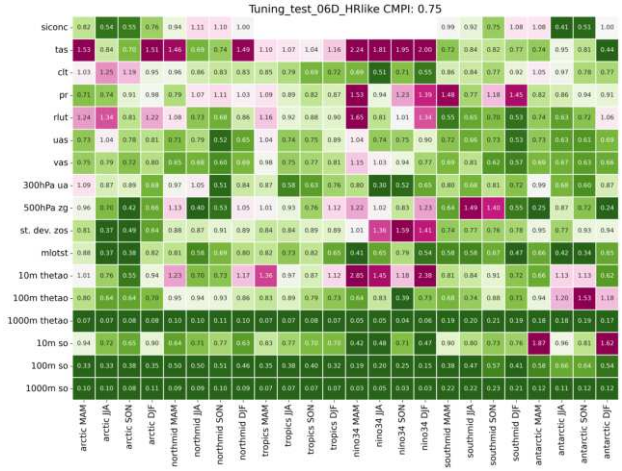
Run	CMPI	TOA imb.	2m RMSD
06_Baseline	0.745	1.531	1.717
06A_albpnd028	0.748	1.457	1.811
06D_HRlike	0.750	1.377	1.892
06H_combo_g_rvice018	0.757	1.220	2.082
06O_1hcpl_mospp	0.772	0.978	2.389
06T_1hcpl_mospp_kpplow	0.773	0.952	2.385
06V_.._entstpc3_1	0.803	0.655	2.671



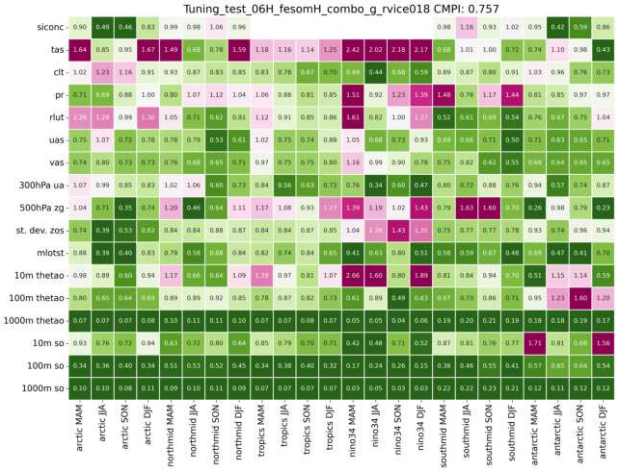
Baseline (CMPI 0.745)



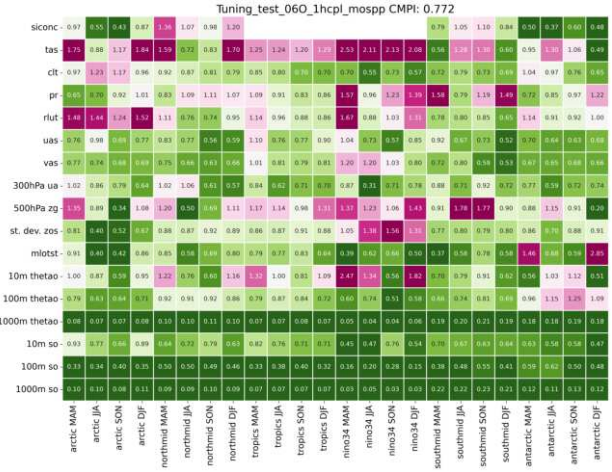
06A (CMPI 0.748)



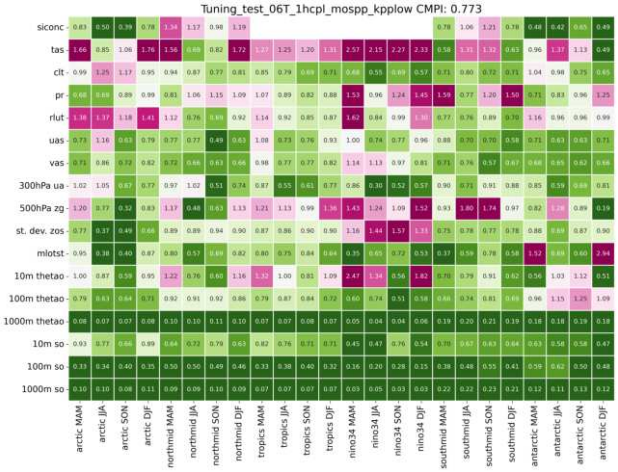
06D HRlike (CMPI 0.750)



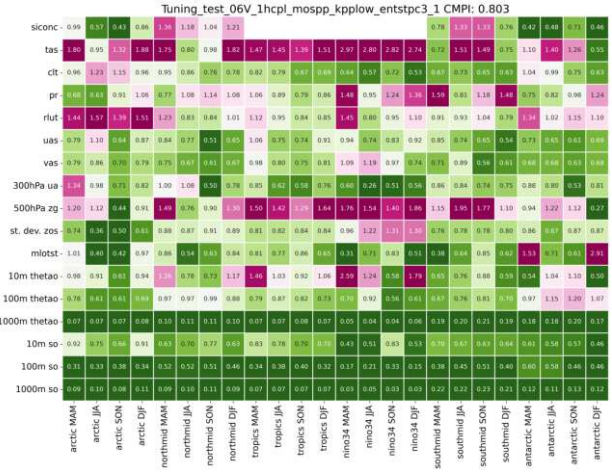
06H (CMPI 0.757)



06O (CMPI 0.772)



06T (CMPI 0.773)



06V (CMPI 0.803)

Figure 6: `part4` CMPI heat maps (per variable/region/season, normalised to CMIP6).

4.3.1 Where does the error change? (per-variable / per-region)

The maps below show each run's CMPI fractional error *minus baseline* (red = worse, blue = better). This is where the PI-vs-PD caveat earns its keep: the global CMPI rises monotonically ($0.745 \rightarrow 0.803$), but the *pattern* of change tells a much more favourable story for the targeted tunings.

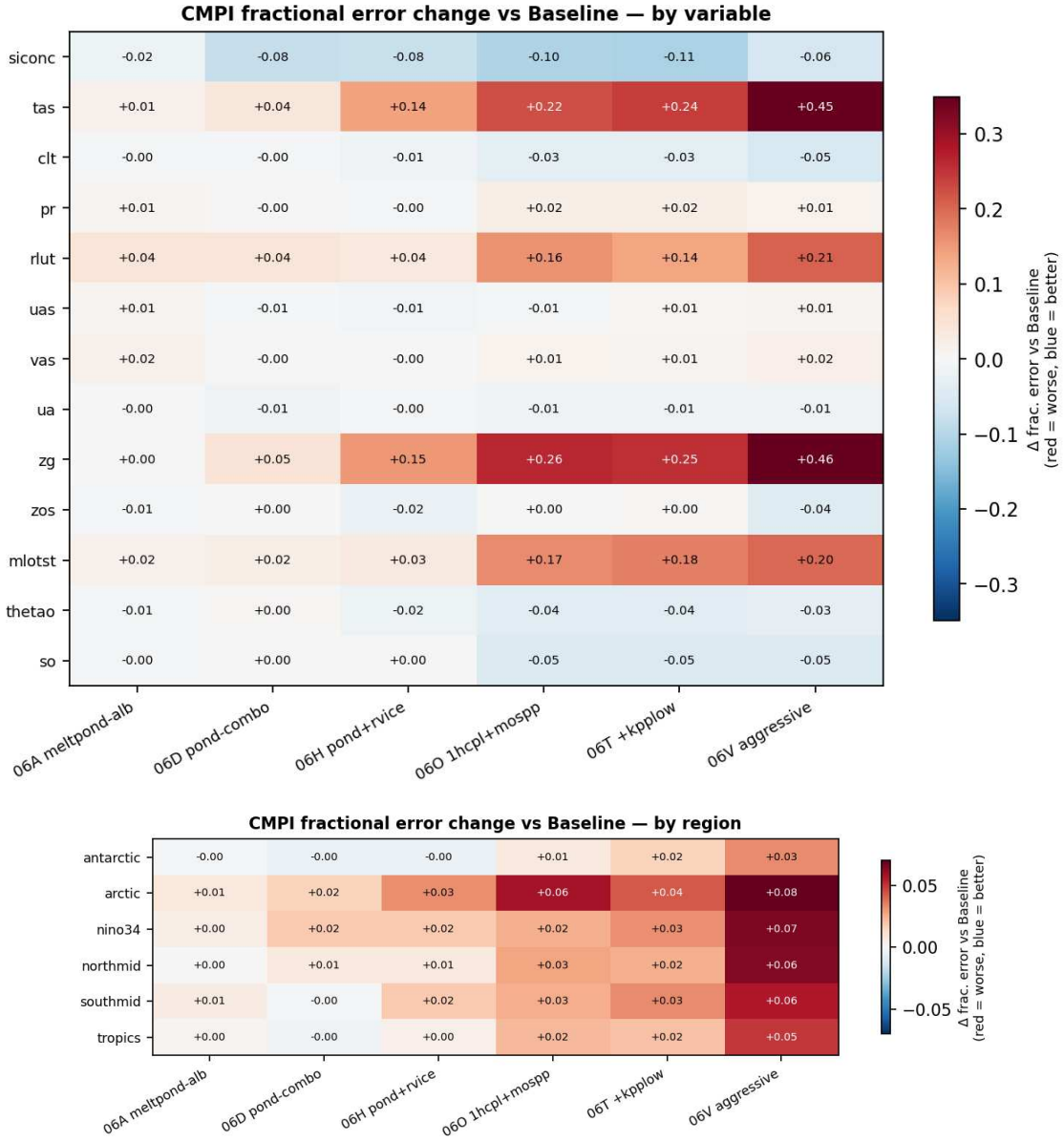


Figure 7: Change in CMPI fractional error vs Baseline, by variable (top) and region (bottom).

Reading the breakdown:

- **06A / 06D (melt-pond):** improve `siconc` ($-0.02/-0.08$) with essentially *zero* collateral elsewhere and a flat region map — a cleanly targeted change. These should be regarded as "no degradation where it matters," despite the tiny global CMPI uptick.
- **06O / 06T (mospp + kpplow):** genuinely *improve* the ocean interior (`thetao`, `so`: $\approx -0.04/-0.05$), `siconc` and `clt` — exactly where ocean-mixing should act. But they *degrade* `tas`/`zg`/`rlut` ($+0.16$ to $+0.26$) and even their own target `mlotst` ($+0.17$), and the region map reddens *everywhere incl. tropics and Niño-3.4* — i.e. the cooling leaks well beyond where an ocean-mixing parameter has any business acting. *That* is the warning sign, more than the global number.
- **06V (+entstpc3):** largest, most widespread atmospheric degradation (`tas` $+0.45$, `zg` $+0.46$); the deep-convection change destabilises the whole atmospheric column.

5 Southern-Ocean mechanism (adapted from a270234's notebooks)

To see *why* the ocean-mixing/coupling levers act as they do, two of a270234's Weddell-Sea diagnostics (`mld_sic_sit_compare`, `temp_fesom_core3_WWS`) were re-pointed at the coupled `Tuning_test_06` runs (same CORE3 mesh). Panels show the Baseline absolute field plus each run *minus* Baseline, austral winter (September), yr1370–79. (The `temp_fesom_core3_WWS` tripyview transect could not decode these runs' pre-1582 `gregorian` calendar, so an equivalent pyfesom2 depth–latitude section was used.)

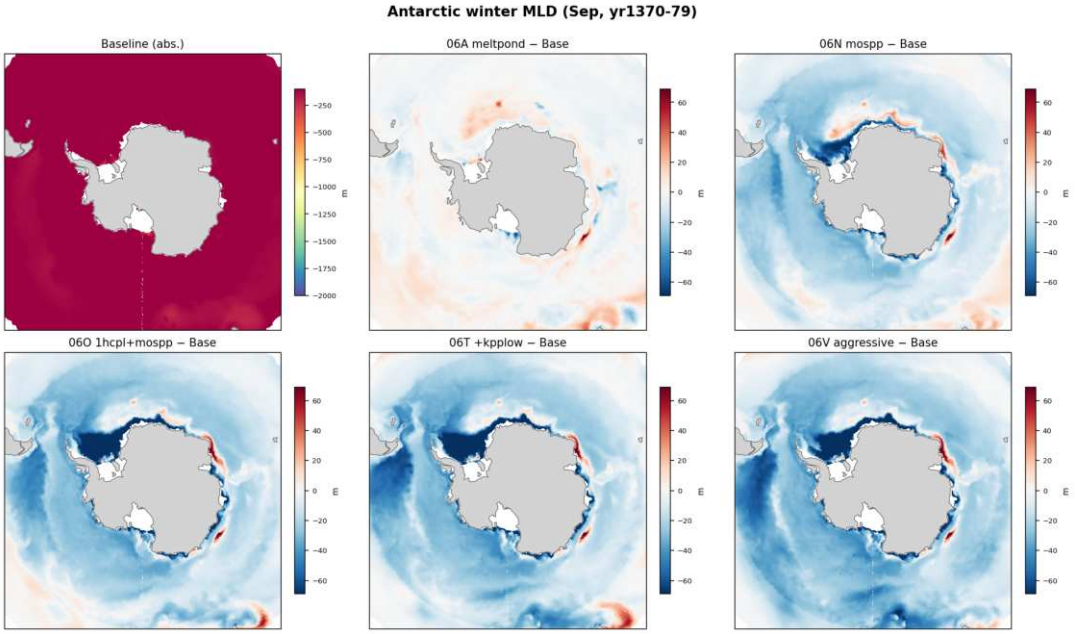


Figure 8: Antarctic September mixed-layer depth (MLD3): Baseline *absolute* (top-left, own scale), others = *anomaly* vs Baseline on a symmetric ±70 m diverging scale (98th-pctile auto-scale; the intense Weddell/Ross convection core saturates it). The mixing runs (06N/O/T/V) restructure the Weddell/Ross winter mixing relative to Baseline, whereas melt-pond (06A) barely changes it. Deep open-ocean winter mixing there is FESOM "open-ocean convection" that ventilates warm deep water — a known driver of the Southern-Ocean warm bias (Heuzé et al. 2013; Heuzé 2021) and of TOA imbalance.

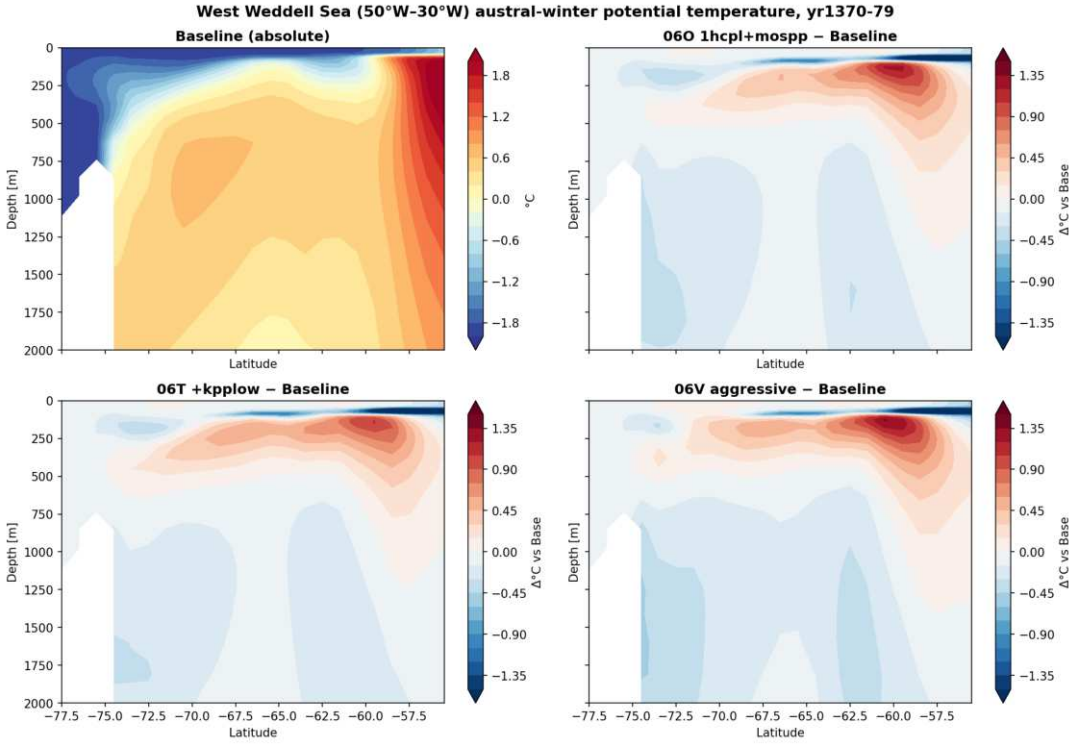


Figure 9: West Weddell Sea (50–30°W) September potential-temperature section: Baseline absolute and anomalies. The ventilated mid-column cools in the mixing runs (less convective entrainment of warm deep water to the surface), consistent with suppressed open-ocean convection; note that suppressing convection should leave the *deepest* water trapped/warmer rather than colder, and a near-surface edge warm anomaly is also present — i.e. the signal is a redistribution between layers, not uniform subsurface cooling.

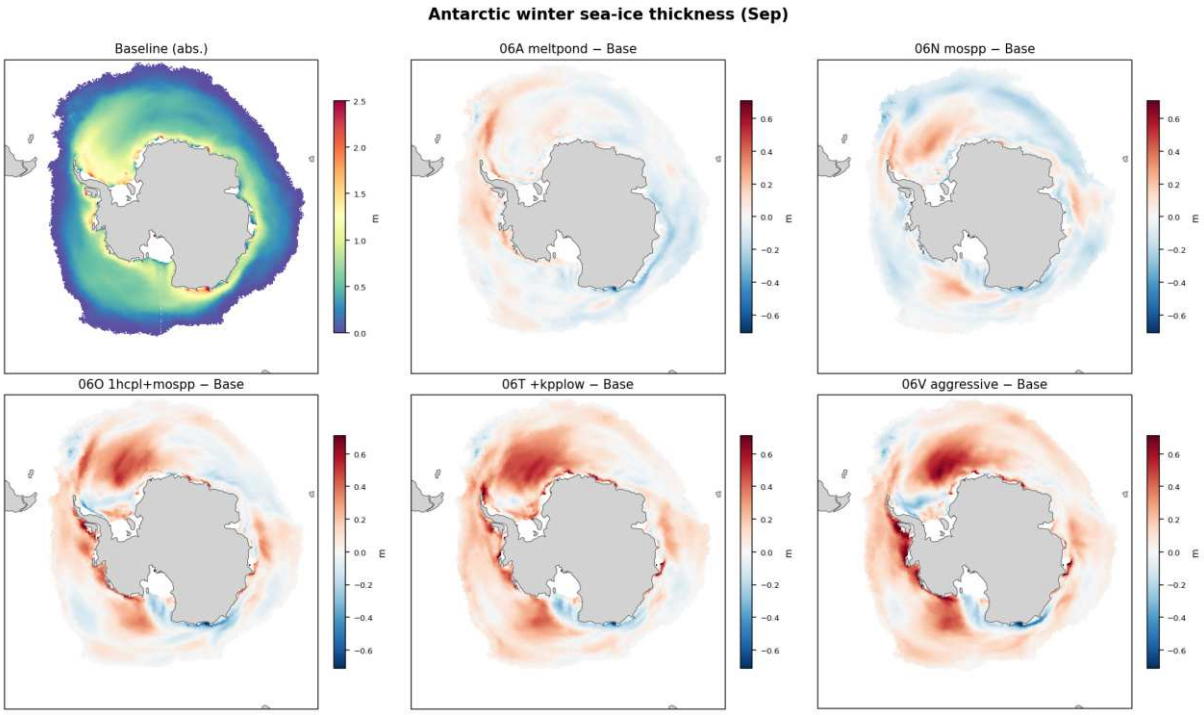


Figure 10: Antarctic September sea-ice *thickness* (Baseline absolute, top-left; others = anomaly vs Baseline). Thickness — not concentration — is the honest growth diagnostic here: winter concentration is already near 1 over most of the pack, so it *saturates* and cannot register added ice, whereas thickness is unbounded. The mixing runs (06O/06T/06V) thicken the pack broadly (positive/red anomaly over the Weddell and Ross sectors), consistent with suppressed open-ocean convection insulating and stratifying the surface; melt-pond (06A) barely changes it. (The concentration version, `sic_winter_antarctic.png`, shows only the ice-edge shifts for the reason above.)

Mechanism. The mospp / weaker-KPP changes reduce Weddell-Sea open-ocean winter convection, cooling the

subsurface and letting sea ice grow — improving `thetao/so/siconc` (consistent with the CMPI breakdown) and removing the Southern-Ocean surface warm bias. Spurious open-ocean deep convection in the Weddell/Ross gyres is the canonical CMIP Antarctic bias (Heuzé et al. 2013; Heuzé 2021, which evaluates AWI–CM in the multi-model set; Mohrmann et al. 2021 for the polynya surface expression), and open-ocean Weddell convection is a real-but-rare mode that models over-express (de Lavergne et al. 2014) — so damping it is physically the right direction. The *proximate radiative* closure is defensible: more winter ice → higher albedo → less absorbed SW → lower imbalance. We are more cautious about the "reduced ocean heat uptake" half of the chain: suppressing convection also *traps* heat at depth, which can *increase* deep ocean-heat-content drift, so the net uptake change is not established here — it should be pinned to an OHC-tendency diagnostic before being asserted. The cost is that the surface cooling propagates globally into the atmosphere (`tas/zg`), the broad cold bias seen in `part8`.

Counter-literature (a compensating-error risk this campaign must own). §5 attributes the SO warm bias and its removal almost entirely to *ocean* convection, but Hyder et al. (2018, *Nat. Commun.*) trace the leading 40–60°S warm-SST bias primarily to *atmospheric* net surface-heat-flux errors dominated by shortwave/cloud biases, and warn that a small *net* flux bias is often achieved by *compensating* component errors. The two coexist at different scales — broad atmospheric-SW warmth vs. localised Weddell/Ross deep-convection anomalies — and our mixing levers act only on the ocean branch. If part of the bias is atmospheric-SW in origin, the ocean-mixing lever is partly compensating for an atmospheric error, which is *exactly* the "cooling leaks beyond where the parameter should act" failure mode flagged for these runs in §4.3 (the region map reddens into the tropics/Niño-3.4). This tension is the report's own evidence pointing at Hyder; it strengthens, not weakens, the case for treating aggressive ocean-mixing tunings as provisional.

Part II

Counter-tuning the NH cold bias — the structural floor

The cold bias that makes Part I's aggressive lever expensive is largely ***not*** *the tuning: it is a structural LPJ-GUESS boreal-forest collapse that every run inherits. That makes it a* ***counter-lever*** *— fixing it warms the NH high latitudes at essentially unchanged imbalance, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. This part attributes the bias with a clean WITH-vs-WITHOUT-LPJG experiment, resolves its two mechanisms and a residual. Feedback-free spin-ups then show that the* source *of the low-tree state is strongly forcing-sensitive: the complete CRUNCEP–AMIP transfer matters greatly, whereas replacing CRUNCEP net radiation by CERES alone has only a small equilibrium effect.*

6 The NH high-latitude cold bias is a structural LPJ-GUESS problem

The runs carry a severe near-surface **cold bias over Siberia and northern Canada** (visible in every `part8` map). This chapter documents a working investigation into whether the coupled dynamic vegetation (LPJ-GUESS) is responsible — motivated by the idea that fixing it could be a *warming* lever to offset the imbalance-tuning cold bias, and by prior EC-Earth experience that LPJG drives such boreal biases.

6.1 Symptom: boreal needleleaf replaced by grass

Dominant-PFT maps (tool `part26`) for the HTESSEL spin-up (the reference, run with the same atmosphere but *without* atmosphere↔LPJG feedback) vs the coupled runs show boreal needleleaf giving way to C3 grass over Siberia / N. Canada. Tree-dominant fraction north of 50°N: **20.1 % (spin-up) → 16.8 % (baseline) → 11.4 % (06V)** — progressively worse with the more aggressive (colder) runs. *Caveat on the prior:* DGVMs disagree substantially on boreal tree-vs-grass partitioning, and real boreal forest is genuinely open with substantial non-tree ground cover (Jiang et al. 2012), so "grass over Siberia" is not *automatically* a bug. What makes the establishment-failure reading defensible here is the *magnitude*: the coupled state reaches cvh ≈ 0.02–0.05 with ~~no closed canopy anywhere (FORESTFPC= 0)~~[1] (§10), an outlier well below the ∼ 0.4–0.7 closed-taiga range (MODIS VCF), not merely a grass-leaning partition.

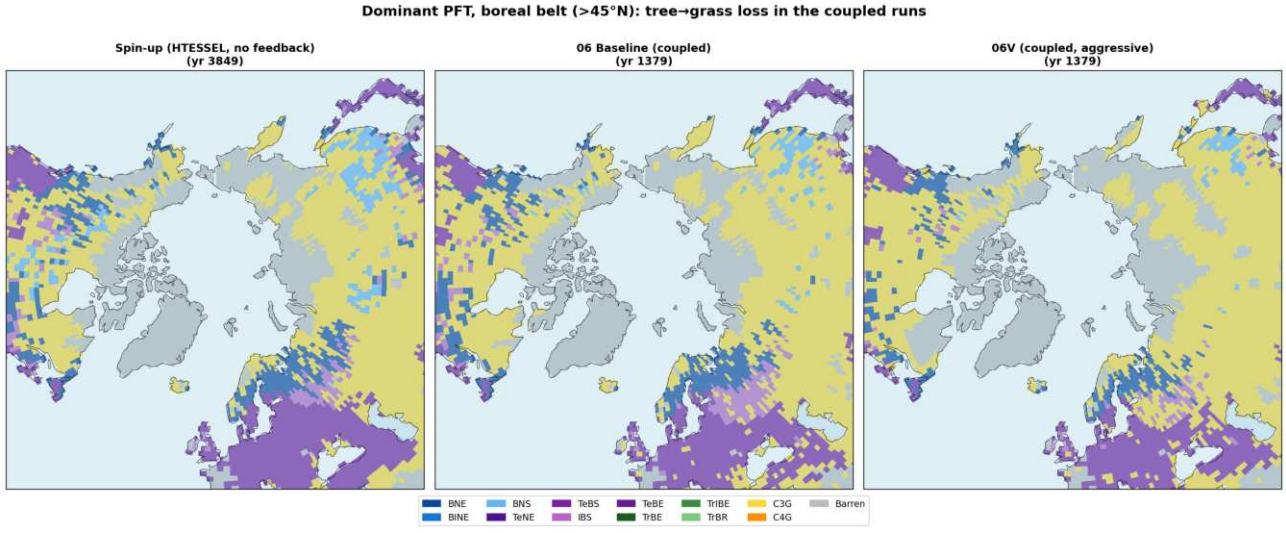


Figure 11: Dominant PFT, boreal belt: HTESSEL spin-up (reference) vs coupled baseline vs 06V. Needleleaf (blues) thins and C3 grass (yellow) expands across central/eastern Siberia in the coupled runs.

[1] **Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC`≡ 0 *by construction* while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

6.2 Is it grass growing, or trees dying? And is it a year-1 jump or a slow drift?

Belt-mean (55–75°N) needleleaf LAI is essentially *conserved* over the baseline run (0.40→0.37, trend ≈0), while C3 grass LAI *rises* (0.65→0.83, +0.10/dec): the signal is **grass growing**, with needleleaf loss only regional (E. Siberia) and offset by gains in the milder west. The coupled run is initialised directly from the spin-up's year-3850 LPJG state, and baseline year-1350 boreal LAI (tree 0.407, grass 0.649) is *identical* to the spin-up end-state (0.401 / 0.669) — so there is **no wholesale year-1 collapse**; the degraded, grass-heavy state is *inherited from the spin-up.* Within 1350–1379 the boreal 2 m temperature is at its (cold) level from year one and does *not* drift down (trend +0.42 K/dec, noisy) — consistent with a memory-less atmosphere that equilibrates to the surface state within a year rather than a slow vegetation runaway. (These are belt-mean *LAI* diagnostics; as §6.3 shows, the quantity that actually matters is not LAI-per-tile but the forest *cover fraction.*)

6.3 The decisive test: a clean WITH-vs-WITHOUT-LPJG pair (CORE3, 30 yr)

The earlier control was confounded by the ocean (core2 vs core3). We removed that confound: a purpose-built **no-LPJG AWI-CM3-develop reference** — *same* CORE3 ocean, *same* TCO95 atmosphere and matched component branches, HTESSEL + prescribed satellite vegetation, a full 30-yr PI-control (model years 1350–1379, `awicm3_noLPJG_CORE3_30y`) — differs from the LPJG Baseline in **LPJG alone**, so the 2 m-T difference is now a clean attribution. Its net TOA imbalance is +1.73 W/m^2 (`part2`), near the warm/high-imbalance end of Fig. 1 — confirming the LPJG signal is a near-surface *land* effect, orthogonal to the TOA tuning of §2. **Crucially, it beats the LPJG Baseline on *both* skill metrics: global 2 m-T RMSD vs ERA5 1.514 K (bias −0.76 K, `part8`) vs 1.717/**−0.94**, and CMPI 0.738 (`part4`) vs 0.745** — at higher (warmer) imbalance. Removing LPJG improves the global temperature fit *and* the multivariate skill score *while* the imbalance changes by only +0.2 W/m^2 (+1.73 vs +1.53) — within the inter-run drift of these unequilibrated 30-yr segments (§8), so the warming lever is real *and costs essentially no imbalance to within that uncertainty* (not, strictly, "zero"). Over the final decade, boreal DJF:

- **Forest cover collapses.** Siberian high-veg cover `cvh`: **0.63 (no LPJG) → 0.02 (LPJG)** (pan-boreal 0.49 → 0.07). LPJG passes essentially no forest to the atmosphere.
- **And it is genuinely colder — confound gone.** With the ocean held fixed, the LPJG run is unambiguously −3.1 **K** colder over Siberia in DJF (−30.4 vs −27.4°C).
- **Snow-masking (spring albedo)** operates as before: forest removal brightens the snow season (MAM **+0.077**). This is the canonical boreal-forest climate mechanism — a masked snowpack lowers albedo and warms high latitudes, and its removal cools most in spring (Bonan, Pollard & Thompson 1992; Betts 2000; Bonan 2008), with tree/canopy cover setting the snow-albedo-feedback magnitude that many models misrepresent (Loranty et al. 2014; Chapin et al. 2005) — so a model that loses its forest gets exactly this feedback wrong. But it is a *sunlight* effect and cannot explain the equally-cold polar night (§6.4).

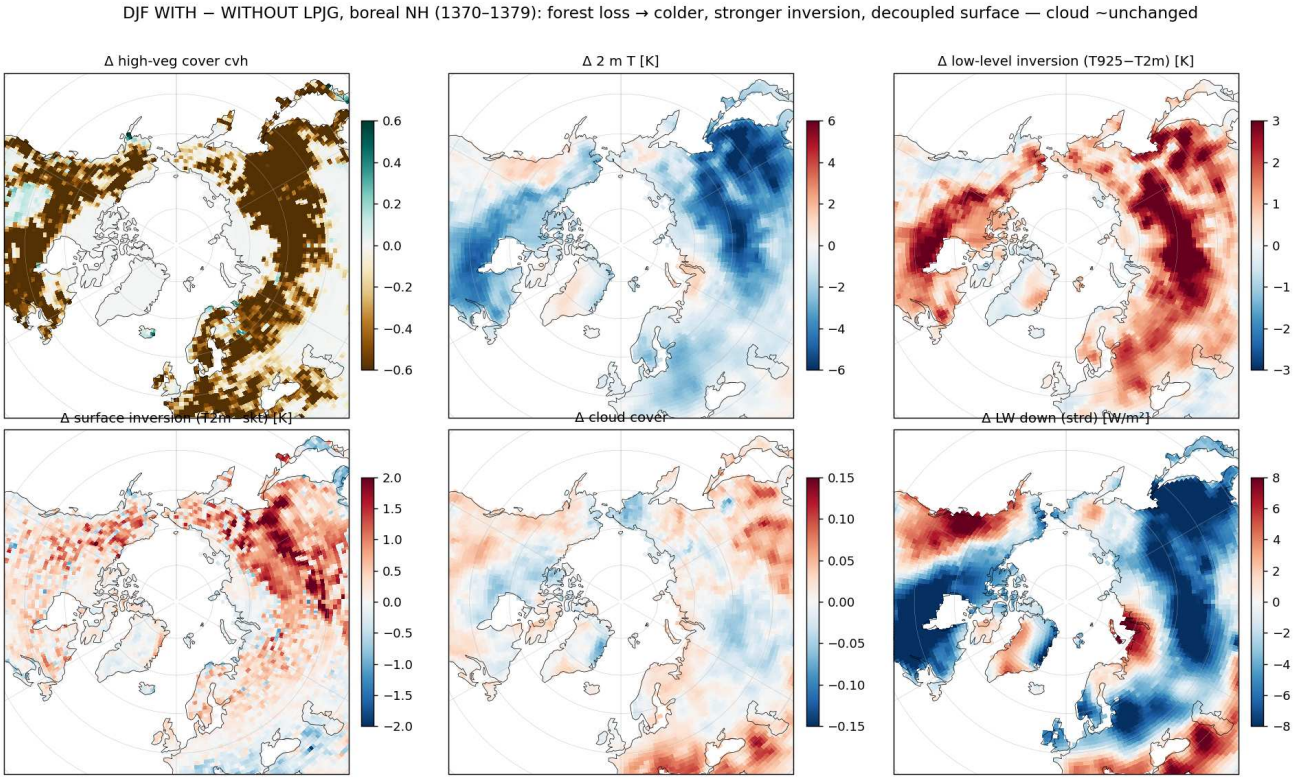


Figure 12: DJF WITH−WITHOUT LPJG, boreal NH (final decade, land-masked). Forest-cover collapse (`cvh`, top-left, brown) is spatially co-located with the cold anomaly (top-mid), a *stronger* low-level inversion ($T_{925}-T_{2m}$, top-right) and surface decoupling ($T_{2m}-T_{skt}$, bottom-left), at *unchanged* cloud (bottom-mid). LW-down (bottom-right) is modestly reduced over the cold interior.

6.4 Seasonality: which months carry the cold effect

Resolving the same clean CORE3 WITH−WITHOUT pair by *month* (boreal box 55–70°N, Siberia + N. America, final decade) shows the LPJG cold effect is **present in every month except high summer**. The WITH−WITHOUT 2 m-T difference is negative from September through June and only vanishes in July–August (−0.1/−0.2 K), peaking in the *shoulder/dark* seasons: **November −3.9 K, April −3.3 K**, with DJF −2.4 K and MAM −2.5 K (annual box-mean −2.0 K). **This is the key seasonal fingerprint**: a pure snow-albedo (sunlit) mechanism would confine the cold to the bright spring, yet the effect is *as large in near-sunless November/December* — which is exactly what motivates the winter roughness→inversion mechanism of §6.5. The near-zero July–August signal also confirms the growing-season atmosphere is unaffected (consistent with JJA gates being cleared, §6.3). Overlaid is the independent **OIFS–AMIP** run (§10.2: observed-SST, no-LPJG, PI 1870–79) — it tracks the *warm* WITHOUT curve (annual −7.6 vs −6.3 °C; DJF −25.6 vs −24.9; MAM even milder), confirming that a feedback-free, realistically-forced boreal climate is warm, and that the forcing we drive the decisive offline re-spin with (§10.2) is *not* itself cold-biased.

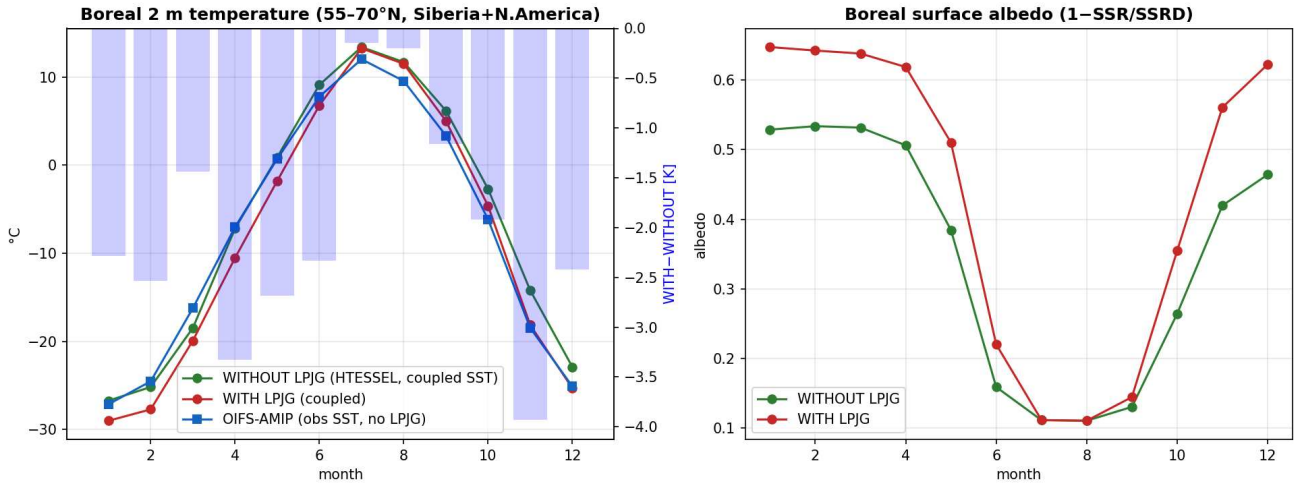


Figure 13: Boreal (55–70°N, Siberia+Canada) monthly climatology. *Left:* 2 m-T annual cycle — WITHOUT LPJG (green, clean CORE3 reference), WITH LPJG (red, coupled), and the OIFS–AMIP obs-SST no-LPJG forcing run (blue); blue bars = WITH−WITHOUT [K, right axis], negative (LPJG colder) in every month bar July–August, strongest in Nov/Apr. *Right:* surface albedo (1−SSR/SSRD); LPJG brightens the snow season (spring/autumn), the sunlit part of the bias. (AMIP emits no down-welling SW, so it appears on the temperature panel only. WITH/WITHOUT are model years 1370–79; AMIP is obs-SST 1870–79 — both ∼PI.)

6.5 A second, winter mechanism: roughness → inversion

Snow-albedo needs sunlight, yet the cold bias is as strong in DJF as in MAM. The missing forest acts a second way, in the dark. OpenIFS roughness length is cover-weighted by `cvh`, and needleleaf forest (z_{0m} = 2.0 m) is 20–60× rougher than the tundra/grass (0.034–0.10 m, `susveg_mod.F90`) that replaces it. A smoother surface couples less efficiently to the atmosphere; in polar night the stable boundary layer (SBL) decouples and the surface radiatively cools, deepening the near-surface inversion. That surface–atmosphere coupling strength in the SBL controls screen temperature and inversion depth is well established (Holtslag et al. 2013; Sandu et al. 2013); we assemble the specific vegetation-roughness → SBL-decoupling → screen-T chain from the land-surface and SBL literatures rather than a single published result, so it is presented as a mechanism the data then test, not a settled citation. **The data show exactly this fingerprint** (DJF Siberia, LPJG − no-LPJG): a **+2.1 K stronger** low-level inversion ($T_{925}-T_{2m}$), a **+0.7 K stronger** surface inversion ($T_{2m}-T_{skt}$), **unchanged cloud** ($\Delta \approx 0.00$) and -4.7 **W/m² less** down-welling longwave — a decoupled, radiatively-cooling stable surface, not a cloud/radiation artefact. Spatially over boreal land the forest-loss pattern *is* the inversion pattern: area-weighted $r(\Delta\texttt{cvh}, \Delta\text{inv}) = -0.60$, $r(\Delta\texttt{cvh}, \Delta\text{sfc-decouple}) = -0.68$, $r(\Delta T_{2m}, \Delta\text{inv}) = -0.83$, while the negative control $r(\Delta\texttt{cvh}, \Delta\text{cloud}) = -0.05$ is ≈ 0 — a real mechanism, not a shared trend. Regression lever: **restoring `cvh` by** $+0.4$ **warms DJF screen-T by** $\sim +1.2$ **K** through roughness alone and weakens the inversion by ~ 0.7 K — additive to the spring/albedo warming. This slope-based estimate inherits the scatter of the underlying correlation ($r = 0.63$, so $< 40\%$ of grid-cell variance explained); read it as $\sim +1.2 \pm 0.5$ K, an order-of-magnitude lever rather than a precise prediction. The *direction* and existence of the effect rest on the WITH−WITHOUT experimental design (§6.3), not on this regression.

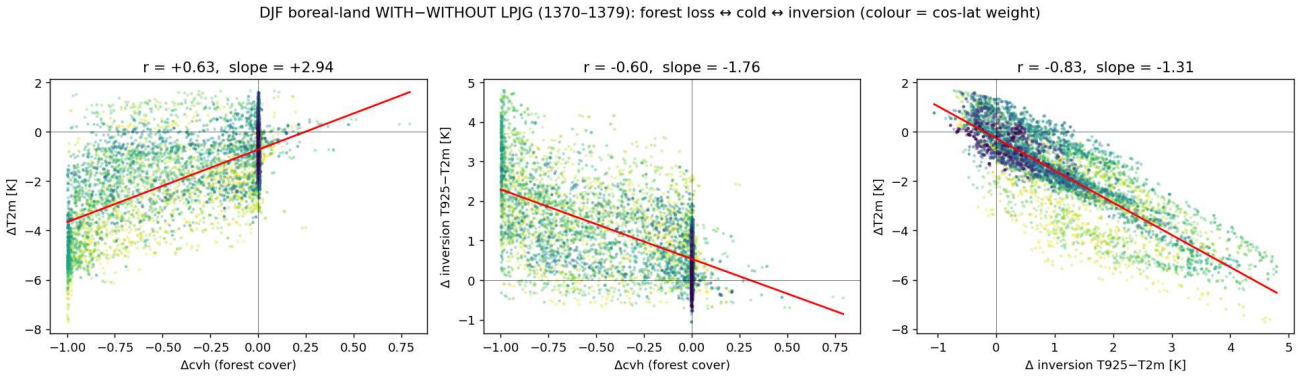


Figure 14: DJF boreal-land, WITH−WITHOUT LPJG (final decade). Forest loss (negative Δ`cvh`) tracks the cold (ΔT_{2m}, left), the stronger inversion (middle) and, transitively, cold ↔ inversion (right, $r = -0.83$). Colour = cos-lat weight.

6.6 How much does the LPJG fix buy? (vs ERA5)

Against ERA5 the current Baseline is $-4.5\,\mathbf{K}$ too cold over Siberia in DJF. The lever helps but does not close it: a realistic partial fix (+1 K) leaves −3.5 K, and *full* forest restoration (the no-LPJG run itself — the ceiling) reaches $-1.4\,\mathbf{K}$. So vegetation/roughness accounts for ∼3 K, roughly two-thirds of the Siberian DJF cold bias, and boreal-land RMSD falls $4.4 \rightarrow 3.3$ K — but a residual ∼1.5 K survives *even with perfect forest*.

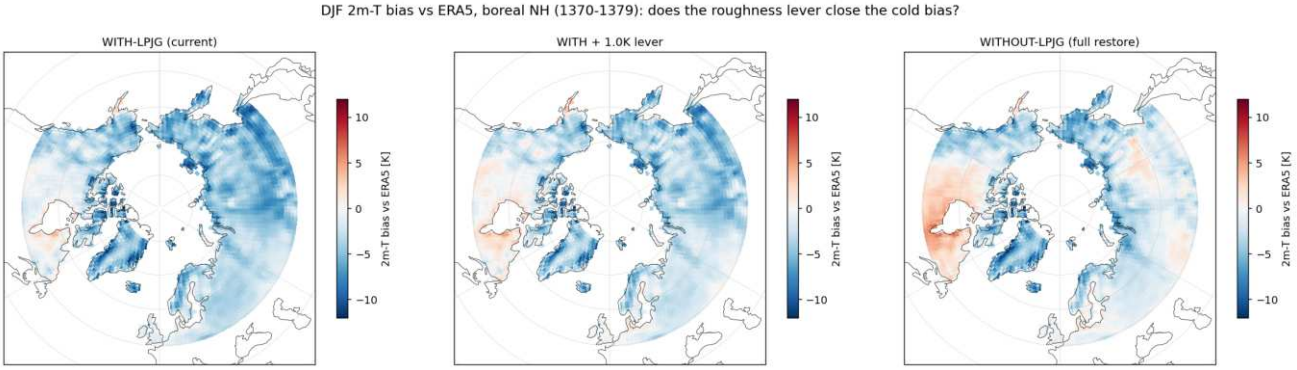


Figure 15: DJF 2 m-T bias vs ERA5, boreal NH (final decade): current LPJG Baseline (left, model output), an **illustrative** +1 K roughness lever (middle — a uniform +1 K offset applied to the Baseline, *not* a simulation), and full forest restoration = the no-LPJG run (right, model output). The lever panel only visualises the magnitude of a plausible partial fix; the load-bearing comparison is left vs right, which shows a residual cold survives even full forest restoration.

6.7 The residual driver: an over-strong stable-boundary-layer inversion

Diagnosing that residual in the *full-forest* run (vegetation thereby ruled out) against ERA5 (DJFM, boreal land) points to a single process. The model's low-level inversion is **+1.4 K too strong** ($T_{925}-T_{2m}$: model +5.6 vs ERA5 +4.2 K), and the excess is **entirely surface-trapped** — the 925–1000 hPa inversion bias is +0.07 K (≈ 0), so all the extra cooling sits in the lowest layer. Spatially this inversion-excess explains $r = +0.76$ of *where* the model is too cold; the Siberian High (MSLP bias +0.15 hPa, $r = 0.27$) and cloud-*fraction* (bias +0.07, $r = 0.00$) are ruled out. A too-strong wintertime polar inversion is a real, observationally-benchmarked model bias (Medeiros et al. 2011; Zhang et al. 2011), which supports reading the "+1.4 K too strong" residual as genuine. That surface-trapped signature, with circulation and cloud-fraction eliminated, is also the textbook fingerprint of a **stable boundary layer that under-mixes** — the IFS-family "too-stable" winter cold bias (Sandu et al. 2013). Its lever is the OpenIFS stable-BL turbulent mixing — the "long-tail" stability functions / minimum stable diffusion whose lineage runs Louis (1979) → Beljaars & Viterbo (1998) — *orthogonal* to the vegetation fix.

Two counters this diagnosis must own (both align with the report's own "compensating error" worry). *(a) The mixing fix is a documented compensating error:* enhanced long-tail stable mixing is stronger than LES/observations justify and is retained only because reducing it degrades the large-scale flow — it improves 2 m temperature but deepens cyclones and harms low cloud (Sandu et al. 2013; Holtslag et al. 2013), so Phase 1B must check the storm-track/low-cloud climatology (which CMPI/`part8` will not catch cleanly). *(b) The inversion may be a cloud-phase problem:* Pithan et al. (2014) attribute much of the modelled Arctic wintertime cold bias to missing mixed-phase low cloud (a LW, cloud-*phase* effect) — our null control ruled out cloud *fraction* but not phase, so we scope the residual claim to "given cloud fraction is unbiased" and flag a cloud-phase / LW↓ test as a prerequisite. Likewise the $r = +0.76$ residual-to-inversion match is suggestive, not proof that the "fourth lever" is truly orthogonal.

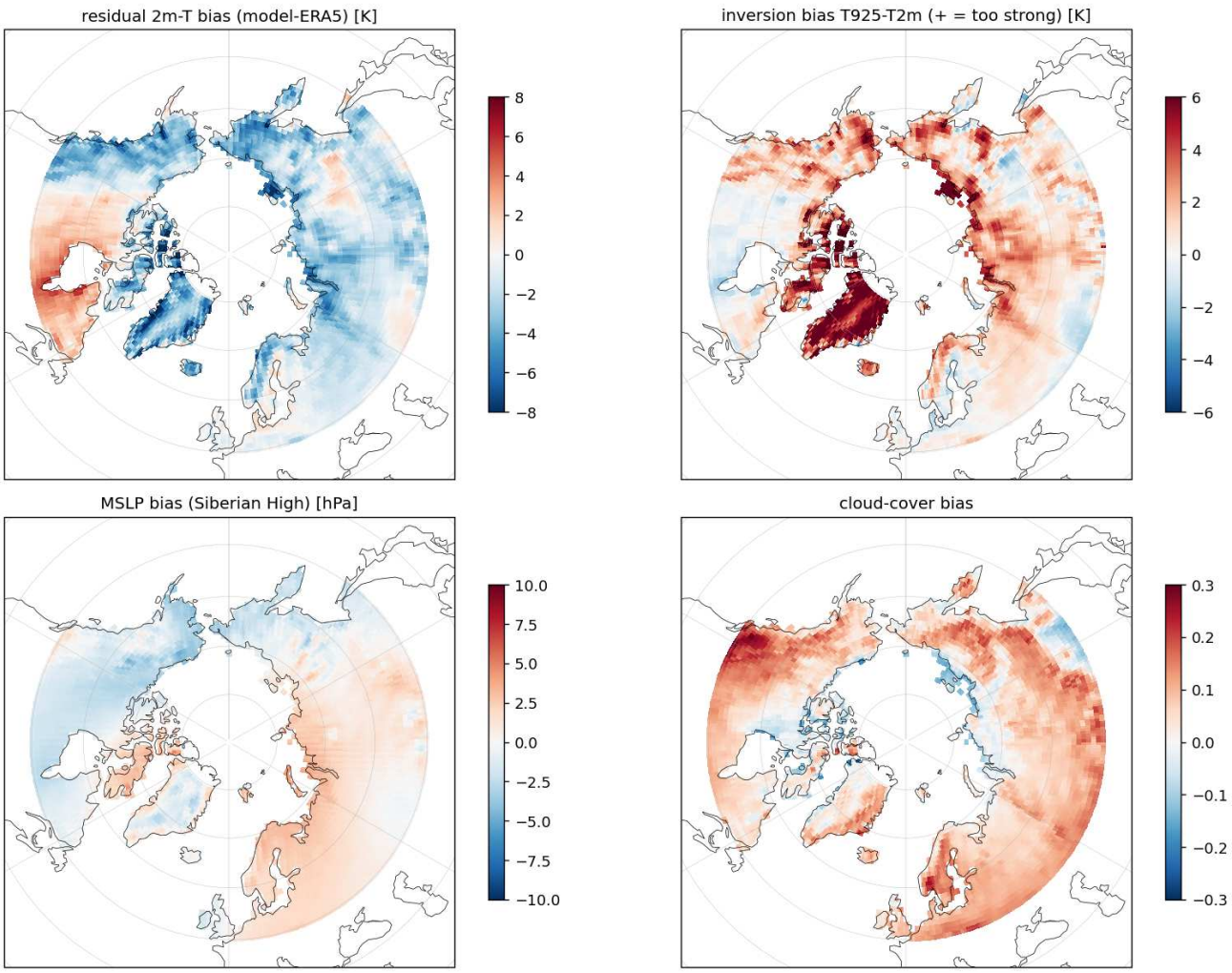


Figure 16: Second-driver search: full-forest (no-LPJG) run DJFM vs ERA5, boreal NH. Residual cold (top-left) matches the inversion-excess (top-right, $r = +0.76$); MSLP (bottom-left) and cloud (bottom-right) biases are diffuse and uncorrelated.

Near-complete attribution. The two mechanisms close the boreal DJF cold bias between them: of the $\approx -4.5\,\mathrm{K}$ Siberian bias vs ERA5, the LPJG forest-cover collapse (snow-albedo in spring, roughness in winter) contributes $\approx -3\,\mathrm{K}$, and the over-strong stable-BL inversion the remaining $\approx -1.4\,\mathrm{K}$. Two *orthogonal* levers — restoring a forcing-consistent forest state and enhanced stable-BL mixing — therefore target essentially the entire bias. (The veg cooling is also a *stable* offset, not a runaway: over 20 yr the LPJG run carries more/thicker Arctic ice as a bounded mean offset with no amplifying sea-ice drift.)

6.8 Working conclusion & lever

LPJ-GUESS sends essentially no boreal *forest cover* to OpenIFS (cvh $\sim 0.6 \rightarrow 0.02$ over Siberia, §6.3), so the canopy no longer masks snow (spring albedo, §6.3) *nor* roughens the surface (winter inversion, §6.5) — a physically robust driver of the Siberia/N. Canada cold bias, consistent with the EC-Earth family in which LPJ-GUESS is interactively coupled (Döscher et al. 2022) and in which dynamic vegetation-cover / effective-LAI improves high-latitude spring climate through the snow-albedo coupling (Alessandri et al. 2017). Source trace (verified): LPJG computes the high-veg fraction and exports it over OASIS as `GUE_FRAH`→`cvh` (`framework.cpp::computeDailyLAIandFPCforIF` `namcouple` line 120). With the AWI setting `iftreefracca=1` (`global.ins`:151), cvh is the summed tree *crown area*×density (no 0.05 clamp — that constant is dead code; `natural_frac`= 1 over PI land), so cvh≈0.05 means boreal tree cover is genuinely ∼5%. The establishment gates (`gdd5min_est=500`, `twmin_est=5`) are *not* the blocker — they are cleared across the boreal — so the degraded state is **inherited from the LPJG spin-up**, not a coupled runaway. It was initially read as a tree-vs-grass competition failure, but the controlled forcing-transfer test (§14.3) reproduces most of the boreal loss without changing LPJG parameters; competition must therefore be diagnosed only after a forcing-consistent re-spin. Seasonally (§6.4) the LPJG cold effect spans September–June

(DJF −2.4, MAM −2.5 K over the boreal box) and vanishes only in July–August — i.e. it is a cold-season, not a growing-season, effect. The co-equal *sunless* late-autumn/winter part — which pure snow-albedo cannot explain — is now resolved (§6.3–6.7): it is the roughness→inversion effect of the lost forest (§6.5) plus a model-intrinsic over-strong stable-BL inversion (§6.7).

Fixes: (1) correct the atmospheric forcing, regenerate the daily LPJG forcing and re-do the spin-up with one verified binary; only the residual against that forcing may justify tuning `greff_min`, C3-grass competitiveness or `pstemp_low` (§14.3, §16); (2) if a structural residual remains, revisit `iftreefracca`/crown allometry; (3) a woody stem-area floor on the snow-masking fraction in OpenIFS (`surfbc_ctl_mod.F90`:389–391 / `surfrad_ctl_mod.F90`:650–655) — but this only bites once (1) raises cvh, since the masked-snow tile is weighted by cvh itself. Restoring forest cover is the physically-grounded *warming lever* of §9 and buys ∼3 K (§6.6), but its cause must not be prejudged as a parameter error. **A fourth, orthogonal lever** attacks the residual ∼1.4 K (§6.7): *enhanced stable-boundary-layer mixing* in OpenIFS (VDF stability functions / minimum stable diffusion), which would relax the over-strong surface inversion. Together (1)–(4) target essentially the entire boreal DJF cold bias.

6.9 Feedback-free AMIP spin-up — substantially qualified by §14.3 and §7

Read this section with the later controlled tests. The argument below turns on a standalone LPJG re-spin driven by the OIFS–AMIP forcing under-vegetating despite the establishment gates being cleared, which was read as an LPJG-internal competition failure. A later controlled experiment (§14.3) shows that *moving between the CRUNCEP and AMIP forcings alone*, with byte-identical parameters, costs −54 % of Siberian and −66 % of East-Siberian `TREEFPC`. The under-vegetation seen here is therefore largely what the forcing change predicts on its own, and cannot by itself demonstrate a competition failure. The four-arm radiation test (§7) further shows that replacing only CRUNCEP `rsns`/`rlns` by CERES removes only a small part of the NH contrast: the strong full-forcing effect is not a net-radiation effect alone. `FORESTFPC` is in fact a *void diagnostic* here: it counts only the managed-forest land-use class, which is empty in this PI configuration, so it is identically zero whatever the vegetation does (§14.3). There is no separate closed-canopy failure.

Is the boreal forest deficit a *climate* problem (fix the atmosphere) or an *LPJG-internal* tree-vs-grass competition problem (fix the vegetation)? We answered it directly with a **feedback-free** experiment: a clean OpenIFS–AMIP run (prescribed observed SST, no coupled ocean/LPJG cold feedback; ∼PI 1870–79) generated a realistic daily forcing, which drove a 2000-yr standalone LPJG re-spin on a *dense* 11 538-cell gridlist (build detail in §10.2). Fig. 17 puts that forcing (top) beside the equilibrium vegetation it produced (bottom). **The forcing is *not* growing-season-cold**: over the Siberian box the warmest-month T2m is 14.7 °C (100% of cells clear the `twmin_est`= +5 gate) and GDD5 is 730 (76% clear 500) — establishment is climatically permitted essentially everywhere trees should be. **Yet the spin-up under-vegetates and grass wins**: cvh= 0.21 (far below the 0.4–0.7 closed-taiga range), `GRASSFPC`= 0.37 > `TREEFPC` = 0.21, and ~~no closed canopy forms anywhere (FORESTFPC= 0)~~[2], with grass filling exactly the East-Siberian interior. On a dense gridlist this cannot be dismissed as under-sampling. **Verdict: clearing the establishment gates does not prove that competition is the dominant error.** The controlled transfer test shows that the complete forcing change reproduces most of the regional tree loss (§14.3), while the new four-arm experiment shows that replacing net radiation alone explains only a small part of the NH contrast (§7). Competition remains a possible *residual* target, but it must be measured after a forcing-consistent re-spin rather than assumed from this AMIP arm. One climate co-limit visible here is the deep-winter cold (−28.3 °C box mean, < −40 in NE Siberia) that gates out evergreens on ∼ 40% of cells — but climatically-permitted *larch* (no cold gate) fails there too (BNS cover 0.035), so the winter warming of Phase 1B is only a *secondary* aid.

[2] **Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC`≡ 0 *by construction* while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

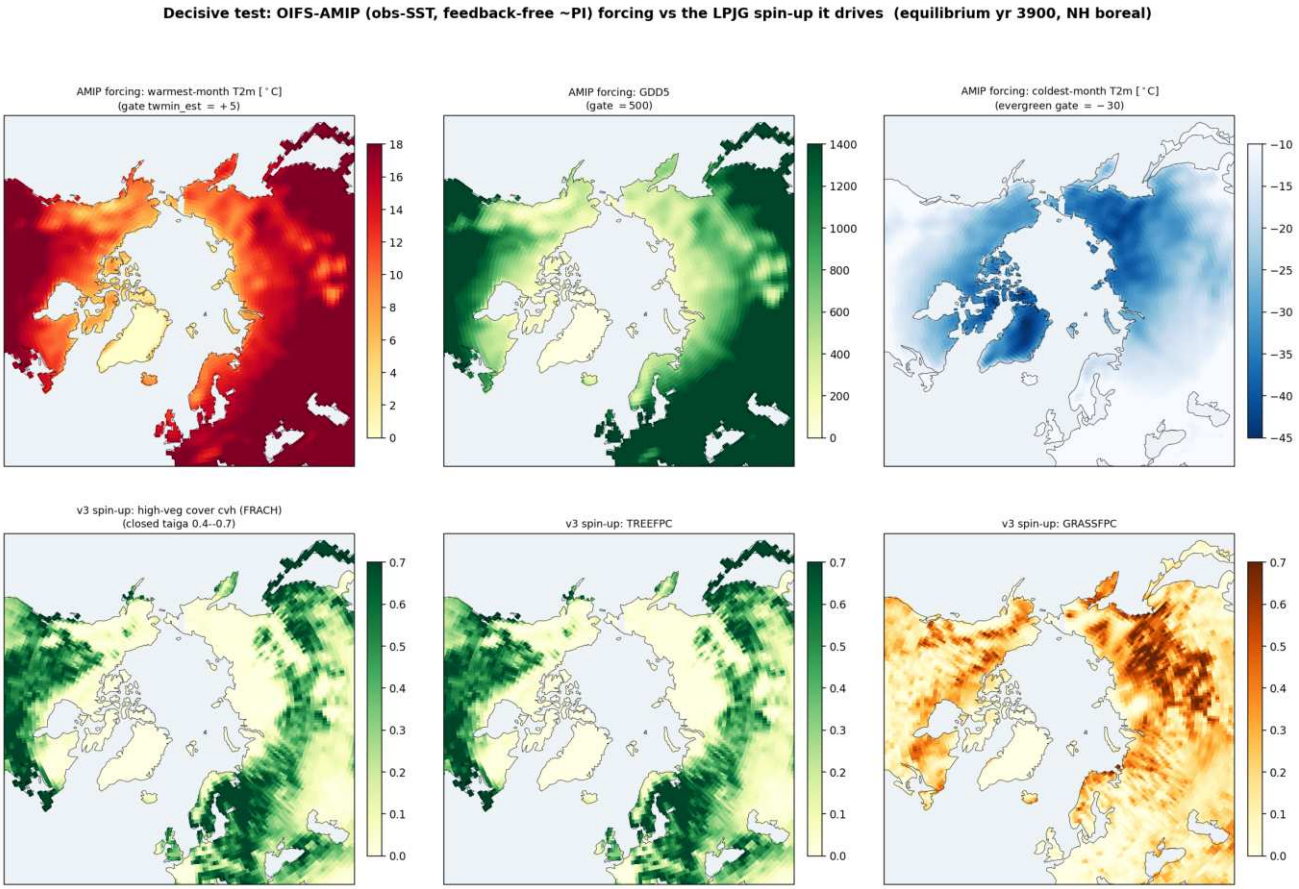


Figure 17: Decisive climate-vs-competition test (NH boreal, equilibrium yr 3900). *Top:* the feedback-free obs-SST ∼PI OIFS–AMIP forcing that drove the standalone LPJG re-spin — warmest-month T2m (vs `twmin_est`= +5), annual GDD5 (vs 500), coldest-month T2m (vs evergreen `tcmin_est`= −30). *Bottom:* the equilibrium vegetation it produced — cvh, `TREEFPC`, `GRASSFPC`. The growing-season gates are largely cleared, yet the AMIP-forced equilibrium has cvh≈ 0.2 and extensive East-Siberian grass. This panel diagnoses the AMIP-arm state; the controlled transfer test (§14.3) shows that it cannot by itself attribute the state to LPJG competition.

7 Forcing sensitivity: CRUNCEP radiation constrained by CERES EBAF

Question and controlled design. The full forcing-transfer test in Part IV shows that moving from CRUNCEP to AMIP strongly reduces equilibrium boreal tree cover, but it does not isolate which forcing fields cause that response. We therefore completed four 2000-yr standalone LPJ-GUESS spin-ups on TCO95 and compared their final year (model year 3900). The two additional arms isolate the narrower effect of replacing only CRUNCEP surface net shortwave (`rsns`) and net longwave (`rlns`) by CERES-constrained radiation:

experiment	daily forcing cycle	radiation treatment
AMIP	OpenIFS–AMIP, 1870–1879	corrected native OpenIFS `rsns`/`rlns`
CRUNCEP v3	calibrated CRUNCEP v7, 1901–1910	calibrated CRUNCEP net radiation
CRUNCEP + CERES direct	CRUNCEP v3 non-radiation fields	CERES EBAF Ed4.1 monthly 2001–2010, bilinear to TCO95 and linearly interpolated between month midpoints
CRUNCEP + CERES daily variability	CRUNCEP v3 non-radiation fields	CERES official 12-month climatological means, retaining CRUNCEP relative daily SW and additive daily LW anomalies

In both CERES experiments `tas`, `tasmin`, `tasmax`, precipitation, humidity and wind are byte-for-byte unchanged from CRUNCEP v3. Runtime files carry the generic name `AM04_atm_cmip6_1d_1990-1999_fast_lpjgforcing.nc`; checksums and full variable comparisons verified that they are exact runtime copies of the intended direct and daily-variability source files.

CERES provenance and what the comparison means. The observational reference is NASA Langley CERES EBAF Edition 4.1 on its native 1° grid, specifically `sfc_net_sw_all_clim` from `CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. It is the official July 2005–June 2015 climatology (product DOI). Figure 18 puts all forcings on a common 1° land grid and shows JJA plus the Siberian seasonal cycle. AMIP–CERES and CRUNCEP–CERES are *forcing differences across products and epochs*, not strict pre-industrial biases. The small differences for the two CERES-derived forcings are *construction checks*, not independent validations: the direct case uses 2001–2010 monthly CERES data, whereas the daily-variability case is explicitly constrained to the same climatology used as the reference.

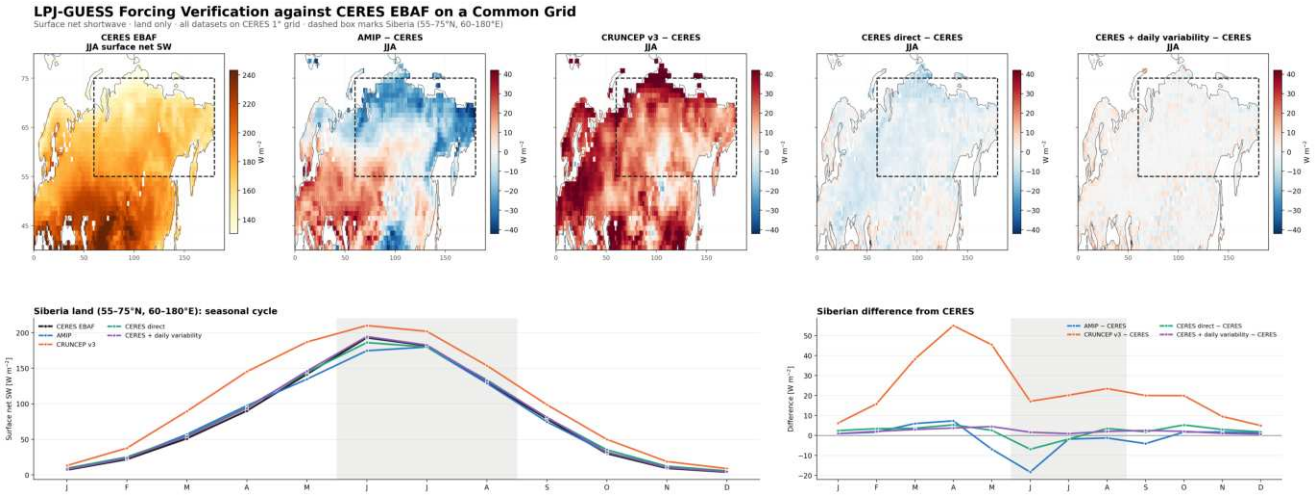


Figure 18: Surface net-shortwave forcing check against CERES EBAF Ed4.1. Top: CERES JJA net SW and each forcing minus CERES after common-grid remapping over land. Bottom: Siberian land seasonal cycles and their differences from CERES (55–75°N, 60–180°E). Near-zero CERES-derived panels confirm the forcing construction; they are not an independent observational test.

Equilibrium vegetation response. In the annual `treeFrac` diagnostic, the NH-land display-grid mean is 0.186 under AMIP, 0.269 under CRUNCEP v3, 0.262 under direct CERES radiation and 0.263 when CRUNCEP daily variability is retained. Thus the full CRUNCEP–AMIP difference is +0.083, whereas the two CERES substitutions remain +0.076 above AMIP. Replacing CRUNCEP radiation reduces its mean tree fraction by only 0.006–0.007 and leaves about 92% of the CRUNCEP–AMIP difference. The almost identical direct and daily-variability results show that this choice of monthly-to-daily radiation treatment has negligible influence on the hemispheric mean at equilibrium, although regional differences remain visible.

Interpretation. Net radiation alone is not the dominant cause of the AMIP–CRUNCEP equilibrium tree-cover difference: changing only `rsns`/`rlns` removes roughly 8% of this hemispheric `treeFrac` contrast. Together with the regional `TREEFPC` transfer result of §14.3, this localises most of the full-forcing penalty to other meteorology (especially the documented temperature/GDD5 change), field covariability and nonlinear LPJG response. It does *not* establish a residual competition error. The appropriate conclusion is: *the vegetation equilibrium is strongly forcing-sensitive, but within the CRUNCEP family a CERES net-radiation correction alone is insufficient; competition parameters should be considered only after a forcing-consistent same-binary re-spin.*

Part III

The path forward: raise the floor, then tune

Parts I–II give the strategy: the imbalance can only be cut by cooling, and cooling is only affordable once the structural NH cold floor is raised. So first fix the floor (vegetation competition + stable-BL mixing), ***then*** *spend a moderate cooling lever to reach surface balance — now without the RMSE blow-up. First the caveats that condition this, then the recommendation and a phased plan.*

8 Caveats

1. **06D "HRlike" has no HR-specific namelist change** — only the common pond shift (identical to 06C); its metrics confirm this. If a resolution/timestep/viscosity change was intended, it is not in the namelists.

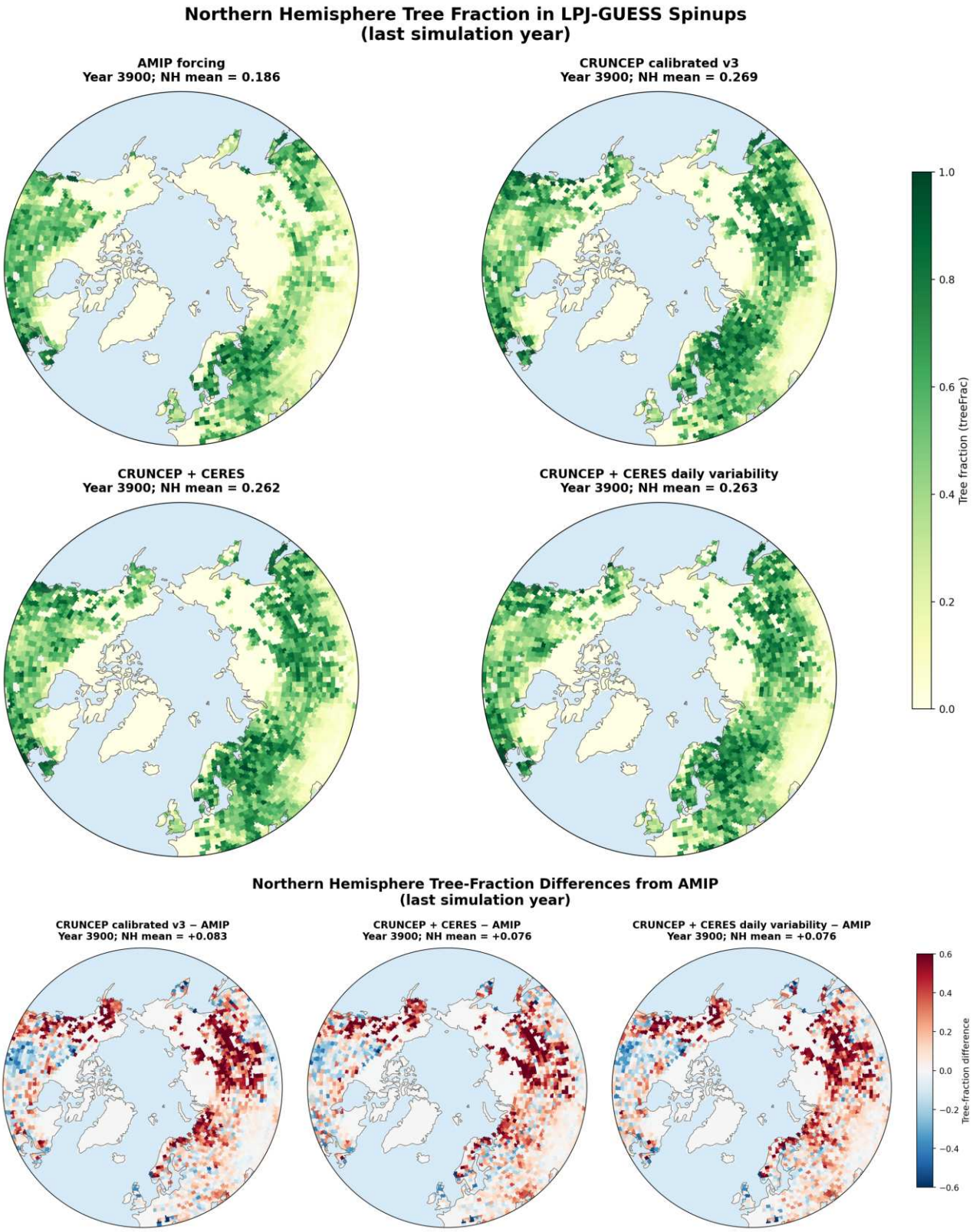


Figure 19: Final-year tree fraction in the four 2000-yr offline spin-ups. Top: absolute year-3900 tree fraction. Bottom: CRUNCEP-family differences from AMIP. CERES-constrained radiation modestly reduces CRUNCEP tree cover, but does not remove the much larger AMIP–CRUNCEP contrast.

2. **Filename convention.** These runs write `atm_remapped_1m_<var>_YYYY-YYYY.nc` (no `_1m_` infix). I added backward-compatible fallbacks to `part2`, `part8` and the CMPI preprocessing; existing configs are unaffected.
3. **Accumulation period** for these XIOS runs is **3600 s** (not the 21600 s of the old HR config); configs use `accumulation_period=3600` (verified: tsr $\rightarrow \approx 240\,\mathrm{W/m^2}$).
4. None of the runs is at equilibrium within 30 yr — numbers are a ranking, not final values.
5. **CERES is a modern, monthly constraint**, not a pre-industrial daily observation (§7). The CERES-derived panels test that the forcing construction worked; only the AMIP–CERES and CRUNCEP–CERES panels compare independent products, and even those span different epochs.

9 Recommendation & synthesis

Reading global CMPI/RMSD as a guide rather than a gate (the PI-vs-PD offset is expected), the picture is:

- The **melt-pond / albedo family (06A–06D, etc.)** are *clean*: they improve sea ice where they should and leave everything else essentially untouched. Their imbalance gain is modest (5–20%) but it comes with no unexpected collateral — safe building blocks.
- The **ocean-mixing / coupling family (06O/T)** delivers the only large imbalance reductions (30–38%) *and* genuinely improves the ocean interior and sea ice — but the cooling leaks into the atmosphere globally (incl. tropics), which is degradation in places the parameter should not reach. That makes them promising but not yet "free."
- **06V** pushes too far: the `ENTSTPC3` convection change degrades `tas`/`zg` everywhere — not recommended as-is.

Best-balanced single steps: **06P** (−29%, well-localised) and **06H** (−20%).

Synthesis — and where the warming lever comes from. The two arcs join here. The imbalance can only be cut substantially by cooling levers (ocean mixing/coupling), so every combo tried so far overshoots into cold bias. But §6 shows the shared Siberia/N. Canada cold lobe is *not* a tuning artefact, and a purpose-built clean no-LPJG CORE3 reference (§6.3) now *quantifies* it: of the $\approx -4.5\,\mathrm{K}$ Siberian DJF bias vs ERA5, $\approx -3\,\mathrm{K}$ is the LPJG forest-cover collapse (via snow-albedo in spring *and* surface roughness in winter, §6.5) and $\approx -1.4\,\mathrm{K}$ is a model-intrinsic over-strong stable-BL inversion (§6.7). The clean run is *better* than the Baseline globally (RMSD 1.514 vs 1.717 K) at *near-unchanged* imbalance ($+0.2\,\mathrm{W/m^2}$, within drift) — indicating both are *warming levers largely orthogonal to* the TOA tuning. That reframes the path forward:

1. First correct the atmospheric forcing and re-spin LPJG with a verified common binary (§14.3, §16). The CERES experiment (§7) shows that net-radiation correction alone is insufficient; tune tree–grass competition only if a residual remains under the corrected forcing. Do not relax the already-adequate summer establishment gates.
2. Attack the residual with *enhanced stable-BL mixing* in OpenIFS (§6.7) — an independent knob that relaxes the surface-trapped winter inversion.
3. On that warmer baseline, apply a *moderate* mixing/coupling setting (06O-like, not 06V) for the imbalance, plus the clean melt-pond changes — the cooling it adds is then offset rather than additive.
4. Re-judge on the *pattern* of CMPI change (§6.3 method), not the global value, since the PI-vs-PD offset is expected.

In short: don't chase the imbalance with ever-colder atmospheres; remove the structural cold bias first, then the moderate cooling needed for the imbalance becomes affordable.

9.1 2026–07–24 coupled follow-up from the CRUNCEP3-initialized branch: the year-50 land–ocean split refines the path

After the earlier report structure was drafted, the coupled CRUNCEP3-initialized branch (alias "080a") reached 50 years, which adds an important constraint on the tuning strategy. The new result is *not* a simple hemisphere-wide cold drift. By year 50 the system shows a marked **warm-ocean / cold-land split**, while boreal tree cover continues to retreat — most strongly over Siberia / East Siberia.

Two points matter:

1. The older LPJG story remains necessary: the spin-up / competition state is still part of the problem, and the vegetation levers are real.
2. But the 50-year coupled adjustment shows that a *regional continental-land* climate failure also develops, so vegetation-only retuning is not sufficient.

The finished year-50 diagnostics make the asymmetry explicit. Relative to year 0, NH45+ `TREEFPC` declines from 0.336 to 0.215, but the strongest loss is concentrated over Siberia / East Siberia. In the Siberian box (55–75°N, 60–180°E) `TREEFPC` drops by −0.188, and in East Siberia (55–75°N, 90–160°E) by −0.232, accompanied by a strong drop in `AGDD5` (−89 and −133 respectively). At the same time, NH45+ aggregate `AGDD5` actually *increases*, and Scandinavia / Canada also warm in thermal-growing metrics. So the problem is not "the NH is too cold"; it is that **boreal continental land, especially East Siberia, remains too cold relative to the oceanic state**. This is exactly what one sees in the year-50 seasonal-temperature maps against the CRUNCEP3 reference: ocean-adjacent sectors are warm-biased while large boreal land sectors remain cold.

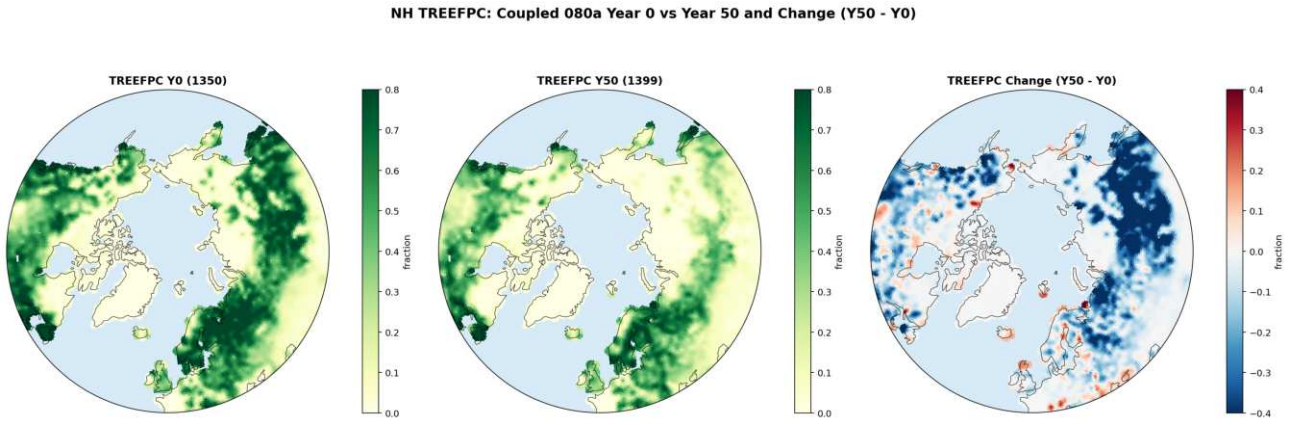
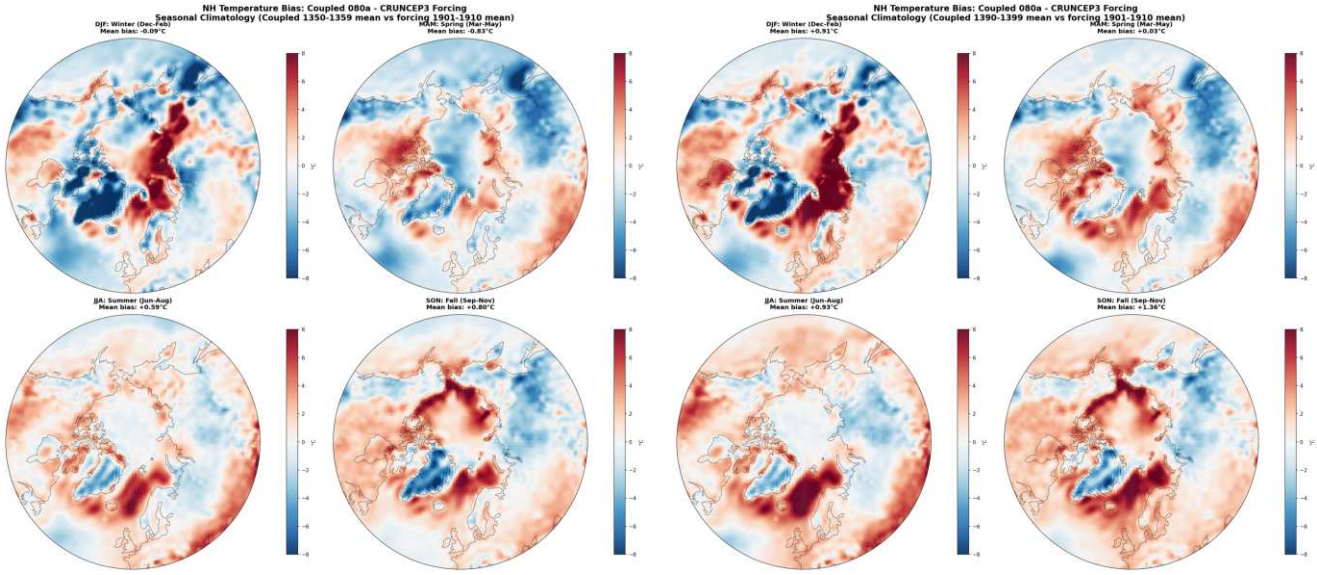


Figure 20: Coupled CRUNCEP3-initialized branch year-50 follow-up: NH `TREEFPC` at year 0 (1350), final year (1399) and change (Y50−Y0). The retreat is broad, but the largest loss is concentrated over Siberia / East Siberia.



(a) Early coupled decade: 1350–1359 vs CRUNCEP3 forcing **(b)** Final coupled decade: 1390–1399 vs CRUNCEP3 forcing

Figure 21: Seasonal NH T2m bias of the coupled CRUNCEP3-initialized branch against CRUNCEP3 forcing. The year-50 state is not a uniform NH cooling; instead it shows a land–ocean split, with boreal continental land still cold while oceanic sectors are relatively warm.

This follow-up changes the recommendation in one practical way: the next runs should **fix the physical coupled branch first, and only then add LPJ compensation**. In other words, the "warm-floor" concept of this report still holds, but the relevant floor is now more specific: **a warmer boreal land floor**, not merely a warmer NH mean. The cleanest next matrix is therefore:

1. **06T-only** physical branch: `oasis3mct.time_step=3600`, `use_momix=True`, `momix_kv=0.01`, `kpp_av0=0.003`, `kpp_kv0=0.003`, `kpp_kvbckg=5e-06`.
2. **06V-only** physical branch: 06T plus `ENTSTPC3=1`.
3. Best of 06T/06V plus **07A** (`BNS greff_min=0.035`).
4. Best of 06T/06V plus **07A+07C** (`C3G pstemp_low=12`).

The year-50 evidence also explains why **LPJ-only tuning should not be the first move**. The all-tuned coupled NH bias panels (seasonal and annual) show that the coupled atmosphere response already differs strongly among branches even before the LPJ compensation is evaluated. If the physical branch leaves East Siberia cold, then LPJ-only changes risk compensating the wrong climate state rather than fixing it. Conversely, if a 06T/06V-type branch can hold the global net-radiation target while reducing the East-Siberian land cold pool, then the LPJ levers can be used for their intended role — retaining boreal trees — rather than masking a regional coupled-climate defect.

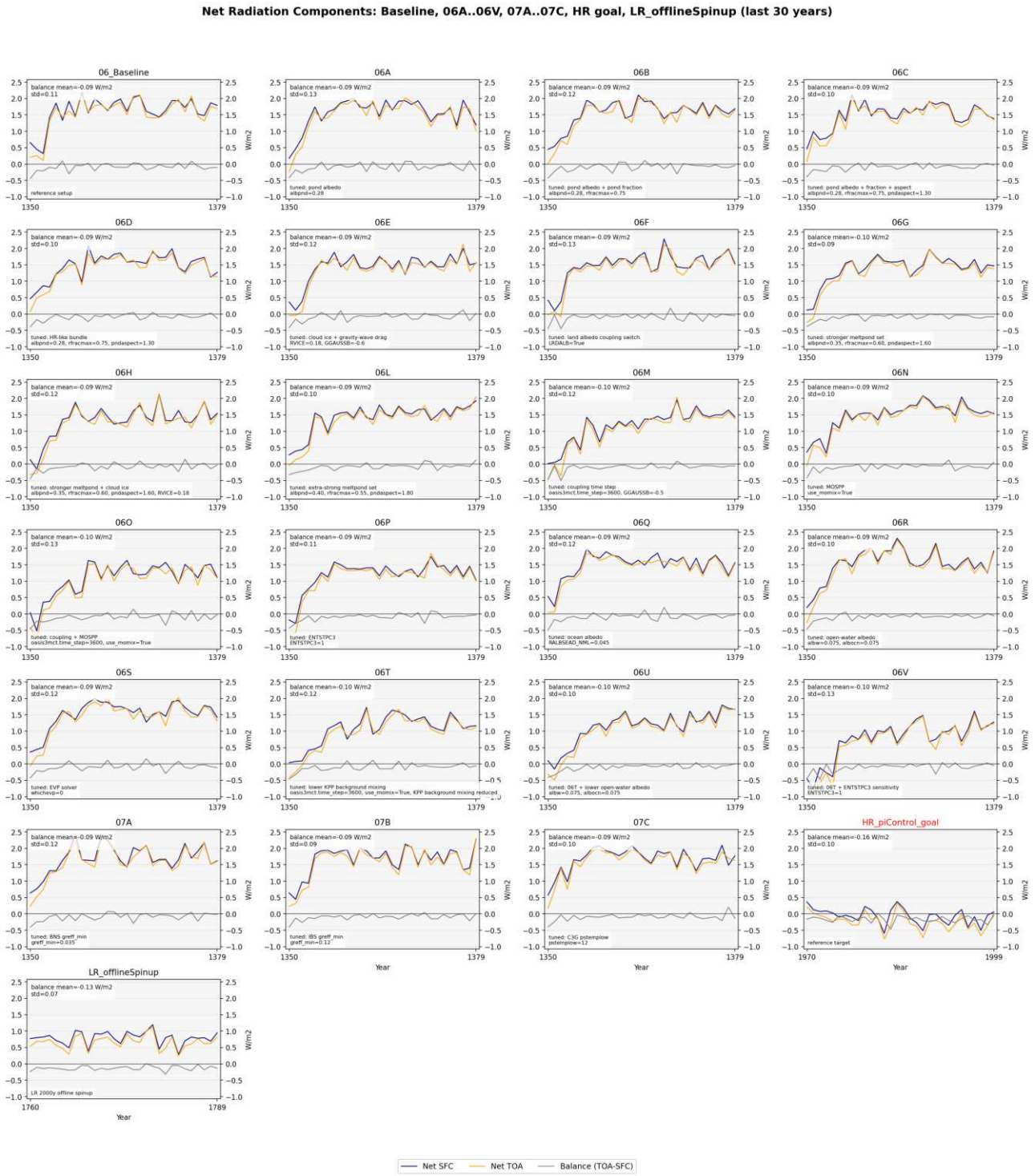
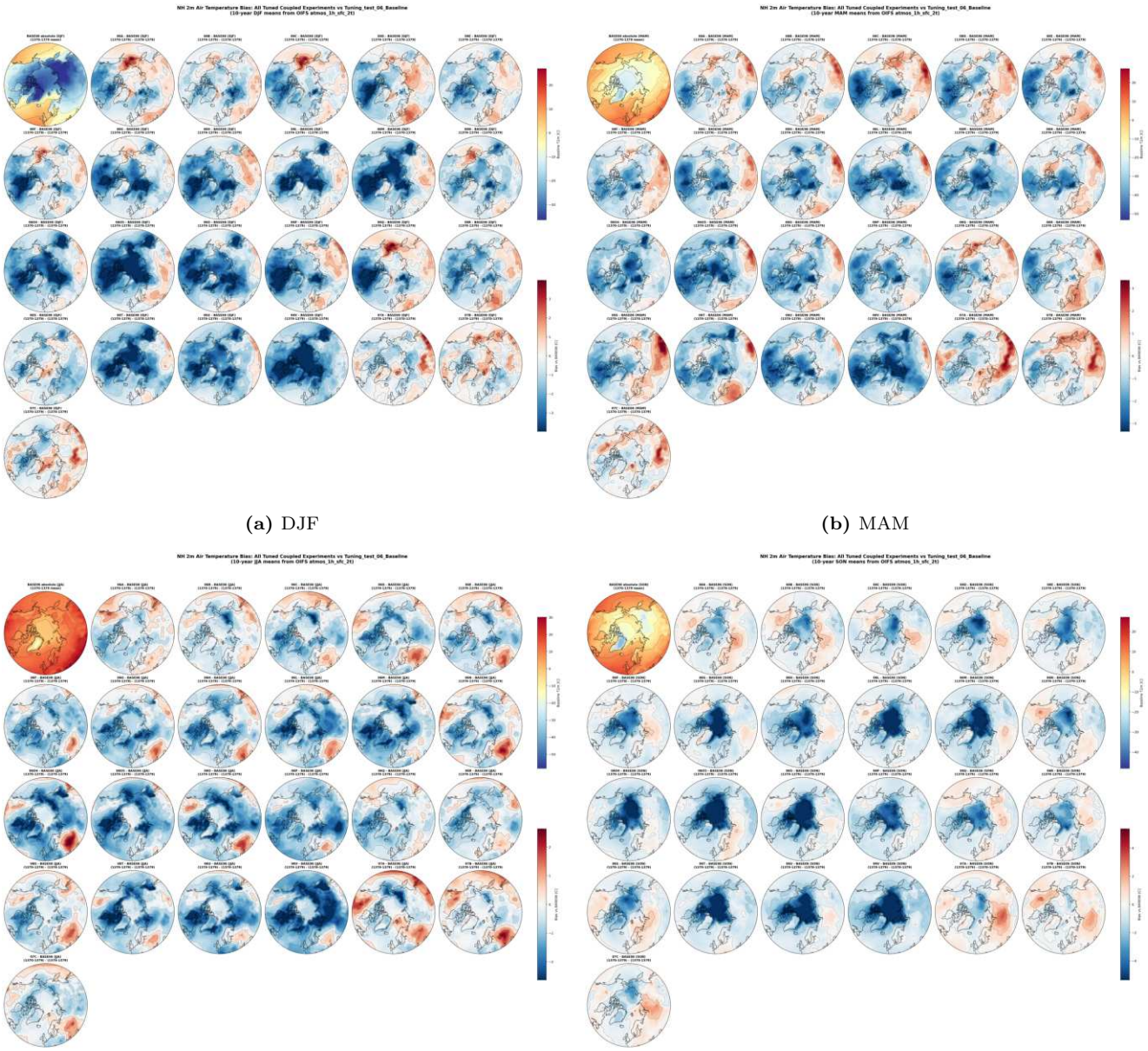


Figure 22: Global net-radiation history used to define the next physical branch. The preferred candidates remain the late 06-series branches that move closest to the target while staying within a plausible tuning envelope; the follow-up now adds the requirement that they must also improve boreal land climate, not only the global radiation budget.



(a) DJF **(b)** MAM **(c)** JJA **(d)** SON

Figure 23: All tuned coupled experiments relative to 06_Baseline, resolved by season (10-year means). These panels help separate branches that reduce the global radiation problem by globally cooling everything from branches that have a more usable boreal-land signal.

10 Work plan: a systematic path to a warm-floor, surface-balanced PI control

Two revised targets. (i) net *surface* energy balance $|F_{\mathrm{sfc}}| \lesssim 0.3\,\mathrm{W/m^2}$ — *not* TOA. Surface and TOA net flux track to within $\approx 0.1\,\mathrm{W/m^2}$ in every run (§4.1); they are one problem, so a surface-zero target is equivalent to TOA $\approx -0.1\,\mathrm{W/m^2}$, marginally stricter than TOA$= 0$. (ii) global 2 m-T bias vs ERA5 $\approx -1.3\,^{\circ}\mathrm{C}$ — the physically-expected PI-minus-present-day offset, *not* zero: a run that reaches $-1.3\,^{\circ}\mathrm{C}$ is *correct*, not cold-biased.

Ordering principle. The imbalance can only be cut by cooling, and every cooling lever over-cools if applied to today's cold-biased base. So we *first* raise the warm floor with two structural fixes that are *orthogonal to the TOA balance* (they change the land surface and BL mixing, not the radiation budget — §6.3 showed the no-LPJG floor sits at *near-unchanged* imbalance, $+0.2\,\mathrm{W/m^2}$ within drift), *then* spend a *moderate* cooling lever to reach surface balance, now affordable because the floor is ~ 3–$4\,\mathrm{K}$ warmer where it was worst. Phases 1A/1B are independent (parallel); Phase 2 depends on both; Phase 3 gates.

10.1 Phase 1A — Restore boreal tree cover (forcing first; LPJG residual second)

Ordering updated by Part IV. This phase was originally written as an LPJG-competition tuning step. The controlled transfer test now requires the opposite order: correct/regenerate the atmospheric forcing, re-spin with one LPJG binary, and tune competition only against the remaining vegetation residual (§14.3, §16).

Literature & observational basis — what realistic trees require (so we don't green the tundra). The observed arctic/boreal treeline sits near the July 10 °C isotherm (Köppen), refined by Körner & Paulsen (2004) to a growing-season mean of ∼ 6–7 °C over ∼ 90 days, with the northern boreal limit at $\lesssim$ 30 days > 10 °C. **Crucially, LPJG's current gates are already the standard, treeline-calibrated values** (Smith et al. 2001; Sitch et al. 2003 — byte-identical to the canonical `global.ins`): BNE/BINE `twmin_est`= 5 (already *below* the 10 °C isotherm) and `gdd5min_est`= 500 sit essentially *at* the observed treeline — so loosening them risks trees north of the real treeline. Two further realism facts constrain the fix: (i) NE Siberia's forest is ∼80 % *larch* (deciduous `BNS`), because evergreen conifers cannot survive its extreme winters (Kharuk et al.; Herzschuh 2020, *Glob. Ecol. Biogeogr.* 29, 198–206, the deciduous-vs-evergreen distribution paper) — LPJG already encodes this (BNE/BINE `tcmin_surv`= −31 excludes them; BNS has *no* cold gate and `gdd5min_est`= 350), so the realistic tree there is larch, and it is *already permitted*. (ii) Closed taiga tree cover is ∼ 0.4–0.7 (MODIS VCF), falling below 0.2 at the taiga–tundra ecotone — the spatial shape cvh must reproduce.

Larch caveat. The realistic NE-Siberian tree is *deciduous* larch, whose leafless winter canopy masks little snow — so the spring *albedo* mechanism is genuinely weaker over the larch zone (though it retains full force over the evergreen west Eurasia / N. America). This only re-weights the two mechanisms, it does not soften the conclusion: the *roughness* mechanism survives (even leafless, larch is ≫ rougher than grass/tundra), and the seasonality fingerprint (§6.4) shows the LPJG cold effect is co-equal in the *sunless* Nov–Jan months — i.e. the dark-season roughness/inversion part is what carries the larch zone.

The coupled winter cold is a self-inflicted trap, not the trigger — *for the evergreens.* **This paragraph applies to BNE/BINE only**: as §31.1 shows, larch (BNS) has no cold gate at all, clears its GDD gate on 84 % of NE-Siberian cells, and is lost to *growing-season competition with grass* (5.8× its LAI) rather than to any gate. Since NE Siberia is ∼80 % larch, the mechanism below does not explain the region that matters most. At the cells where prescribed veg has forest but LPJG does not (1785 boreal-land cells), the growing-season gates are already cleared in every run (JJA ≥ 12 °C, GDD5 > 500); only the winter `tcmin_est`= −30 evergreen gate binds. But against the *forested* WITHOUT-LPJG climate it excludes evergreens on just 23% of cells (≈ERA5's 22%), vs 43% in the degraded WITH-LPJG run — the missing forest's own −3.3 °C winter cooling doubles the exclusion. So it is a **bistable trap** seeded by the under-forested spin-up, not a climate trigger.

The feedback-free tests now constrain the order. The full CRUNCEP–AMIP forcing change reproduces most of the regional loss with unchanged parameters (§14.3); the four-arm experiment shows that CERES net-radiation replacement removes only a small part of the NH `treeFrac` contrast (§7). Trees do establish and fire is not involved, but those facts do not isolate a competition error. Boreal `greff_min`, C3-grass competitiveness and cold-tree `pstemp_low` remain candidate levers only for a residual measured after a forcing-consistent re-spin. Phase-1B winter warming (§10.3) remains an independent test on the coldest evergreen-gated cells.

Why not just relax the gates? The summer gates are not binding (above), so loosening them cannot help; and the *one* binding gate — the winter `tcmin_est` — correctly excludes evergreens from the extreme-cold larch zone, so relaxing it would plant *unrealistic* evergreens where even ERA5 says larch. The fix is therefore the winter *climate*, not the limits. (Keep `iftreefracca 1`, `global.ins`:150: cvh is the summed tree *crown area*, so it tracks establishment success directly.)

Screen & deploy. First run the CRUNCEP and corrected-AMIP arms with the *same* LPJG binary and unchanged parameters. Only then screen any competition knobs on the cheap standalone spin-up, targeting the MODIS-VCF pattern (∼ 0.4–0.7 closed taiga, < 0.2 at the ecotone, larch winning the cold NE) and ~~a closed canopy (FORESTFPC> 0)~~[3] — tree NPP > grass NPP — *not* merely a higher global mean. On a passing screen, re-do the coupled LPJG spin-up, regenerate the restart, and carry it into Phase 2.

10.2 Execution: the clean AMIP–noLPJG forcing generator

The decisive test (§6.9) needs a *feedback-free* climate to drive it — reusing coupled-LPJG output would carry the very cold bias we are testing. We therefore built a dedicated **OpenIFS–AMIP** generator at TCO95 (`oifsamip-cy48`, git-aligned to `awiesm3-develop`): AMIP prescribes observed SST/sea-ice, so the atmosphere has no coupled ocean/LPJG feedback. Building and running it required fixing three esm-tools bugs (a FESOM-less crash, missing XIOS/coupling compile flags, a missing WAM date-preprocessing recipe — all committed &

[3] **Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC`≡ 0 *by construction* while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

pushed; detail in App. B). A 10-yr ∼PI decade (1870–79 observed SST, GHG fixed 1850) produced the daily forcing. *One methodological catch worth recording:* the first re-spin came out barren everywhere because the XIOS field-def de-accumulated fluxes over the 6 h `freq_op` cadence instead of the true `NFRHIS`×`TSTEP`= 1 h period, making `rsns`/`rlns`/`pr` exactly 6× too low (net SW 28 vs 166 W/m^2) — fixed by a validated ×6 patch and, at root, by templating the divisor from esm-tools. **Expected payoff** of the whole Phase 1A (§6.6): ≈ −3 K off the Siberian DJF bias, ≈ −0.2 K global RMSD, at near-unchanged imbalance.

10.3 Phase 1B — Enhance stable-boundary-layer mixing (OpenIFS)

Change. The residual ≈ 1.4 K surface-trapped inversion (§6.7) is relaxed by strengthening stable-BL turbulent mixing in the EDMF K-diffusion scheme via the "long-tail" (LTG) stability functions — lengthening the tail keeps mixing non-zero at large Richardson number and weakens the over-strong surface inversion. There is no namelist for the stable branch, so this is ∼2 single-line OpenIFS source edits (minimum stable mixing length `ZLMIN` and the LTG coefficient `ZCB`) plus recompile; the exact file:line targets and secondary companions are in App. B. *Bear the documented side-effects in mind* (§6.7): long-tail mixing beyond the observational range can deepen cyclones and degrade low cloud, so verify the storm-track/low-cloud climatology, and pre-test cloud phase / LW↓ (Pithan et al. 2014) since the residual may be partly a cloud-microphysics rather than turbulence error.

Test. Change *one* knob first (`ZLMIN`) to keep attribution clean; scan 2–3 values with short (1–5 yr) coupled sensitivity runs of the no-LPJG reference. **Gate:** the DJFM low-level inversion bias (`djf_second_driver.py`) moves from +1.4 K toward ERA5's +4.2 K level and the residual boreal cold shrinks, *without* eroding legitimate inversions or warming the tropics (judge globally via `part8`/CMPI pattern). **Risk:** over-mixing warms too broadly / degrades low cloud — mitigate with small steps and global CMPI-pattern judging.

10.4 Phase 2 — Moderate imbalance tuning on the warm floor

With the floor raised (1A+1B) and global skill already better than Baseline (RMSD 1.514, CMPI 0.738; §6.3), spend a *moderate* cooling lever to bring the *surface* budget to zero. Build the combined config in layers, cleanest-first:

layer	lever(s)	ΔTOA [W/m^2]	raw bias [K]	rationale
base	melt-pond (06A–06C; opt. 06G)	−0.08 to −0.26	−1.0 to −1.3	collateral-free
localised	06P (`ENTSTPC3`)	−0.44 (−29%)	−1.36	best efficiency ∼ 1.05 W/m^2/K
moderate	06O (`1hcpl`+`mospp`)	−0.55 (−36%)	−1.77	improves ocean/ice; *not* 06V
(spare)	+`kpplow` = 06T	−0.58 (−38%)	−1.79	if more cut needed

Why this stops short of 06V. 06V (`entstpc3`+`kpplow` all stacked) gives the biggest cut (−57%) but overshoots to −2.1 K and degrades `tas`/`zg`/`rlut` in *every* region including the tropics. The base+06P+06O stack cuts ∼ 1.0–1.1 W/m^2 of surface imbalance (reference surface +1.52 →∼ 0.4–0.5; iterate the coupling strength or add `kpplow` to close the last ∼ 0.4). **Cold-bias accounting:** the ∼ −1.7 K raw cooling of the coupling lever, applied on a floor lifted ∼ 3–4 K where the bias was worst (and ∼ 0.2–0.4 K globally), lands the global mean near the target −1.3 °C rather than the −1.8 to −2.1 °C the same lever produces today.

10.5 Phase 3 — Verification protocol and decision gates

Run the combined config 30 yr (evaluate the last decade). Judge with the `reval` toolchain *plus* the DJF driver diagnostics:

target	metric (tool)	accept
surface balance	`part2` "Surface total"	$\lvert F_{\mathrm{sfc}}\rvert \lesssim 0.3$ W/m^2 (TOA ≈ −0.1)
cold bias	`part8` 2 m bias vs ERA5	−1.3 ± 0.3 °C; RMSD ≤ ref. 1.51
multivariate skill	`part4` CMPI	≤ Baseline 0.745; *pattern*, not just global
no new regionals	CMPI by region (esp. tropics)	no family-typical global cooling leak
driver check	`djf_second_driver.py` / `...roughness...`	inversion bias → ERA5; cvh restored

Iteration & critical path. Each axis adjusts independently (the floor fixes are orthogonal to the imbalance lever): too cold → back off coupling; imbalance too high → add `kpplow`; inversion too strong → raise `ZLMIN`. Phases 1A and 1B run in parallel now (both cheap/offline); 1A's coupled re-spin is the long pole, and the Phase-2 run starts once the re-spun restart and the chosen stable-BL build are both in hand.

The submission copy of all scripts, data, figures and this report lives in `/work/ab0995/a270270/analysis/investigation_awiesm3_high_lat_cold_bias_global_toa_positive/`: `data/MASTER_metrics.csv`, `data/cmpi_by_variable.csv`, `data/cmpi_by_region.csv`, `configs/`, `plots/`, `run_figures/<run>/`, `notebooks/`. Tool: `release_evaluation_tool2` (reval env); per-run diags via `part2_rad_balance.py` / `part8_t2m_vs_era5.py` / `part4_cmpi.py`.

Part IV

The atmosphere alone — attribution, and where the remaining error lives

Parts I–III argued from coupled runs, in which atmosphere, ocean and vegetation errors are inseparable. This part removes that ambiguity with a ***feedback-free OpenIFS–AMIP reference****: prescribed observed SST/sea-ice, prescribed vegetation, same OIFS branch as the coupled campaign. Two results follow.* ***(i)*** *With SST prescribed, the model still runs +0.67 W/m² out of TOA balance — that fraction of the imbalance cannot be ocean heat uptake and is a genuine atmospheric radiative error, which places a* floor *beneath what any ocean-side lever can achieve.* ***(ii)*** *The boreal* summer *cold bias survives in full without ocean or LPJ-GUESS, so it is the atmosphere's — and it is a surface* shortwave *deficit, not the boundary-layer problem of §6.7. Together these redirect the work plan: the next tuning should happen on AMIP, where both remaining errors live and where an iteration costs ∼7× less than a coupled run.*

11 The feedback-free AMIP reference (2026–07–27/28)

`amip_pi_base`, `/work/bb1469/a270092/runtime/oifsamip-cy48/`: OpenIFS TCO95L91, GHG/ozone/ aerosol fixed at 1850 (`NCMIPFIXYR=1850`), driven by the earliest observed AMIP SST decade 1870–1879 — a self-consistent ∼PI atmosphere directly comparable to the PI-control coupled runs. 10 yr complete in 2:44 on 14 nodes.

Two configuration decisions make it a valid counterpart to the coupled model, and both were wrong in the earlier `_lpjgforcing` run that Part II used (§10.2):

- **OIFS branch.** The oifsamip tree was on `local_combined_fixes` @ `1303a90`; the coupled campaign runs esm-tools version `48r1v5` = `movcav-landice+co2-concdriven`. The branches differ in *atmospheric physics*, not merely coupling plumbing (`suecozv`, `updecozv`, `suecrad`, `suphec`, `srfi_mod`, `surfece`, `voskin_mod`). Rebuilt at `f3ccacb` — the last non-debug commit, and the same state as our coupled tree.
- **Sea albedo.** The old runscript carried `TMPRALBSEAD`= 0.075. The coupled runs never set it, so they use the source default `RALBSEAD`= 0.06 (`susrad_mod.F90`:107); OASIS exchanges only the *ice* albedo (`sia_feom`→`A_Ice_albedo`) so OpenIFS computes open-water albedo itself in coupled mode too. A 0.015 albedo offset over ocean is worth order 1 W/m² of reflected SW — comparable to the quantity being attributed. Removed.

11.1 Which budget the target applies to: snow enthalpy

`part2_rad_balance.py`:255–268 computes

$$F_{\mathrm{sfc}} = \mathrm{ssr} + \mathrm{str} + \mathrm{sshf} + \mathrm{slhf} - \mathrm{sf}\cdot\rho L_f, \qquad F_{\mathrm{TOA}} = \mathrm{tsr} + \mathrm{ttr}.$$

The snowfall-enthalpy term enters the *surface* budget only, and correctly so: it is an internal atmosphere↔surface transfer and cancels at the top of the atmosphere. Every net-TOA number quoted in this report is (tsr+ttr)/Δt_{acc}, i.e. exactly part2's TOA total.

	TOA	SFC w/o snow	snow enthalpy	SFC (part2)	TOA−SFC
AMIP 1872–79	0.667	1.441	0.925	**0.516**	+0.152
09C 1370–79	0.918	1.897	0.944	**0.953**	−0.035
080a 1370–79	1.682	2.588	0.823	**1.765**	−0.083

The term is 0.82–0.94 W/m² — *larger than the entire remaining gap to target.* Without it the surface budget misses TOA by ≈ 1 W/m²; with it, 080a closes at −0.083, reproducing the ≈ −0.09 quoted in §4.1. **Convention warning:** the work plan's acceptance target is on the *surface* (§10), where the same runs read +0.52 (AMIP), +0.95 (09C), +1.77 (080a). State which convention is in use whenever these numbers are compared. *Open item:* AMIP's closure (+0.152) is 2–4× worse than the coupled runs'. Small against the +0.67 signal, so the attribution is unaffected, but unexplained — suspect the prescribed SST/sea-ice surface tile. Chase it before quoting the AMIP floor to better than ∼ 0.15 W/m².

11.2 Result I — roughly 45 % of the imbalance is atmospheric

AMIP net TOA = +0.67 **W/m²** (1872–79, after discarding two soil spin-up years; per-year 0.13–0.92; `tsr`= 240.3 W/m² confirms the 3600 s accumulation divisor). With SST prescribed this *cannot* be ocean heat uptake.

Against a coupled baseline of +1.44 (080a), that is ≈ 45 % of the coupled imbalance. The best branch in the entire campaign (09C, +0.92) now sits only 0.25 W/m² above this floor: of the 1.08 W/m² still separating it from the HR goal, ≈ 0.83 is atmospheric and only ≈ 0.25 is reachable by ocean-side tuning. **This is the campaign's most consequential number** — it says the lever family of Part I is nearly exhausted.

Caveat. AMIP runs over *observed* SST/ice while the coupled model runs over its own biased state, so +0.67 is "what this atmosphere does over a correct ocean", not a strict lower bound on the coupled value. But for a ∼PI configuration, where the real world was near equilibrium, the expected AMIP TOA is ≈ 0 — so most of the +0.67 is model error.

11.2.1 Where does it live? Nowhere in particular — the global budget is already right

Decomposing the +0.67 against CERES-EBAF (`amip_toa_decomp.py`) was expected to localise it. It does the opposite:

AMIP global	model	CERES	diff	band	net TOA	contribution
absorbed SW	240.13	240.5	−0.37	90S–60S	−90.5	−6.06
OLR	239.46	240.2	−0.74	60S–30S	−22.1	−4.04
planetary albedo	0.2947	0.293	+0.0017	tropics 30S–30N	+42.6	+21.30
SW CRE	−44.40	−45.4	+1.00	30N–60N	−21.8	−3.98
LW CRE	+24.88	+25.8	−0.92	60N–90N	−97.7	−6.55
net CRE	**−19.52**	**−19.6**	**+0.08**	**sum**		**+0.67**

The cloud radiative effect matches observations to 0.08 **W/m²** and the planetary albedo to 0.002. The AMIP and 09C zonal profiles lie almost on top of one another: the coupled model's radiation *is* the atmosphere's.

~~The band contributions reproduce the textbook meridional structure with no band carrying an anomalous imbalance, so the Hyder et al. atmospheric-SW hypothesis does not appear as a gross cloud error and there is no region to fix.~~ **That conclusion was an artifact of the comparison.** The bands above were compared to *each other*, and only the *global* CRE to CERES. Differencing each band *against CERES* (§11.2.2) shows the global agreement is a **cancellation of large regional errors**, and that the Southern Ocean carries most of the imbalance.

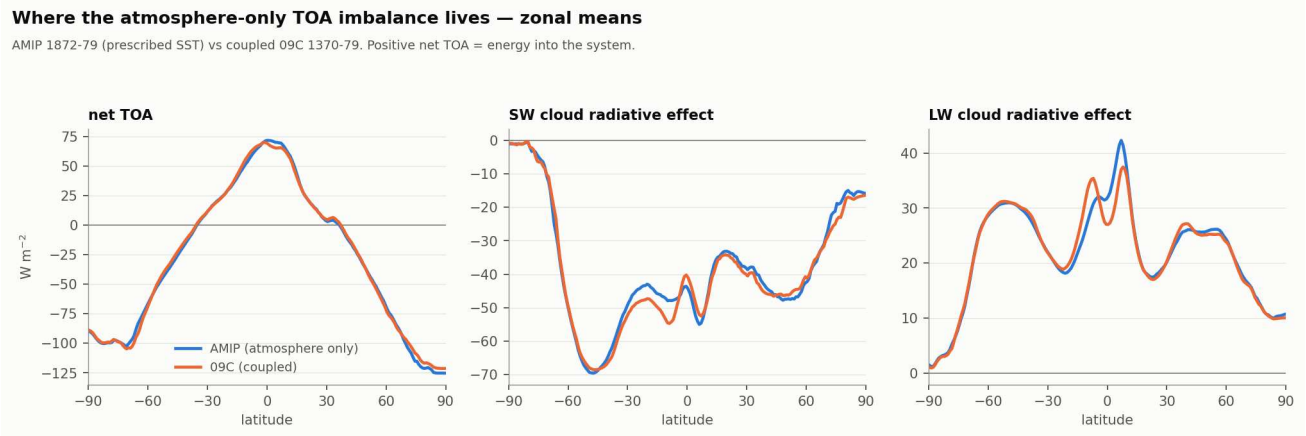


Figure 24: Zonal-mean net TOA and cloud radiative effect, AMIP (atmosphere only) vs coupled 09C. The curves are nearly indistinguishable, and the imbalance is the small residual of a large tropical surplus against extratropical deficits — not a localised error.

11.2.2 Differenced against CERES per band: the Southern Ocean carries it

band	model	CERES	diff	area	**contribution**
90S–65S	−97.0	−104.0	+7.03	4.8 %	**+0.34**
SO 65S–45S	−49.5	−55.3	**+5.81**	9.8 %	**+0.57**
45S–30S	−5.6	−1.4	−4.20	10.4 %	−0.44
tropics 30S–30N	+42.6	+45.1	−2.51	50.0 %	−1.26
30N–45N	−4.8	−5.7	+0.89	10.4 %	+0.09
45N–65N	−50.1	−54.1	+4.02	9.8 %	+0.40
65N–90N	−106.3	−106.1	−0.15	4.8 %	−0.01

The global budget is right by cancellation, not by being right. The Southern Ocean (65–45°S) runs +5.8 W/m^2 too much absorbed energy and contributes **+0.57** W/m^2 to the global mean — on its own, comparable to the entire +0.67 imbalance. The cause is a cloud *deficit*: cloud area 83.1 % against CERES's 89.7 % (−6.6 pp) with SW CRE −60.3 against −68.1 (+7.8 W/m^2 too little reflection). **This is precisely the Hyder et al. (2018) signature**, and it vindicates the compensating-error worry of §5: the ocean-mixing levers of Part I have been removing a Southern-Ocean warm bias that is substantially *atmospheric* in origin.

Consequence for the plan. The +0.67 is not irreducible after all — but neither is it fixable by a global knob, since the global mean is already correct. It is fixable *regionally*, and the two largest regional errors have *opposite* sign at TOA, which is what makes the round-10 design work (§16). Re-framing the campaign arithmetic:

	net TOA	= atmosphere +	coupled-state
080a (untuned coupled)	1.44	0.67 +	**0.77**
09C (06V + newSeaIce)	0.92	0.67 +	**0.25**

The ocean-side tuning has already delivered about two-thirds of what was available to it. Whether the HR piControl goal of −0.16 is even the right target for this configuration should be revisited, given that ≈ 0.67 of any coupled value is an atmosphere matching observations. *Revisited and answered on 2026–08–07: it is not the right target, and the goal is now* 0 *(§37).*

11.3 Result II — the boreal *summer* cold bias is atmospheric

Monthly 2 m-T bias against CRUNCEP3 (1901–10), land only, cos-lat weighted:

series	Boreal JJA/ann	Siberia JJA/ann	E. Siberia JJA/ann
AMIP (atmosphere only)	−1.64 / −1.05	**−2.16** / −0.89	**−2.24** / −0.49
080a (baseline, old sea ice)	−0.36 / −0.90	−1.41 / −1.36	−1.80 / −1.48
09A (baseline + newSeaIce)	−1.24 / −2.06	−2.19 / −3.00	−2.47 / −2.89
09B (06T + newSeaIce)	−1.56 / −2.33	−2.50 / −3.11	−2.72 / −3.04
09C (06V + newSeaIce)	−1.93 / −2.64	−2.54 / −3.08	−2.69 / −3.05

The atmosphere alone reproduces essentially all of 09A's Siberian JJA bias (−2.16 vs −2.19) and most of 09B/09C's. **Removing the ocean and LPJ-GUESS does not remove the summer cold bias.** The cold-season half behaves oppositely — AMIP's Siberian *annual* bias is only −0.89 against −3.0 coupled — so September–April remains the coupled/vegetation problem of Part II, and JJA is now a separate, atmospheric one.

11.3.1 Correction (2026–07–29), itself superseded 2026–07–30: the reference period matters, but far less than estimated

Read §17.2 first. The indirect estimate in this subsection put the Siberian JJA period offset at ≈ 1.12 K and concluded the genuine boreal bias was ≈ −1.0 K. The `amip_presentday` run has since *measured* that offset: **+0.42 K**. The conclusions below are therefore **withdrawn** — the boreal bias is ≈ −2.0 K, close to the original figure. The reasoning is retained because the method error is instructive: it applied an *observed* epoch change to a model that does not reproduce that change. The number above is scored against **ERA5 1990–2014**, while these runs use **1870s observed SST with 1850 GHG**. Roughly 130 years of greenhouse warming therefore sits inside a quantity the campaign has been treating as model error — and tuning against. Measured from the DKRZ ERA5 pool (`scripts/analysis/era5_period_offset.sh`), Siberian JJA over the same box used for evaluation:

ERA5 period	offset vs 1990–2014
1979–1989	**−0.670** K
1940–1969	**−0.790** K

The interannual sd is 0.578 K, giving SE = 0.209 K on the 1979–89 vs 1990–2014 difference ($t \approx 3.2$): the offset is real. ERA5 stops at 1940, so chaining through HadCRUT5's global series (1870s → 1990–2014 = +0.756 K) scaled by ERA5's own Siberian-JJA-to-global amplification over the overlapping window (×1.48) implies a Siberian JJA period offset of ≈ 1.12 K, leaving a **genuine model cold bias of ≈ −1.0 K rather than −2.2 K**.

This changes the target, not the diagnosis. The attribution above is unaffected — it is a model-versus-model comparison (AMIP against coupled 09A) in which any reference offset cancels, so "the JJA bias is atmospheric" stands. What changes is *how much* of it we should try to remove: closing the full 2.2 K would manufacture a ≈ 1 K *warm* bias in the coupled pre-industrial state.

Caveats. The ERA5 box is unmasked whereas the model figures are land-masked; land warms faster than ocean, so the true land-only offset is if anything larger and this correction is conservative. The ×1.48 amplification is extrapolated back from a single window, and HadCRUT5's 1870s global mean rests on sparse coverage. **The same flaw applies to the CERES-based energy targets** (§13.1, §11.2.2): CERES EBAF here is the 07/2005–06/2015 climatology scored against an 1870s model, so the +8.08 W/m² SO SW CRE gap and the +0.53 W/m² imbalance carry an analogous, *as yet unquantified*, period offset. The decisive fix for both is a single present-day AMIP leg (1989–2015, transient GHG) scored directly against ERA5 1990–2014 and CERES 2005–2015; the AMIP SST forcing already covers 187001–201512.

11.3.2 Attribution (2026–07–29): the boreal forest is a *temperature* problem

The forcing-transfer test (§14.3) showed that swapping CRUNCEP → AMIP forcing alone costs −54 % of Siberian `TREEFPC`, but could not say *which* variable carried it: radiation and temperature both feed NPP, and therefore both feed the `greff_min` growth-efficiency mortality that §6.9 identifies as the actual killer (§6.9). A colleague's spin-ups separate them by holding CRUNCEP temperature fixed and swapping *only* the radiation to CERES (`LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPandCERES`). **`AGDD5` is identical between those two arms** (775.8 Siberia, 748.0 E. Siberia, 1157.5 NH) — the control confirming that only radiation moved.

region	radiation only (CRUNCEP→CERES)	full swap (CRUNCEP→AMIP)	radiation share
NH 45N+	−1.7 %	−25.1 %	**7** %
Siberia	−8.9 %	−54.3 %	**16** %
E. Siberia	−10.5 %	−66.0 %	**16** %

The radiation arm moves −21 W/m² while the full swap moves ≈ −32, so scaling linearly puts radiation at ≤ 25 %. The daily-variability variant is indistinguishable (−7.8 vs −8.9 %), so it is not a sub-daily distribution effect either. **Temperature carries ≈ 75–85 % of the boreal tree collapse.**

Consequence: prong A's radiation work will not restore the forest. Closing the *entire* Siberian SW deficit recovers at most a quarter of the tree loss. The forest depends on the ≈ 1.0 K temperature bias established in §11.3.1, not on the Southern-Ocean/energy target. These are two separate jobs and should stop being described as one.

The period mismatch also sits on the LPJG side. CRUNCEP3 spans **1901–2015** — twentieth-century climate — yet was used to spin up the *pre-industrial* vegetation state that the coupled 1850 model restarts from. Part of the 131-`GDD5` Siberian gap is therefore a real PI-versus-20thC climate difference rather than model error: 131 `GDD5` over a ≈ 120-day season is ≈ 1.1 K, of which the 1870s → 1901–2015 offset plausibly accounts for 0.3–0.45 K, leaving 0.6–0.8 K genuine — consistent with the ≈ 1.0 K derived against ERA5 1990–2014, since CRUNCEP's period is cooler than ERA5's. *Caveat:* the AMIP arm used the superseded forcing build, ≈ 0.9 K warmer over Siberian JJA, so the temperature share is if anything understated.

Does this shift the residual onto LPJ-GUESS's CRUNCEP calibration? Partly, but the corroboration attempted on the *temperature* axis fails: NCEP2 Siberian JJA is −0.118 K *colder* than ERA5 over 1990–2014 (paired by year, $t = -2.6$) — real but negligible, and the wrong sign for a story in which LPJ-GUESS grew its trees under an over-warm forcing. That is consistent with CRUNCEP3's construction: temperature derives from CRU, bias-corrected against stations, while *radiation* derives from NCEP, which is precisely where the +21 W/m² excess against CERES was measured (§13.1). The calibration mismatch sits on the radiation axis. The direct evidence remains the forcing-transfer test (§14.3), which does not depend on which variable carries it.

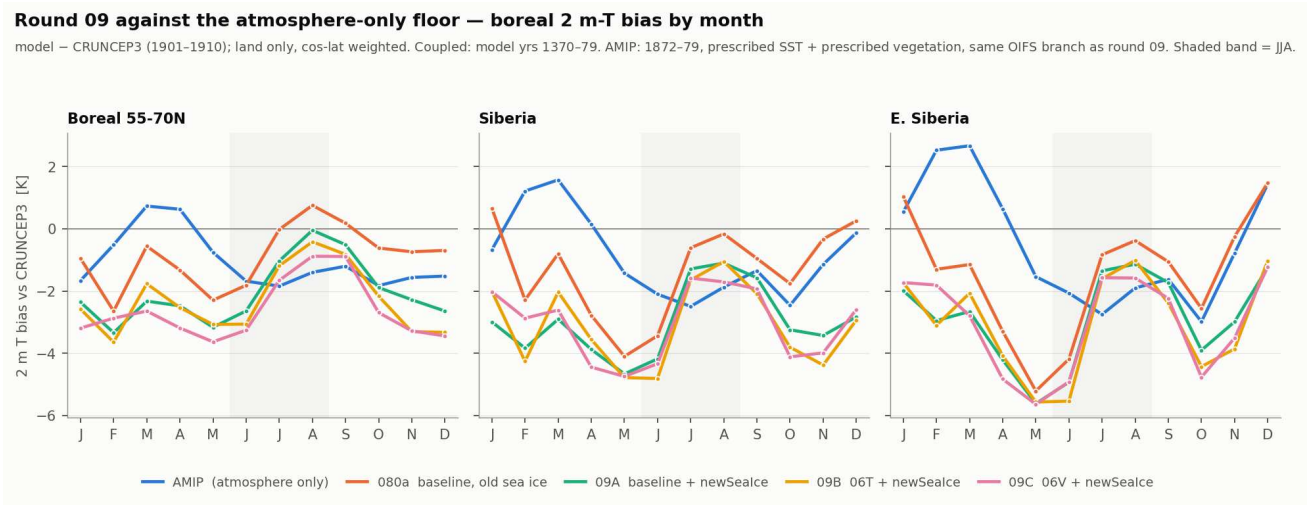


Figure 25: Boreal 2 m-T bias by month against CRUNCEP3, land-masked. The atmosphere-only AMIP curve (blue) sits at or below the coupled runs through JJA (shaded) but far above them in winter: **the summer bias is atmospheric, the cold-season bias is not**. *Caveat:* the "Boreal 55–70°N" panel is contaminated by the Greenland ice sheet (JJA snow depth 320 mm w.e. in that box vs 5.5 mm for Siberia); the Siberia and E. Siberia panels are clean and carry the argument.

Superseded numbers. ~~AMIP Siberia JJA −1.23, E. Siberia −1.59, i.e. indistinguishable from the coupled baseline, so coupling adds nothing in summer.~~ Those came from the earlier `amip_pi_forcing` run, built on the wrong OIFS branch and with `TMPRALBSEAD`= 0.075. The corrected build is ≈ 0.9 K colder in Siberian JJA; the conclusion strengthens (the atmosphere is now *colder* in summer than the coupled baseline) but the numbers changed.

12 Round 09 — new sea-ice thermodynamics (2026–07–27/28)

FESOM 2.7 `awiesm3-implicit-ice-surftemp` @ `db7dd9af` makes the ice surface-temperature solve implicit ($\mathrm{d}Q/\mathrm{d}T$-linearised about the last transmitted `ist`) and hands ownership of the ice surface energy balance to OpenIFS, which now runs a slab-conduction ice tile using FESOM-supplied thickness; FESOM ice growth becomes $Q_{\mathrm{atmice}} = -a2ihf$. The pairing is verifiable at runtime in the OASIS `namcouple`: the 09 runs carry a sixth o2a field, `sit_feom→A_Ice_thickness`, absent from every 08 run. Run identities and the 09B parameter block are catalogued in the 2026–07–28 update at the head of this report; the quantitative comparison is below. `09C` was built and run on our own tree (`Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`) to close the missing 06V corner.

run	net TOA [W/m^2]	global T2m [°C]	Siberia JJA [K]	Siberia ann [K]
080a baseline, old sea ice	1.442	13.606	−1.41	−1.36
09A baseline + newSeaIce	1.286	13.148	−2.19	−3.00
09B 06T + newSeaIce	1.171	12.794	−2.50	−3.11
08B 06V, old sea ice	0.935	12.347	−3.11	−4.09
09C 06V + newSeaIce	**0.919**	12.492	**−2.54**	**−3.08**

On the clean old-vs-new 06V pair (08B → 09C) the radiation barely moves (0.935 → 0.919) but the boreal **warms by +1.0 K in the Siberian annual mean**, +0.57 K in JJA, +0.15 K globally. **09C dominates 08B on both objectives simultaneously** — the first time in the campaign that has happened, and the "free warming" sought in §9. But the effect is *not* an offset: on the baseline (080a → 09A) the same code change goes the other way, cooling 0.46 K globally and 1.6 K in the Siberian annual mean. The sign depends on the ocean-mixing branch; no mechanism is established here, and pinning it down needs the surface energy budget over ice.

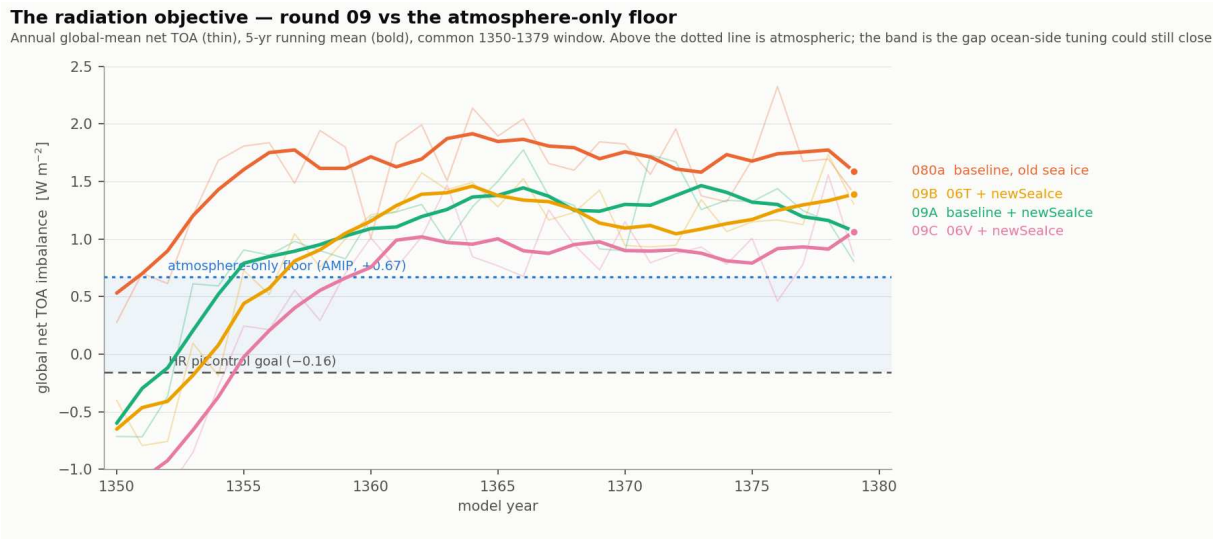


Figure 26: Annual global-mean net TOA (thin) with 5-yr running mean (bold), common 1350–1379 window. The dotted line is the **atmosphere-only floor** measured by AMIP (+0.67 W/m^2); the shaded band is the only part of the remaining gap that ocean-side tuning can still close. Everything above the dotted line is atmospheric.

Bookkeeping caveat. `sacct` shows a `Tuning_test_08A_06T_1hCPL_MOSPP_KPPLOW_CRUNCEPinit` completing all three chunks on 24–25 July, but no 08A directory exists on disk. 08A was the 06T-only physical control — Run 1 of the 2026-07-24 design note's own sequence — so within round 08 every surviving 06T-family entry (08C, 08E, 08G) carries an LPJ lever and the intended 06T-vs-06V comparison cannot be made cleanly. Round 09 supersedes this, but conclusions drawn from round 08 alone inherit the hole.

13 Why the summer bias exists — a shortwave deficit (2026–07–28)

Siberia JJA surface energy budget, land-masked, cos-lat weighted (AMIP 1872–79 vs CRUNCEP3 1901–10, which carries `rsns`/`rlns` and so permits a direct radiative comparison):

	AMIP	CRUNCEP3	Δ
2 m temperature	9.74 °C	11.92 °C	**−2.18** K
SW net at surface	155.1	188.3	**−33.2** W/m^2
LW net at surface	−49.2	−64.5	+15.2 W/m^2
net radiation			**−18** W/m^2

Supporting AMIP diagnostics: SW↓ 188.6, surface albedo 0.178 (`fal` 0.172), LW↓ 315.6, SH/LH −24.6/ − 56.8 (**Bowen** 0.43), cloud `tcc` 0.778 / `lcc` 0.505 / `mcc` 0.386, `skt`−`2t` +0.43 K, `swvl1` 0.343, snow depth 5.5 mm w.e.

The LW term is *compensation* (a colder, cloudier surface loses less longwave), not cause. It is *not* albedo: a realistic boreal-summer 0.13–0.15 instead of the model's 0.178 buys only 5–9 W/m^2. So SW↓ itself is too low. This is a different mechanism from the winter stable-BL inversion of §6.7, and it is not addressed by any lever proposed so far.

13.1 Re-measured against CERES: cloud, not clear-sky — and three times smaller

CRUNCEP3 is a reanalysis-derived forcing product, so the deficit was re-measured against **CERES EBAF Ed4.1** (satellite), which supplies all-sky *and* clear-sky net SW at the surface and so permits the decomposition $\Delta(\mathrm{SW_{net}}) = \Delta(\mathrm{SW_{net,clear}}) + \Delta(\mathrm{SW\ CRE_{sfc}})$ with no assumption about surface albedo or the direct/diffuse split.

JJA	Siberia land			NH 55–75°N ocean		
	model	CERES	diff	model	CERES	diff
SW net at surface, all-sky	155.1	166.3	**−11.2**	142.4	155.4	−13.0
SW net at surface, clear-sky	246.6	241.4	+5.3	241.9	252.6	−10.7
SW CRE at surface	−91.6	−75.1	**−16.5**	−99.5	−97.3	**−2.2**
cloud area [%]	77.8	69.6	**+8.2**	84.6	83.0	**+1.6**

Three results, all of which revise §13:

(i) It is unambiguously a cloud error. The decomposition closes exactly ($-11.2 = +5.3 - 16.5$), and clear-sky surface SW is if anything slightly *too bright*. Aerosol, water vapour and surface albedo are therefore all exonerated — and since PI aerosol is lower than present-day, the PI-vs-CERES period mismatch works *against* this finding, making it conservative.

(ii) The deficit is 11 W/m^2, not 33. ~~Surface net SW over Siberian land in JJA is 33 W/m^2 too low.~~ That figure was measured against CRUNCEP3, which is itself ≈ 21 W/m^2 brighter than satellite — **CRUNCEP3 was the outlier, not the model**. An 11 W/m^2 summer deficit remains the right order to explain a ≈ 2 K land cold bias, so the mechanism survives; only its magnitude changes.

Verified independently, because the original comparison had assembled the two sides on different grids with different land masks — a way to manufacture exactly this kind of difference. Redone with all three sources on the CERES 1° grid under a single land mask (`verify_ceres_cruncep.py`): CERES 166.7, AMIP 155.8 (−10.8), CRUNCEP3 187.6 (+21.0). Raising the land-mask threshold 0.5 → 0.8 changes these by < 1 W/m^2, and CERES's own JJA-mean file gives 168.4 against 166.7 from the monthly climatology. The grid inconsistency was therefore not the cause.

Why CRUNCEP3 is bright splits cleanly by season: in JJA it is the downward flux (234 W/m^2 in July against CERES's 209, the NCEP high-latitude brightness bias), whereas in spring it is the forcing file's fixed albedo (§14.1). CRUNCEP3 blends CRU *station* temperature with NCEP *reanalysis* radiation, which is the natural reading of why its T2m is usable while its SW is not.

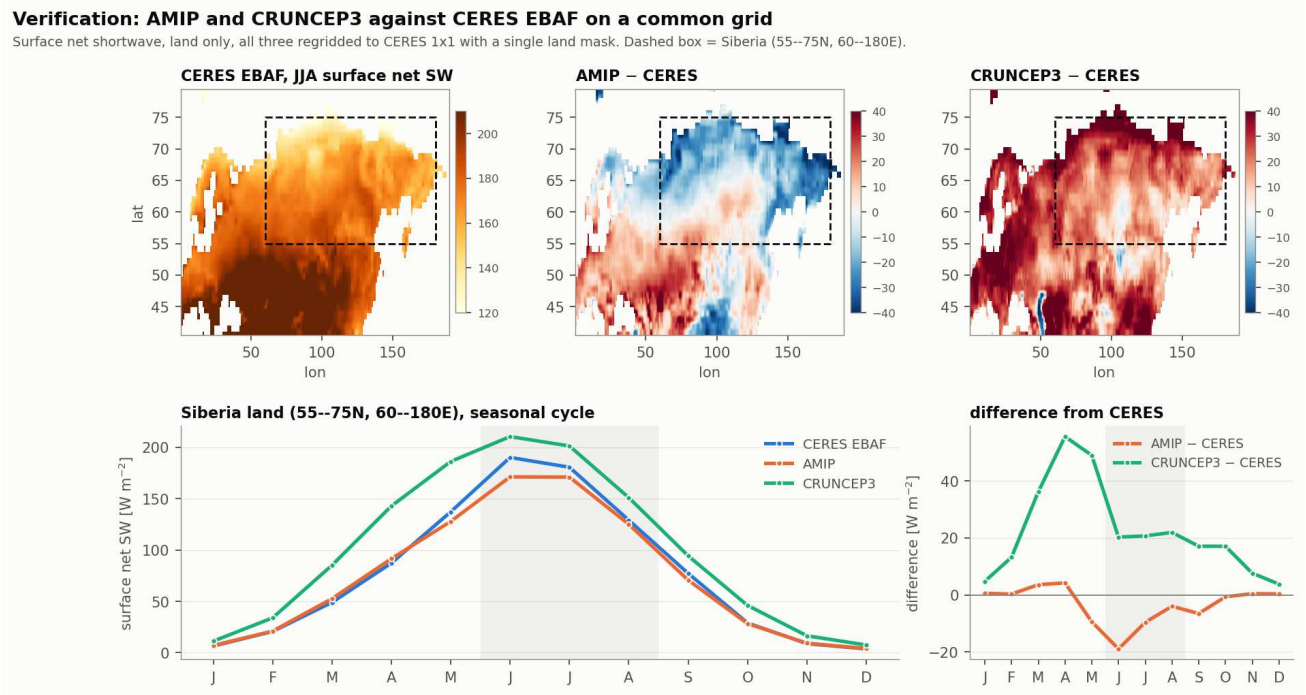


Figure 27: Surface net shortwave against CERES EBAF, all sources on a common 1° grid with one land mask. AMIP (orange) is ≈ 11 W/m^2 *dark* in JJA; the CRUNCEP forcing (green) is ≈ 21 W/m^2 *bright* in JJA and far more so in spring, where its fixed albedo does not mask snow. The two forcings differ from each other by ≈ 32 W/m^2 in the growing season — the transfer step of §14.

(iii) It is specific to boreal land. Over ocean in the same latitude band the cloud error nearly vanishes (-2.2 W/m^2 CRE, +1.6 pp area) while over Siberian land it is −16.5 and +8.2. A *global* cloud knob would therefore move the ocean, where nothing is wrong, and spend TOA budget for no benefit. The error is over land, in summer — pointing at surface-driven convection and boundary-layer cloud rather than the large-scale cloud scheme.

Amount or opacity? Both, roughly half each. At the observed CRE per unit cloud area ($-75.1/0.696 \approx -108$ W/m^2), the +8.2 pp excess accounts for ≈ -8.8 of the -16.5 W/m^2 CRE error; the residual ≈ -7.7 is opacity or vertical placement. A cloud-*amount* lever alone therefore reaches at most half of it.

13.2 Does the cloud fix conflict with the energy target? Less than first thought

~~Burning off that cloud stops reflecting it at TOA, costing $\approx +0.18\,\mathrm{W/m^2}$ on the annual global TOA — a 27 % worsening — so a pure cloud fix cannot meet a fixed-TOA constraint, and only a radiatively neutral surface-flux lever can.~~ Two things falsify that framing:

- **The target is the *surface*, not TOA** (§11.1), and the drift it controls is driven by the flux into the *ocean*. The cloud error is over *land* (§13.1), where annual-mean energy storage is ≈ 0 — land returns what it receives. A boreal-land cloud correction therefore barely touches the quantity that governs spin-up drift.
- **The error is land-specific**, so the correction need not be a global knob at all (§13.1) — which was the whole reason a TOA cost was incurred in the first estimate.

The two objectives are consequently far more separable than the original framing implied, and the cloud error can be attacked at its diagnosed cause rather than through a proxy.

13.3 The stomatal-resistance sensitivity test (experiment A)

A second, radiatively neutral lever was nonetheless identified and is being tested, because the measured Bowen ratio over Siberian JJA land is SH/LH $= -24.6/-56.8 =$ **0.43**, low against the general observed picture for boreal conifer forest. Minimum canopy resistance for boreal needleleaf (`susveg_mod.F90`:266–267) is 250 s/m, with 500 retained just above as a comment.

Provenance — and a correction. ~~500 is the IFS default and has been halved in this build.~~ Git refutes this: both the active 250 and the commented 500 are present in commit `cf82033`, "initial commit of 48r1 released version", and no AWI commit has ever touched `RVRSMIN`. 250 **is what ECMWF ships in 48r1**; 500 is the older HTESSEL-era value they themselves revised. Setting 500 moves *away* from the released configuration, so this is a **sensitivity test, not a default restoration**.

Status: `amip_expA_rvrsmin500`, 6 yr, paired against `amip_pi_base` on 1872–1875. Verified compiled in by inspecting the rebuilt object file (the value appears twice for types 3–4; 250 survives only for Desert and Mixed Forest). Note this tests *partitioning*, not the diagnosed cause: if the cloud error is real then a resistance change compensates for it, and with 11 $\mathrm{W/m^2}$ less available energy there is also less to repartition. *Caveat:* "Bowen 0.43 is too low" is inferred from the literature's general picture, not measured — neither CRUNCEP3 nor CERES supplies SH/LH here, and FLUXNET boreal sites would settle it.

A connection worth testing in the same run. Excess summer evapotranspiration over boreal land would *both* cool the surface directly *and* moisten the boundary layer, feeding the excess low cloud (model `lcc`= 50.5 % over Siberian land in JJA). If cutting transpiration also reduces low cloud, the partitioning error and the cloud error are the same problem, and experiment A tests both at once.

14 Forcing transfer: the spin-up was grown under brighter, warmer conditions than the model delivers

The vegetation state every coupled run in rounds 08 and 09 restarts from is the standalone offline spin-up `LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPcalibrated_v3` (`/work/bb1469/a270270/runtime/lpjg-spinup/`), driven by `CRUNCEP_noLPJG_1d_1901-1910_TCO95_calibrated_v3.nc` (verified at `lpj_guess_levante_spinup_..._2000y.yaml`:38). **That forcing and the atmosphere the coupled model provides differ substantially, and in the same direction for both light and temperature.**

Siberia land, JJA	surface net SW [$\mathrm{W/m^2}$]	vs CERES
CERES EBAF (observed)	166.7	—
CRUNCEP forcing (what LPJG grew under)	**187.6**	**+21** (+13 %)
AMIP (what our atmosphere delivers)	**155.8**	**−11** (−7 %)
gap between the two forcings		**32** $\mathrm{W/m^2}$ (≈ 20 %)
plus 2 m temperature		**2.2** K (11.9 vs 9.7 °C)

The errors point *opposite* ways and therefore compound: CRUNCEP is too bright (the NCEP high-latitude brightness bias — its downward SW is 234 $\mathrm{W/m^2}$ in July against CERES's 209, ≈ 12 % high), while our atmosphere is too dark by the cloud error of §13.1. Vegetation equilibrated to the first and then handed the second experiences a ≈ 20 % drop in growing-season radiation and a 2.2 K cooling — a step change in exactly the direction that pushes

a boreal system from trees toward grasses. All numbers here are computed on a common CERES 1° grid with a single land mask (`verify_ceres_cruncep.py`); mask threshold 0.5 → 0.8 moves them by $< 1\,\mathrm{W/m^2}$.

14.1 Ruled out by measurement

Several components were suspected along the way and cleared. Recording them so the same ground is not recovered:

suspect	the observation that cleared it
LPJ-GUESS core radiation handling	~~PAR is inflated because no albedo is applied.~~ Reading `rsns` as `NETSWRAD_TS` and ap input (`driver.cpp`:1435–1449); a second correction would double-count. Model beh
EC-Earth spin-up interface	Reads the right variable and sets the right type (`framework.cpp`:2501–2507). No fa
The forcing script's fixed albedo	~~`rsns`=`FSDS`×0.861 is a fixed 0.139 albedo, so spring PAR is inflated ×1.66 and the~~ True that the albedo is fixed (ratio 0.861 in every cell, April and July alike). But i 0.139 against CERES's **0.142** — right to 0.003. The large spring error (0.139 vs 0. is **biologically inert**: measured over 687 Siberian cells at equilibrium, April carrie GPP and April LAI is 0.02 (no canopy), while **JJA carries 80 %**. Weighted by wh is actually assimilated, the PAR excess is ≈ 13 %, not 66 %.
~~...and it is an unusual, campaign-specific defect~~	Standalone LPJ-GUESS assumes a constant albedo natively: `BETA`= 0.17 (`driver.` in both the sunshine and `SWRAD` paths. The script's 0.139 is a slightly darker versio standard practice, and switching to its `bulk_albedo` option would give 0.195 in JJ *worse* than the fixed value. **Do not "fix" the script; it is not the problem.**

14.2 What survives, and how to settle it

What remains is not a defect in any component but a **transfer problem**: a ≈ 20 % growing-season radiation step and a 2.2 K temperature step between the forcing the vegetation equilibrated to and the forcing it is then asked to live in. **Whether LPJG's *parameters* were tuned against CRUNCEP — the "calibrated" in the directory name — is not established**; the 07-series did change `greff_min` and `pstemp_low`, but no evidence was found of what was calibrated against what, and the folder name is not proof.

It still weakens the "competition, not climate" verdict of §6.9, though less than first argued. That test moved a standalone re-spin from the CRUNCEP forcing to the OIFS–AMIP forcing and read the resulting under-vegetation as an LPJG-internal competition failure. But that move alone removes ≈ 20 % of the growing-season light and 2.2 K, so under-vegetation is partly what the forcing change predicts on its own. The verdict should be read as provisional. (The forcing used there was also the *superseded* AMIP build, §11.3.)

14.3 The transfer test, measured: forcing alone halves boreal tree cover

The test needs no knowledge of what was tuned — take the *same* LPJG parameters and spin to equilibrium under each forcing; the difference *is* the transfer penalty. **Both arms already existed** and required no new computation:

CRUNCEP arm	`lpjg-spinup/LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPcalibrated_v3`
AMIP arm	`lpjg-spinup-develop/LR_2000y_PIforcing_v3`

Controlled: `global.ins` **byte-identical** between the two trees, `firemodel "GLOBFIRM"` in both `.ins` files, `nyear_spinup`= 2000 / `freenyears`= 100, fixed 1850 CO_2/Ndep/LU, both reaching year 3900, and **10 043 gridcells in common** (99.8 % of the AMIP gridlist). The material difference is the forcing file.

region		CRUNCEP-forced	AMIP-forced	diff	%
Siberia	`TREEFPC`	0.357	**0.163**	−0.194	**−54**
	`GRASSFPC`	0.288	0.330	+0.042	+15
	`AGDD5`	776	645	−131	−17
E. Siberia	`TREEFPC`	0.359	**0.122**	−0.236	**−66**
	`GRASSFPC`	0.316	0.398	+0.082	+26
	`AGDD5`	748	568	−180	−24
NH 45°N+	`TREEFPC`	0.383	0.287	−0.096	−25

Changing nothing but the forcing halves the Siberian tree fraction and cuts East Siberia's by two-thirds, with grass taking up the space. For scale, the coupled 080a run lost `TREEFPC` 0.310 → 0.121 over Siberia in 50 years (§9.1); the forcing swap alone reproduces essentially that entire loss at equilibrium, with no parameter changed. `AGDD5` falls 17–24 % in the same direction, so this is not a light effect alone — the 2.2 K also does real work through the growing-degree-day gates.

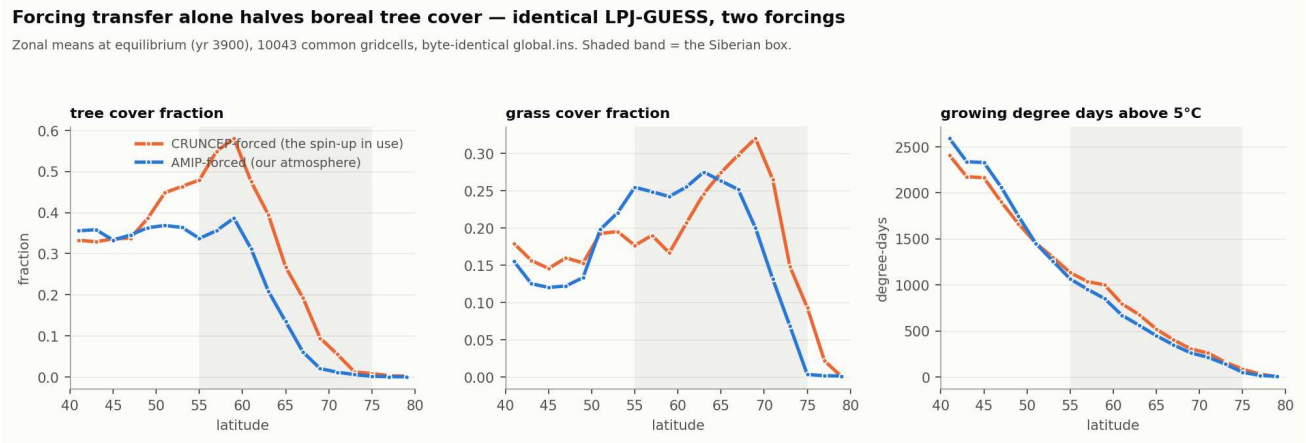


Figure 28: Zonal means at equilibrium (yr 3900), 10 043 common gridcells, byte-identical `global.ins`. South of ≈ 52°N the two arms are indistinguishable; northward they diverge sharply and the AMIP-forced treeline retreats several degrees south. The deficit the campaign has been attributing to competition is reproduced by the forcing change alone.

Three caveats, all material.

1. The arms were built from *different binaries* (a270092 vs a270270) of the same LPJG version with identical `global.ins`. A compiled difference cannot be excluded; re-running both arms from one build (≈ 3 h) would close it, and should be done before this is published.
2. The AMIP forcing is the *superseded* build (§11.3), ≈ 0.9 K warmer over Siberian JJA land than the corrected one — so this is a **lower bound** on the penalty against today's atmosphere.
3. ~~`FORESTFPC`= 0 in both arms, so the absence of closed canopy survives as a separate unexplained problem.~~ `FORESTFPC` is a *void diagnostic* in this configuration — it counts only the managed-forest land-use class, which is empty here, so it is identically zero regardless of what the vegetation does. There is no separate closed-canopy failure to explain.

15 Revised plan: tune the atmosphere on AMIP

Both errors that remain — ≈ 0.83 W/m^2 of TOA imbalance (§11.2) and ≈ 2.2 K of Siberian summer cold (§11.3) — are atmospheric, and both are now *measured* in a configuration where neither the ocean nor LPJ-GUESS can confound the answer. They should therefore be tuned on AMIP, not in coupled mode. This does not retire Phases 1A/1B of §10 — the LPJG competition fix and the stable-BL mixing fix still target the *cold-season* bias — but it inserts a cheaper, better-posed stage ahead of Phase 2.

Two objectives, now known to be largely separable (§13.2):

1. **Surface energy balance** → 0 to cut spin-up drift. This is an *ocean* surface-flux problem; the relevant number is F_{sfc} over ocean, which has *not* yet been computed — all figures quoted so far (+0.52 AMIP, +0.95 09C, +1.77 080a) are global. **Open item.**
2. **Boreal summer warmth** to restore GDD5 and let trees compete. This is a *land* problem, caused by an 11 W/m^2 cloud-driven SW deficit that is absent over ocean (§13.1).

	change	where	status / expectation
A	`RVRSMIN(3,4)`: 250 → 500	`susveg_mod.F90`:266–267	*running*; radiatively neutral; tests partitioning, not the diagnosed cause
B	`RVLAI(3,4)`: 5.0 → 4.0	`susveg_mod.F90`:133–134	same sign, smaller
C	~~`RCLDIFF` ↑, global~~	~~`NAMCLDP`~~	~~+surface SW, but +TOA~~
C′	a *land*-acting cloud lever	convection / BL cloud	the error is land-only, so a global knob would move the ocean where no
D	A + C′		joint target

Gates: Siberian JJA bias toward 0 *and* SW CRE toward CERES; ocean F_{sfc} not degraded; no new error outside the boreal (judge on the global `part8`/CMPI pattern, §4.3). Since the cloud error splits roughly half amount, half opacity (§13.1), a lever acting only on cloud amount can close at most half of it, and the diagnostic to watch is SW CRE against CERES rather than temperature alone.

Why AMIP is the right testbed. 2.3 node-hours per model year against 15.8 coupled — a $\approx 7\times$ saving — and 10 yr completes in one job. Measured interannual scatter of the boreal JJA mean is $\sigma = 0.25$ K (Boreal), 0.59 K (Siberia), 0.72 K (E. Siberia), so with 8 usable years the standard error is 0.09–0.26 K against biases of 1–2 K: sampling is not the binding constraint. *Land spin-up is* — with `lresume: false` the deep soil (`stl4`, `swvl4`) carries initialisation shock, so discard the first 1–2 years. For paired tuning experiments 5 yr (1 discarded + 4 used) suffices, since identical prescribed SST makes most interannual variance cancel in the difference.

16 Round 10 design: a two-pronged, targeted attack (2026–07–28)

Original design as proposed. Operator notes to be appended in §16.4.

The structural fact the design rests on. The two largest regional radiation errors have *opposite* sign at TOA (§11.2.2, §13.1), so fixing them together is self-balancing, whereas fixing either alone is not:

	error	fixing it	ΔTOA
A1 Southern Ocean	too *little* cloud (−6.6 pp, SW CRE +7.8)	adds cloud, reflects more	**−0.5** to −0.6
A2 boreal land	too *much* cloud (+8.2 pp, surface SW −11 in JJA)	removes cloud, absorbs more	**+0.1**

16.1 Prong A — AMIP (atmosphere), A1 before A2

A1 first. It moves the global target in the right direction (+0.67 →≈ +0.1) *and* attacks the error the campaign has suspected since §5. A2 then costs back only ≈ 0.1 of the ≈ 0.55 that A1 buys, so the boreal fix becomes affordable — the "warm floor" logic of Part III, relocated from the vegetation to the atmosphere.

Critical: in AMIP the Southern-Ocean target is *radiation only*, never temperature. SST and sea ice are prescribed, so the SO surface temperature cannot respond to a radiation error — the excess absorbed shortwave is simply exported through the fixed boundary. **A1 must therefore be judged on SW CRE and cloud area against CERES, and it is meaningless to look for an SO temperature response in AMIP.** This is also why the SO *warm* bias appears only in AWI-CM3/AWI-ESM3: it is a *coupled* phenomenon, in which the same +5.8 W/m^2 of excess absorbed energy has somewhere to go and warms the ocean. The causal chain is then explicit and closes the compensating-error argument of §5:

AMIP SO cloud deficit (−6.6 pp) → +5.8 W/m^2 excess absorbed → *(coupled only)* SO warm SST bias → suppressed by the 06T/06V ocean-mixing levers

i.e. Part I was damping Weddell/Ross convection to compensate an *atmospheric* cloud error. It follows that the only test of whether A1 fixed the real problem is a *coupled* run — AMIP can confirm the radiation is right, but not that the SO warm bias is gone.

On levers. OpenIFS has no land-only cloud knob, and that is informative rather than obstructive: the land-specificity of A2 (−16.5 W/m^2 CRE error over Siberian land against −2.2 over ocean at the same latitude) is itself evidence that the cause is **land-surface moisture supply**, not the cloud scheme — which makes the `RVRSMIN` sensitivity test (§13.3) potentially the actual fix rather than a proxy for one. A1, being over ocean, is a genuine cloud-scheme / microphysics target.

Guardrails on every A-run — all four, every time:

1. global net TOA *and* surface flux (state which convention, §11.1);
2. SO 45–65°S SW CRE and cloud area against CERES — *radiation only*, since prescribed SST makes an SO temperature response impossible in AMIP by construction (above);
3. **the tropics** — 50 % of the globe and the largest single contribution (−1.26), and the term the 06O/06T/06V levers degraded (§4.3);
4. CMPI by *pattern*, not the global number.

16.2 Prong B — LPJ-GUESS, and it must go second

Do not tune competition parameters yet. The tree deficit is largely forcing-driven (−54 % Siberian `TREEFPC` from the forcing swap alone, §14.3), so tuning LPJG while its forcing is 11 W/m^2 dim and 2.2 K cold would bake the atmospheric error into vegetation parameters. That is the compensating-error trap, and it is what the 07-series was already doing — which is the natural reading of why it achieved nothing (§12).

1. Complete Prong A.
2. **Regenerate the LPJG forcing** from the corrected atmosphere. Enabling work required: `amip_pi_base` uses the default `file_def` and emits the evaluation fields but *not* the daily LPJG forcing set, so a merged `file_def` (or a second short run) is needed.
3. **Re-spin and re-derive the residual.** Whatever tree deficit survives against a corrected forcing is the real LPJG target. Only then are `greff_min` / `pstemp_low` legitimate levers.

Gates, replacing the void one. ~~FORESTFPC> 0~~ is unachievable by construction (§14.3) and must not be used. Use instead: `TREEFPC` against the CRUNCEP-arm equilibrium (0.357 Siberia, 0.359 E. Siberia); `AGDD5` recovery toward 776/748; and `cvh` toward the MODIS 0.4–0.7 closed-taiga range.

16.3 Ordering

A1 → A2 → regenerate forcing → re-spin LPJG → B. Prong B cannot be evaluated honestly until Prong A has fixed what feeds it.

16.4 Operator notes

(reserved — to be filled in)

17 Round 10 results (2026–07–30): 46-year runs rebuild the ranking

All 19 prong-A runs were extended from 6 to 46 years (evaluated 1872–1915, 44 yr) after a statistical-power check showed the original 4-year window could not resolve the boreal signal at all.

17.1 The 4-year window was resolving nothing in the boreal

A run × year ANOVA (`scripts/analysis/noise_floor.py`) decomposes each per-year value into a tuning signal, the SST-forced excursion common to every run, and internal noise. At 4 years the 95 % detection threshold on Siberian JJA T2m was ±0.89 K and **1 of 18 runs cleared it** — A1a, the overshoot already rejected. Siberian JJA surface SW: 1 of 18. Siberian JJA cloud area: **0 of 18**. The energy target was by contrast well resolved (±0.23 W/m^2 on global TOA, 9 of 18 significant), so the campaign's Southern-Ocean and global conclusions were safe while *every boreal ranking was noise.*

At 44 years the threshold falls to ±**0.240** K, and the ranking inverts:

run	4-yr Δ	**44-yr Δ**	t	verdict
B5 `RCAPDCYCL` 2 → 0	+0.001	**+0.407**	+3.32	**significant — best**
B2 `RCLDIFF_CONVI`= 25	+0.290	**+0.344**	+2.80	**significant**
B3 `RCLDIFF`= 1.5E-5	+0.366	**+0.306**	+2.50	**significant**
A1a `ovl`= 0.10	−1.149	−0.749	−6.11	significant, wrong way
B8 `RVLAMSK`= 5	**+0.502**	**−0.038**	−0.31	**noise**
AB, B7, C2, E1, A2, A1c, B6	—	≤ +0.12	—	noise

Two reversals matter. **B8, promoted to "best lever of the campaign" on its +0.502 K, is noise**; it had already been retracted on 07–29 when the noise floor was first measured, and the 46-year runs confirm that retraction. **B5, dismissed as "inert" at +0.001 K, is the strongest lever in the campaign.** The caution recorded at the time — *unresolved, not proven inert* — is what kept it in play.

Method note for future rounds: B8 was promoted and then built upon twice (ABB8, E1) before its uncertainty was ever estimated. Three of that round's four runs were spent following a result inside the noise. The noise floor cost one post-processing script and no node-hours.

17.2 Period offset, measured rather than estimated

`amip_presentday` is the same model and configuration run over 1989–2015, so the epoch offset is measurable directly instead of chained through observations (§11.3.1).

arm	Sib JJA T2m	vs ERA5 90–14	SO SW CRE	vs CERES	net TOA
control (1872–1915)	9.73	**−2.45**	−60.29	**+7.85**	0.64
presentday (1990–2014)	10.14	**−2.04**	−60.36	**+7.78**	2.20
measured offset		**+0.42**		**−0.07**	+1.56

Both 07–29 corrections were too large. The boreal offset is +0.42 K, not +1.12, so the period-clean bias is **−2.04** K; the SO SW CRE offset is −0.07, not −1.43, so that gap is **+7.78** — essentially the original +8.08, and the "≈ 6 W/m^2" claim is withdrawn.

The real finding is underneath the error. The model warms Siberian JJA by only +0.42 K between the epochs where observations imply ≈ +1.1 K, so **the model under-warms the historical period by ≈ 0.7 K** — a *trend* error distinct from the mean-state bias. Taking observations as correct for the epoch change puts the 1870s bias at ≈ −1.3 K; taking the model's own offset puts it at ≈ −2.0 K. The honest target range is **−1.3 to −2.0 K**.

Caveat: `presentday` global net TOA is +2.20 against CERES's +0.97. Aerosols were verified correct (MACv2-SP transient, years 1989→2002 logged), so this is a genuine radiative bias rather than a missing-forcing artefact, consistent with the SO cloud deficit being identical across epochs.

17.3 The runs were never 1850 GHG — a dead namelist block

`NCMIPFIXYR: 1850` never took effect in any `oifsamip` run. All 24 runscripts hand-wrote a `&NAMECECMIP6` block; OpenIFS 48r1 renamed that namelist to `&NAMECECMIP` (`ece_cmip.F90`:68 `POSNAM`s `'NAMECECMIP'` only, and `NAMECECMIP6`/`SSPNA` appear *nowhere* in the source). The legacy block survives in esm-tools' `fort.4` template, so `f90nml` patched it without complaint and the generated namelist looked correct — while the model read only `NAMECECMIP` and left `NCMIPFIXYR` at its default −1. Since `LFIXYEAR = NCMIPFIXYR > 0`, forcing was **transient**. From the model's own output:

```
UPDRGAS: greenhouse gas concentrations for YEAR/MONTH  1870/01
CO2 = 286.94 -> 287.12 -> 287.31 -> 287.49 ...   (rising through the run)
SP_SETUP: IYR = 1870, 1871, 1872, ...            (MACv2-SP aerosols advancing)
```

So every run is a transient CMIP7 *historical* run at its actual calendar year, Krakatoa (1883: −0.98 K in Siberian JJA) included, not a fixed-1850 PI run.

The tuning results are unaffected: all runs share that identical forcing trajectory, the perturbations live in `NAMCLDP`/`NAMVDF`/source which *are* read, and the ANOVA absorbs the shared trend and the volcanoes in g[year]. **The absolute targets are affected**, since they are all read off the control: CO_2 drifts 287 → 301 ppm (≈ +0.15 W/m^2 mid-run vs 1850, partly offset by aerosols) and the volcanoes depress the 44-yr mean by ≈ 0.05 K. Small — but the global TOA target is only +0.53 W/m^2, so ≈ 0.1 W/m^2 of spurious forcing is not negligible *there*. `amip_presentday` is unaffected: `historical` is *correct* for 1990–2014.

The fix is the setup's own switch, not a hand-patch: `oifs.scenario: "piControl"` routes through `configs/components/oif` which writes `NCMIPFIXYR` into the live block and sets the CMIP7 stratospheric-aerosol switches in `NAERAD`. The dead block has been removed from all 24 runscripts and `amip_picontrol` is running as a true PI reference; `amip_pi_base` is deliberately untouched because all 19 lever deltas reference it. **The CMIP7 production spin-up is not affected** — its `NCMIPFIXYR`= 1850 is in the live block and CO_2 is pinned at 284.30 ppm at calendar year 2018. *Residual code gap:* nothing pins MACv2-SP tropospheric aerosols (`updtim.F90`:812–818 takes the calendar year unconditionally).

17.4 Can the levers be combined? Not as they stand

Targets from the 44-year control, and the significant per-lever responses:

	SO SW CRE	Sib JJA T2m	global TOA	tropics TOA
required Δ	**−7.85**	**+1.3** to **+2.0**	**−0.64**	**+2.50**
A1a `ovl`= 0.10	**−6.46**	−0.749	−1.90	−1.07
A1b `ovl`= 0.35	−2.17	−0.135	−0.58	−0.29
B5	−0.23	**+0.407**	−1.00	**−1.94**
B2	+0.11	+0.344	+0.39	+0.65
B3	+0.67	+0.306	+1.34	**+2.02**

Note the tropics are 2.5 W/m^2 *too low*, not too high, so B5's −1.94 is a real cost and only B3 helps there. Added linearly:

stack	SO	Sib	global TOA	tropics
A1b+B5	−2.40	+0.27	−0.94 ✗	40.4 ✗
A1b+B2+B5	−2.30	+0.62*	−0.55 ✗	41.0 ✗
A1b+B2+B3+B5	−1.63 ✗	+0.92*	+0.79 ✗	43.0
A1a+B2+B3+B5	**−5.91**	+0.31 ✗	−0.53	42.3 ✗

*optimistic: AB measured −0.053 K where additivity predicted +0.21, so A1b *cancels* B2's boreal gain.

Two structural walls, not bad luck.

- **The Southern Ocean is reachable only through A1-strength mixed-phase overlap, which cools Siberia.** That is the same process on both ends: more supercooled liquid brightens SO cloud *and* boreal mid-level cloud. A1a buys 82 % of the SO target and spends −0.75 K of Siberia doing it.
- **Siberia's levers are too small and do not add.** The three significant ones sum to +1.06 K against a +1.3–2.0 K need, before the non-additivity that AB demonstrated.

The LPJ-GUESS fallback is closed: §11.3.2 showed the forest is 75–85 % temperature-driven, so prong B cannot absorb a boreal residual the atmosphere does not deliver. The boreal problem must be solved in OpenIFS.

18 Round 11 design (2026–07–30): attack the mechanisms, not the parameters

Round 10 closed on two structural walls (§17.4). Round 11 therefore targets the underlying mechanisms. Two proposals were written and then **withdrawn for being tuning-to-fit**, and recording why is part of the result:

- ~~Land/sea split on RCL_OVERLAPLIQICE.~~ **Anti-physical.** That parameter is the sub-grid *overlap fraction of supercooled liquid with ice* (range $[0, 1]$, default 0.65, EC-Earth4 uses 0.1), entering $\texttt{ZDEPOS} = \max(\texttt{ZOVERLAP_LIQICE} \cdot \texttt{ZA} \cdot (\texttt{ZINEW} - \texttt{ZICE0}), 0)$. The code's own comment states the intended selectivity — *"Reduce in shallow convection because assume SLW in active updraught is less overlapped with ice in less active part"* — i.e. convective segregation *lowers* overlap. But ocean mixed-phase cloud is predominantly stratiform (well mixed, so *high* overlap) and land cloud more convective (segregated, *low* overlap), the **opposite** of what the tuning wants.
- ~~RCAPDCYCL mode 3 = disable the CAPE diurnal-cycle correction over land.~~ **A regression.** That correction is Bechtold et al. (2014), which exists to fix the documented error of land convection triggering too early in the day.

18.1 D1 — RCAPDCYCL mode 4: reformulate the land closure

B5 was misdiagnosed. RCAPDCYCL is a *mode selector*, not a magnitude (cumastrn.F90:773,777): 1.0 = land only, scaled on ZKHVFL (surface kinematic virtual heat flux, :464); 2.0 = everywhere, land via $\texttt{ZCAPPBL} \cdot \texttt{ZTAU}/\texttt{ZTAURES}$ and ocean via a wind-based ZTAUPBL; 0.0 = off entirely. So B5's 2.0 → 0.0 **deleted the correction globally**: its +0.407 K boreal gain came from the land branch, its $-1.94\,\mathrm{W/m^2}$ tropical cost from losing the ocean branch. The two are separable.

The physical argument is in the scaling. Mode 1 is proportional to the *actual surface buoyancy flux* and vanishes where that forcing is weak; mode 2's land branch uses boundary-layer CAPE instead. Boreal summer has weak insolation, a shallow boundary layer and small surface fluxes, so the buoyancy-flux form self-limits while the BL-CAPE form need not — a real reason to suspect **mode 2's land branch is mis-scaled for weak-forcing high-latitude convection**, and it predicts the observed sign. **Mode 4** (added this build) is mode-1 land physics with mode-2's ocean branch verbatim; both are existing code. *Prediction:* recovers part of +0.407 K with little of the tropical cost. If the boreal gain vanishes too, the effect was never about the land closure formulation and the B5 result needs re-reading.

18.2 D2 — ice nuclei, not overlap

The Southern-Ocean supercooled-liquid deficit is an **ice-nucleation** problem, and the scheme has no aerosol dependence whatsoever:

$$\texttt{ZICENUCLEI} = 1000 \cdot \exp(12.96 \cdot \texttt{ZSUPERSATICE} - 0.639) \qquad (\texttt{cloudsc.F90} : 2516,\ : 2618)$$

That is **Meyers et al. (1992)** — a function of ice supersaturation *only*, documented to overestimate INP in clean environments, which is precisely the remote Southern Ocean; DeMott et al. (2010/2015) make INP a function of aerosol number instead. Reducing INP over ocean therefore has a real mechanism (INP sources are continental: mineral dust, biological) with the correct sign in both regions at once, and strong leverage since $\texttt{ZCVDS} \propto \texttt{ZICENUCLEI}^{0.666}$ — a factor 0.1 cuts deposition to ≈ 0.22.

Three better routes are closed. (i) PNICE is already a cloudsc argument (:265) and already used for autoconversion (:2855), but it is filled only when NAERCLD > 0 (callpar.F90:1155, else zeroed at :1166) and NAERCLD defaults to 0; that configuration is several times slower and **would not finish CMIP7 in time**. (ii) onlinedust_v4_rmp.nc is staged and *opened* unconditionally, but its arrays (POTSRC, SOILTYPE, CULT) are

allocated only by TM5M7_SRC_DUST_INIT via tm5m7_init.F90 — **only under m7** — so its contents are never populated here (suecrad's YDUSTCLIM is a different object, dust *optical properties* for the radiation climatology). (iii) MACv2-SP plumes are anthropogenic sulphate/BC/OC: poor INP, sulphate coating *deactivates* mineral dust INP, and in a piControl anthropogenic aerosol is near zero, giving no land/ocean contrast.

18.2.1 Two physical caveats that shape the test

The tropics will pay, and "ocean" is the wrong discriminator. The scaling acts only inside the mixed-phase window (RTHOMO $\leq T <$ RTT $- 5$, i.e. 235.15–268.15 K, with liquid present), which in the tropics sits at $\approx$ 500–300 hPa and *is* populated — deep convective updraughts carry supercooled liquid well above the freezing level, and anvil detrainment feeds it. Reducing INP there retains more supercooled liquid, brightening tropical mid-level cloud and pushing tropical TOA *further below* CERES. Expect a cost of order A1a's $-1.07\,\mathrm{W/m^2}$ or worse. More fundamentally, tropical and subtropical oceans are *not* dust-remote (Saharan dust over the tropical Atlantic, Asian dust over the North Pacific); the Southern Ocean is uniquely far from continental sources. The property that distinguishes it is **remoteness**, which a local land-fraction mask cannot encode.

Sea salt sets an INP floor. Sea salt itself is a poor INP (highly soluble, inefficient in the immersion mode at these temperatures), but sea spray carries **marine biogenic organics** — sea-surface microlayer material, phytoplankton exudates — which are a modest INP source (Wilson et al. 2015; DeMott et al. 2016). So the ocean INP factor **must not approach zero**: ≈ 0.1–0.3 is defensible, not 0.01. Sea salt's larger role is as CCN, which is moot here since PCCN is zeroed with NAERCLD $= 0$ and droplet number is the constant RCL_KK_CLOUD_NUM_SEA. A tempering caveat: marine biogenic INP peaks with the austral-summer bloom (DJF), exactly when SO insolation and our SW bias peak, so the SO is *least* INP-starved in the season we care most about.

18.3 The three runs

All on the same forcing as the 19 round-10 levers (oifs.scenario defaults to historical), so deltas are directly comparable to B5's; 46 yr, evaluated 1872–1915.

run	change	purpose
D1	RCAPDCYCL= 4 (source + namelist)	separate B5's boreal gain from its tropical cost
D2a	ZICENUCLEI $\times f$(PLSM), no remoteness	does the INP route have leverage at all?
D2b	tile-based two-term INP + remoteness gate	the physically honest version

D2a is run *knowing* it should damage the tropics: the question it answers is whether INP has Southern-Ocean leverage, which decides whether D2b is worth building. D2b computes the INP field in callpar from SURFL%ZFRTI (tile fractions, already used at callpar.F90:2006) and PAUX%PGEMU (sine of latitude, available at :1801), passing one new array to cloudsc:

$$\mathrm{INP} \;\propto\; a \cdot [\text{bare soil (tile 8)} \times \text{wind}] \;+\; b \cdot [\text{low veg (4)} + \text{high veg (6)}]$$

Tile convention from srfi_mod.F90:52. The two terms matter: a *dust-only* proxy would give the **wrong sign** over boreal forest, which is a weak dust source — lowering its INP would brighten boreal cloud and cool it further. Vegetated land is instead a strong **biological** INP source in summer (pollen, bacteria, fungal spores, litter), peaking in exactly our season, while snow tiles (5/7) collapse that term in winter and leave the cold-season bias untouched.

All three are source edits on the single oifsamip-cy48 tree and therefore **serialise**: edit, esm_master recomp-oifsamip-cy48, verify the binary md5 changed, submit, wait for staging, repeat.

19 Round 11 results: D2b adopted for the Southern Ocean, D1 falsified

	SO SW CRE	SO SW RMSE	Sib JJA T2m	global TOA	tropics
control	−60.29	6.88	9.73	0.64	42.61
piCTRL 1850	−60.65	6.83	9.74	**0.79**	42.69
D1 RCAPDCYCL= 4	−60.38	6.51	**9.56**	0.61	42.53
D2a INP, no gate	**−65.13**	5.07	9.56	−0.28	42.23
D2b INP + p700	−62.93	**4.96**	9.51	0.36	**42.70**

D2b is adopted as the Southern-Ocean lever. It gives the best SO surface-SW RMSE of any run (6.88 → 4.96, −28 %) at *zero* tropical cost (+0.09), beating both A1b (5.54) and the ungated D2a (5.07) — the latter despite

a larger mean-CRE change, because D2b improves the spatial *pattern*, which is what the priority metric exists to measure. The ice-nucleation hypothesis (§18.2) is therefore supported, and the pressure gate did what it was designed to do: D2a's tropical cost (−0.38) vanishes (+0.09) while 55 % of the SO gain is retained.

D1's prediction is excluded, but as a NULL, not a sign reversal. It was predicted to recover part of B5's +0.407 K by switching the land CAPE closure to mode-1's buoyancy-flux scaling; it delivered **−0.170** K — which does *not* clear the ±0.242 K detection floor (§17.1), so the predicted recovery is excluded but the sign is unresolved. Quoting it as "the wrong sign" was the very error §17.1 exists to prevent. So B5's boreal gain requires *removing* the land correction, not rescaling it — the effect is not about which quantity the closure scales on. Stating the prediction in advance is what made the refutation unambiguous.

`amip_picontrol` settles two things. The energy target is +0.79 W/m^2, not +0.64 — slightly *larger*, so A1b does not overshoot. And Siberian JJA T2m differs by only +0.008 K, so the dead-namelist forcing bug (§17.3) never affected the boreal target and every round-10 boreal conclusion stands.

20 Round 12: boreal surface exchange, and the best lever of the campaign

Round 11 left the boreal unsolved. Round 12 changed target from convection and cloud to **boreal surface exchange**, anchored on one observation: the measured Siberian JJA Bowen ratio is **0.43** against ≈ 0.5–1.5 observed for boreal conifer, so the model puts too much surface energy into latent heat. Four levers, each raising Bowen by an independent route, all indexed on vegetation types 3/4 so that types 5/6/18 — and hence the tropics — are untouched *by construction*. That is exactly the property B5 lacks.

run	change	ΔT2m	t	verdict
F4	`RVRSMIN(3,4)` 250 → 1000	**+0.749**	+6.06	**significant, best ever**
F5	all four combined	**+0.746**	+6.03	significant, ≡ F4
B5	(previous best)	+0.407	+3.29	significant
F2	`RVLAI(3,4)` 5 → 3	+0.321	+2.59	significant
F1	`RVZOH(3,4)`=`z0m`/10	+0.185	+1.49	marginal
F3	`RVCOV(3,4)` 0.9 → 0.7	≈ 0	—	noise

F4 costs the tropics nothing (42.68 vs control 42.61) where B5 wrecked them (40.67), and it gives the best global T2m RMSE of any run (1.54). Bowen moves 0.44 → 0.60, into the observed range *without* overshooting — the risk named in advance did not materialise.

F5 equals F4 to 0.003 K. Against a naive sum of ≈ 1.25 K, adding F1, F2 and F3 on top of F4 contributes nothing. The advance prediction — F5 ≪ $\sum$, closer to the largest single lever — held, and more strongly than expected: near-total saturation of a single shared latent-heat pathway with a floor at zero. *Use F4 alone.*

A prediction that failed. F1 was called strongest a priori because $kB^{-1} = 0$ is the largest departure from observation (every closed-forest type sets $z_{0h} = z_{0m}$, i.e. $kB^{-1} = 0$, where observations give $kB^{-1} \approx 2$). It is marginal and barely moved Bowen (0.44 → 0.43). "Largest departure from observation" is not a reliable guide to leverage.

B5 carries a hidden −0.72 K DJF cooling (F4: +0.06), invisible for the whole campaign because only JJA was ever evaluated. Given that the coupled model's original complaint is a *cold-season* bias, that is disqualifying for B5, and DJF now belongs in the standard guardrails.

20.1 Is F4 fixing a cause or a symptom?

Two tests, both partly exonerating. *Seasonal selectivity:* DJF +0.061, MAM +0.047, JJA +**0.749**, SON +0.110 — exactly the signature a transpiration mechanism must have, since stomata are shut and the canopy snow-covered in winter. *Vertical targeting:* F4 warms 1000/925/850 hPa by +0.83/ + 0.84/ + 0.83, closing 56–83 % of the *Siberia-specific* layer bias (§21.1), and only +0.53 aloft.

What remains true is that `RVRSMIN` 250 → 1000 is an unanchored 4× excursion from what ECMWF ships, and that the vertical profiles show it removes moisture that is already deficient. **G1 = F4 + D2b** tested this: the first pairing whose components are disjoint in both process and geography, hence expected not to conflict — but AB and ABB8 both got superposition wrong in sign, so it was measured, not assumed. It superposes (§20.2).

20.2 G1 = F4 + D2b: the best configuration of the campaign

G1 is the first combination to improve *both* targets at once, and the first whose superposition is additive rather than wrong in sign.

metric	control	G1	Δ	target
SO SW RMSE (priority)	6.877	**4.809**	−2.067	—
SO TOA SW CRE [W m^{-2}]	−60.29	−63.13	−2.84	−68.14
SO cloud area [%]	83.07	83.59	+0.52	89.72
Siberia JJA T2m [°C]	9.73	**10.25**	**+0.521** (t=4.22)	≈ 12.2
Siberia sfc net SW [W m^{-2}]	153.78	159.31	+5.54	166.26
global net TOA [W m^{-2}]	+0.64	+0.45	−0.195	∼ 0
tropics net TOA [W m^{-2}]	42.61	42.71	+0.096	45.11
global T2m RMSE [K]	1.579	1.553	−0.026	—
Nordic Seas SW RMSE	9.058	9.358	**+0.300**	—

Superposition, measured: Siberia JJA T2m F4 +0.749 + D2b −0.222 = +0.527 predicted vs **+0.521** actual; SO SW RMSE −2.131 vs −2.067; SO CRE −2.719 vs −2.840; global net TOA −0.188 vs −0.195. Additive to within noise on every metric. The plausible reason it holds here and failed for AB/ABB8: F4 acts on a land surface flux in boreal summer, D2b on ice nucleation over ocean below 700 hPa, whereas AB and ABB8 combined levers modifying the *same* cloud scheme in the *same* regime.

The one cost is the Nordic Seas, +0.300 SW RMSE against a ±0.052 threshold, and worse than additive (+0.184 expected). No other lever in the campaign has ever moved Nordic RMSE; this one moves it the wrong way. Small box, not the deep-water priority, but it must not be allowed to grow.

G1 closes 26 % of the boreal bias and 36 % of the SO CRE gap. The SO *cloud-area* deficit of ∼6 pp remains untouched by every lever tried.

20.3 Seasonal audit: the noise floor is not the same in every season

The campaign evaluated JJA only for eleven rounds, while the coupled model's original complaint is a *cold-season* bias. `scripts/analysis/seasonal_by_run.py` applies the run×year ANOVA of §17.1 separately to each season. The detection threshold varies by a factor of 2.4 across the year:

season	control	sd(ϵ)	95 % threshold
DJF	−29.35 °C	1.407 K	±0.588 K
MAM	−9.28 °C	0.923 K	±0.386 K
JJA	+9.73 °C	0.580 K	±0.242 K
SON	−10.41 °C	1.031 K	±0.431 K

Applying the JJA threshold to a winter delta overstates significance by 2.4×, and a first pass through these results did exactly that, briefly recording D2b (−0.455) and B3 (−0.486) as winter-damaged. Both are *within* the DJF noise floor. Corrected:

- **B5 `capdcycl0` is the only lever with genuine winter damage** (DJF −0.720, clears ±0.588) and the only one that significantly warms JJA while significantly cooling DJF. Its rejection stands, now properly thresholded.
- **D2b's real seasonal cost is spring, not winter**: MAM −0.467 clears ±0.386; its DJF −0.455 does not clear ±0.588.
- **F4 is seasonally clean** — DJF +0.061, MAM +0.047, SON +0.110, all within noise, the entire signal in JJA. A further argument for F4 over the cloud levers.
- **G1 is seasonally clean**: DJF +0.070, MAM −0.296, SON +0.280, none significant; D2b's MAM cooling is diluted below threshold.
- A1a warms DJF/SON and cools MAM/JJA by large amounts — a global cloud change, not a boreal lever. B6, B7 and F3 cool MAM significantly; F5 inherits F3's MAM −0.507.

21 Vertical structure vs ERA5: where the biases live

21.1 The Siberia-specific cold bias is confined below 850 hPa

Model minus ERA5 on the model's 19 pressure levels, `amip_presentday` vs ERA5 1990–2014 so the comparison is period-clean. ERA5 comes from the DKRZ pool; all 19 model levels exist exactly in ERA5's 37, so no interpolation is needed.

hPa	Siberia	Global	**Siberia-specific**
1000	−1.90	−0.73	**−1.17**
925	−2.17	−0.65	**−1.52**
850	−1.99	−0.99	**−1.00**
700	−1.22	−1.15	−0.07
500	−1.67	−1.49	−0.18
300	−2.32	−2.22	−0.10

This separates two problems that had been conflated. Above 700 hPa Siberia merely shares a **global tropospheric cold bias of** 0.7–2.9 **K** that no boreal lever will touch; the boreal-specific excess lives entirely in the bottom three levels, which is the signature a surface-exchange problem should have and which validates the F-series framing.

The excess boreal cloud is thermally driven, not moisture-driven. Siberian RH is +5.6 to +8.2 % too high while q is *too low* (−0.2 to −0.3 g/kg): a 2 K cold bias cuts q_{sat} by ≈ 14 % while q falls only ≈ 6 %. So a large F-response must not be read as confirming the cause.

Independent support for D2: the Southern Ocean has RH +5.8 to +6.6 % too high through the mixed-phase layer while reflecting *too little* SW — humidity present, liquid absent, exactly the supercooled-liquid deficit the INP hypothesis predicts.

Broken diagnostic, recorded: the model's pressure-level relative humidity `r` is **identically zero in every run checked** (0 nonzero of 17 510 400 values) while `q` is fine. The XIOS `r` output yields silent garbage and must not be used; RH above is recomputed from `t` and `q`.

21.2 Extending the column downward: the soil is the least-biased part

§21.1 stops at 1000 hPa. Continuing into the skin and the four soil layers (`scripts/analysis/vertical_bias_column.py`), JJA, model `amip_presentday` minus ERA5. Low-lying land (< 500 m) is shown alongside because the all-land figure is inflated by orographic mismatch (§21.3):

JJA	global land	land < 500 m	Siberian box
700 hPa	−0.83	−0.89	−1.22
850 hPa	−0.79	−0.92	−1.99
925 hPa	−0.82	−1.02	−2.17
2 m	−1.34	**−1.00**	−2.04
skin	−0.83	**−0.53**	−1.66
soil L1 (7 cm)	−0.40	**−0.36**	−1.73
soil L2 (28 cm)	−0.42	−0.38	−1.90
soil L3 (1 m)	−0.78	−0.72	−2.14
soil L4 (2.9 m)	−1.09	−0.89	−1.02

Over ordinary low-lying land the **soil and skin are the least-biased parts of the entire column**, and 2 m sits about 0.5 K *below* the skin under it. The ground is not what holds the near-surface cold; if anything it is doing better than the air above it. Siberia is the exception that §21.1 already identified — there the soil is cold too (−1.7 to −2.1 K), which is the surface-confined excess the F-series was built to attack.

Caveats. The 1000 hPa row is unreliable over land, where much of that level is below ground and both datasets extrapolate; 925/850 are the lowest honest levels. ERA5's soil and skin are HTESSEL output, so those rows are model-versus-sibling rather than observation — the 2 m and pressure-level rows are the observation-constrained ones.

21.3 The near-surface cold is not boreal: every surface type is cold

The campaign has only ever scored one box, so it had never checked whether its levers address a local fault or a global one. Resolving the 2 m bias by dominant surface type (`scripts/analysis/bias_by_tile.py`) gives an unambiguous answer: **all 20 surface types are cold, in every season. Not one is warm.**

dominant type	area	DJF	MAM	JJA	SON
bare soil	25.3 %	−1.86	−2.12	−2.16	−2.03
Evgr Broadleaf (tropical)	8.7 %	−1.78	−1.79	−1.46	−1.42
Tundra	5.2 %	−2.51	−0.18	−1.43	−1.84
Evgr Needleleaf	4.7 %	−2.00	−1.84	−1.16	−1.05
Dec Needleleaf	3.0 %	−1.22	−1.55	−1.87	−1.69
ALL LAND	100 %	**−1.73**	**−1.60**	**−1.35**	**−1.39**

Two controls bound how much of this is real. The land figure is *elevation-dependent* — −0.70 K below 200 m rising to −2.43 K above 2000 m, the expected signature of the model's smoothed orography against ERA5's 25 km — so the all-land number overstates it and the low-lying value is the honest one. And over **prescribed-SST ocean** 2 m is −0.72 K: about 0.7 K of near-surface cold exists where the land surface is absent entirely.

Consequence for the campaign. Siberia's JJA −2.58 K decomposes roughly as ∼0.7 K universal, ∼0.3 K orographic, and 1.3–1.6 K boreal-specific. G4 has taken ∼0.95 K of that boreal budget — which is a far better explanation of the F-series saturation (F5 = F4 to within noise, §20) than any parameter limit. *The boreal budget is nearly spent, and the remainder is not reachable by a boreal-indexed lever.*

One benign consequence. Because every type is cold, a lever indexed by vegetation type does not damage the regions it leaks into: the non-boreal areas of types 4 and 9 are cold too (JJA −0.68 and −1.06). That is what licenses the round-17 restriction to those two types (§28).

22 Surface albedo: half the Siberian SW deficit is not cloud

Every boreal lever so far has attacked cloud. Decomposed against CERES on the identical box and mask, the 12.5 W/m^2 Siberian JJA surface-SW deficit is only half cloud: **7.0** W/m^2 from too little SW reaching the surface, and **5.5** W/m^2 from **surface albedo too high** — 0.1753 model versus 0.1461 CERES, entirely independent of cloud. F4 does not touch it (0.1749).

22.1 Not a global albedo error, and not the forest

Temperate and tropical land is essentially perfect (Europe, Amazon, Great Plains, steppe, Australia all within ±0.004), so the albedo *scheme* is sound and a global fix would be wrong. *This held up, with one amendment.* §29.1 later measured land albedo globally and by type with snow masked per cell per month: every forest type is inside +0.006 and tropical broadleaf is at −0.0003, confirming the scheme — but a residual +0.0080 does exist on crops, semidesert and bare soil, so "a global fix would be wrong" is right about the *vegetation table* and too strong about the *soil background*. The error is concentrated at high latitude: Siberian tundra +0.094 (−16.0 W/m^2), Canada boreal +0.032, Siberia boreal +0.029, Sahara +0.020 — but **Fennoscandia, the same needleleaf type, is correct at** +0.002. Splitting Siberia by latitude gives +0.011 in the forest core and +0.087 at 70–75N.

22.2 How the albedo is built, and two hypotheses that died

All configurations — AMIP, coupled round-09, *and the CMIP7 production piControl* — run `LRDALB = .false.` with `NALBEDOSCHEME = 3`, which **discards the MODIS albedo present in the input file** and constructs albedo each step (`surfbc_ctl_mod.F90`:512) as $\alpha = \texttt{RVVEGALB}(t_{vl})c_{vl} + \texttt{RVVEGALB}(t_{vh})c_{vh} + \alpha_{soil}(1 - c_{vl} - c_{vh})$.

~~Hypothesis 1: the soil-albedo field dominates tundra.~~ The bare fraction is only 0.01–0.03, so the soil term is negligible. ~~Hypothesis 2: the soil fields are corrupt.~~ GRIB 117–120 do carry an inconsistent header (`N=128` with a 256-row `pl` but only 40 320 values, the O96 count), which **breaks `cdo` and `grib_get_data`** on those fields and should be reported upstream — but the values are O96-ordered and correct: zero over 99.2 % of ocean cells, $\mathrm{corr}(\alpha_{soil}, \mathrm{lsm}) = +0.894$. Not our bias.

22.3 The real signature is snow-melt timing

The blend at 70–75N yields 0.158 while the model runs at 0.303; the extra ≈ 0.145 is snow. Monthly, the model is **too dark in April** (−0.06 to −0.07), **near-perfect in July/August** when snow-free (+0.004), and **too bright in June** (+0.086 to +0.116) **and September** (+0.08 to +0.10). So `RVVEGALB` is essentially right and the *snow season is too long at both ends.*

Cause and effect. Through February–April the model is too cold *and* too dark simultaneously; if bright snow caused the cold those signs would agree, so the winter cold is **not** albedo-driven. The albedo bias turns positive only in June, after the surface crosses freezing. But the June snow is a genuine defect in its own right: **June is**

+6.2 °C with 213 W/m² of insolation and 0.020 m w.e. still lying, and the implied snow-cover fraction is ≈ 17 % where CERES implies ≈ 2 %. That points at the **snow-cover-fraction formulation**, not snow albedo (too *low* in April) and not melt energy (ample).

22.4 Growing season: June is albedo, July is cloud

Siberia 55–75N, W/m²	Jun	Jul	Aug	JJA
cloud	6.9	**9.9**	4.4	7.0
albedo / snow	**13.4**	1.7	1.3	5.5
total	20.2	11.6	5.7	12.5

Almost cleanly separated in time. The residual snow term is 5.5 W/m² on the JJA mean, essentially all in June, worth ≈ 0.2–0.7 K depending on which sensitivity applies — potentially another F4-sized gain, and *orthogonal* to F4, which fixes SW-down while this fixes the absorbed fraction.

But albedo does not explain the land cold bias. Insolation-weighted over land 60S–75N excluding Greenland, the mean albedo-induced SW loss is −3.00 W/m² against a mean T2m bias of −1.35 K, with spatial correlation $r = +0.145$ and slope +0.033 K per W/m². Decisively, **cells where the model is too *dark* are still −1.16 K cold**. A ≈ 1.2 K land-wide cold bias therefore has no albedo signature at all, consistent with the global tropospheric cold bias of §21.1. Fixing albedo is worth doing; it is not the root cause.

23 Round 13: snow cover fraction, the June lever

23.1 Design and prediction

The albedo decomposition (§22) splits the 12.5 W m^{-2} Siberian JJA surface-SW deficit into ∼7.0 cloud and ∼5.5 **surface albedo**, with the albedo half concentrated almost entirely in **June** (§22.4; bias +0.086, July and August near-perfect). That is a snow-*melt-timing* signature, not a snow-*brightness* one. Cause and effect were established before touching anything: winter cold is *not* albedo-driven, since February–April is simultaneously cold and dark (§22.3), which rules out the albedo → cold direction for those months.

Snow cover fraction (`surfbc_ctl_mod.F90:317-322`) is

$$\mathrm{ZCVS} \;=\; \min(1,\; d_{\mathrm{snow}}[\mathrm{cm}] \cdot \texttt{RQSNCR}), \qquad \texttt{RQSNCR} = 1/10 \;\text{ under }\; \texttt{LESNO9=T}.$$

A grid box is therefore fully snow-covered at **10 cm**, linearly below that, with no dependence on vegetation height or orographic roughness. For boreal forest this is wrong in an obvious direction: 10 cm of snow does not hide a spruce canopy, and the observed masking depth over forest is several times larger. The H-series sets `RQSNCR` = 1/30.

Why the ice sheets are safe: Antarctica and Greenland carry snow far deeper than 30 cm, so $\min(1, \cdot)$ saturates at 1 for any value in this range and they are untouched. Only shallow, marginal, melting snow responds — which is the target.

H1 applies the change alone against control (clean attribution); **H2** applies it on top of G1's F4 source change and D2b namelist (the deployable stack). H2 − G1 measured against H1 − control gives the interaction directly.

Falsifiable prediction, recorded before the runs finished: ~~H1 recovers a good part of June's 13.4 W m^{-2} albedo term, worth ∼ + 0.2 to +0.7 K on Siberian JJA T2m, and does little in July and August.~~ **Falsified** (§23.2): JJA came out at +0.020 K, $t = 0.17$. The failure mode was not the anticipated one — it did not warm July as much as June, it failed to warm *any* summer month.

23.2 Results: prediction falsified, lever rejected

Predicted +0.2 to +0.7 K on Siberian JJA T2m; measured +0.020 K, $t = 0.17$**.** And the lever inflicts the largest winter cold bias in the campaign, −1.233 K DJF against a ±0.588 threshold, beating the −0.720 for which B5 was rejected.

	DJF	MAM	JJA	SON
H1 `snowcr30`	**−1.233***	−0.473*	**+0.020**	+0.428
H2 = G1 + snow	−0.973*	−0.342	+0.662*	+0.863*
(G1, reference)	+0.070	−0.296	+0.521*	+0.280

Why it failed: the mechanism was pointed at the wrong month. The lever did exactly what it was built to do to the albedo — but not where the reasoning assumed. H1 all-sky surface albedo change (×100, * = clears that month's threshold):

	Jan	Feb	Mar	Apr	May	**Jun**	Jul	Aug	**Sep**	**Oct**	**Nov**	Dec
H1	−0.51*	−0.21	−0.42*	−0.84*	−1.43*	**−0.95***	−0.10*	−0.32*	**−3.08***	**−8.16***	**−4.66***	−1.60*

The response concentrates in **September–November**, the *snow-onset* season, and is small in June. The design error is plain in hindsight: ZCVS = $\min(1, d_{\mathrm{cm}} \cdot$ `RQSNCR`$)$ can only respond where snow is **shallow**, and snow is shallow while it is *accumulating in autumn*, not while it is melting in June — by June the pack either still saturates $\min(1, \cdot)$ under both settings or is already gone. **Shallow snow is an autumn phenomenon, not a melt phenomenon.** Consistently, June/July/August T2m move $-0.076/-0.067/+0.205$ (none significant) while **October T2m is** $+1.135$ **K**.

The winter cooling is not radiative. February and March are dark yet cool by −1.65 and −1.19 K. Reducing the snow-covered fraction moves area from the exposed and sheltered **snow tiles (5, 7)** onto the vegetation tiles, and the snow tile is what buffers the surface against radiative cooling in polar night. The lever trades a small autumn warming for a large winter cooling through **tile fractions, not albedo**.

Verdict: rejected, and `RQSNCR` reverted. Three things survive:

1. The June surface-albedo bias is **not** dominated by the snow-cover-fraction formulation. Its cause — snow-albedo decay, vegetation masking of snow albedo, or the snow-free surface albedo itself — remains unidentified.
2. `RQSNCR` has large leverage on *autumn* albedo (−8 points in October). If an autumn or snow-onset bias is ever diagnosed, this is the knob.
3. H2 posts the campaign's best SO SW RMSE (4.787), best subpolar N Atl (4.713) and best global T2m RMSE (1.530) — but that is D2b's doing, and the winter damage disqualifies the combination. **G1 remains the configuration to carry forward.**

24 Round 14 (2026–08–03): the June albedo bias is a SPRING MELT bias

Round 13 established what the June albedo bias is *not* (§23.2): not the snow-cover formulation. This section identifies what it is, entirely from output already on disk plus ERA5 — no new integration was needed to get here.

24.1 Against ERA5: too much snow, and snow that is too DARK

`scripts/analysis/albedo_decompose.py`, model `amip_presentday` vs ERA5 1990–2014, both land-masked to the Siberian box, June:

June	model	ERA5	Δ
surface albedo `fal`	0.2147	0.1730	**+0.042**
snow cover fraction	0.380	0.261	**+0.119**
snow albedo `asn`	0.658	0.759	**−0.102**

Both `f_snow` columns are the *same* HTESSEL formula $\min(1, d_{\mathrm{cm}}/10)$ applied to each dataset's snow mass, so the gap is **snow AMOUNT, not the cover formula** — which is precisely why round 13, which changed only the formula, could not move it. The model carries roughly twice ERA5's June snow mass.

Corroborated independently by CERES (§25.1), which puts the June excess slightly higher at +0.046; and the cover excess is larger still against satellite, +0.176 rather than +0.119 (§25.2).

And the model's snow is too dark, not too bright (−0.10 in June, −0.11 in May). Making snow albedo more realistic would make June *worse*. Any lever that brightens snow is pushing the wrong way; the excess reflection comes from snow *extent*, not snow *brightness*.

Two things this analysis does not claim. The naive two-term split $\alpha_{\mathrm{box}} = f\,\alpha_{\mathrm{snow}} + (1-f)\,\alpha_{\mathrm{free}}$ does **not** close: backing out α_{free} gives negative values over the box, because HTESSEL does not apply `asn` to the whole snow fraction — tile 7 is snow *sheltered* under high vegetation and carries a canopy-shaded albedo far below `asn`. That the residual is large and negative in exactly the melt months is direct evidence the canopy-masking term already

does most of the work there, so no snow-free albedo is reported. Separately, ERA5's albedo scheme is a relative of HTESSEL, so `asn` agreement is not independent evidence; only ERA5's snow *extent* is observationally constrained, via assimilated IMS snow cover.

24.2 Accumulation is right; the melt is a month late and 30 % too slow

Snow water equivalent over the box, model vs ERA5 [mm], with the month-to-month increment:

	model	ERA5	Δ	δ(model)	δ(ERA5)
Jan	89.8	82.3	+7.5		
Feb	106.5	97.6	+9.0	+16.7	+15.3
Mar	122.1	114.5	+7.7	+15.6	+16.9
Apr	131.6	112.8	+18.7	**+9.4**	**−1.6**
May	98.1	65.3	**+32.8**	**−33.5**	**−47.5**
Jun	14.1	9.6	+4.5	−84.0	−55.7
Nov	49.2	41.5	+7.7	+25.9	+25.4
Dec	70.7	63.8	+6.9	+21.5	+22.3

Accumulation is essentially correct. Through the accumulation season the model is only 7–9 % high and the monthly increments track ERA5 almost exactly (+16.7 vs +15.3, +15.6 vs +16.9, +25.9 vs +25.4, +21.5 vs +22.3). The divergence is entirely in spring:

- **April**: the model is still *gaining* (+9.4 mm) while ERA5 has already peaked and begun to melt (−1.6). **The snowpack peaks a month late.**
- **May**: the model melts 33.5 mm against ERA5's 47.5 — **30 % too slow** — leaving it with 50 % more snow by the end of May.

Both figures come from ERA5. The satellite record confirms the timing directly and makes it worse: melt-out is 13 days late, and June cover is nearly double observed (§25.2).

24.3 May is albedo, not cloud — and it is a runaway

Model `amip_presentday` vs CERES on the same box:

	SW_{net} mod	SW_{net} CERES	Δ	$SW_{\downarrow}$ mod	$SW_{\downarrow}$ CERES	Δ
Apr	87.8	86.6	+1.2	176.9	187.7	−10.8
May	122.5	136.4	**−13.9**	220.7	224.2	**−3.5**
Jun	167.6	189.7	−22.2	215.7	228.2	−12.5
Jul	165.9	180.4	−14.5	193.8	208.7	−14.9

In May the downwelling flux is nearly right (−3.5) while the net is −13.9: the May deficit is *almost pure surface albedo*, i.e. the excess snow reflecting. June is roughly half cloud (−12.5 of downwelling) and half albedo. July is essentially all cloud (−14.9 down, −14.5 net), consistent with §22.4.

This closes a **self-reinforcing loop**: late melt → more snow in May → higher albedo → $-13.9\,\mathrm{W\,m^{-2}}$ less absorbed → less melt energy → snow survives into June. It also explains why the model melts slower *despite* having darker snow than ERA5: the brightness term is working against the bias, and the extent term is winning.

24.4 The gap F4 leaves: tundra is the largest cover type in the box

Area-weighted vegetation cover over land in the evaluation box, from the model's own `tvh/tvl/cvh/cvl` output:

	type	cover	
low	9 Tundra	**25.6 %**	**F4 does not touch this**
high	4 Deciduous Needleleaf	19.2 %	F4
high	18 Mixed Forest	5.2 %	
low	13 Bogs/Marshes	3.9 %	
high	5 Deciduous Broadleaf	3.6 %	
high	19 Interrupted Forest	3.5 %	
high	3 Evergreen Needleleaf	3.4 %	F4
high	16 Evergreen Shrubs	3.0 %	

Tundra is the single largest cover type in the box, and the campaign's best lever misses it entirely. F4 reaches about 24 % of the box (types 3, 4); tundra is another 26 %.

Its stomatal resistance is a table-internal outlier. `RVRSMIN(9)` $= 80\,\mathrm{s\,m^{-1}}$ is the *lowest value of any vegetated type* in HTESSEL:

```
crops 100  short grass 100  tall grass 100  semidesert 150  decid. broadleaf 175
interrupted forest 175  evergreen shrubs 225  decid. shrubs 225  evergr. broadleaf 240
bogs/marshes 240  desert 250  needleleaf 250  mixed forest 250  TUNDRA 80
```

The model therefore has arctic tundra transpiring *more freely than tropical grassland.* Tundra's closest physiognomic analogues in the same table — evergreen and deciduous shrubs — are both 225, and bogs/marshes, which co-occur with tundra in this very box, are 240. This is a **table-consistency** argument, checkable against the table itself and independent of our bias; it is not a value chosen because it helps. Contrast the standing criticism of F4, whose 250 → 1000 remains an unanchored 4× excursion.

24.5 Round 14 runs, and their predictions

All three are **namelist-only** (§24.7) — one shared binary, no rebuild, no serialisation against the model tree. That is what the `NAMSURFTUNE` work bought.

- **G2** — `RVRSMIN(3,4)` = 500. Half-strength F4.
- **G3** — `RVRSMIN(3,4)` = 2000. Double-strength F4.

  G2/G3 bracket G1 and map the response curve, answering the standing criticism that 1000 was never chosen from a measurement. *Prediction:* G2 keeps > 60 % of G1's gain ($\geq +0.31$ K) ~~and G3 adds little ($< +0.15$ K beyond G1)~~. First half held (64 %); second half **failed** — G3 added +0.355 (§24.6). *The response is a ramp, which was flagged in advance as an argument* **against** *the approach, not for a larger number.*
- **G4** — G1 plus `RVRSMIN(9)` 80 → 225 (tundra), keeping 1000 for types 3 and 4. *Prediction:* +0.2 to +0.5 K beyond G1's +0.521 (held: +0.431), ~~accompanied by a measurable drop in May/June snow water equivalent~~ — **falsified**, §24.6. As stated in advance, **T2m rose with no change in May–June snow mass, so the melt mechanism is unsupported** and the gain is the plain sensible-heat route. Guardrails: tundra is confined to high latitudes so tropical net TOA should stay near 42.6; check DJF against its own ±0.588 K threshold and the Nordic Seas SW RMSE that G1 already degraded.

24.6 Round 14 results: G4 adopted, mechanism falsified, `RVRSMIN` capped

G4 is the best configuration of the campaign and closes 48 % **of the boreal bias**, against G1's 26 % — and its stated mechanism is falsified. Both matter.

metric	control	G1	G2 (500)	G3 (2000)	**G4**
Siberia JJA T2m [°C]	9.73	+0.521	+0.336	+0.876	**+0.952** (t=7.68)
Siberia sfc net SW	153.78	+5.54	+3.21	+7.27	**+8.10**
SO SW RMSE	6.877	4.809	4.816	4.903	**4.800**
subpolar N Atl SW RMSE	5.007	4.872	4.866	4.728	**4.738**
Nordic Seas SW RMSE	9.058	+0.300	+0.098	+0.304	**+0.089**
tropics net TOA	42.61	+0.096	+0.041	+0.082	+0.124
global T2m RMSE [K]	1.579	1.553	1.558	1.546	**1.543**
DJF	—	+0.070	−0.325	**−0.798***	−0.385
MAM / SON	—	−0.296/+0.280	−0.462*/−0.030	−0.217/−0.012	−0.360/+0.185

G4 is seasonally clean on every season's own threshold, and it **largely repairs G1's Nordic Seas damage** (+0.300 → +0.089) — which nothing predicted and nothing yet explains. An unexplained improvement is as much a loose end as an unexplained degradation.

The melt mechanism is falsified

The prediction on record was a measurable drop in May–June snow water equivalent. Siberian land box SWE [mm], with run×year ANOVA thresholds:

	Mar (±3.53)	Apr (±3.85)	May (±5.25)	Jun (±2.72)
control	121.98	131.07	97.12	15.39
G1	+0.88	+0.37	**+5.58***	**+5.68***
G4	−1.44	−2.39	+2.75	+1.60

G4's snow mass did not move. T2m rose +0.952 K with May and June SWE unchanged within noise, so the gain is the **plain sensible-heat route**, exactly as for F4 — *not* the melt-timing route that motivated the run. The tundra lever stands on its independent table-consistency justification (§24.4) and it works; the mechanism claimed for it does not.

And F4 makes the snow bias worse. G1 *significantly increases* May–June SWE (+5.58, +5.68, both clearing threshold). The campaign's best boreal lever improves temperature while **aggravating** the very snow bias it was meant to address. Leading suspect: cutting evapotranspiration also cuts **sublimation**, a genuine snowpack sink. Untested.

`RVRSMIN` does not saturate — cap it at 1000

500/1000/2000 give +0.336/ + 0.521/ + 0.876: increments per doubling of +0.185 then **+0.355**, i.e. *accelerating*. **There is no knee.** 1000 is defensible only as *the largest value before the winter damage starts* — G3 at 2000 costs DJF −0.798, clearing ±0.588 and joining B5 and H2 in the "warms JJA, cools DJF" list. F4's magnitude is a fitted ramp, not a corrected bias, and §24.5 flagged this outcome in advance as an argument against pushing it further.

Two mechanisms for the June albedo bias are now dead: the snow *cover formulation* (round 13) and the snow *melt rate* (round 14). Untried candidates: the snow-albedo decay timescale; the canopy-masking tile-7 formulation, which the round-13 residual showed already does most of the work in the melt months; and the `RVVEGALB` vegetation albedo table, which the campaign has never touched.

24.7 Tuning without rebuilding: `NAMSURFTUNE`

The ECMWF `surf` library deliberately ships no namelist, so every HTESSEL lever in this campaign was a source edit plus a rebuild. That serialises experiments against the single model tree, and it puts AWI tuning inside upstream ECMWF files where it collides with EC-Earth's tuning of the same tables at every merge.

`surf/module/surfece.F90` is already an AWI module *inside* `surf` with its own namelist, so the overrides live there: `ECE_TUNE_RVRSMIN`, `_RVLAI`, `_RVCOV`, `_RVZ0H`, `_RVLAMSK` (all 0:20, by vegetation type) and scalar `ECE_TUNE_RQSNCR` — covering the whole F-series, the H-series and the skin-conductivity levers. They are applied by `SURFECE_APPLY_TUNING` from `susurf.F90` *after* `SUSURF_CTL` has filled the tables; nothing in the setup path derives from these entries, so a late override is equivalent to editing the table in place. Every element defaults to a sentinel meaning "keep the as-released value", so a run setting none of them is bit-identical to untouched code, and every applied override is logged to `NULOUT`.

They occupy their own `&NAMSURFTUNE` group, *not* `NAMECECFG`: `NAMECECFG` is read twice from the same `fort.4`, by `ECE_CONFIG` in `arpifs/ecearth.F90` and by `SURFECE_CONFIG` in `surf`, and a Fortran namelist read aborts with *invalid reference to variable* on any name the reading module does not declare — so adding the tune entries there killed the arpifs read at `su0yoma.F90:152`. Found by a one-day test run, which is the argument for always doing one. Exactly **one line** changes in an ECMWF file. G1 is now

```
NAMCLDP: RCL_INPSEA=0.2, RCL_INPPMIN=70000.0   NAMSURFTUNE: ECE_TUNE_RVRSMIN(3)=ECE_TUNE_RVRSMIN(4)=1000.0
```

25 Observational grounding: what ERA5 can and cannot settle

Rounds 13 and 14 leaned on ERA5 for the snow diagnosis. That is a methodological weakness worth stating plainly: **ERA5's land surface is the same HTESSEL family as ours**, so comparing our snow and flux fields to ERA5's is partly comparing a scheme against a sibling of itself. This section replaces ERA5 with genuine observations wherever one exists, and marks the places where none does.

25.1 Surface albedo: CERES corroborates ERA5, and the model bias is real

CERES supplies `sfc_sw_up` and `sfc_sw_down` separately, so their ratio is a surface albedo derived from a *radiative retrieval* rather than a land model. Siberian land box, 1990–2014, against model `amip_presentday` and ERA5 `fal`:

	CERES	ERA5	model	ERA5−CERES	**model−CERES**	$SW_{\downarrow}$
Mar	0.5540	0.5325	0.5055	−0.021	**−0.049**	109
Apr	0.5386	0.5045	0.4990	−0.034	**−0.040**	188
May	0.3918	0.3722	0.4285	−0.020	**+0.037**	224
Jun	0.1685	0.1730	0.2147	+0.005	**+0.046**	228
Jul	0.1357	0.1362	0.1487	+0.001	+0.013	209
Oct–Feb	—	—	—	+0.10 . . . + 0.13	—	6–42

ERA5 and CERES agree to ±0.02 in every month with enough sun to retrieve (Mar–Sep), which is what licenses ERA5's albedo for this purpose. The June excess is confirmed *independently* at +0.046 — slightly larger than the +0.042 measured against ERA5 (§24.1). The two diverge badly Oct–Feb, but at 6–42 $W\,m^{-2}$ downwelling no albedo retrieval is trustworthy at these latitudes, so that is expected rather than disqualifying.

25.2 Snow cover and melt date: satellite, and ERA5 was flattering us

The Rutgers NH 24 km Weekly Snow Cover Extent CDR (NSIDC G10035, 1980–2024; NOAA visible-band snow charts to 1999, NOAA/NIC IMS thereafter) gives snow *extent* and *melt timing* from observation. `scripts/analysis/snowcove`

snow cover fraction	**Rutgers (sat.)**	ERA5	model	**G4**
Apr	0.950	0.991	0.964	0.961
May	**0.647**	0.796	0.890	0.893
Jun	**0.203**	0.261	0.380	**0.447**
Sep	0.075	0.178	0.257	0.249
Oct	0.612	0.755	0.825	0.834
melt-out (50 % crossing)	**11 May**	∼17 May	**24 May**	**28 May**

Three consequences.

(i) ERA5 was flattering the model. ERA5 itself runs +0.058 high in June and +0.149 in May against satellite, so the model excess quoted in §24.1 as +0.119 is really **+0.176** — June snow cover is close to *double* the observed value.

(ii) The model melts about 15 days late (control; `amip_presentday` is 13), now measured against an observation rather than inferred from a sibling model. *Note the two runs differ* — June cover 0.413 / melt-out 26 May for the control, 0.380 / 24 May for `amip_presentday` — and earlier drafts of this report quoted them interchangeably. This is round 14's "peaks a month late" (§24.2) confirmed independently and given a number.

(iii) G4 makes the snow field worse. June cover 0.380 → 0.447; melt-out 24 May → 28 May. The campaign's best *temperature* configuration pushes the spring snow cycle *further* from satellite, consistent with G1/F4 significantly increasing May–June SWE (§24.6). **This matters for the coupled goal:** a later melt is a later growing-season onset for LPJ-GUESS, which is a plausible route to losing boreal forest even while AMIP T2m improves. It should be weighed before G4 is committed to the PI spin-up.

Comparability caveat: Rutgers SCE is a **binary** classification per 24 km cell, whereas HTESSEL's `ZCVS` is a *subgrid* fraction within a ∼100 km cell. The two agree on "how much of the region is snow" but differ in how a partially covered cell is counted, which matters most mid-melt. The **crossing date** is the robust comparison; absolute mid-melt fractions are not over-read here.

25.3 Where no observation exists: the turbulent fluxes

A claim is withdrawn here. The surface energy budget (`scripts/analysis/snow_energy.py`) appeared to show the model venting 3–5 $W\,m^{-2}$ more upward sensible heat than ERA5 in February–April, which would have made turbulent exchange over snow the round-15 target. ~~The model loses its March–April melt energy to the atmosphere as sensible heat.~~ **That comparison has no observational authority**: ERA5's `sshf`/`slhf` are pure HTESSEL output, unconstrained by any assimilated observation, so it compares our scheme to a sibling of itself. It survives only as a hypothesis. The same caveat applies to the cold-content test on `tsn` (model within a few tenths of a K of ERA5), which is likewise model-versus-model — though as a *null* result it is the less dangerous of the two.

No satellite product measures surface turbulent fluxes. Only eddy-covariance towers can, and the Siberian candidates (Spasskaya Pad, ZOTTO) are point sites, not a gridded field. Any round-15 lever aimed at turbulent exchange over snow would therefore be tested against its *consequences* — melt date and snow cover, both now observationally anchored — rather than against the flux itself.

25.4 Summary: what each reference can support

quantity	reference	status
surface albedo	CERES up/down	**observational**; confirms model +0.046 June
snow cover extent, melt date	Rutgers/IMS	**observational**; model 13 days late
TOA and surface SW fluxes	CERES	**observational**
snow water equivalent	ERA5	partly constrained (SYNOP, IMS); forest bias
snowfall, snowmelt	ERA5	model output, weakly constrained
`sshf`, `slhf`, `str`, `tsn`	ERA5	**unconstrained — HTESSEL output**

ERA5 assimilates IMS snow cover, so ERA5 agreeing with Rutgers on extent is *expected* and is not independent corroboration — but it is what justifies having treated extent as ERA5's trustworthy field throughout.

26 Route B resolved: HTESSEL has no sub-grid snow depletion

Rounds 13 and 14 both failed on Route B, and the observational work (§25) sharpened the target without naming the cause. It is now named, and it also explains *why both rounds failed in the way they did*.

26.1 The arithmetic that settles it

Snow cover fraction is set from the **cell-mean** snow depth alone (`surfbc_ctl_mod.F90:317-322`):

$$\mathrm{ZCVS} = \min\bigl(1,\ \bar{d}_{\mathrm{cm}} \cdot \texttt{RQSNCR}\bigr), \qquad \texttt{RQSNCR} = 1/10.$$

So cover can only fall below 1 once the *cell mean* drops under 10 cm. The model's actual depth distribution over the box:

	mean d	p_{10}	p_{50}	p_{90}	frac <10 cm	ZCVS
Mar	51.1	31.4	48.7	75.6	0.037	0.964
Apr	48.9	24.8	47.2	75.8	0.044	0.964
May	28.8	4.5	**27.1**	56.5	0.181	**0.890**
Jun	4.8	0.0	2.7	13.5	0.842	0.380

The median May cell still holds 27 cm, so the formula returns essentially full cover — while the satellite record shows 0.647. Real 100 km regions at 27 cm mean depth are patchy: ridges blow bare, hollows hold metre drifts, canopies intercept. The model has no representation of that variance.

26.2 Why both previous attempts had to fail

This is the part that makes the diagnosis convincing rather than merely plausible.

- **Round 13 was doomed by arithmetic.** Changing `RQSNCR` $1/10 \to 1/30$ moves the threshold from 10 to 30 cm — which does nothing at all where depths are 27–51 cm. It could only bite where depths sit *near* the threshold, i.e. 13–27 cm, which is October and November. That is precisely where the response appeared (-8.2 albedo points in October, -0.95 in June, §23.2). The lever was not weak; it was arithmetically incapable of acting in spring.
- **The commented-out `tanh` and `sqrt` variants** at `surfbc_ctl_mod.F90:325` share the defect — still functions of the cell-mean depth. $\tanh(27/10) = 0.99$.
- **Snow brightness is the wrong direction** (model snow is darker than both ERA5 and CERES) and **energy supply is not short** in Mar–Apr (§25.1).

26.3 The fix, validated offline before implementation

A depth-only function cannot work; the depletion must respond to something that tracks the *state* of the pack. Niu & Yang (2007), as used in Noah-MP and CLM:

$$\mathrm{SCF} = \tanh\left(\frac{d}{2.5\, z_{0g}\, (\rho_{\mathrm{bulk}}/\rho_{\mathrm{new}})^{m}}\right)$$

The density ratio supplies exactly the missing behaviour: aged, melting snow is denser, so it patches out at a given depth. It needs **no new prognostic** — depth and density are already carried.

Tested on existing output *before* writing any code, Siberian box, `amip_presentday` vs Rutgers 1990–2014:

	satellite	as-released	$z_0 = 0.01$	$z_0 = 0.02$
Jan–Mar	0.999	0.964	0.964	0.952
Apr	0.950	0.964	0.948	0.893
May	**0.647**	0.890	0.772	**0.588**
Jun	**0.203**	0.380	0.321	**0.180**

$z_0 = 0.02$ matches the melt season closely at a small cost in midwinter cover; $z_0 = 0.01$ is conservative, leaving winter untouched and closing about half the May–June gap.

Provenance. The problem and the two standard remedies are well established in the land-surface literature: Liston (2004) sets out why a cell-mean depth cannot represent depletion; Niu & Yang (2007) give the density-dependent tanh form used here, calibrated against North American basin observations and adopted in Noah-MP and CLM; Swenson & Lawrence (2012) give the main alternative, separate accumulation and depletion curves driven by a snow-depth standard deviation, which is more faithful but needs an extra prognostic; Roesch et al. (2001) take the third route of adding sub-grid *orographic* variance in a GCM. Douville et al. (1995) is the classic depth-threshold form that HTESSEL's `RQSNCR` formulation resembles, and Dutra et al. (2010) documents the `LESNO9` scheme this campaign has been modifying. That CMIP-class models share the bias found here is documented by Thackeray et al. (2015), and its consequence for the snow-albedo feedback spread by Qu & Hall (2014) and Loranty et al. (2014). Betts & Ball (1997) remains the reference for boreal canopy masking, which is why the tile-7 residual behaves as it does (§24.1).

Implemented as `ECE_SNOW_SCF` in `&NAMSURFTUNE` (§24.7), with `ECE_SNOW_SCF_Z0`, `_M`, `_RHONEW`. **Default is** 0 **= the as-released formula**, so the change is inert unless switched on, consistent with every other entry in that namelist. Verified by a 1-day run: the switch logs to `NULOUT`, no namelist errors, the model steps normally.

26.4 What this still does not tell us

- **The temperature leverage remains unproven** (§76.4). G4 gained a full kelvin with snow mass unchanged and cover *worse*. Fixing the cover may buy much less T2m than its $-5.6\,\mathrm{W\,m^{-2}}$ suggests. Where it should matter is *coupled*, through LPJ-GUESS growing-season onset.
- **The March–April mass deficit is a separate, weaker claim.** It rests on ERA5 SWE (forest-biased) and may be a second effect or partly an artifact (§24.2).
- **Comparability.** Rutgers is a *binary* 24 km classification, so its 0.647 is "65 % of cells classed snow", not a true areal fraction. The melt-out *date* is the robust comparison; mid-melt absolutes are softer, and the z_0 choice should not be over-fitted to them.

27 Round 15 (2026–08–04): the snow fix works, buys no summer, and cools winter

Three runs, all namelist-only on one binary: **I1** = G4 + depletion, **I2** = depletion alone (isolating the snow route from the vegetation route), and **I3** = G4 + the SDOR-scaled variant. Sections §27.1–§27.5 were written on I1 and I2 alone; **I3 is evaluated in §27.6**, and it changes the round's verdict — the snow route *does* buy summer temperature, but only in the scale-aware formulation, and it presents a bill in the Arctic energy budget.

27.1 The scheme does exactly what it was built to do

Siberian snow cover	satellite	control	G4	**I1**	**I2**
Apr	0.950	0.962	0.961	0.931	0.931
May	0.647	0.891	0.893	**0.703**	**0.721**
Jun	0.203	0.413	0.447	**0.160**	**0.175**
melt-out	**11 May**	26 May	28 May	**12 May**	**13 May**

Melt-out went from **15 days late to 1 day late**, against the satellite record.

27.2 And it delivers no summer temperature

	surface SW gain	JJA T2m	K per $\mathrm{W\,m^{-2}}$
G4 (vegetation route)	+8.10	**+0.952**	0.12
I2 (snow route alone)	+2.91	**+0.094** (noise)	**0.03**

This settles §76.4: the link is weak. Route B was a genuine, satellite-measured radiative bias with almost no temperature leverage — per watt it is **four times less efficient** than the vegetation route. The reading is clean: energy added over a *melting* snowpack goes into melt, not sensible heat, which is precisely what it did. Fixing it remains right — the snow field is now correct, which is what matters for LPJ-GUESS growing-season onset — but Route B is not where the remaining boreal kelvin lives.

27.3 The winter response is a compensating error being removed

Scored against ERA5 rather than against the control (period-clean, `amip_presentday`):

bias vs ERA5 [K]	DJF	MAM	JJA	SON
control	−1.95	−1.97	−2.58	−2.69
G4	−2.33	−2.33	**−1.63**	−2.51
I1	**−4.71**	−2.28	**−1.33**	−3.46
I2	−4.63	−2.36	−2.49	−3.77

The control is too cold in every season, so there was never a warm bias to protect — scoring against observations rather than against the control is what makes that visible, and eleven rounds had not done it.

What this does NOT establish. ~~The excess snow cover had been propping DJF up by ∼2.7 K, masking a ∼4.7 K bias that was there all along.~~ That reading was asserted here and is **not supported**: winter cover is *unchanged* between the two schemes (0.963 vs 0.964, both saturating at 43 cm, §27.4), so excess *winter* cover cannot have been doing the propping. What is measured is only that **I1 cools DJF by −2.76 K by an unidentified route**. Whether that unmasks a pre-existing bias or creates a new one is **untested**.

The point is sharpened by a precedent this report got the other way round: H1 produced the same signature (−1.233 K DJF by cutting snow-tile fraction, §23.2) and was *rejected* for it. Identical evidence cannot be disqualifying in one run and revelatory in another. Until the route is identified, the honest position is that the snow fix and the winter response are two findings, not one story — which is also why round 16 tests the winter on its own terms rather than assuming it is recoverable.

27.4 Ruled out by measurement: it is not a radiation-supply problem

Downward longwave over the box, model vs CERES [W m^{-2}]: Oct **−12.9**, Nov **−11.9**, Dec −3.1, Jan +0.1, Feb +5.6 — **DJF mean +0.9, essentially exact**. That kills the Pithan et al. (2014) supercooled-liquid route for midwinter. The Oct/Nov deficit is real and is a *separate* autumn problem. The +30 pp polar-night cloud excess against CERES is retrieval failure over snow at night, not evidence.

Still unresolved: winter cover is **unchanged** (0.963 vs 0.964 — both formulas saturate at 43 cm) and winter SWE is slightly *higher*, so the DJF response is not a direct cover or mass effect. `str` and `sshf` shifts are the surface being colder, i.e. responses. ~~`stl1` and `skt` are unusable (∼ 0.07 K — the same dead-field trap as pressure-level `r`).~~ **That was wrong**: both fields are sound (210–319 K). The 0.07 was 273/3600 — an analysis-side bug that divided an *instantaneous* temperature by the accumulation period. Fixed in `boxcache.py`; unlike pressure-level `r`, which really is identically zero, these fields were never broken. Using them (§27.5) shows I1 and I2 cool `stl4` by ∼2.1 K, so the unexplained winter signal reaches the *deep soil* — which matters because LPJ-GUESS is fed `ST1L`–`ST4L`. H1 cooled DJF by −1.233 through a different route to the same tile fractions, so this is the second time cutting snow-tile fraction has cooled winter hard.

27.5 Does the warming reach the skin and the soil?

A lever that raises 2 m air temperature by mixing heat *away* from the surface would improve the metric we score while degrading the surface itself. That is not a cosmetic concern: **LPJ-GUESS is fed `ST1L`–`ST4L_LPJ` (soil temperature, layers 1–4) and `SM1L`–`SM4L_LPJ`** alongside `T2M_LPJG` (`lpj_guess/framework/OasisCoupler.cpp:17-95`), so a warmer-air/colder-ground configuration would hand the vegetation model an incoherent state. Skin temperature itself is not coupled.

Siberian land box, JJA, deltas against control [K]:

	T2m	`skt`	`stl1`	`stl4`	Δ(`skt`−T2m)
control (absolute)	9.73	10.18	9.31	−1.31	0.45
G4	+0.952	**+0.990**	**+0.883**	+0.157	+0.037
I1	+1.250	+1.318	+1.160	**−2.092**	+0.068
I2	+0.094	+0.100	+0.046	**−2.190**	+0.006

G4's warming is genuine, not a redistribution: skin and layer-1 soil warm *as much as* the air and the skin-to-2 m gradient barely moves (+0.037 K). What LPJ-GUESS would receive from G4 is coherent.

But I1 and I2 cool the deep soil by ∼2.1 K, I2 doing so with essentially no surface change at all. The unexplained winter response of the snow scheme therefore reaches `stl4`, which *is* a coupled field. That is a second, independent reason to keep the snow-depletion scheme out of a coupled run until round 16 explains the winter behaviour.

This subsection exists because the diagnostic was wrongly believed impossible: the soil and skin fields had been written off as dead (§27.4) on the basis of an analysis-side scaling bug.

27.6 I3 (2026–08–04): the strongest summer lever, and the Arctic bill

I3 = G4 + mode 2, the SDOR-scaled depletion, calibrated to match I1 *over the Siberian box* so that any remaining difference would be spatial structure (§26.3). It does not behave like I1, and the differences are the point.

What it wins.

	control	G4	I1	**I3**	CERES
Siberian JJA T2m [°C]	9.728	+0.952	+1.250	**+1.322** (t=10.61)	—
Siberian surface net SW [W m^{-2}]	153.78	+8.10	+12.77	**166.58**	**166.26**
global T2m RMSE vs ERA5 [K]	1.579	−0.036	**+0.048**	**−0.023**	—
global net TOA [W m^{-2}]	0.643	−0.128	−0.021	+0.056	—
SO SW RMSE	6.877	−2.076	−1.906	−2.065	—

+1.322 K is the **largest JJA gain of 36 evaluated tuning runs**, and the Siberian surface SW deficit that §22 opened this whole thread with **closes to within** 0.3 **W m^{-2}** of CERES. Global TOA does not move, and where I1 *degrades* global T2m RMSE (+0.048), I3 improves it. Per season it dominates I1 outright — more JJA, 0.32 K less DJF damage, and I1's significant autumn penalty (−0.766*) collapses to a null:

Δ vs control [K]	DJF	MAM	JJA	SON
threshold (own season)	±0.601	±0.394	±0.244	±0.441
G4	−0.385	−0.360	+0.952*	+0.185
I1	−2.759*	−0.306	+1.250*	−0.766*
I3	**−2.439***	−0.214	**+1.322***	**−0.105**

What only the full guardrail set revealed. The Siberian-box scripts alone would have promoted I3. They were run first, and the verdict "best lever of the campaign" was drafted on them — prematurely, and caught only because the guardrails were demanded before writing. Four guardrails move together, all one way:

guardrail	control	I1	**I3**	CERES
net TOA 60–90N [W m^{-2}]	−97.74	+0.87	**−96.02** (**+1.72**)	**−97.98**
net TOA NH [W m^{-2}]	0.280	+0.245	**0.952** (**+0.672**)	**0.08**
NH−SH albedo	−0.003	−0.002	**−0.007** (**−0.004**)	—
Nordic Seas SW RMSE	9.058	+0.196	**9.557** (**+0.499**)	—

This is the mechanism *working as designed*: removing high-latitude snow cover admits more shortwave at high latitude. The Arctic did not have that energy spare. The control sat almost exactly on the observation at 60–90N (−97.74 vs −97.98); I3 leaves it 1.96 away. The NH net TOA change (+0.672) and the NH−SH albedo change (−0.004) are each the **largest of all 37 runs**, and Nordic Seas is the **only significant SW RMSE degradation in the campaign** — `rmse_significance.py` flags exactly two runs there and I3 is one (+0.07 against a ±0.051 threshold).

Caveat against these four, stated so it is not over-read: these are PI runs scored against present-day CERES, and this report's own rule (§76.7) is that the PI target is net TOA ≈0, **not** the CERES column. That weakens the "away from CERES" reading of the two regional TOA rows. It does *not* weaken the NH−SH albedo or Nordic Seas results, which are model-versus-model and model-versus-RMSE.

Why this matters more for the coupled run than for AMIP. All four are Arctic energy-*gain* guardrails, and AMIP structurally hides their consequence: prescribed SST and prescribed sea ice absorb 1.7 W m^{-2} without responding. In the coupled PI spin-up — the configuration I3 is a candidate for — they would not. This is the same structural blindness as §76.5, in the opposite direction.

27.7 The winter penalty is not radiative

§27.4 left the DJF route unidentified. It can now be excluded as radiative. Siberian land, DJF, all four configurations:

Siberian DJF	snow albedo	SWE [m]	absorbed SW [W m^{-2}]
control	0.6789	0.6933	8.16
G4	0.6760	0.6901	8.18
I1	0.6800	0.6926	8.16
I2	0.6796	0.6958	8.16
I3	0.6797	0.6944	8.22

Albedo, snowpack and absorbed shortwave are **unchanged to three decimals**, and there are only ∼8 W m^{-2} of insolation in the box in midwinter regardless. **A −2.4 K penalty cannot be constructed from that budget.** This generalises the finding that killed the compensating-error story (§27.3, winter cover 0.963 vs 0.964) from cover to the whole radiative chain, and it holds for I2, which carries the penalty with no vegetation change at all. The route must be **thermal — the snow/exposed-tile coupling** — which is precisely what round 16 (§28) tests, so the diagnosis and the running experiment agree.

27.8 Direct observations at last: RIHMI-WDC station soil temperature (2026–08–05)

Every soil claim in this report had rested on ERA5, whose land surface *is* HTESSEL — the same snow scheme, the same soil thermal conductivity — so it is a sibling, not a witness. **RIHMI-WDC (Sherstyukov v3)** ends that: daily soil temperature at Russian meteorological stations, 1963–2024, twelve depths from 2 to 320 cm, measured **under natural cover** so the snowpack is intact. 43 of the 110 stations lie in the campaign box. Model and ERA5 are sampled *at the station points*, not box-averaged, which is the one representativeness control available without land-cover weighting.

What the archive cannot do: depths 2/5/10/15 cm carry *no* winter data — the Russian network withdraws the shallow Savinov thermometers for the cold season. So `stl1` (0–7 cm), where the damage is largest, **has no observational counterpart**; `stl2` is the shallowest verifiable layer and bounds it. Only `tsoil_qc == 0` is used.

	obs depth	OBS	model	bias	OBS offset	model offset
`stl2`	20 cm	−6.07	−5.03	**+1.04**	**+17.78**	+20.11
`stl3`	40/80 cm	−3.58	−3.19	**+0.39**	**+20.28**	+21.95
`stl4`	120–240 cm	+0.34	−0.23	**−0.57**	**+24.18**	+24.85

Model is `amip_presentday` (1990–2014), period-clean. Offset = soil minus 2 m air, each dataset using its *own* air, which is what makes it insensitive to the 2.8 K air-temperature change the I-series also carries.

Two results, both DJF-only. In *winter* the control soil is correct to ∼1 K at every verifiable depth — there was no pre-existing *winter* soil bias to remove — and the model is mildly *over*-insulated, by +0.7 to +2.3 K of offset.

~~The control soil is correct to ∼1 K at every verifiable depth.~~ **That was measured in DJF only and does not hold in summer** (§27.9): the same control is −3.4 to −5.4 K too cold in JJA, a bias *larger* than the 2.58 K Siberian air bias this campaign has been chasing. The winter statement stands; the unqualified one does not.

~~The winter soil-air offset under a deep Siberian snowpack is +10 to +20 K.~~ That was quoted from memory of the permafrost literature and is too low: the station record gives +17.8 to +24.2 K. The correction strengthens rather than weakens the conclusion below.

Scored on offset (the tuning runs are 1872–1915, so absolutes are not period-comparable):

	`stl1`	`stl2`	`stl3`	`stl4`
control	19.87	20.39	22.24	25.13
G4	20.27	20.81	22.72	25.82
I1	**1.53**	**2.61**	**6.36**	**15.25**
I3	1.41	2.54	6.46	15.63
OBSERVED	—	**+17.78**	**+20.28**	**+24.18**

I1 is **−15.2 K wrong at 20 cm**, −13.9 at 40–80 cm, −8.9 at 1.2–2.4 m. In raw terms its soil sits at −25.8 °C where stations measure −6.1 °C. The snowpack has stopped insulating at every depth.

27.9 The summer soil bias, and why it dwarfs the air bias

The DJF validation above was extended to JJA at the same 43 stations, and it changes the picture completely.

	DJF			**JJA**		
`amip_presentday`	OBS	model	bias	OBS	model	**bias**
boreal `stl2` 20 cm	−6.07	−5.03	+1.04	13.68	10.33	**−3.35**
boreal `stl3` 40/80 cm	−3.60	−3.19	+0.42	10.17	5.84	**−4.33**
larch `stl2` 20 cm	−11.05	−7.76	+3.29	12.52	8.33	**−4.19**
larch `stl3` 40/80 cm	−7.80	−5.54	+2.26	7.64	2.29	**−5.35**

The soil is 3–5 K too cold in summer while being slightly too warm in winter, and the summer soil bias is *larger than the* 2.58 *K Siberian air bias* the campaign has spent 45 runs on. It was never looked at because no soil diagnostic was scored until §27.8.

It is far worse coupled. LPJ-GUESS's `soiltemp25cm` is the OIFS soil temperature received through OASIS, so it measures the same quantity:

larch zone, ≈25 cm	DJF	**JJA**
RIHMI obs (20/40 cm bracket)	−11.4 / −8.3	**+12.5 / + 10.0**
06 Baseline	−15.96	+1.80
080a CRUNCEP	−16.35	+1.38
09A newSeaIce	−18.09	+0.72
10A (G4)	−21.04	**−0.37**
10B (G4 + snow depletion)	−20.85	**+2.31**

Against an observed ≈ + 11.8 °C at 25 cm, every coupled run sits at +0.7 to +2.3 °C: a ∼10 **K summer soil cold bias**. **Larch cannot build cover in soil near freezing all summer**, which makes this a more plausible driver of the forest collapse than either the evergreen gate or grass competition (§31) — those describe *which* PFT wins, this describes why none of them grow.

Representativeness caveat, and it is a real one. RIHMI plots are open, bare or short-grass sites in full sun; a forested cell's summer soil is legitimately cooler under canopy shade. Part of AMIP's −4 K is therefore representativeness and cannot be separated without land-cover weighting. What is *not* representativeness is the ∼6 K gap between AMIP and coupled at the same stations with the same comparison — that is real degradation contributed by the coupled vegetation feedback.

27.10 The snow scheme, reconsidered on coupled evidence (2026–08–05)

~~`ECE_SNOW_SCF` is switched off and stays off; no production run should set it.~~ **Retracted the same day.** That verdict was reached on AMIP alone, and **AMIP was the wrong testbed for it**: it has no vegetation response, which is precisely the thing the scheme was helping. On the coupled 10A/10B pair — identical binaries, identical settings (`ECE_SNOW_SCF=1`, `Z0=0.016`), differing *only* in the scheme:

larch zone	BNS LAI	C3G LAI	grass : larch	TREEFPC
09C reference	0.096	0.400	4.2×	0.048
10A (G4)	**0.019**	0.388	**20.4**×	**0.019**
10B (G4 + scheme)	**0.134**	0.830	6.2×	**0.065**

10B carries 7× the larch and 3.4× the tree cover of 10A, and G4 alone (10A) is *worse than the 09C reference* — it actively degrades the forest. NH total LAI bias against the Lund CRUNCEP reference halves, −0.62 (−28.1 %) → −0.28 (−12.9 %).

And the winter soil damage does not appear coupled. 10A and 10B are −21.04 and −20.85 in DJF — identical — while JJA goes −0.37 → +2.31. The ∼5 K winter cooling relative to the 06/070/080a baselines came in with G4 + CERES-init, *not* with the scheme. And 10A's summer soil is the outlier against every earlier baseline (+1.4 to +3.0); **the scheme restores the baseline range rather than departing from it**, recovering 2.7 K of the ∼12 K deficit.

~~Standing position: the scheme is not rejected. The AMIP soil error is real, but it is not disqualifying for coupling, where the scheme is the difference between a badly degraded and a substantially recovered boreal forest.~~ **RETRACTED the same day — see §32.** The claim that the winter damage does not appear coupled was an *artefact of where I sampled*: I measured only the larch zone, where 10A is already saturated near −21 °C, so both runs looked alike. Across the full station network 10B is −16.2 K in DJF and −18.3 K in January. The AMIP testbed had predicted exactly this.

Why. Verified against *direct observations* — RIHMI-WDC station soil temperature, 43 stations in the box, under natural cover (§27.8) — the scheme destroys the winter snow insulation. Observed thermal offset (soil minus 2 m air) is +17.8 K at 20 cm, +20.3 at 40–80 cm, +24.2 at 1.2–2.4 m. The control reproduces those to within 0.7–2.3 K. I1 gives +2.6, +6.4, +15.3: an error of −15 K at 20 cm. The control soil itself is correct to ∼1 K at every depth the Russian network reports in winter, so this is a **new error, not the removal of an old one**. Since `ST1L`–`ST4L` feed LPJ-GUESS, that alone disqualifies it for coupling.

What we still do not know, and it matters. The mechanism is *unidentified*. The monthly decomposition rules out the obvious candidates: Siberian snow cover changes by −0.02 in October and SWE not at all, so it is neither a cover nor a mass effect. The soil *leads* the air (October: soil −4.19 K, air −0.02), so it is a surface-coupling change rather than an atmospheric one. Winter snow density falls 20–27 $\mathrm{kg\,m^{-3}}$, which with unchanged SWE means a *deeper, less dense* pack — that insulates *better*, so density is a consequence of a cold pack, not its cause. ~~The tanh formula bites hardest in October when snow is shallow, cutting cover and exposing the soil.~~ Measured: cover is essentially unchanged. That explanation was written into `soil_temp_vs_obs.py` before it was checked and is wrong.

Reactivation therefore needs three things, in order:

1. **Diagnose `ZCVS`.** It is not in the output stream, which is why the mechanism is still open — every field we do write exonerates itself. Nothing should be rebuilt before the snow-tile fraction can actually be seen.
2. **Accumulation/ablation hysteresis.** Our implementation applies one depletion curve year-round. Both Niu & Yang (2007) and Swenson & Lawrence (2012) use *different* curves for accumulation and ablation, and that omission is the most likely structural defect. It would also match the measured timing: the benefit is May–June, the damage begins in October, so the two are cleanly separable in season.
3. **Re-verify on the SOIL OFFSET, not on snow cover.** The scheme was accepted in round 15 because it fixed melt-out to within a day of satellite. That was true and it was not sufficient. §27.8 is the test any successor must pass.

And weigh the prize before spending the effort. I2 — the scheme alone, no vegetation levers — bought +0.094 **K JJA, inside the noise floor**. The value of the snow fix is not temperature; it is that the snow field becomes correct, which matters for LPJ-GUESS growing-season onset. That is a real but narrow benefit, and it does not justify a source change plus a run family while a 20 K soil error remains unexplained.

28 Round 16 (2026–08–04): the snow-tile skin conductivity

`vlamsk_mod.F90` hardcoded the snow-tile skin conductivities, so they were untunable. λ_{sk} sets how tightly the skin couples to the medium below it; a small value lets it radiatively decouple and crash in polar night. Physically $\lambda_{sk} \approx k_{\mathrm{snow}}/d_{\mathrm{skin}}$ with k_{snow} 0.1–0.4 $\mathrm{W\,m^{-1}\,K^{-1}}$ and d_{skin} 0.01–0.05 m, so **roughly 2–40**. As released:

constant	value	tile
`ZSNOW`	**7**	5, **exposed snow** — the weakly coupled one, and the tile covering tundra (25.6 % of the box)
`ZSNOW_GLACIER`	8	5, where SWE > 9000 mm
`ZSNOWHVEG`	20	7, snow under high vegetation
`ZLARGESN`	50	5, when melting

The as-released 7 sits at the decoupled end of the physical range, on precisely the tile whose *fraction* round 15 corrected. Now exposed as `ECE_LAMSK_SN/_SNGL/_SNHV/_SNMLT`, all defaulting to the shipped values.

run	setting	prediction
J1	I1 + `ECE_LAMSK_SN: 15`	DJF +1 to +3 K, JJA barely moves
J2	I1 + `ECE_LAMSK_SN: 25`	same, stronger

Falsifier on record: if DJF does not respond, skin conductivity is not the route, and the winter bias lies in boundary-layer mixing instead (Holtslag et al. 2013; Sandu et al. 2013). Also watched: melt-out and May/June cover must *stay* fixed — drift back toward the control would mean the two changes interact and the pair is not separable.

28.1 Round 16 result (2026–08–05): falsified, and the falsifier was on record

Δ vs control [K]	DJF	JJA	SON	global T2m RMSE
I1 (the base)	−2.759*	+1.250*	−0.766*	1.627
J1 (λ_{sk} 15)	**−2.834***	+1.313*	−1.132*	1.581
J2 (λ_{sk} 25)	**−3.226***	+1.246*	−0.879*	1.597
control	—	—	—	1.579

The prediction was **DJF +1 to +3 K**. Measured: DJF got **worse**, and *monotonically with the knob* — −2.834 at λ_{sk}=15, −3.226 at 25, against I1's −2.759. Both runs carry a global T2m RMSE *worse than the control*.

The pre-registered falsifier fires: skin conductivity is not the winter route. By §27.7 the route is non-radiative, and λ_{sk} was the leading non-radiative candidate; with it excluded, **boundary-layer mixing** (Holtslag et al. 2013; Sandu et al. 2013) is what remains. The round is doubly moot in practice, since its base I1 is now deactivated (§27.10) — but the falsification stands on its own and the diagnosis it leaves behind is the useful part.

Both J runs were built on I1, before §27.6 showed I3 to be the better base in every season. That was the correct decision with the information available and the round is still diagnostic — it tests whether λ_{sk} is the winter route at all, and §27.7 shows I1 and I3 share the same non-radiative penalty, so an answer on I1 transfers. But if λ_{sk} does prove to be the route, **the production pair must be re-run on I3**, and the Arctic guardrails of §27.6 re-checked on the combination rather than assumed to superpose — superposition has been wrong in sign twice in this campaign (§76.7).

29 Land albedo (2026–08–04): half of it is the snow tile

With the boreal budget nearly spent (§21.3) the target moved to the ∼0.7 K *universal* land bias, and §21.1 localised its energy to a clear-sky shortwave deficit — cloud is the wrong sign. Decomposing that deficit assumption-free (each dataset divided by its *own* downward flux) gives an all-sky land surface albedo +0.0154 too high against CERES.

29.1 Snow is a tile, not a vegetation type

The first attribution split that +0.0154 by *dominant vegetation type*, built from `tvh/cvh/tvl/cvl`. That map has no snow in it: in HTESSEL snow is a separate tile (5 on low vegetation, 7 under high) that **overlies** every vegetation type. So each type was part-time snow, and the ranking the split produced was largely a map of *snow duration* — evergreen needleleaf is snow-covered 53.9 % of the year, tundra 63.2 %, "bare soil" 49.2 % (a bucket holding the Sahara and high-Arctic bare ground together). An annual-mean $T_{2m} > 5$°C filter does not fix this: a cell averaging 6°C still has months of snow.

Masking snow **per cell per month** (SWE < 1 mm, sunlit months only) splits the error almost exactly in half:

Δ albedo vs CERES	all sunlit months	snow-free only
ALL LAND	**+0.0154**	**+0.0080**
⇒ *snow contributes* +0.0074, *the snow-free surface* +0.0080		

and the per-type ranking *inverts*:

type	snow-covered	all months	**snow-free**
Evgr Needleleaf	53.9 %	+0.0250	**+0.0032**
Bogs/Marshes	37.1 %	+0.0143	+0.0039
Evgr Shrubs	41.2 %	+0.0150	+0.0058
Evgr Broadleaf (tropical)	6.9 %	+0.0017	**−0.0003**
Crops	25.9 %	+0.0215	**+0.0204**
Semidesert	22.5 %	+0.0342	**+0.0169**
bare soil	49.2 %	+0.0240	**+0.0145**
Irrig Crops	24.6 %	+0.0253	**+0.0138**
Tundra	63.2 %	+0.0064	**+0.0105**

Every forest type falls inside +0.006 **once snow is removed**, tropical broadleaf to −0.0003. **This vindicates §22.1 and §22.3**: `RVVEGALB`'s high-vegetation entries are sound and the snow season really is the story. What is *new* is the residual +0.0080, which sits on the sparse and cultivated surfaces where the soil background shows through — and on tundra, which gets **worse** under masking (+0.0064 → +0.0105) because snow was hiding a genuine snow-free summer error. Nothing in 41 runs has touched that half.

29.2 The snow half already has a lever

Land albedo over snow-covered sunlit months, model-internal:

run	snow-month albedo	vs G4	all-land
G4	0.5185	—	0.2586
I1 = G4 + mode 1	0.5011	−0.0174	−0.0024
I2 = mode 1 alone	0.4996	−0.0189	−0.0018
I3 = G4 + mode 2	**0.4689**	**−0.0496**	**−0.0052**

I3 removes ∼70 % of the snow half, I1 ∼32 % — and I3 was calibrated to match I1 *in the Siberian box*, so globally it is twice as strong as intended. That is new information about mode 2 and it is the same over-reach that §27.6 bills at 60–90N: the SDOR scaling does most of its work outside the box it was tuned in.

29.3 Why the two halves cannot simply be stacked

The obvious next move — keep I3 and add a vegetation-albedo correction on top — is **not additive in the direction that matters**. A darker land surface at high latitude adds absorbed shortwave to precisely the region I3 has already pushed 1.72 W m^{-2} past CERES. The two changes compete for one energy budget. Consequences for the plan:

- **The mid-latitude and subtropical entries are the safe ones.** Crops (+0.0204), semidesert (+0.0169), bare soil (+0.0145) and irrigated crops (+0.0138) lie outside the Arctic, so correcting them attacks the ∼0.7 K *universal* bias without adding to the 60–90N surplus.
- **Tundra (+0.0105) is the dangerous one.** It is the single entry that *would* stack with I3, being boreal and in-season — which makes it both the most attractive for the summer target and the one most likely to break the Arctic guardrails.
- **The snow-free half is not reachable by `RVVEGALB` alone** where bare fraction dominates; there the background soil albedo climatology carries it, and that is an ancillary field, not a namelist parameter (§22.2).

Assets: `land_albedo_snow_split.py`, chained into `evaluate.sh -obs`.

30 Round 17 (2026–08–05): K1 adopted, and land albedo closed out

Two families on the G4 base, run concurrently and deliberately orthogonal: **L** puts a moss/organic layer on the boreal types, **K** corrects the snow-free land albedo diagnosed in §29.

30.1 L: the moss proxy is falsified, monotonically the wrong way

λ_{sk} 10 → 2.9 (L1) and 10 → 1.7 (L2) on types 4 (larch) and 9 (tundra) only — the two most boreal-confined types, 62 % and 70 % of their area poleward of 50N. Gaillard (2025) attributes most of a >2 K high-latitude summer gain to a surface organic layer, which HTESSEL lacks entirely.

	JJA	vs G4	verdict (threshold ±0.244)
G4	+0.952*	—	the base
L1 (λ_{sk} 2.9)	+0.784*	−0.168	inside → null
L2 (λ_{sk} 1.7)	+0.638*	**−0.314**	**outside → significant degradation**

More moss is *worse*, the opposite of the prediction, and the monotonic ordering makes a noise explanation implausible. **Caveat that limits the claim**: λ_{sk} is only a *proxy* for an organic layer — it changes the skin-to-substrate coupling but not the heat capacity, not the moisture retention, and not the summer evaporative regime. This falsifies *the proxy*, not necessarily the Gaillard mechanism. A real organic layer would need a soil-column change, which is a much larger piece of work.

30.2 K: adopted

K1 = G4 + `RVVEGALB` on crops (×0.887), irrigated crops (×0.910) and semidesert (×0.906) + `ECE_TUNE_SOILALB` 0.95. K2 is the soil background *alone*, isolating the ∼25 % of land that `RVVEGALB` cannot reach. **Tundra was deliberately excluded** so K stayed orthogonal to L and did not add absorbed shortwave to the Arctic (§29.3).

	Siberian JJA	DJF	net TOA 60–90N	**global T2m RMSE**
control	—	—	−97.737	1.579
G4	+0.952*	−0.385	−97.603	1.543
I3	+1.322*	−2.439*	**−96.017**	1.556
K1	**+1.036***	−0.496	−97.463	**1.524**
K2	+0.913*	−0.144	−97.525	**1.518**
CERES	—	—	−97.98	—

Both are clean — every season inside its own threshold — and they are the **two best global T2m RMSE of the campaign**. The tundra exclusion did its job: 60–90N stays near the control, against I3's $+1.72\,\mathrm{W\,m^{-2}}$ excursion. In the Siberian box neither K is significant against G4 (+0.084, −0.039, both inside ±0.244), *which was predicted in advance* — K's mass is crops, semidesert and the subtropical desert belt.

30.3 What K bought, measured on the metric it was designed for

global LAND, vs G4	land T2m	land SW_{net}	land albedo
K1	**+0.089** K	$+0.77\,\mathrm{W\,m^{-2}}$	−0.0059
K2	+0.076 K	+0.48	−0.0038

The lever did what it was built to do. The snow-free residual was +0.0080; K1 removed −0.0059 of it, **74 %**. The measured sensitivity is 0.089/0.77 = **0.116** K per $\mathrm{W\,m^{-2}}$ against the 0.12 assumed — essentially exact. The geography is right too: land equatorward of 60N warms +0.109 K while land poleward of 60N *cools* 0.055 K, exactly what excluding tundra was meant to achieve.

~~Predicted ~1.1 $\mathrm{W\,m^{-2}}$ and 0.1–0.2 K of land warming.~~ Measured 0.77 $\mathrm{W\,m^{-2}}$ and 0.089 K — the SW gain was 70 % of the prediction and the temperature at or below its low end. The sensitivity assumption was right; the forcing estimate was optimistic.

THE NEGATIVE RESULT IS WORTH MORE THAN THE 0.089 K. Three quarters of a *directly measured* land albedo bias has now been removed, and it recovered 0.089 K of the ~0.7 K universal bias. Extrapolating, a *perfect* albedo fix buys ~0.12 K — about **one sixth** of the target. **Land surface albedo is therefore closed out as the explanation for the universal cold bias**, which is a real constraint on where the remaining ~0.6 K lives.

~~K3 should push the soil-albedo scale further, since K2 alone gives the best global RMSE.~~ **Retracted before it was run.** The remaining 26 % of the residual is worth roughly 0.03 K, and the 0.95 scale was calibrated to the *measured* −0.0163 bias — pushing past it means tuning *against* the observation, which is the same overfitting-to-present-day trap that ruled out editing the soil ancillary (§29.3). **K1 is adopted and albedo tuning stops here.**

31 The boreal forest fails by TWO different routes (2026–08–05)

The report has carried a single story for the missing forest: the winter evergreen cold gate (§6.6). Evaluating the `Tuning_test_07*/08*` LPJ-GUESS families (`lai.out`, `fpc.out`, `est_limits.out`) shows that story is **correct but only half the picture**, and that it does not apply at all to the region that matters most — NE Siberia, which is ~80 % larch.

31.1 BNE is gate-excluded; BNS is permitted and outcompeted

Larch zone, 60–72°N × 100–160°E, 2520 cells, last decade:

	080a (CRUNCEP base)	09A (newSeaIce)
`mTmin20`	−38.94 °C	−41.89 °C
BNE gate `tcmin_est`= −30	**0.7 % clear**	**0.0 % clear**
BNS cold gate	**none, 100 %**	**none, 100 %**
`agdd5` vs BNS `gdd5min_est`= 350	**83.6 % clear**	79.0 %
BNE LAI	**0.000**	**0.000**
BNS LAI	0.129	0.103
C3G LAI	0.751	0.510
grass : larch	**5.8**×	5.0×
cells with C3G > BNS	65 %	60 %

The two PFTs fail for completely different reasons.

- **BNE/BINE: a climate gate.** 0.7 % of larch-zone cells clear `tcmin_est`= −30, and their LAI is *exactly* 0.000. The winter-cold story of §6.6 is right, and this is where it applies — the evergreen zone of west Eurasia and North America, *not* NE Siberia.
- **BNS (larch): no gate at all, and it loses.** It has no cold limit and clears its `gdd5min_est`= 350 on 84 % of cells, so it is **climatically permitted** across the zone — yet grass carries 5–6× its LAI, grass exceeds larch on 65 % of cells, and larch is absent outright from half the cells where it is allowed. **Larch is not excluded; it is outcompeted in the growing season.**

This is a correction to this report, not a new hypothesis. §6.6 already states that BNS has no cold gate and is "*already permitted*" — and then never explains why it is absent anyway. A reader was left applying the winter-gate mechanism to a zone where it explains nothing, with only the roughness leg (§6.5) for the 80 %-larch region.

31.2 Reconciling this with the falsified "competition, not climate"

§78 carries *"the boreal failure is an LPJG-internal competition error"* as dead, killed by the forcing-transfer test reproducing most of the loss with byte-identical parameters. **That falsification stands and is not weakened here**, because it and the result above are different claims:

claim	status	
competition is an LPJG *parameter error*	**dead**	forcing reproduces the loss unchanged
competition is the *pathway* by which climate removes larch	**supported**	permitted on 84 %, loses 5.8:1

Forcing is the *driver*; grass-versus-larch is the *mechanism*. Both hold at once. As written, the report recorded only the falsification, so a reader would reasonably conclude that growing-season competition had been ruled out — it had not been tested.

No contradiction with §6.4 either, though the two look opposed. That section measures vegetation→climate (the LPJG cold effect on T2m, which vanishes in July–August). This measures climate→vegetation (how the growing season sets larch cover). Opposite directions of the same coupling; both can be true.

Weak evidence, flagged as such. Weakening grass (`C3G pstemp_low`= 12, run 08G) moves BNS 0.129 → 0.156 and TREEFPC 0.060 → 0.072 while C3G falls 0.751 → 0.583 — directionally exactly what competition predicts. But the 08 runs sit on *different atmospheric bases* (06T vs 06V), so this is not a clean attribution and should not be quoted as one. The gate/permission evidence above is the load-bearing part; the knob evidence is suggestive only. On the old forcing, **07A (BNS `greff_min` relaxed) did not recover BNS at all** (0.024 → 0.021), which is why a forcing-consistent re-spin remains the precondition for any competition tuning (§14.3).

32 The snow scheme, settled on coupled soil (2026–08–05)

A colleague's figure — OpenIFS `stl2` against RIHMI-WDC at 105 stations, coupled gen-10 runs — overturned §27.10. Reproduced independently to within rounding:

stl2 − RIHMI 20 cm [K]	DJF	MAM	JJA	SON	annual
AMIP K1 (adopted, tuned)	+0.49	−1.60	−1.44	−1.30	−0.96
COUPLED 10A (G4)	−4.33	**−6.57**	−3.62	−3.61	−4.53
COUPLED 10B (+scheme)	**−16.19**	−6.04	**−2.44**	−5.95	−7.66

January reaches −18.3 K. The scheme buys 1.2 K of summer and pays 12 K of winter. **It is rejected for coupling.**

Why the earlier verdict was wrong, recorded because the error is repeatable. I sampled the larch zone alone (60–72°N, 100–160°E), where 10A already sits near −21 °C in DJF — *saturated*. Both runs looked identical there. The damage is only visible across the wider, more southerly station network where 10A still retains its snow insulation. **I measured in the one place the signal was flat.** Worse, the AMIP run had already predicted it: the I1 delta is DJF −13.57 K, composing to ≈ − 13.3 against the −16.2 measured. When the coupled LAI figure appeared I treated it as overturning the AMIP evidence instead of asking why the two disagreed.

32.1 Reconstructing `ZCVS`, and what it excludes

`ZCVS` is not an output field, and neither are the tile fractions (`ZCVS` = `FRTI`(5)+`FRTI`(7) exactly, from `surfbc_ctl_mod.F90`:421,4
since the three cover fractions sum to one). But it is a pure function of `sd` and `rsn`, both of which *are* written, so it can be reconstructed offline from each run's own snowpack:

Siberian land	Sep	**Oct**	Nov	Dec	Jan	May
ΔZCVS	+0.026	**−0.075**	−0.008	−0.001	−0.000	−0.188
Δsoil [K]	+0.0	**−4.2**	−16.2	−22.2	−23.8	+2.4

Midwinter cover is identical (−0.0004 in January): at 0.43 m depth the tanh is saturated. So the −19 K January soil **cannot be a cover effect**. The cover difference *peaks in October exactly as the soil begins to diverge*, then vanishes while the soil runs on to −24 K — autumn seeding plus thermal memory, with nothing in midwinter needing to differ at all.

A mechanism proposed and falsified within the hour. ~~The snow fraction is applied twice on the conduction path: ZHFLUX is already area-weighted while ZSNCONDH is multiplied by PFRSN again, so the pack drains through a conductance scaled by area a second time.~~ **Wrong.** ZHFLUX is W per m^2 of *grid box* (which is why the /ZFRSN at srfsn_driver_mod.F90:271 is commented out), ZDSN uses *grid-mean* SWE, and ZSNCONDH is therefore also per m^2 of grid box. Flux, conductance and heat capacity are all in the same units. There is no second division for the factor to pair with.

Supporting evidence for the surviving reading: the snowpack cools only 3.9 K and the air 2.6 K while the soil cools 23.8 K. In January the control holds a +21.7 K gradient between snow surface and soil; I1 holds +1.8 K. The insulation has gone, with SWE identical and density *lower* — deeper, less conductive snow. No physical field changed in a way that could do it, which is what pointed at autumn seeding rather than a winter process.

32.2 Rounds 19: what is running, and what would falsify it

Four AMIP runs on the K1 base at **full campaign length** (1870–1916, evaluate 1872–1915), with daily sd/rsn/tsn/asn/stl1/stl2 added through a per-run add_config_sources override — so all four difference directly *and* are scorable against the standard thresholds. An earlier 10 yr version was cancelled: at that length they could not be judged against the 44-yr detection floors, which would have made them incomparable with the other 43 runs.

run	role	z_0	ρ_{new}	m	SCF Oct	SCF May
N1	reference, scheme off	—	—	—	—	—
N2	current parameters	0.016	100	1.6	0.894	0.817
O1	re-parameterised	0.018	170	**4.0**	**0.995**	**0.495**
O2	re-parameterised, milder	0.018	170	3.0	0.995	0.764

The O series rests on one piece of arithmetic. With SCF = $\tanh(d/L)$ and $L = 2.5 z_0 (\rho/\rho_{new})^m$,

$$\frac{x_{Oct}}{x_{May}} = \frac{d_{Oct}}{d_{May}} \left(\frac{\rho_{May}}{\rho_{Oct}}\right)^m = 0.469 \times 1.853^{\,m}.$$

Depth works *against* us — October's pack is half as deep. Density works for us, and $m > 1.23$ is needed merely for October to hold more cover than May. $m = 1.6$ **clears that bar but gives a ratio of only** 1.23, hence a separation of 0.077: the −0.075 October deficit that seeds the winter collapse buys a spring depletion only 0.08 deeper. $m = 4$ gives 5.5. Raising m alone makes October *worse* (0.383), because $\rho_{Oct} = 170$ already exceeds $\rho_{new} = 100$ — so ρ_{new} must be recentred on autumn density first.

~~Density cannot substitute for melt state; the two regimes are inseparable by density at their respective depths.~~ **Wrong, corrected the same day**: that was argued from the current parameters rather than from what the formula can do.

Why this replaced the hysteresis it was meant to be. A ripe/not-ripe state test is *not* hysteresis — it is reversible, so cover would flip back every night and every cold snap, injecting a spurious diurnal oscillation into the tile fractions. Real melt-out is irreversible within a season. Swenson & Lawrence (2012) avoid this with a prognostic seasonal SWE$_{max}$, which is a new surface prognostic. **Density sidesteps the problem entirely**: it is a genuine physical memory variable, ripening monotonically and not walking backwards. And it is namelist-only.

Pre-registered falsifiers. (i) Winter soil must return to the N1 reference; if O1 still shows the −16 K DJF collapse the density route is dead and only the SWE$_{max}$ rewrite remains. (ii) May/June cover must still deplete, or the summer gain is lost and O1 is all cost. *And this is curve-fitting until the daily data supports it*: $m = 4$ is far from Niu & Yang's calibrated 1.6 and was derived from two box-mean monthly numbers, while a steep exponent amplifies cell-to-cell spread.

33 Round 19 results and Round 20 (2026–08–06): the `tanh` is falsified on observations, and the curve is rebuilt

The O series failed its own pre-registered falsifier, and so did every mechanism proposed to explain the failure. Three hypotheses died in sequence before the real one was found, and *two of the deaths were caused by my own analysis errors* — both recorded in §33.4 because both are easy to repeat.

33.1 The measurement: cover is identical, the soil is 23 K apart

All four runs at 46 yr on the K1 base, Siberian box, differenced against N1:

	Δcover DJF	ΔSWE DJF	Δdepth DJF	**Δsoil DJF**	Jan f_{full}
N1 reference	—	—	—	—	**0.960**
N2 current m=1.6	−0.002	+0.65	+0.064	**−22.77**	**0.773**
O1 m=4	+0.001	−0.60	+0.047	−18.72	0.947
O2 m=3	+0.001	+0.12	+0.043	−16.93	0.958

DJF snow cover is identical to within 0.002 **across all four runs while the soil differs by 23 K.** October soil damage is $-4.94/-5.02/-4.95$ K — identical to 0.08 K — while October cover spans 0.10. Mean cover is not the winter channel, and no re-parameterisation of a cover curve could ever have been.

33.2 The mechanism: same mean, opposite state

The exposed area is $1-\overline{\text{cover}}$ by definition, so it is *identical* between the schemes. Where it sits is not, and the soil response is nonlinear in it.

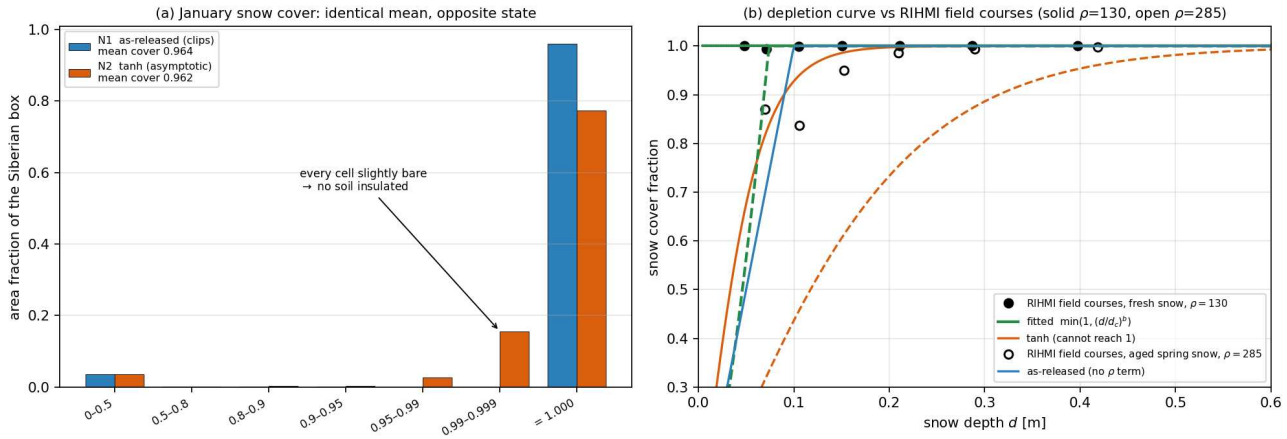


Figure 29: (a) January snow cover across the Siberian box. N1 and N2 have the same box-mean cover to within 0.002 and the same total exposed area, but N1 (clipping ramp) puts 93.5% of the box at cover *exactly* 1.0 while N2 (`tanh`) dumps a quarter of it into 0.99–0.999. A sliver of bare ground in *every* cell instead of a few genuinely bare cells: each sliver couples that cell's soil to the air through exposed tiles at $\lambda_{sk} = 10\,\text{W m}^{-2}\,\text{K}^{-1}$, about 16× stiffer per unit area than the whole snow column (0.61), so no soil is left insulated. **(b)** The depletion curve against 36 492 RIHMI snow-course surveys. `tanh` has range $(0, 1)$ *open at 1* and so cannot represent complete cover at any parameters; the as-released ramp reaches 1 but ignores density and so has no spring depletion; the fitted form does both.

The signature in the model is unmistakable: **N1 holds a 22 K gradient from soil** (265.0 **K) to snowpack** (242.7 **K); N2 holds 2 K** (240.4/238.6). A 2 K gradient under half a metre of snow cannot be produced by conduction — N2's soil sits at air temperature under a deep pack. `RTHRFRTI` = `EPSILON` (`su0phy.F90:1308`), so no tile is ever culled and every sliver is fully active.

f_{full}, the area fraction at cover ≥ 0.999, is the only quantity found that predicts the damage: per-cell correlation with Δsoil is $+0.75/+0.52/+0.52$ against $+0.32/+0.19/+0.10$ for mean cover.

33.3 RIHMI-WDC settles it: the as-released cover was right

Snow-course transects (`snmar`, 1–2 km, the direct observational analogue of `ZCVS`): **14 331 of 14 369 Siberian DJF surveys report cover of exactly** 10/10 — 99.74%, mean 0.9995. Complete cover is real, not an artefact

of clipping. Soil temperature (`tpg`, 43 stations, 174 676 DJF observations at 0.2 m against `stl2`):

	f_{full} DJF	soil at 0.2 m	**bias**
observed	**0.997**	**−6.1** °C	—
N1, scheme off	0.990	−5.3	**+0.8**
N2, `tanh`	**0.678**	−26.3	**−20.2**
O1	0.932	−22.2	−16.1
O2	0.965	−20.8	−14.7

Field January f(complete cover): **observed** 0.996**,** `tanh` 0.290. Period caveat: observations are 1963–2024 against a PI model, so the PI-equivalent observation is nearer −9 °C, making N1 ∼ +3 K warm — the scheme is still −13 to −18 K wrong. **This retracts §27.3**: for soil temperature the as-released cover was correct and the scheme replaced a good field with a broken one.

33.4 Two analysis errors, and four dead mechanisms

~~The soil collapse is seeded by an October cover deficit and carried by thermal memory.~~ Killed by its own pre-registered test: O1/O2 produced *positive* autumn cover anomalies and still lost 18–20 K.

~~A melt-state (ripe/not-ripe) gate can separate spring from autumn.~~ May ripe-day fraction is 0.04 in every run; the gate would fire in June, after melt-out.

~~The residual exposed fraction at saturation explains it — `tanh` never reaches 1 while the ramp clips to exactly 1.~~ Right about the codomain, wrong about the box: N1's own January cover is 0.963, not 1.0, so both schemes leave the *same* exposed area. The distribution, not the total, is what differs.

~~Drying the soil before freeze-up removes the latent-heat buffer.~~ The freeze-up buffer differs by only $-0.51/-0.72/-0.68$ K against 17–25 K to explain, and is not ordered with the damage — O1 has the largest water deficit and not the largest cooling.

~~More cover means a colder soil, because `ZSNCONDH` ∝ `PFRSN`.~~ $\mathrm{d}G/\mathrm{d}f = 0.61 - 10 \approx -9.4$: the exposed route swamps the snow route, so more cover means a *warmer* soil. The blanket intuition holds; the exposed fraction is what matters.

Both of my errors were the same error: averaging before applying a nonlinear function. `snow_daily_diag.py` averaged depth and density over 40 years and *then* applied the `tanh`; its Δcover numbers are void and the conclusions drawn from them were withdrawn. The first monthly fit check evaluated $\mathrm{SCF}(\overline{d}, \overline{\rho})$ against $\overline{\text{cover}}$, which made a good fit look like it saturated in May. $\mathrm{SCF}(\overline{x}) \neq \overline{\mathrm{SCF}(x)}$ — **apply the formula per sample, always.**

33.5 Round 20: `ECE_SNOW_SCF` = 3, fitted to observations

A source change this time (`surfbc_ctl_mod.F90` branch, `surfece.F90` parameters); modes 1/2 retained so N/O stay reproducible.

$$\mathrm{SCF} = (1 - c_{vh})\,\min\!\big(1, (d/d_{cl})^{b_l}\big) + c_{vh}\,\min\!\big(1, (d/d_{ch})^{b_h}\big)\,, \qquad d_c = \min\!\big(D_{max},\ S \cdot D \cdot (\rho/\rho_{ref})^{M}\big)$$

$\min(1, \cdot)$ **reaches 1 exactly** — that is the entire fix; everything else is calibration. It nests the as-released ramp at $D = 0.1$, $M = 0$, $b = 1$. Split by vegetation **type and fraction only** (via `PCVH`), never gridpoint or hemisphere, so it stays portable to other climate states and follows the vegetation if LPJ-GUESS moves the treeline. **No new prognostic and no new input**: `PSNM1M`, `PRSNM1M` and `PCVH` are already arguments of `SURFBC_CTL`, and **nothing knows the date** — the seasonal cycle is an output of the snow physics, not an input to the formula.

Parameters are fitted to 36 492 snow-course surveys, each measuring depth, density and covered fraction on the same transect, split by course type (1 = *pole*/field → low vegetation, 2 = *les*/forest → high). Constrained fit: hold Oct–Feb saturated, then minimise Mar–May error.

	Oct	Nov	Dec	Jan	Feb	Mar	Apr	**May**
field observed	0.998	0.999	0.999	0.999	0.999	0.994	0.974	**0.914**
field fit	0.999	1.000	0.998	1.000	1.000	0.996	0.970	**0.915**
forest observed	0.999	1.000	1.000	1.000	1.000	1.000	0.987	**0.931**
forest fit	0.999	1.000	1.000	1.000	1.000	1.000	0.987	**0.931**

Mar–May RMSE 0.0025/0.0001 against 0.0229/0.0156 (as-released) and 0.1207/0.0964 (`tanh`). **The frontier is flat** — the same parameters win at every winter floor from 0.000 to 0.998, so the winter/spring trade-off that dominated rounds 15–19 was an artefact of the `tanh` being unable to do either, not a real tension. Kept: $D_{cl} = 0.014$, $D_{ch} = 0.026$ m at $\rho_{ref} = 200$, $M = 4.70$, $b_l = 1.46$, $b_h = 0.40$, $D_{max} = 0.30$.

The physical reading is that d_c runs from ∼2 mm at $\rho = 130$ to 8–14 cm at $\rho = 285$: fresh low-density snow drapes over everything, while old snow is dense *because* it has been wind-redistributed and melt-metamorphosed, and that same history is what leaves bare patches. **Density is the proxy for age and melt state** — the hysteresis the campaign spent two rounds trying to build, delivered by a state variable.

33.6 The runs, and what would falsify them

P1 (`amip_P1_scffit`, $S = 1$) takes the observations literally. P2 (`amip_P2_scffit_x3`, $S = 3$) scales d_c for 100 km sub-grid variance. Both on the K1 base at 46 yr with the daily snow/soil output retained so f_{full} can be measured directly.

S is the one uncalibrated knob and is labelled as such. A snow course is 1–2 km; a TCO95 box is ∼100 km, and sub-grid variance grows with scale, so course-fitted d_c is a *lower bound* and $S > 1$ is the physically expected direction. At the Siberian box-mean May state ($d = 0.31$ m, $\rho = 313$) $S = 1$ gives $d_c = 0.118$ m, i.e. $d/d_c = 2.6$ and **no depletion at all** — P1 is expected to behave much like as-released in spring. Rutgers/IMS 24 km cover aggregated to the model grid is what should set S, and **that calibration is not done**; P2 is an exploratory bracket, not a candidate for adoption.

Pre-registered falsifiers. (i) The DJF soil must return to the N1/K1 reference; if it does not, the distribution diagnosis is wrong and this whole section fails. (ii) January f_{full} must return to ∼0.96 from N2's 0.773. (iii) If the scaled variant recovers the JJA gain *but* reopens the winter soil bias, then spring depletion and winter insulation are genuinely coupled at grid scale and the flat frontier found on station data does not survive.

33.7 P1 crashed: the fit extrapolated where it had no data (2026–08–06)

P1 aborted at 1888–03–22 with `ABOR1: Very snow cold temperature` (`srfsn_webal_mod.F90:451`, the PTSN < 100 K sanity guard):

```
Tsn 91.93613   SWE-1 5.6058891E-02   SWE 0.0   Snow frac 1.000000   heat 10.88281
```

A pack of 0.056 kg m^{-2} — half a millimetre — was given *full* snow cover. The snow tile then absorbed the entire grid-box flux into essentially zero heat capacity and the snow temperature collapsed. The cause is that the fitted power law was extrapolated below any depth the snow courses ever sampled: there is no survey under ∼5 cm, and at $\rho = 100$ the formula gives $d_c = 0.014(0.5)^{4.7} = 5.4 \times 10^{-4}$ m, so a 0.56 mm dusting reaches $d/d_c = 1.04$ and saturates. The governing ratio is SCF/SWE: as-released gives 0.100 at that state, mode 3 unfloored gives 17.86 — **180× worse**. I had written down that the low-depth end was unconstrained and then extrapolated into it with no guard.

The fix is `ECE_SNOW_SCF_SWEMIN`, which floors d_c at SWEMIN/ρ: a minimum snow *mass* before cover may saturate, which is the right currency because the abort is a heat-capacity failure, not a depth one. SWEMIN = 3 kg m^{-2} is the *smallest* value that beats the as-released margin (SCF/SWE = 0.053 against 0.100) while barely touching the calibration — October field cover 0.999 → 0.996 against an observed 0.998, and Dec–May unchanged to three decimals. SWEMIN = 5, which I had picked by guess first, would have cost 0.983 in October for no extra safety that matters.

P2 was cancelled un-crashed and is also withdrawn: SCALE = 3 only makes d_c 1.6 mm instead of 0.5 mm, so it carries the same defect and its output is TAINTED wherever thin snow occurs, which includes every autumn onset and spring melt-out. **P3** (SCALE = 1) and **P4** (SCALE = 3), both with SWEMIN = 3, supersede them.

~~P1 was stopped by an unexplained `scancel` from the account UID rather than by a model failure.~~ Wrong, and the reasoning behind it was worse than the conclusion: the model aborted normally on `PROC=48` and esm_tools tore the job down afterwards, exactly as designed. I searched `NODE.001_01` (rank 1 only) for `forrtl/nan/ abort` and for the esm_tools trigger strings, none of which match `ABOR1` or `MPL_ABORT`, concluded "no model error", and then constructed an elaborate infrastructure story on top of that gap. **Method note: an OpenIFS abort is `ABOR1/MPL_ABORT` and can land on any rank — grep the whole compute log for those tokens before concluding a job died externally.**

One tension, found while making Figure 29 and not resolved. The constrained fit optimises the *monthly climatology*, but the model applies the curve *pointwise*. Scored on the pointwise (d, ρ) bin surface instead, the adopted field parameters give RMSE 0.0624 against 0.0384 for a fit made directly to that surface ($b \approx 0.1$) — though both beat as-released (0.0771). For forest the adopted parameters are better on *both* metrics (0.0420 vs 0.0483; as-released 0.0644). So the field curve is a compromise: optimal on the seasonal cycle, second-best on the surface, and an improvement on the incumbent either way. Whether the climatology or the surface is the right objective is not obvious — the model integrates the curve over a cell distribution, which is closer to the climatology aggregation — and it should be settled alongside the Rutgers calibration of S.

Residual hysteresis is real but small. At matched (d, ρ) the ablation season is lower in *24 of 24* non-zero bins — systematic, $p \approx 10^{-4}$ — but the weighted mean is only -0.005 (field) and -0.001 (forest). Capturing it would mean plumbing `PWSNM1M` through the caller chain for $\sim$0.005 of cover; not done. The matched test can only compare bins populated in both seasons, so spring's extreme states ($\rho > 300$) are structurally untestable this way.

34 Round 20 results (2026–08–07): P3 adopted, and the fix costs the autumn

Both pre-registered falsifiers passed, and by more than they required.

	DJF soil vs N1	Jan f_{full}	soil→snow gradient, Jan
N1, scheme off	—	0.961	22.4 K
N2, `tanh`	**−22.71** K	0.755	**1.8** K (impossible)
P3	**+0.30** K	**0.965**	**22.9** K
P4	+0.16 K	0.965	21.8 K

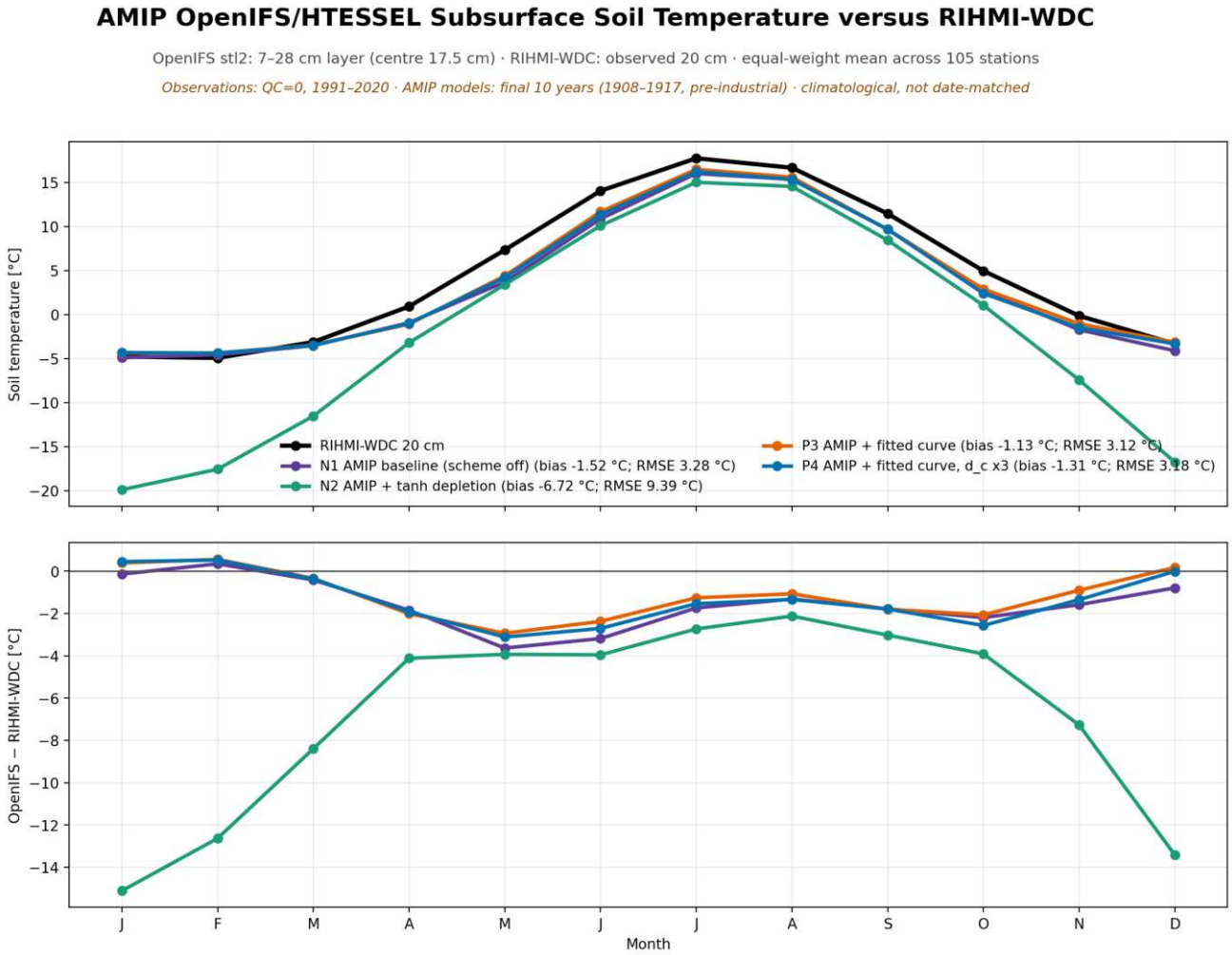


Figure 30: AMIP soil temperature against 105 RIHMI-WDC stations, built with the same methodology as the coupled 09C/10A/10B figure so the two are directly comparable (`stl2` 7–28 cm vs observed 20 cm, QC=0, 1991–2020, final model decade). The `tanh` run (green) peels away from October and reaches −15 °C in January — the same catastrophe 10B shows coupled. **P3 and P4 lie on top of the scheme-off baseline all winter, and slightly above it.** P3 beats that baseline on every metric: bias −1.13 vs −1.52, RMSE 3.12 vs 3.28, JJA −1.56 vs −2.08. The residual bias is now entirely a growing-season deficit that no snow scheme can touch. Absolute offsets are not comparable with the coupled figure (prescribed vs interactive SST); the spread between lines is.

Seasonal ANOVA against the control, each season on its own threshold: P3 −0.584 / +1.341* and P4 −0.132 / +1.092* (DJF / JJA). **Neither appears on `evaluate.sh`'s "warms JJA significantly but cools DJF**

significantly" list, which contains I1, I3, J1, J2, N2, O1 and O2. The winter penalty that has followed this scheme since round 15 is gone.

Marginal over the K1 base, which is the honest accounting: P3 +0.305 JJA / −0.088 DJF; P4 +0.056 / +0.364. Most of the +1.3 is K1's. P3 beats P4 despite SCALE = 1 being the conservative choice, which *falsifies my own prediction* that P3 would show no spring depletion — cutting cover *slows* box-mean melt, because the snow tile receives energy in proportion to its area.

~~SCALE is the parameter that controls the spring depletion, and the uncalibrated one that matters.~~ Varying it $1 \to 4$ moves the offline Sep–May RMSE by 0.002. P4 was a wasted run; `SWEMIN` is the whole lever.

34.1 Rutgers: the fix made the autumn 2.4× worse

Rutgers 24 km, Siberian box, area-weighted, model minus observed:

	Sep	Oct	Jan	Apr	May	Jun
observed (absolute)	0.067	0.599	0.999	0.956	0.655	0.202
N1 as-released	+0.107	+0.103	−0.035	−0.012	+0.118	−0.011
P3	**+0.259**	**+0.249**	−0.034	−0.019	+0.074	−0.035

The fitted d_c for fresh low-density snow is ∼2 mm, so a dusting saturates. That is *correct* at 1–2 km, which is what the snow courses measure, and wrong at grid scale, where September Siberia is 7 % covered rather than 33 %. The uniform −0.035 in midwinter is the model's permanent lake/water bare fraction, present in every run including scheme-off — not the scheme.

An offline `SWEMIN` sweep scored over the whole Sep–May season gives an *interior* optimum at 30 (RMSE 0.045 against P3's 0.139); past it the September gain is paid for by November and December going deficient. **P5** (SWEMIN = 15) and **P6** (= 30) bracket the albedo→melt→SWE feedback the offline sweep holds fixed, so those numbers are lower bounds on the sensitivity.

34.2 Round 21: resolution awareness, and a cap that disabled the floor

Three source changes (`562df81`), one of them a real bug.

`DCMAX` was capping the mass floor. The order was $d_c = \min(D_{max}, \max(\text{fitted}, \text{floor}))$, so the 0.30 m cap killed the floor above SWEMIN $= D_{max}\rho$ — 30 kg m^{-2} at $\rho = 100$. Every grid coarser than TCO95 therefore got no floor beyond the reference value, and it is *also* part of why the sweep flattened past 30 and reversed past 40, which I had attributed entirely to November. Now $d_c = \max(\min(D_{max}, \text{fitted}), \text{floor})$: the cap guards the steep $M = 4.7$ extrapolation of the *fit*, the floor is a separate physical statement about how much box-mean mass fills a box of this size. The two were nested and should not have been.

A resolution preset. `ECE_SNOW_SCF_DXKM` rescales `SWEMIN` from its TCO95 reference, $\text{SWEMIN}_{\text{eff}} = \text{clamp}(\text{SWEMIN} \cdot (\Delta x/100)^{0.624}, 3, 120)$, off by default so every earlier run reproduces. The exponent $\ln 10/\ln 40$ comes from the only two *measured* anchors: 3.0 at 2.5 km (the snow courses are 1–2 km transects, i.e. the high-resolution limit observed directly) and 30 at 100 km (Rutgers). Presets: TL21 94, TL63 48, TL255 26, TCO95 30, TCO399 13, TCO1279 6, TCO4000 3.

Below 2 km the scheme disables itself, falling back to the as-released ramp with a loud log line. At that scale the model resolves individual patches, so a subgrid curve double-counts them, and 2 km is the finest scale any calibration exists for.

~~The scale dependence can be measured as a power law by coarse-graining Rutgers and the model together.~~ It cannot, and the attempt is kept because the negative result is the useful part. The ladder confirms the *sign* — `SWEMIN` grows with box size, 41.5 at 66 km against ∼114 at ≥ 133 km — but the values **plateau** beyond 133 km rather than following a power law; the per-cell fit implies ∼80 at 100 km where the seasonal fit gives 30, a 3× disagreement that is the same climatology-versus-pointwise tension as the (d, ρ) surface fit; its own extrapolation to 2.5 km gives 9.4 against the course anchor of 3; and absolute skill is poor at every scale (RMSE 0.20–0.25 on a 0–1 field), which indicts the functional form rather than the parameter. **The preset is an interpolation between two trusted anchors, not a measured law, and a new resolution needs validation rather than a `DXKM` value.**

35 Round 18 results (2026–08–07): the aerosol term is six times smaller than inferred

M1 removes the anthropogenic MACv2-SP plumes entirely; M2 gives the same CAMS mass its 3D vertical distribution. Both sit on the present-day base, so the comparison is against `amip_presentday` over the same years.

Units first, because they invalidated the first pass. IFS TOA fluxes are *accumulated* $\mathrm{J\,m^{-2}}$ over the 3600 s step, not $\mathrm{W\,m^{-2}}$; the raw numbers come out ∼3600× too large. Dividing by the timestep reproduces the campaign's own recorded decomposition almost exactly — global +2.68 clear-sky and −1.18 cloud against the recorded +2.68 / −1.17 — which is the check that the pipeline is right.

band	M1 clear-sky	t	M1 **cloud**	t
60–90N	+0.381	+3.8*	**−1.014**	−7.7*
30–60N	+0.699	+9.4*	**−1.397**	−13.3*
tropics	−0.037	−2.0	−0.439	−5.9*
SO 45–65S	**−0.116**	**−18.9***	+0.022	+0.1
60–90S	+0.010	+0.3	+0.079	+1.0
GLOBAL	+0.126	+11.1*	**−0.542**	−8.9*

Paired annual differences over 25 yr; the runs share prescribed SSTs and years, so the pairing removes the forcing variability. * clears two-sided 95 % ($|t| > 2.06$).

~~Aerosol contributes ∼0.7 $\mathrm{W\,m^{-2}}$, 13 % of the Southern Ocean's +5.10 clear-sky excess, inferred from a +0.033 AOD excess against MISR.~~ **Measured, it is $-0.116\,\mathrm{W\,m^{-2}}$ ($t = -18.9$) — six times smaller.** Removing *all* anthropogenic aerosol moves SO clear-sky reflection by a tenth of a $\mathrm{W\,m^{-2}}$. Aerosol is ∼2 % of the SO excess, not 13 %, and **the unexplained residual grows from ∼1.3 to ∼1.9 $\mathrm{W\,m^{-2}}$.**

The two reflection columns are not separable. M1's effect is overwhelmingly on *cloud* — −1.40 against +0.70 at 30–60N, −0.54 against +0.13 globally — through `LMACV2SP_CCNF`. Any future aerosol lever must be scored on both columns; a clear-sky-only argument is invalid. M2 moves the tropical clear-sky column by +0.553 ($t = +79$) at *fixed column mass*, so where the aerosol sits is a first-order term of its own.

Open and untraced: the M1 clear-sky deltas are mostly *positive* — removing aerosol *increased* clear-sky reflection (+0.126 global, $t = +11.1$). That is backwards for a direct effect and is significant, so it is not noise; most plausibly a surface-albedo or water-vapour adjustment to the changed surface SW, but it has not been chased down.

36 Round 21 results (2026–08–07): P5 adopted, and the summer ceiling

run	DJF	MAM	JJA	SON
K1 (base)	−0.496	−0.265	+1.036*	+0.238
N2 `tanh`	−2.783*	−0.599*	+1.350*	−0.888*
P3 SWEMIN = 3	−0.584	−0.333	+1.341*	+0.217
P5 SWEMIN = 15	**−0.060**	**−0.140**	**+1.149***	**+0.004**
P6 SWEMIN = 30	**−0.850***	−0.288	+1.096*	**+0.487***

P5 is the cleanest run in the campaign — every season inside its own threshold bar a significant summer gain. Against Rutgers it cuts the autumn excess P3 introduced by ≈40 % (Sep +0.259 → +0.151, Oct +0.249 → +0.148) while leaving Nov–Jan intact, and its DJF soil is +0.22 K against the scheme-off reference.

~~Raising SWEMIN cannot reopen the winter, because at midwinter SWE≈93 the floor is 0.15 m against a 0.43 m pack — safe by construction.~~ True in *January*, and I never checked **November**, where the pack is still shallow. P6's November f_{full} falls to 0.752 against N1's 0.884 and its DJF soil to −1.00 K. The floor binds in early winter; that is the interior optimum the offline sweep predicted, arriving by the mechanism I had dismissed.

36.1 The JJA ceiling: the lever was nearly exhausted before round 15

Marginal over the K1 base, against the ±0.242 threshold: P3 +0.305 (marginal), P5 +0.113 (noise), P6 +0.060 (noise). Only P3 buys a measurable summer gain, and P3 is the version with the worst autumn cover. The reason is structural. May cover against Rutgers' 0.655:

	May cover		vs observed
N1 as-released	0.773	**+0.118**	(the entire available error)
N2 `tanh`	0.615	**−0.040**	(overshot *past* observations)
P3 / P5	0.729 / 0.732		+0.074 / +0.077

The whole spring cover error is 0.118**.** P5 removes 0.044; a perfect correction removes 0.118, worth roughly +0.2 K of JJA over K1 on a crude scaling of N2's response. **And N2's larger gain was never legitimate** — it took 0.158 out of a 0.118 error, overshooting into a −0.040 deficit, so that extra warmth came from having too little May snow. It is the same over-depletion that destroyed the winter soil, and any comparison against N2's +1.350 is a comparison against an error.

Honest accounting for the whole depletion line: worth ≈0.1–0.3 K of Siberian JJA, not the ≈1 K the early rounds implied. What it delivered instead is a scheme that is no longer *wrong*. **The missing summer is elsewhere**: every AMIP run including scheme-off tracks RIHMI soil through Dec–Mar and falls 2–3.5 °C short from April to September (Fig. 30), a deficit common to all of them, and that is the global tropospheric cold bias of §21.1 — still untouched after 50 runs, and now the sharpest remaining target. **Further `SWEMIN` optimisation is not worth runs**: P3 and P5 differ by 0.19 K in JJA, at or below noise.

37 The coupled radiation objective, and why the goal moves to zero (2026–08–07)

The 10 series branches from the same point as round 09 and runs twenty years longer, so it extends the campaign's central radiation figure rather than replacing it. 10A is 09C plus G4; 10B adds the `tanh` depletion since falsified (§34).

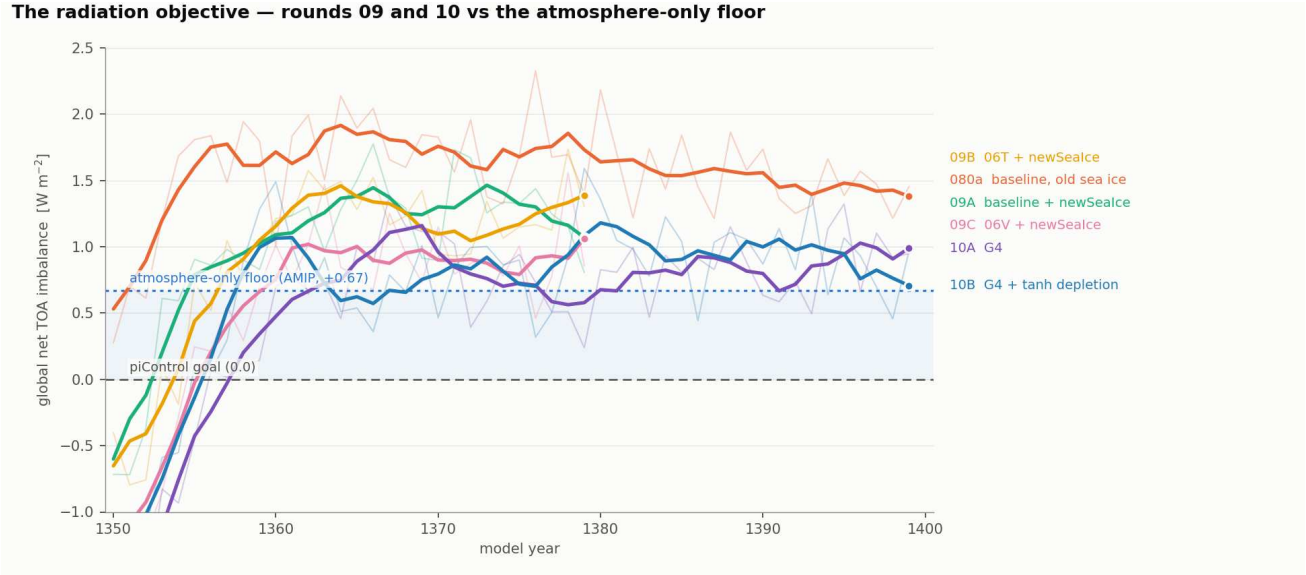


Figure 31: Global-mean net TOA by model year, annual (thin) and 5-yr running mean (bold). **G4 is worth ≈0.25 W m^{-2} at top of atmosphere**: 10A settles at +0.854 against 09C's ≈1.06, the best of the previous round, and both 10-series runs spend 1362–1380 at or below the +0.67 atmosphere-only floor. **10A and 10B are indistinguishable** (+0.854 vs +0.899, curves interleaving throughout), so the depletion scheme that destroyed the winter soil by 20 K cost essentially nothing at TOA — consistent with its damage being a distributional surface effect invisible to box-mean energy diagnostics. The dashed goal is now 0, not the inherited −0.16; the shaded band is the part of the gap ocean-side tuning could still close, and everything above the dotted line is atmospheric. 11C is absent: it aborted in its first leg on the crash of §33.7.

37.1 Why the goal is 0 and not −0.16

−0.16 was never derived. It is whatever the HR piControl settled at, adopted as a matching target. A target displaced from zero is only defensible if the atmosphere fails to conserve energy: with a spurious atmospheric source S, equilibrium gives SFC = TOA + S, so zero *ocean* drift would require TOA = $-S$. Whether the two conditions differ is measurable, not arguable.

Measured on 10A over 1390–1399, global, cos-latitude weighted, using the release tool's own formula (`ssr` + `str` + `sshf` + `slhf` − `sf` · 3.3355 × 10^8, each divided by the accumulation period):

term	$\mathrm{W\,m^{-2}}$
`ssr` + `str` + `sshf` + `slhf`	+1.843
snow enthalpy (`sf` = 1.042×10^{-5} m/step)	−0.966
net surface	**+0.877**
net TOA (`tsr` +238.749, `ttr` −237.895)	**+0.854**
residual	**−0.023**

The atmosphere conserves to $0.02\,\mathrm{W\,m^{-2}}$. There is no leak for an offset target to absorb, so "the ocean is not drifting" and "the system is not drifting" are the same condition, and both put the goal at **0**.

Snow enthalpy is doing nearly all of the reconciliation — $0.97\,\mathrm{W\,m^{-2}}$ of a 0.99 discrepancy. Omitted, the surface reads +1.84 against a TOA of +0.85, and that artefact is exactly the kind of apparent imbalance an offset target gets invented to absorb. Anyone comparing a raw surface flux against TOA without it will see a spurious $\approx 1\,\mathrm{W\,m^{-2}}$ gap.

Re-framing the campaign arithmetic against the corrected goal:

	net TOA	= atmosphere +	coupled-state	gap to 0
080a (untuned coupled)	1.38	0.67 +	**0.71**	1.38
09C (06V + newSeaIce)	1.06	0.67 +	**0.39**	1.06
10A (+G4)	**0.85**	0.67 +	**0.18**	**0.85**
10B (+`tanh`)	0.90	0.67 +	0.23	0.90

The conclusion is unchanged in kind and smaller in size: the ocean-side coupled-state term is now only $0.18\,\mathrm{W\,m^{-2}}$ in 10A, so **almost the entire remaining imbalance is the atmosphere** — the same +0.67 the AMIP testbed measures directly. Chasing −0.16 would have meant over-tuning by that margin against a target with no physical basis in this configuration.

~~The HR piControl goal of −0.16 is the target for the coupled spin-up.~~ Inherited, not derived; the budget closes to $0.02\,\mathrm{W\,m^{-2}}$, so the goal is 0.

38 Round 22 (2026–08–09): the energy target re-based, and three targets relocated

Four diagnostics, no new model time except the coupled runs a colleague had already finished. Three of them move a target, and two of them falsify something this report was recommending.

38.1 Coupled round 11: the first runs carrying G4 *and* D2b

The report has said the coupled branch carries only G4 out of the whole AMIP campaign. **That is out of date.** `11A`, `11B` and `11D` all carry `ECE_TUNE_RVRSMIN` 1000/1000/225 *and* `RCL_INPSEA`= 0.2 with `RCL_INPPMIN`= 70000 — G4 plus D2b — on top of a new `useIFSsoiltemp` coupling. `110Baseline` is 09C plus that coupling change alone and is therefore the correct control; comparing against 10A would confound the snow scheme with the soil-temperature coupling.

Δ vs 110Baseline	net TOA	glob T2m	Sib JJA	Sib DJF	**soil DJF**	SO cloud
11A +G4+D2b	−0.233	−0.150	−0.16	+0.13	−0.41	+0.84
11B +`tanh` snow	−0.060	−0.019	+1.81	−1.62	**−19.40**	+0.76
11D +fitted, `SWEMIN`= 30	+0.023	−0.031	+0.16	+0.05	**−0.45**	+0.41

The `tanh` falsification reproduces coupled on a clean pair: 11B minus 11A is −18.99 K of Siberian DJF soil, against the −13 to −16 K the AMIP run predicted. This is a better-controlled version of the 10B result, because 11A and 11B differ *only* in the snow scheme.

11D is not catastrophic, but it has not tested what it looks like it tested. 11D runs the fitted scheme at SWEMIN = 30 — that is P6, the setting AMIP *rejected* (DJF −0.850*, November f_{full} 0.752, soil −1.00 K), not the adopted P5 at 15. Against 11A it gives soil −0.04 K and DJF T2m −0.08 K. Before reading that as vindication, the detection threshold was measured from the interannual scatter of these very runs:

	interannual sd	95 % threshold, 10-yr pair
Siberian DJF soil (`stl2`)	0.80 K	±0.75 K
Siberian DJF T2m	1.33 K	±1.25 K

So the soil half of the P6 falsifier (−1.00 K) is marginally testable and comes out null; **the DJF T2m half (−0.850 K) is not testable at all on ten coupled years**, and a null there was the expected outcome whether the scheme is safe or not. What 11D *does* establish is that this diagnostic has ample power for damage of the kind that mattered — it sees 11B's −19 K without difficulty. **The outstanding coupled run is 11D at** SWEMIN = 15 **plus K1**, which would bring the coupled configuration level with the atmosphere-only one for the first time.

38.2 The energy target, period-clean

Every band target in this report was measured on `amip_pi_base` against CERES 07/2005–06/2015 — an 1870s atmosphere against a present-day satellite record. `amip_presentday` overlaps CERES directly and its forcing is *correct* for 1990–2014 (unlike the PI arm, whose `NCMIPFIXYR` never took effect, §17.3), so the epoch offset is measurable per band rather than caveated. Band means over full latitude circles are exact on each source's own grid, so nothing is remapped. The CERES read is validated against the published EBAF globals first: ASR 241.18 (published ∼240.5), OLR 240.21 (∼240.2), net +0.970.

band	PI vs CERES	**PD vs CERES**	offset	verdict
90S–65S	+7.24	+8.11	−0.87	robust to the epoch
SO 65S–45S	+6.32	**+7.46**	−1.15	**robust, and larger**
45S–30S	−4.46	−2.49	−1.97	partly epoch
tropics 30S–30N	−2.51	**−0.67**	−1.83	**epoch-dominated**
30N–45N	+0.81	+2.08	−1.27	partly epoch
45N–65N	+4.40	+5.35	−0.96	robust to the epoch
65N–90N	+0.39	+1.53	−1.14	partly epoch
GLOBAL	−0.33	**+1.23**	−1.56	

Two targets move. **The tropics are epoch-dominated** — period-clean the band is only $-0.67\,\mathrm{W\,m^{-2}}$ from CERES, not −2.51, so most of what §76.8 has carried as a tropical deficit was never model error. **The Southern Ocean survives the correction and grows**, +6.32 → +7.46, and its contribution to the global mean rises to $+0.735\,\mathrm{W\,m^{-2}}$. It is now unambiguously the largest single regional term.

A small discrepancy, recorded. §11.2.2 quotes +5.81 for this band against a CERES value of −55.3; reading the gridded climatology directly gives −55.81 and hence +6.32 on the same PI arm. The new number comes with a validated global check and is preferred, but the difference is not chased further — it does not change any conclusion.

The OLR test, which does not close. A cold troposphere under-emits, so the measured OLR deficit and the measured cold bias should be quantitatively consistent. Period-clean, ΔOLR = −2.529 and ΔASR = $-1.299\,\mathrm{W\,m^{-2}}$, which is where the +1.229 net comes from. But the effective emitting temperature from CERES OLR is 255.1 K, giving a Planck slope of $3.77\,\mathrm{W\,m^{-2}\,K^{-1}}$, and the model is −1.62 K through 700–300 hPa: a purely radiative response would be $-6.10\,\mathrm{W\,m^{-2}}$. **Measured minus predicted is $+3.57\,\mathbf{W\,m^{-2}}$**, and §38.4 identifies what is offsetting it.

38.3 The Southern Ocean deficit is cloud AMOUNT — and no lever moves it

The band error has been tracked by CRE, which cannot distinguish too little cloud from cloud that is not bright enough. Splitting it at the *observed* CRE per unit area (`amip_presentday` vs CERES, period-clean):

SO 65–45S	model	CERES	diff	
cloud area [%]	83.06	89.49	**−6.43**	
SW CRE [$\mathrm{W\,m^{-2}}$]	−60.39	−67.84	**+7.45**	
CRE per unit area	−72.71	−75.80	+3.10	
decomposition of the +7.45:				
AMOUNT (missing area × observed CRE/area)			**+4.87**	(65.5 %)
OPACITY (residual)			+2.57	(34.5 %)

Amount-dominated, and **zonally uniform** — Atlantic −5.40, Indian −6.82, Australian −6.55, Pacific −6.81 pp, a sector spread of 1.42 pp, which points at microphysics rather than circulation or the ice edge. It is worst in austral spring and summer (SON −7.03, DJF −6.94 pp), when there is shortwave to reflect: DJF alone carries $+11.67\,\mathrm{W\,m^{-2}}$ of net TOA error in the band. That seasonality is what the supercooled-liquid hypothesis predicts.

Then the part that falsifies the plan. Scored on cloud *area* for the first time, against the +6.43 pp that needs closing:

lever (44-yr AMIP, Δ vs control)	Δ area [pp]	Δ SW CRE	
D2a `inpsea.2`	+0.63	−4.80	9.7 % of the gap
D2b `inp+p700`	+0.43	−2.61	6.7 %
A1a `ovl=0.10`	+0.62	−6.44	9.7 %
G4 `tundra225`	+0.48	−2.73	**a LAND lever**
K1 `landalb`	+0.45	−2.62	**a LAND lever**
B7 `rvice.22`	−0.60	+0.66	wrong sign

~~G4 and K1 are boreal-land levers that cannot physically change Southern Ocean cloud area, and they move it by +0.48 and +0.45 pp, so ≈0.5 pp is the noise level of this diagnostic and every INP run sits inside it.~~ **Wrong, and corrected the same day.** G4 and K1 are not independent of D2b — they *contain* it (G4 = F4 + D2b + tundra, K1 = G4 + land albedo), so their +0.48/+0.45 is D2b's +0.43 inherited, not a null. Differenced against D2b the pure land contribution is +0.050 and +0.022 pp, which is the clean control and is genuinely negligible.

The real threshold, measured. Interannual scatter of SO cloud area in the 44-year control is 0.232 pp, giving a 95 % detection threshold of ±0.098 pp on a paired difference — five times smaller than the figure asserted above. **Every lever in the table is therefore statistically significant.** The conclusion survives, but it rests on MAGNITUDE rather than on significance:

lever	Δarea	% of the 6.43 gap	ΔCRE	of which opacity
D2a `inpsea 0.2`	+0.626	10 %	−4.80	90 %
A1a `ovlliqice 0.10`	+0.623	10 %	−6.44	93 %
D2b `inp+p700`	+0.432	7 %	−2.61	87 %
A1b `ovlliqice 0.35`	+0.331	5 %	−2.17	88 %
G4 minus D2b	+0.050	—	−0.12	*the land control*

The best lever in the campaign closes one tenth of the area gap and takes ≈90 % of its CRE from opacity. Two independent mixed-phase routes agree — ice nuclei (D2a/D2b) and the liquid/ice deposition overlap (A1a/A1b, the latter pushed to EC-Earth4's 0.1) — so this is a property of the mixed-phase branch, not of one parameter. **The amount deficit, two thirds of the error, is 90 % intact.**

~~The Southern Ocean cloud-area deficit should be attacked by pushing the INP branch harder, since D2b is the only lever that has moved the SO.~~ Written into the plan on 2026–08–09 and falsified the same day. The INP branch acts on opacity, which is 34.5 % of the error; its area response closes only one tenth of the gap, and the same is true of the deposition-overlap route even at EC-Earth4's setting. A cloud-*amount* knob is required, and it is a different knob.

One more pointer. CERES puts the SO effective cloud pressure at 656 hPa, i.e. just *above* the 700 hPa ceiling of D2b's pressure gate, with optical depth 7.41. The gate was designed to protect the tropics and it does, but it may also be excluding the layer where the observed cloud actually sits.

38.4 The cold bias is extratropical and UTLS — and the atmosphere is dry as well as cold

§21.1 established that above 700 hPa Siberia merely shares a global cold bias. Where that bias lives has never been asked. `amip_presentday` vs ERA5 1990–2014, annual, all levels exact in both so nothing is interpolated:

hPa	90S–60S	60S–30S	**tropics**	30N–60N	60N–90N	GLOBAL
850	+0.02	−1.17	−1.10	−0.90	−0.87	−0.99
700	−0.53	−1.40	−1.23	−1.00	−0.86	−1.15
500	−1.81	−1.75	−1.31	−1.56	−1.57	−1.49
300	**−3.65**	−2.48	**−1.72**	−2.34	**−3.43**	−2.22
200	**−4.20**	−3.28	**−2.09**	−3.53	**−4.45**	−2.87
100	+0.25	−1.52	−2.07	−1.30	+0.26	−1.51

The tropics are the *least* biased band, −1.72 at 300 hPa against an extratropical mean of −2.97. That excludes the hypothesis a reader would reach for first: a cold tropical upper troposphere over prescribed, correct SST would have meant a convective heating-profile error propagating globally. It is not that. The bias peaks at 200 hPa over *both* poles (−4.20 and −4.45) and **reverses sign at 100 hPa** (+0.25, +0.26) — a tropopause-region signature, not a tropospheric one. The seasonal tables behave the same way, so it is not a summer or winter effect.

And the atmosphere is dry, increasingly so with height:

hPa	ΔT [K]	q model	q ERA5	$\Delta q/q$ [%]
850	−0.99	5.7843	5.9887	−3.4
700	−1.15	2.9203	3.1596	−7.6
500	−1.49	1.0082	1.1018	−8.5
300	−2.22	0.1454	0.1695	**−14.2**
200	−2.87	0.0208	0.0260	**−20.2**
100	−1.51	0.0014	0.0026	−46.9

This is the missing $3.57\,\mathrm{W\,m^{-2}}$ of §38.2. The two errors oppose each other in OLR: the cold depresses emission, while the dryness lowers the effective emission level into warmer air and raises it. Over 700–300 hPa the model is −1.62 K and −10.1 % relative humidity content — both large, both real, and their sum is small.

The consequence is a constraint on all future radiation tuning. Fixing either one alone moves net TOA by several $\mathrm{W\,m^{-2}}$ in the *wrong* direction. The global budget that §11.2.1 found "right by cancellation" is cancelling in the vertical as well as in the horizontal, and any lever scored on a global mean is being scored on top of that. *Caveat:* ERA5 upper-tropospheric humidity is itself weakly constrained by observation, so the 100 hPa row is indicative only; the 300–200 hPa rows carry the argument.

38.5 What round 22 changes about the plan

- **The Southern Ocean needs a cloud-*amount* lever**, not more ice nucleation. Two thirds of the error is missing area; the INP family moves area at the noise level. Whatever is chosen must be scored on area, and the D2b gate ceiling should be revisited against the observed 656 hPa cloud pressure.
- **The tropical TOA deficit is largely not model error** and should come off the open list at its old size.
- **The cold-bias investigation starts at the extratropical tropopause**, and it must treat cold-and-dry as one problem. A lever that warms without moistening will make the energy target worse.
- **The next coupled run is 11D at** SWEMIN = 15 **plus K1.** Ten years cannot resolve a 1 K DJF signal, so if the P6-versus-P5 question is to be settled coupled at all, it needs either a longer run or the AMIP answer taken on trust — and the AMIP answer says 15.

39 Round 23 (2026–08–09): the reference check, and a round withdrawn

Round 23 was designed to close the Southern Ocean cloud-*amount* deficit. Three of its four runs were withdrawn before completion — one as a duplicate, two on a design error — and the round's lasting product is a reference check that decides which observation the campaign is allowed to tune against.

39.1 Prior work this campaign had not incorporated

Two AWI-ESM project issues (#169, #170, 2025–03 to 2026–04) cover the same Southern Ocean warm bias from the coupled side and contain results this report was working without.

- **The origin is cy48's moist-physics upgrade.** Cycle 48r1 raised high cloud cover everywhere relative to 43r3, strongest in the SH high latitudes, and lowered SO low cloud (#170). The exact diff is the ECMWF Newsletter 164 moist-physics patch, which was evaluated mainly at ∼5 km and mainly in low latitudes.
- **The lead/lag test, which orders the causal chain.** Scored on *year one* rather than the 50-year mean, the hcc offset is already present while the T2m bias and the SO lcc dip are *not*. So hcc leads and lcc lags: `fix hcc` → `reduces t2m` → `fixes lcc`. In the coupled model the SO low-cloud deficit is partly a *consequence* of the warm bias.
- **`RVICE` is already the hcc lever.** The v3.4 tuning raised it from the 0.13 source default to 0.16–0.19 (TUNE1–TUNE5, with `NDECOLAD` 2 → 0 and `RALBSEAD` 0.06 → 0.05) precisely to suppress SO high cloud. Every run in this campaign carries `RVICE`= 0.16 for that reason.
- **The compensating-bias trap is documented.** "All further tests regarding reducing HCC resulted in Arctic and Tropics becoming too cold", and `RALBSEAD` was found "not a strong enough counter tuning method for the combined `RVICE` + `ENTSTPC3`".

~~Lowering RVICE below 0.16 would add tropical cirrus and warm the tropics, providing the counter-lever the Southern Ocean levers need.~~ Mechanically correct — B7 at 0.22 gives Δhigh −2.31 pp and ΔLW CRE −1.35 — but it would undo the v3.4 SO tuning, which raised RVICE for exactly this reason. RVICE is global and the two regions need opposite signs, so it cannot serve both.

39.2 Four-way attribution: what kind of error each band has

Net TOA cannot distinguish a surface-albedo error from a cloud-reflection one. Splitting each band against CERES into clear/cloud × SW/LW, period-clean, positive = the model gains energy:

band	net	SW clr	**SW cld**	LW clr	**LW cld**	contribution
90S–65S	+8.11	+1.53	+3.98	+4.35	−1.75	+0.390
SO 65S–45S	+7.46	−3.79	**+7.45**	+4.21	−0.40	**+0.735**
45S–30S	−2.49	−2.54	−2.82	+4.66	−1.79	−0.257
tropics	−0.67	−1.39	−0.40	+3.28	**−2.16**	−0.336
30N–45N	+2.08	−4.90	+4.67	+4.36	−2.05	+0.215
45N–65N	+5.35	−2.67	+2.45	+5.11	+0.45	+0.527
65N–90N	+1.53	−3.79	−1.45	+6.30	+0.46	+0.073

The two problems are different cloud types. The Southern Ocean loses its energy through **SW cloud** (+7.45): the low cloud does not reflect enough. The tropics lose theirs through **LW cloud** (−2.16): the high cloud does not trap enough. With SST prescribed and correct, a tropical LW-cloud deficit is an anvil/cirrus problem — too low, too thin, or radiatively too weak — not a surface or a convective-heating one.

That is why no single microphysics knob has ever satisfied both, and it is also the shape of a solution: a lever that makes ice cloud *fewer but optically stronger* would move the Southern Ocean and the tropics in opposite, both-correct directions. RVICE changes amount alone, which is why it trades one for the other.

Scanning all 15 NAMCLDP-family levers for tropical response, only three warm the tropics at all: B3 clddiff= 1.5e-5 (+2.02, via SW not LW), B2 clddiff_convi (+0.65) and K1 (+0.37, clear-sky, its subtropical crop/semidesert albedo). **Nothing in 50 runs warms the tropics through the longwave**, which is the term that is actually deficient.

39.3 Which reference is right about Southern Ocean cloud?

The round-22 amount deficit rests on one source. Bringing ERA5 in appeared to contradict it outright, so before spending runs the references were tested against each other.

SO 65–45S	model	ERA5	CERES	MODIS
cloud area [%]	83.1	81.6	**89.5**	**89.3**
SW CRE [W m^{-2}]	−60.39	−61.52	**−67.84**	—
vs CERES	**+7.45**	**+6.31**	—	—

The test is what the cloud DOES, not how much is counted. Cloud area is not one quantity: CERES cldarea is a MODIS *mask* fraction that flags optically thin cloud in full, while ERA5 and model tcc are radiative covers built by overlapping layer fractions. Comparing them directly is a category error. Shortwave CRE is a difference of two broadband fluxes and *is* identically defined across all three.

If ERA5's lower area were a definition artefact, ERA5 would reflect like CERES. **It does not: ERA5 has low area *and* low reflection, in proportion** — 6.31 W m^{-2} short where the model is 7.45 short. ERA5 is reproducing the model's bias because ERA5 *is* IFS. **So CERES is the arbiter and the Southern Ocean amount deficit is real.**

~~The Southern Ocean cloud-area deficit may be an artefact of comparing a satellite mask against a model tcc, since ERA5 puts SO cloud *below* the model's.~~ Raised as a concern and refuted the same day by the CRE test above. **ERA5 cannot be used as a cloud reference for an IFS-derived model at all** — its global cloud area is 62.8 against CERES's 67.4, and in the tropics it reads 55.3 where the model's 61.6 matches CERES's 61.7 almost exactly. On cloud, ERA5 is the outlier, not the model.

A limit on CERES, in the other direction. At 90S–65S CERES reads 67.5 and MODIS 59.8 — a 7.7 pp disagreement between two products of the same instrument. Passive cloud masking over bright ice is unreliable. ~~The Arctic has +11.49 pp too much cloud (§38.2 band table).~~ Not supportable: that number is a cloud mask over sea ice and snow, exactly where the two retrievals disagree most. **CERES cloud area is trustworthy over dark ocean and should not be used over ice.**

39.4 The runs, and why three were withdrawn

Four levers were designed on the P5 base (which already carries G4, K1 and D2b):

run	lever	outcome
S1	`RCLDIFF` 6e-6 → 3e-6	**cancelled** — would cool the tropics
S2	`RCL_OVERLAPLIQICE` 0.65 → 0.35	**withdrawn** — this is A1b, already run
S3	`RCLCRIT_SEA` 2.5e-4 → 6e-4	**cancelled** — same reason as S1
S4	`RCL_INPPMIN` 70000 → 50000	**running**

~~`RCL_OVERLAPLIQICE` 0.65 → 0.35 is an untested mixed-phase lever.~~ It is **A1b**, run for 44 years in round 3; A1a is the same knob at EC-Earth4's 0.1. The A-series label "ovl" denotes the liquid/ice *deposition* overlap, not the radiative cloud overlap, and it was read as the latter. The duplicate is not a total loss: A1a/A1b give a *second, independent* mixed-phase route, and it behaves exactly like the INP route (+0.62 pp of area for −6.44 of CRE, i.e. 93 % opacity), which is far stronger evidence that opacity-not-amount is a property of the whole mixed-phase branch than one parameter could be.

S1 and S3 were cancelled for the tropics, not for the Southern Ocean. Both add low cloud, and B3 — the same parameter as S1 moved the other way — *warms* the tropics by $+2.02\,\mathrm{W\,m^{-2}}$. S1 would therefore cool a band that is period-cleanly only −0.67 from CERES, overshooting it roughly threefold. The deficit they were built to close is real; a *global* low-cloud knob is the wrong instrument for it. Their runscripts are kept under `runscripts/oifsamip/withdrawn/`.

`RCLCRIT_SEA` and `RCLCRIT_LAND` were exposed to `NAMCLDP` for S3 (`sucldp.F90`: moved from the `ASSOCIATE` block onto the `POINTER` mechanism every other `NAMCLDP` variable uses, with the association placed before the defaults are assigned so a silent namelist still reproduces). The exposure is kept — it works, and the lever may be wanted later with better targeting.

39.5 What round 23 changes about the plan

- **The Southern Ocean deficit is confirmed and stays the largest single term**, but it needs a *regime-selective* reflection lever. Global low-cloud knobs overshoot the tropics; the D2b pressure gate is the template for what selective looks like. *Answered in §40*: marine CCN is that lever, it is selective by construction, and it reaches the opacity third only.
- **The tropics need longwave cloud *effectiveness*, not more cloud.** The deficit is $-2.16\,\mathrm{W\,m^{-2}}$ of LW CRE, and no lever in 50 runs has moved the tropics through the longwave at all.
- **Reference hygiene is now a stated rule.** ERA5 is not an independent cloud reference for this model; CERES cloud area is valid over dark ocean and not over ice.
- **Check the prior issues before designing a round.** Both errors this round — the A1b duplicate and the `RVICE` proposal — were already answered in #169/#170.

40 Round 24 (2026–08–09): the marine CCN path, traced end to end

No runs. The question was whether cloud condensation nuclei offer the *regionally selective* Southern Ocean lever round 23 concluded was needed. The answer is a qualified yes, by a route nobody had looked at, and the trace corrected one of my own claims on the way.

40.1 The path, and where it is decided

Marine CCN reaches cloud albedo through four hops, each gated by a switch whose *runtime* value had to be read from `fort.4` rather than a source default:

where	what	runtime state
`surfrad_ctl_mod.F90:772`	Genthon (1992) sea salt from wind	`LCCNO`=**.TRUE.**
`liquid_effective_radius.F90:153`	CCN → N_d → r_{eff}	`NRADLP`=**2**
`cloudsc.F90:1023`	MACv2-SP CDNC factor	`LMACV2SP_CCNF`=**.true.**
`cloudsc.F90:2050`	CCN → autoconversion (lifetime)	`LAERLIQAUTOLSP`=**.FALSE.**

With `LCCNO` true the code takes the `PCCN_SEA` branch, so **`RCCNSEA` is inert** — its own source comment ("now irrelevant") is correct, but only because of a switch three files away. The live formula is

$$Q = e^{0.16u+1.44}, \quad \mathrm{CCN} = 10^{\,1.2+0.5\log_{10} Q}, \quad N_d = -1.15{\times}10^{-3}\mathrm{CCN}^2 + 0.963\,\mathrm{CCN} + 5.30.$$

~~Marine CCN is pinned at a uniform 125 cm^{-3} because `NAERCLD`= 0 switches the aerosol–cloud path off.~~ Read from a *default assignment* in `sucldp.F90` and reported as the runtime state, while `fort.4` carries `LMACV2SP_CCNF=.true.`, `PCCN` had already been printed as a 2-D field, and §35 of this report already documented that CCN path. Three independent disproofs, two of them on screen at the time.

40.2 The correction that matters: N_d is right

Evaluating Genthon on *monthly-mean* winds gives a Southern Ocean wind of 6.54 m s^{-1} and N_d = 55, which against an observed 60–120 cm^{-3} looks like a clear deficit. **That was wrong.** Monthly *vector* averaging cancels the transient eddies that are the entire Southern Ocean wind field, and the parameterisation acts on instantaneous speed. On 6-hourly output:

SO 65–45S	wind [m s^{-1}]	CCN	N_d
monthly-mean wind (wrong)	6.54	55.6	55.3
6-hourly wind	**10.28**	**77.6**	**73.5**

~~Southern Ocean droplet number is too low, so the cloud is too dark, and raising marine CCN is the opacity lever.~~ $N_d \approx 73$ cm^{-3} sits squarely inside the observed 60–120. **The mean is correct.** This is the same error class as $\mathrm{SCF}(\bar{d}) \neq \overline{\mathrm{SCF}(d)}$ from round 19 — a nonlinear operator applied to a mean — committed hours after writing it into a skill as a rule.

What survives is the phase. The model's seasonal cycle is inverted:

SO 65–45S	DJF	MAM	JJA	SON
N_d [cm^{-3}]	**68.5**	74.9	**76.1**	74.5
SW CRE error [W m^{-2}]	**+10.86**	+3.71	+2.09	**+13.11**

Model N_d is *lowest* in DJF because it is wind-driven and Southern Ocean winds peak in winter. Observations put the maximum in DJF, driven by biogenic sulphate. So the model has its fewest droplets exactly when the sun is up and the measured error is largest. Crudely — $\tau \propto N_d^{1/3}$, two-stream albedo $A \approx \tau/(\tau + 7.7)$ on CERES's $\tau = 7.41$ — lifting DJF N_d from 68.5 to an observed ∼95 is worth about −6 W m^{-2} of DJF SO CRE, roughly half that month's error.

40.3 The amount two-thirds is unreachable from here

N_d acts through r_{eff}, i.e. **opacity**, which round 23 measured as 34.5 % of the SO error. The cloud-*lifetime* route that would reach the amount two-thirds runs through `LAERLIQAUTOLSP`/`LAERLIQCOLL`, and both are `.FALSE.`

Enabling them would divide by zero. Traced: `callpar.F90`:1165 sets `AUXL%ZCCN`= 0 when `NAERCLD`≤ 0; `cloud_layer.F90`:191 passes that array to `CLOUDSC` as `PCCN`; `cloudsc.F90`:2050 then evaluates (`RCCN`/`PCCN`)$^{0.333}$. Using the lifetime effect requires `NAERCLD`> 0 and prognostic aerosol, which also changes the direct effect — so it stops being one lever. **Recorded so nobody enables the switch.**

40.4 The missing term, and that the data already exists

Genthon is sea-salt-from-wind with *no biogenic term*, and MACv2-SP — the only other live CCN path — is anthropogenic plumes only. Natural marine biogenic sulphate is absent from this model by construction, which is exactly why the seasonal phase is wrong.

The input already ships with OpenIFS and nothing reads it:

`climate.v020/95_4/month_dms`	`emdms`, paramId **210043**
units	kg m^{-2} s^{-1} (DMS surface *emission*)
grid	N96 reduced Gaussian, 40 320 points — **TCO95's own grid**
time	12 monthly fields
land	−99 **sentinel**

It exists at every resolution from TL21 to TCO1279. It is not ingested: `ICMCL` carries only `al`, `aluvp/d`, `alnip/d`, `lai_lv/hv`, and OpenIFS references DMS only in the TM5/M7 chemistry paths, which need `NACTAERO`≠ 0.

Route, via ocp-tool rather than `mkclimr.sh`. ocp-tool already owns `ICMCL` (`config.py`:180–186, `paleo_input.py`:278–320, `run_ocp_tool.py`:189 step 14b) and, decisively, already applies a changed land–sea mask to it. `emdms` is an *ocean* field with the opposite land convention to albedo and LAI, so under a moved coastline new ocean points need infill and new land points need −99. `mkclimr.sh` cannot do that; ocp-tool can, which is what makes the field paleo-safe.

1. ocp-tool: append `emdms` to `ICMCL` in deterministic order — `ece_updclie_climr.F90`:233 aborts on "Inconsistent order of fields".
2. ocp-tool: extend step 14b for the inverted land convention.
3. Reader: one new `paramId` case (210043) beside the existing 174 / 15–18 / 66–67 branches. **`max_fields`= 10 against 7 in use**, so there is headroom for an eighth field without touching the limit.
4. Physics: an additive biogenic term on `PCCNO` in the `LCCNO` branch.

The interface is a DMS surface flux in kg m^{-2} s^{-1}, which is exactly what a REcoM DMS module would produce. So a prescribed climatology now and a prognostic source later are a drop-in swap at the ocp-tool boundary, with no change to items 3–4. That is a better seam than a chlorophyll relation, which would need its own empirical fit on both sides.

Unpriced, all cheap: the provenance of ECMWF's `emdms` (Kettle & Andreae 2000 or Lana et al. 2011 — matters for citation); the flux→CCN relation, which is the one genuinely scientific choice and needs a published basis, not a fitted curve; and whether −99 is the sentinel everywhere or only where it was sampled.

Why this is not scheduled first: the SO band is 9.8 % of the globe, so closing the opacity third perfectly is 2.57 W m^{-2} in-band but only ≈0.25 W m^{-2} global, against a +1.23 error. 11E is a namelist edit on a lever already validated over 44 AMIP years. Do the cheap decisive run first.

~~Prognostic marine biogenic CCN needs the `-cc` setup with REcoM ocean biogeochemistry.~~ REcoM carries no DMS tracer; the only DMS in the tree is Icepack's sea-ice biogeochemistry (`dmspp, dmspd, dms`), which is DMS *within* sea ice and is in a component this configuration does not run (`whichevp`= 1, FESOM's native EVP). And there is no coupling field or atmospheric sulphur chemistry to receive it. The prescribed `emdms` route needs none of that.

40.5 T1 read test: passed, and the `emdms` data characterised

The 1-year read test completed with no abort and the reader's own diagnostics confirm every pass criterion: `Found 12 dates`, `Found 8 fields`, `emdms` discovered as the eighth GRIB message matching `NCLIGC(8)`, and `CLIMR` slot 8 filled. Slot 8 takes **twelve distinct values through the model year**, following a clean seasonal march on rank 1's northern high-latitude strip: 1.13 in January, peaking at 3.04 in June, minimum 0.30 in October, returning to 1.13. **The DMS ingest works.**

The data, characterised at full precision (source file and built `ICMCL` are identical, so the ocp-tool step is also verified):

land sentinel (−99)	29.9 % of points
ocean values	min 0.043, median 2.05, max 53.25
exact zeros	240 of 339 360 (0.1 %)
SO 65–45S, September	0.52
SO 65–45S, December	**5.06**
austral seasonal ratio	**9.8**×
tropics, all months	2.0–3.1 (flat)

The phase is exactly the missing term. Model marine CCN is wind-driven and so minimises in DJF (N_d = 68.5) and peaks in JJA (76.1), anti-phased with observation. `emdms` peaks in DJF at 9.8× its September value — largest precisely when and where the SO cloud error is worst (DJF −6.94, SON −7.03 pp).

~~`emdms` is a DMS surface *emission flux* in kg m^{-2} s^{-1}, which is exactly what a REcoM DMS module would produce, so a prescribed climatology now and a prognostic source later are a drop-in swap.~~ The GRIB carries that unit label, but values of ∼2 kg m^{-2} s^{-1} of DMS are physically absurd. A global mean of 2.36 with maxima of 53 matches **DMS seawater *concentration* in nmol L^{-1}** — Lana et al. (2011) put the global surface mean at ≈2.3. **The unit label is wrong.** That inserts a step the round-24 design did not price: a sea–air flux parameterisation (wind-dependent piston velocity, Nightingale or Wanninkhof) is needed *before* any flux→CCN relation. The interface

argument survives — REcoM would also produce a concentration — but the flux calculation belongs on the atmosphere side.

Two diagnostic traps recorded, both mine. The reader printed paramIds with an `i5` format, so the 6-digit 210043 came out as `*****`; widened to `i8`. And the field was twice declared broken from too small a sample — once from an `awk` column error that reported the ocean maximum as 0, once from reading four *consecutive* timesteps, which all fall in the same month, and concluding the field never varied. Both were wrong; the sampling was.

40.6 T2: the sea–air flux, validated in the model

§40.5 established that `emdms` is a *concentration*, so a flux has to be formed before the field can drive anything — and the flux is what must be validated first. Implemented as a diagnostic in `surfrad_layer.F90`, where the DMS field, the model's own instantaneous wind and the surface temperature are all already in scope:

$$\mathrm{Sc} = 2674.0 - 147.12\,T + 3.726\,T^2 - 0.038\,T^3 \qquad \text{Saltzman et al. (1993)}$$

$$k_{600} = 0.222\,u^2 + 0.333\,u \quad [\mathrm{cm\,hr^{-1}}] \qquad \text{Nightingale et al. (2000)}$$

$$k = k_{600}\sqrt{600/\mathrm{Sc}}\,/\,360000 \quad [\mathrm{m\,s^{-1}}]$$

$$F = k\,C_w \times 10^{-6} \quad [\mathrm{mol\,m^{-2}\,s^{-1}}]$$

with $1\,\mathrm{nmol\,L^{-1}} = 10^{-6}\,\mathrm{mol\,m^{-3}}$. The wind is the model's instantaneous lowest-level speed, *not* a monthly mean — k is quadratic in u, and §40.2 showed monthly vector averaging halves the Southern Ocean wind.

The pass criterion was set before running. Hand-evaluating the chain at $C_w = 2\,\mathrm{nmol\,L^{-1}}$, $u = 7\,\mathrm{m\,s^{-1}}$, $T = 15\,^\circ\mathrm{C}$ gives $k_{600} = 13.2$, $\mathrm{Sc} = 1177$, $k = 2.6 \times 10^{-5}\,\mathrm{m\,s^{-1}}$ and $F = 5.2 \times 10^{-11}\,\mathrm{mol\,m^{-2}\,s^{-1}}$; scaled by an ocean area of $3.6 \times 10^{14}\,\mathrm{m^2}$, $3.156 \times 10^7\,\mathrm{s}$ and $32.06\,\mathrm{g\,mol^{-1}}$ that is $\approx 19\,\mathrm{Tg\,S\,yr^{-1}}$, against a published 18–34 (Lana et al. 2011 give 28.1). So the model should print a mean flux of order 10^{-11}–$10^{-10}\,\mathrm{mol\,m^{-2}\,s^{-1}}$.

The run completed a full year, 8760 steps, no abort, 70 088 diagnostics (8 blocks × 8760).

flux, minimum	3.78×10^{-13}
flux, median	2.33×10^{-11}
flux, mean	$\mathbf{3.52 \times 10^{-11}}$
flux, maximum	3.31×10^{-10}
predicted window	$\mathbf{10^{-11}}$–$\mathbf{10^{-10}}$
indicative global budget	$12.8\,\mathrm{Tg\,S\,yr^{-1}}$
published	18–34 (Lana et al. 2011: 28.1)

The unit chain is correct. The mean flux sits inside the predicted window, and it varies while C_w is uniform on a block — so the variation is coming from wind and temperature through k, which is the intended behaviour.

Three limits, stated so the number is not over-read. The diagnostic covers rank 1's blocks, a northern high-latitude strip, not the globe. Land points enter as zero because the flux clamps C_w at 0 to reject the −99 sentinel, which drags the sampled mean down. And the 12.8 is that non-global mean scaled as though it applied over all ocean. Landing within a factor of ≈2 of 28.1 on so crude an extrapolation is a *pass* for a unit test — the purpose was to catch an error of orders of magnitude, not to measure the budget. A real budget needs a global reduction the diagnostic does not do.

The run changes no physics. The flux is computed and printed, deliberately not wired into `PCCNO`, because the flux→CCN step is a separate scientific choice that needs a published relation rather than a fitted constant.

40.7 T3: DMS-Rev3 ingested through ocp-tool

The shipped `month_dms` is Lana et al. (2011). DMS-Rev3 (Hulswar et al. 2022) rests on 865 109 observations against L11's 47 313, and the paper states the largest revisions are in the polar oceans — the only region this campaign is looking at. Measured on the model's own grid after regridding, SO 65–45S:

SO 65–45S	L11	DMS-Rev3	change
annual	1.93	2.24	+16 %
DJF	4.23	4.36	+3 %
SON	**1.34**	**2.02**	**+51 %**
November	**2.18**	**4.24**	**+95 %**
global maximum	53.25	26.56	−50 %

Rev3 does not raise the summer peak, it broadens it — the bloom starts a month earlier — and it halves the maximum, which is the paper's "reduced patchiness". The model's SO shortwave CRE error is worst in SON (−7.03 pp) and DJF (−6.94), so the upgrade lands in the season the old climatology was weakest.

Units derived, not read. `DMS_REV3.nc` carries no attributes at all. Integrating the shipped `DMSflux` globally gives 28.19 Tg S yr^{-1} (`DMSfluxSI` 28.36) against a published 27.1–28.1, which fixes the flux as mol m^{-2} d^{-1} and the concentration as nmol L^{-1}; every other assumption is out by orders of magnitude. Only the *concentration* is ingested — the shipped flux uses Rev3's own wind climatology, whereas the model must form the flux online from its own wind, which is quadratic in speed. Their flux is retained as a validation target.

Built through ocp-tool (`dms_rev3_to_grib.py`), which is what makes it paleo-safe. Nearest *valid* neighbour in 3-D — 1° source against ≈0.9° target is matched resolution, so nothing is invented, no land NaN bleeds into coastal ocean, and the dateline and poles cannot yield a spurious neighbour. Values are overwritten into the shipped messages, so grid, `dataDate`, `paramId` and all headers stay byte-identical to what the reader accepts. The land sentinel is taken from the template, preserving the model's own mask; a Rev3 hole at a template ocean point would read as land, so L11 is retained there and counted — for TCO95 that count is **zero**, with 12 040 land points held at −99. `icmcl_dms.py` then interleaves it: 84 → 96 messages, 12 dates × 8 fields.

T3, one year, verified. `Found 12 dates`, `Found 8 fields`, no ordering abort, full year, no abort. Against T2 on the identical wind field:

	C_w [nmol L^{-1}]	flux mean	indicative global
L11	1.825	3.515×10^{-11}	12.8 Tg S yr^{-1}
DMS-Rev3	**2.044**	$\mathbf{4.218 \times 10^{-11}}$	15.4
ratio	1.120	**1.200**	

The flux rises by 20 % where the concentration rises by only 12 %. That is not an inconsistency — the flux is $\overline{k\,C_w}$, not $\bar{k}\,\overline{C_w}$, and k varies spatially, so the ratio of means need only agree if the spatial pattern is unchanged. It means **Rev3 places more DMS where the wind is stronger**, a positive C_w–k covariance relative to L11, and the flux is what feeds CCN.

Caveat on the 12.8 → 15.4. That is rank 1's blocks — a northern high-latitude strip with land entering as zero — not a global budget. Rev3 and L11 have near-identical *global* fluxes (28.19 against a published 28.1), so the +20 % is regional sampling, consistent with the polar oceans being where Rev3 revised most, and is not a global statement.

40.8 Screening at one year: measured, with the threshold that makes it safe

Project issue #170 found the cy48 high-cloud offset already present in year 1 while the T2m bias and the SO low-cloud dip were not, and concluded that one-year runs would suffice to test tuning parameters. That claim is about `hcc`; the campaign's Southern Ocean metric is SW CRE, a different quantity with different noise, so the transfer was measured rather than assumed.

lever	44-yr Δ	**year-1 Δ**	ratio	screens at 1 yr?
A1a `ovlliqice 0.10`	−6.44	**−6.56**	1.02	**yes**
D2a `inpsea 0.2`	−4.80	**−5.23**	1.09	**yes**
D2b `inp+p700`	−2.61	**−2.01**	0.77	**yes**
A1b `ovlliqice 0.35`	−2.17	−1.74	0.80	no
B3 `clddiff 1.5e-5`	+0.67	−0.17	**−0.25**	no — *wrong sign*
B7 `rvice 0.22`	+0.66	+1.22	1.85	no

Control interannual scatter in SO SW CRE is 0.712 W m^{-2}, so a *pair* of single years resolves ±1.97 against ±0.30 for a 44-year pair. **The claim holds, with a threshold: a lever screens at one year if its effect exceeds ≈2 W m^{-2}, and is unreliable below it** — B3 comes out with the *opposite sign* at one year.

Why this is worth having. Closing the SO amount deficit requires ≈4.9 W m^{-2}, so any lever large enough to matter would show at one year. A 1-year AMIP leg is ≈11 minutes against 9.5 hours for the 44-year standard, so candidate triage is ≈50× cheaper and only survivors need full length. Round 23 spent four 48-year submissions on levers that could have been screened in 45 minutes together.

What it does NOT license. Temperature. #170's own lead/lag result is that T2m and the SO low-cloud dip lag the cloud change, and the campaign's seasonal thresholds (DJF ±0.588 K on 44 years) are far beyond a 1-year window. **Screen the radiative response at one year; never adopt on one.**

41 Round 25 (2026–08–09): the tropopause is not displaced, and cold-and-dry share one entry point

No runs. Round 22 found the cold bias extratropical, peaking at 200 hPa over both poles and reversing sign at 100 hPa, and read that shape as a displaced tropopause. It is not, and what is there instead is better news.

41.1 The tropopause height is right

WMO lapse-rate tropopause, model and ERA5 on identical levels, annual, period-clean:

band	model [hPa]	ERA5 [hPa]	diff
90S–60S	250	250	0
60S–30S	200	200	0
tropics	100	100	0
30N–60N	200	200	0
60N–90N	250	250	0

~~The sign reversal at 100 hPa is a displaced tropopause: the model is still on a tropospheric lapse rate at 200 hPa where the real atmosphere has turned isothermal.~~ **Falsified.** The tropopause sits on the same level in every band. The vertical resolution here is one level, so a displacement smaller than ∼50 hPa would not be seen — but a displacement large enough to produce a 4 K bias would be.

What the lapse rates show instead is a *sharper* transition, not a moved one: the model cools faster than ERA5 through 300–250 hPa (+1.76 and +1.78 K km^{-1} at the two poles, against +0.21 in the tropics) and then slower than ERA5 above 200 hPa (−1.49, −1.85). Same tropopause, too much curvature at it.

41.2 The cold point, and one caveat on reading it

band	T model	T ERA5	ΔT	p model	p ERA5
90S–60S	208.77	212.03	−3.26	250	70
60S–30S	211.80	213.32	−1.52	100	100
tropics	**193.44**	**195.51**	**−2.07**	100	100
30N–60N	211.54	212.84	−1.30	100	100
60N–90N	213.21	217.55	−4.33	250	30

The two polar rows are not like-for-like and must not be quoted as 4 K cold points. Model and ERA5 locate their profile minimum at different levels there (250 against 70 and 30 hPa), because at high latitudes the upper stratosphere is near-isothermal and the minimum is ambiguous. What those rows actually say is a structural difference: the model's polar stratosphere is too *cold* at 250 hPa and too *warm* above 100 (§38.4 gives +0.25 and +0.26 there). The three rows where both datasets pick the same level are clean, and the tropical one carries the next result.

41.3 Cold and dry are one problem with one entry point

Stratospheric water vapour is set by the temperature of the tropical cold point, which freeze-dries air entering the stratosphere; the Brewer–Dobson circulation then carries that air poleward. So a cold-biased tropical cold point dries the stratosphere *everywhere*, including over the poles. Testing it quantitatively with Clausius–Clapeyron over ice (Murphy & Koop 2005) on the saturation mixing ratio at each dataset's own cold point:

tropical cold point, model	193.44 K at 100 hPa
tropical cold point, ERA5	195.51 K at 100 hPa
saturation mixing ratio	3.575 vs 4.998 (ppmv-mass)
predicted entry-air drying	**−28.5 %**
measured drying at 100 hPa	**−46.9 %**
share explained	**61 %**

A 2 K cold bias at the tropical cold point accounts for three fifths of a 47 % stratospheric dry bias. That reframes §38.4, which recorded cold and dry as a compensating *pair* — two independent errors whose opposing OLR effects happen to cancel. They are not independent. They share an entry point, and the polar UTLS dryness is the tropical cold point seen downstream.

Why that is much better news. Two independent cancelling errors would have to be fixed together or not at all, and fixing either alone moves net TOA several $\mathrm{W\,m^{-2}}$ the wrong way (§38.4). One error with one entry point means **a lever that warms the tropical cold point should move both**, in the right direction, at once. The remaining 39 % of the drying is a second source — transport, or the model's own UTLS microphysics — and is not yet identified.

Caveat carried forward: ERA5's upper-level humidity is weakly constrained by observation, so the −46.9 % denominator is the softest number here. The −2.07 K cold point and the −28.5 % it implies rest only on temperature, which ERA5 constrains well.

42 Round 26 (2026–08–09→10): marine biogenic CCN wired and priced, and the tropical longwave becomes the gate

Round 24 traced the marine CCN path end to end and found the DMS field read but *unused*. This round wires it into the radiation, brackets the one free coefficient across its published range, and reaches a verdict that changes the shape of the remaining campaign: the Southern Ocean lever works, and the tropical longwave deficit is what stops it being adopted. The two are no longer separate items on the open list.

42.1 The wiring, and why it is default-off

In `radpar.F90`, immediately after `SURFRAD` fills `PCCNO` from Genthon (1992) sea salt, ocean points gain a DMS term: Saltzman et al. (1993) for the Schmidt number, Nightingale et al. (2000) for the transfer velocity $k_{600} = 0.222u^2 + 0.333u$, flux $F = kC_w$, and

$$\texttt{PCCNO} \mathrel{+}= S \cdot F_S, \qquad S \equiv \texttt{ECE_DMS_CCN_SENS}, \quad F_S \text{ in } \mathrm{mg\,S\,m^{-2}d^{-1}}.$$

`PCCNO` then flows to `radiation_scheme`, through the Martin et al. (1994) quadratic to droplet number and effective radius — the Twomey effect. The input is DMS-Rev3 (Hulswar et al. 2022), ingested through ocp-tool and verified in T3 (§40.7).

S is a `NAMECECFG` namelist knob defaulting to 0.0, and the ICMCL field is registered only when `ECE_CLIMR_DMS` is set, so `NUMFLDS` is unchanged and **every existing configuration reproduces bit-for-bit with the switch off**. That is what allows one library to serve both arms of every pair below.

Two near-misses worth recording. The first attempt wired CCN into `surfrad_layer`, which would have been a **no-op**: the radiation's CCN comes from `radpar`, where `PCCNO` is `INTENT(OUT)` and any earlier value is overwritten. The second was a build failure, `error #6404`, diagnosed twice as a stale `.mod` and once as a CMake dependency-scanner gap — *all three wrong.* A full clean `esm_master recomp-` reproduced it. `nm` showed the symbol present in the object file: the OpenIFS modules declare `PRIVATE` by default and export an explicit list, so a new module variable is invisible at every `USE` site until it is added to `PUBLIC`. **Check accessibility before blaming the build system, and diagnose with `nm` rather than by theory.**

42.2 U1, the kill test at the top of the published range

Woodhouse et al. (2010) measure $\mathrm{dCCN/d}F_{\mathrm{DMS}}$ anywhere from −43 to +166 $\mathrm{cm^{-3}}$ per $\mathrm{mg\,S\,m^{-2}d^{-1}}$ depending on season and hemisphere. That range is too wide to pick a number from, so the coefficient was made a namelist knob and the range *sampled*, starting at the most optimistic end. Control is T3: identical DMS-Rev3 input, identical winds, $S = 0$.

$S = 166$, 1 yr	DJF	MAM	JJA	SON
SO 65–45S ΔSW CRE	**−7.25**	+0.03	−0.63	−1.14
tropics ΔSW CRE	−1.56	−2.17	−2.44	−2.39

Annual: SO −2.25, tropics −2.14, global $-1.72\,\mathrm{W\,m^{-2}}$. December alone is −8.54 and January −7.34.

The lever is real and it is seasonally selective in exactly the right way. The SO DJF deficit is $+10.86\,\mathrm{W\,m^{-2}}$ and this closes 67 % of it, in the season the deficit is worst, at a DJF tropical cost of only −1.56 — 4.6 : 1 in our favour. **The selectivity is the bloom's, not a tuned parameter**; no hemispheric constant was invented to produce it. That also confirms round 24's phase argument quantitatively: the model's SO droplet number was right in the mean but inverted in phase, and this is the correction.

The annual means look like a wash only because the SO gain is concentrated in one season while the tropical cost is flat year-round.

42.3 U2 and U3: the bracket, and what one year can actually resolve

Two more arms at $S = 43$ (the published floor) and $S = 90$ (the midpoint), to read an operating point off a curve instead of guessing a fourth value. Detection thresholds were computed first, per band, from the control's own interannual scatter as $1.96\,\sigma\sqrt{2}$ for a pair of single years — the construction that gives the already-known ±1.97 for SO SW CRE:

metric	σ (interannual)	1-yr pair, 95 %
SO SW CRE, annual	0.712	±1.97
SO SW CRE, DJF	1.379	±3.82
tropics SW CRE, annual	0.357	±0.99
tropics SW CRE, DJF	0.441	±1.22
global SW CRE, annual	0.402	±1.11

S	SO annual	SO DJF	tropics annual	resolved?
43	−0.32	−1.78	−0.76	nothing
90	−0.38	−1.22	−1.11	tropics only, marginally
166	**−2.25**	**−7.25**	**−2.14**	**all three**

The bracket produced exactly one usable point. SO DJF is not even monotonic in S (−1.78, −1.22, −7.25), which is the signature of the two sub-floor arms being internal variability rather than signal.

~~The tropical response is monotonic and clean at $0.0124\,\mathrm{W\,m^{-2}}$ per unit S, fitted through three arms.~~ **Withdrawn the same day.** Two of those three points sit below the ±0.99 tropical threshold, so the fit was reading a line through noise. The analysis script now refuses to interpolate the SO points and guards the tropical extrapolation explicitly — `np.interp` would have *clamped* to the sampled range and returned a confident wrong answer. The same paragraph also treated the $\pm 0.5\,\mathrm{W\,m^{-2}}$ tropical guardrail as though it were a detection threshold. It is not: it is a *tolerance*, set because the tropics are only $-0.67\,\mathrm{W\,m^{-2}}$ from CERES period-clean.

42.4 The verdict, and why scaling cannot rescue it

From the single resolved arm the tropical cost is $2.14/166 = 0.0129\,\mathrm{W\,m^{-2}}$ per unit S, so the ±0.5 tolerance binds at $S \approx 39$ — **below Woodhouse's published floor of 43**. Twomey scales with $\Delta \ln N_d$, so the response is *sub*-linear and the per-unit cost at low S is if anything larger; the bound holds a fortiori. Third branch of the pre-registered verdict logic: **no coefficient in the published range works as a tuning knob.**

Why the tropics respond more per unit ΔCCN, and why that is not a bug. Twomey goes as $\Delta \ln N_d$, and tropical background CCN ($\sim 52\,\mathrm{cm^{-3}}$) is *lower* than the Southern Ocean's (~ 78), so the same absolute increment is a larger fractional change there. Physically the model has *no* marine biogenic CCN anywhere — Genthon is sea salt from wind, MACv2-SP is anthropogenic plumes — so adding it in the tropics is *more* correct, not less. What fails is a tuning tolerance, not the physics. And because both ends scale together, no choice of S separates them.

42.5 The tropical longwave deficit is now the gate

This reframes the open list. The two remaining regional failures act through *different* channels (§76.6): the Southern Ocean through SW cloud (+7.45), the tropics through LW cloud (−2.16). The tropical cost of the DMS lever is a *shortwave* cost; the tropical bias is a *longwave* one. Those are independent terms, so:

- **The tropical LW deficit is no longer a parallel item — it gates the Southern Ocean fix.** Close it and the SW cost of DMS CCN becomes affordable, and the best SO lever this campaign has found goes in unchanged.
- **The pair is complementary in the global budget.** DMS CCN *removes* $1.72\,\mathrm{W\,m^{-2}}$ globally, which helps the +0.52 imbalance; closing the tropical LW deficit puts roughly +1 back once the tropics are area-weighted. Together they fix both regional biases and partly cancel globally — a better position than either alone, since the DMS lever's problem was never its SO effect but its global reach.
- **The requirement on the LW lever is sharper than "acts through the longwave".** `RVICE` is the counter-example: an ice fall speed applies wherever there is cloud ice, so it *trades* SO high cloud against tropical anvils, and it was raised 0.13 → 0.16 precisely to suppress SO `hcc` (project issues #169/#170). A usable lever must be keyed on **deep convection**, which barely exists over the Southern Ocean, so that selectivity comes from where convection happens rather than from a latitude band — the same structural trick that made the DMS lever seasonally selective.

The cheap check comes first: ∼52 arms have been scored on SO SW and global RMSE and *none* on the tropical LW term. `tropical_lw_scan.py` is written to answer that from data already on disk before any new run is spent; it has not yet been run.

Not Blomquist. The standing todo to switch the transfer velocity from the Nightingale quadratic to Blomquist et al. (2017) should be dropped for this purpose: the DMS-specific scheme is *less* steep in wind, which shifts flux away from the windy Southern Ocean toward the calm tropics. It would make selectivity worse, not better.

42.6 Two coupled runs, on a campaign-branch coupled build

The coupled tree was on `movcav-landice+co2-concdriven`, which carries neither the snow-depletion knobs nor the CCN wiring. The campaign branch is a *strict descendant* of that branch's tip (`f3ccacb` is an ancestor), so it contains every coupled commit, and every campaign feature is namelist-gated off by default — switching the tree in place removes nothing and changes no existing configuration's behaviour. One clean OpenIFS rebuild then serves both runs. Six files completing the DMS ICMCL chain, plus the `RCLCRIT_SEA`/`RCLCRIT_LAND` exposure, had been left uncommitted in the AMIP tree; U1–U3 ran against them, so the library had them and only the commit was missing.

run	change vs. its control	question
11E	11D + `SWEMIN` 30 → 15 + K1	is the coupled configuration level with AMIP?
11F	11E + `ECE_DMS_CCN_SENS`= 166	does coupling change the Southern Ocean answer?

Both are 50 years, both submitted 2026–08–10. **11F is deliberately not a tuning candidate** — AMIP has already said no published S clears the tolerance. It runs at the only setting AMIP could resolve, to answer what AMIP structurally cannot: with SST and sea ice free to respond, does brightening Southern Ocean cloud in bloom season move the coupled SO state and the TOA imbalance by more than the tropical cost? Fixed SSTs suppress exactly the feedback that makes an SO cloud change matter. Fifty years also prices the tropical response with feedbacks and proper statistics instead of from one AMIP year.

Status update, 2026–08–10. Both started the same day (11E job 26837423, 11F 26837527) and are through their first ten-year legs.

11F is reclassified as a MECHANISM PROBE, not a production candidate, and should not later be read as one. Two things changed within hours of submitting it. First, the bracket verdict (§42.4) means $S = 166$ is a configuration we will not adopt under any circumstances. Second, S4 (`RCL_INPPMIN` 70000 → 50000) turned out to deliver $-1.98\,\mathrm{W\,m^{-2}}$ of Southern Ocean SW CRE — nearly the same forcing as 11F's −2.25 — at a tropical cost of 0.10 against DMS's 2.14, i.e. **twenty times the selectivity, namelist-only, and already run**. So the specific lever inside 11F has been superseded before its first leg finished.

What 11F still buys is the *generic* question, and it is worth 50 years for it: what does $\approx 2\,\mathrm{W\,m^{-2}}$ of Southern Ocean SW brightening do once SST and sea ice can respond? That answer transfers to the whole S4/DMS family, and AMIP cannot produce it at any run length — prescribed SST absorbs the perturbation without responding. It is being kept for that reason alone. If compute becomes tight, **11F is the run to cut and 11E is not**.

43 Round 27 (2026–08–10): the tuned stack nobody ran, and a lever that fixes the tropics by breaking the Southern Ocean

The largest single-day gain of the campaign, and the largest single-day correction. A jointly-tuned eight-parameter cloud stack for *this resolution* was found sitting unused in the runscript directory; one of its parameters closes the tropical longwave deficit outright; and scoring it properly showed the same parameter warms the Southern Ocean, putting the campaign's two remaining regional targets in direct conflict.

43.1 The archive has no tropical longwave lever

Round 26 made the tropical LW deficit the gate on the Southern Ocean fix, so the free check came first: score all 50 arms on a term none had ever been scored on.

The term is extraordinarily stiff — interannual sd $0.101\,\mathrm{W\,m^{-2}}$, so the 44-year pair threshold is ±0.042 and the −2.16 deficit is **51 times** the detection limit. Yet the largest positive response in the whole archive is **A1a at** +0.373, 17 % of the deficit, and it costs SO SW −6.44 and global −1.90. Everything else is ≤ 6 %. **No clean tropical-longwave lever exists in anything tried.**

The useful result was B1. It *lowered* DETRPEN 0.75 → 0.45E-4 and tropical LW fell 0.510 — second-largest in the archive — while the Southern Ocean did not move at all (−0.067 against its own ±0.298). Against RVICE, which moves tropical and SO longwave ≈1:1, that is **57:1 selectivity**. B1's recorded verdict was "failed — more SW yet colder; *worst tropics*"; the tropical half of that failure *is* this term, so its reverse is a lever.

43.2 Two levers screened, one of them the wrong way round

1 yr vs control	trop LW ±0.279	trop SW ±0.990	trop net ±1.656	SO LW ±0.584	global net ±1.333
V1 ENTRORG 1.25E-3	+0.288	**+3.583**	**+4.536**	**−0.830**	**+2.734**
V2 DETRPEN 1.11E-4	**+0.583**	**−1.295**	−0.398	+0.171	−0.428

V2 landed on prediction: B1's slope forecast +0.61, measured +0.583, with the Southern Ocean untouched. **V1 failed, and it should not have been run in that direction.** Project issue #170, comment 7, already says "there seems some potential for *increasing* ENTRORG". V1 lowered it on Semmler's 43r3 value and cost +2.73 W m^{-2} globally. Eleven minutes, not 9.5 hours — but the desk check existed and was not done.

43.3 A tuned stack for TCO95L91, written and never run

`awiesm3-v3.4.2-levante-TCO95L91-DARS2_{1d,2y}.yaml` carry eight jointly-tuned cloud parameters at *our exact resolution*:

	NAMCUMF default	DARS2		NAMCLDP default	DARS2
RPRCON	1.4E-3	1.48E-3	RVICE	0.16	0.18
ENTRORG	1.75E-3	2.07E-3	RLCRITSNOW	2.0E-5	1.46E-5
RMFDEPS	0.30	0.48	RSNOWLIN2	0.030	**0.04**
DETRPEN	0.75E-4	**1.32**E-4			
ENTRDD	3.0E-4	1.08E-4			

Three-figure precision — an optimiser's output, consistent with the EPPES table in the logbook. **No `fort.4` on disk sets any of them.**

Two of this campaign's own proposals were superseded on the spot: V2 ran DETRPEN at 1.11E-4 as "the largest defensible value" from the EPPES 1σ bound, and DARS2 uses 1.32E-4, 19 % beyond; and V1's ENTRORG direction is the opposite of DARS2's.

43.4 Why it was invisible: `preflight` looked in one place

~~RPRCON: never set in any run — genuinely untested.~~ ~~RSNOWLIN2: never set in any run — genuinely untested.~~ **Both false, on the same day.** `preflight.py` scanned one runtime tree and one runscript directory, so it could only ever see *this* campaign.

RPRCON was tuned at TCO319/43r3 in the DART line (#87): 1.4E-3 → 0.7E-3, adopted through DARTC/D/E/F/J/K, relaxed to 1.0E-3 from DARTL; and launched once at TCO95L91-CORE2 in #95, where **the result was never posted** and it did not reach the AWI-CM3.1 final configuration. RSNOWLIN2 was set to 0.04 in DARS2 and to 0.025 in `awicm3/MELDPOND`.

`preflight` now globs every runtime tree and every runscript directory, separates "tried here" from "tried elsewhere at AWI", and reports the *value* — because "mentions it" and "sets it" are different facts and only the second is prior art. Its remaining blind spot is recorded in its own docstring: **the GitHub issues**, where the *why* lives, and which issue search does not index because it ignores comment bodies.

~~The logbook's EC-Earth3 ENTRORG row (1.80E-4) is a transcription error; 48r1 defaults to 1.75E-3.~~ **Also wrong.** The EC-Earth3 files genuinely say 1.80E-4 with their own reference column in E-4 as well. It is a real formulation difference between EC-Earth3's IFS cycle and OpenIFS 43r3/48r1 — which matters far more than a typo would, because "correcting" the logbook would invite someone to set 1.8E-4 in 48r1 and cut entrainment tenfold. **EC-Earth3's ENTRORG is not transferable; its DETRPEN and RPRCON are.**

43.5 W: the stack taken apart, and the lever that works

Eight one-variable arms plus the combination, one year, against `amip_pi_base`.

	trop LW	trop SW	trop net	SO SW	SO LW	global net
W4 `DETRPEN` 1.32E-4	+1.025	−2.355	−0.959	−0.087	−0.429	−0.804
W8 `RSNOWLIN2` 0.04	**+2.581**	−2.204	+0.671	−0.884	+1.223	+0.321
W9 whole stack	+1.939	−6.494	**−4.518**	−0.983	+0.854	**−2.659**
interaction W9 − $\sum$	−0.458	−0.449	−1.194	−1.763	**+2.415**	−0.270

`RSNOWLIN2` = 0.04 **gives** +2.581 **W m**$^{-2}$ **of tropical LW CRE** — 120 % **of the deficit, nine times its threshold, and five times the largest response among the other fifty arms** — with tropical and global net TOA both *below* their thresholds.

It works for a reason that was predicted before it was run. It is the *temperature exponent* of ice→snow autoconversion, ZZCO = PTSPHY · `RSNOWLIN1` · $e^{\texttt{RSNOWLIN2}(T-T_0)}$ (`cloudsc.F90:2858`). Raising it slows conversion 31 % at a 200 K cirrus temperature but only 11 % at 250 K, so it builds **cold thin cirrus rather than thickening anvils** — and thin cirrus is longwave-dominant while a thick anvil is bright in both. The contrast is measured: `DETRPEN` gives LW:SW of 0.44, `RSNOWLIN2` 1.17.

~~You cannot buy tropical longwave without paying tropical shortwave; every lever that raises tropical LW does it by adding condensate, which also raises SW reflection.~~ **Withdrawn the same day.** It was generalised from two SW-dominant arms. The archive already contained LW-dominant responses — D1 at 4.0:1, B6 at 1.4:1, B7 at 1.3:1 — and W8 settles it.

The full stack overshoots and is significantly non-additive: SO longwave interaction +2.415, with the sum of parts predicting −1.561 and the combination measuring +0.854. The arms are not a menu.

A near-miss worth recording. `RKOOPTAU`, the Koop homogeneous-freezing timescale, was the first candidate for a temperature-selective cirrus lever: it is in `NAMCLDP`, never set, and physically exactly right. Its only use in `cloudsc.F90` is **commented out**. Setting it is a no-op — caught by tracing the code path rather than by running it.

43.6 X and Y: on the real base, and the seasonal reason DMS survives

Ladder from T3 (adopted stack, DMS field read at $S = 0$), one variable per rung.

cumulative vs T3	trop LW	trop SW	trop net	SO SW	SO LW	global net
X1 +`RSNOWLIN2`	+2.740	−2.631	+0.311	−0.507	+2.626	+0.419
X2 +DMS S=166	+2.547	−4.062	−1.438	−2.765	+2.240	−1.259
X3 +`INPPMIN` 50k	+2.687	−4.299	−1.274	**−5.051**	+2.841	−1.434
X4 +`INPPMIN`, no DMS	+2.563	−2.450	+0.225	−2.595	+2.718	+0.067
Y1 +`ovl` 0.35	+2.677	−2.729	+0.088	−1.963	+2.827	−0.005
Y2 +`ovl` 0.10	+2.852	−3.742	−0.718	−5.029	+3.563	−1.184

`RSNOWLIN2` reproduces on the real base (+2.740 vs +2.581 alone) and DMS reproduces its Southern Ocean effect exactly (−2.258 vs −2.250).

~~`RCL_INPPMIN` buys the same Southern Ocean shortwave as DMS for free (+0.164 tropical cost against DMS's −1.750), so the DMS rung is redundant and the marine CCN line can be retired.~~ **Withdrawn.** Their *annual* means agree by coincidence; their seasons do not:

SO 45–65S SW CRE	DJF	MAM	JJA	SON	ANN
`INPPMIN` 50k (S4−P5, 44 yr)	−3.72	−1.30	−0.51	−2.35	−1.97
DMS S=166 (U1−T3, 1 yr)	**−7.25**	+0.03	−0.63	−1.14	−2.25

DJF is austral *summer* and carries the shortwave: incoming solar is 460 W m^{-2} against 87 in JJA, and the control's SO SW CRE is −115.5 against −18.7. The deficit is +10.86 there. DMS reaches 67 % of it, `INPPMIN` 34 %. **DMS is the only lever that reaches the season the error lives in.**

43.7 Scoring the Southern Ocean on NET — and the correction that follows

~~Southern Ocean levers scored on SW CRE.~~ Used throughout rounds 22–27 and **wrong**: the coupled complaint is a warm SST bias, which responds to ΔSW CRE + ΔLW CRE.

Re-ranked on net, the 44-year archive is clear and consistent — longwave cancels 25–29 % in *every* arm, so SO SW × 0.72 is a good rule:

44 yr	SO SW	SO LW	**SO net**
A1a `ovl`= 0.10	−6.438	+1.871	**−4.567**
D2a `inpsea` 0.2	−4.802	+1.200	−3.602
S4 `inppmin`50k	−4.669	+1.195	−3.474

The new levers break the rule. True net TOA at one year: X3 −2.024; **DMS alone** −1.976 with only 17 % longwave cancellation; Y2 −1.300; X4 +**0.377**; Y1 +**0.764**; and **X1 +2.294, resolved**.

`RSNOWLIN2` warms the Southern Ocean. Its +2.6 of SO longwave swamps its shortwave. On the bare base it was neutral (−0.100); on the adopted stack it flips hard positive — a sign reversal produced by the base, not a small non-additivity. Every configuration built on it inherits the penalty.

So the campaign's two remaining regional targets are in direct tension through one parameter: `RSNOWLIN2` closes the tropical longwave deficit partly by putting cirrus over the Southern Ocean, where the model already traps too much. And DMS — dismissed as redundant that morning — is on net forcing the cleanest new lever there is, and the only reason X3 points the right way.

43.8 S4, evaluated at last

S4 completed 48 years on 2026–08–09 and had *never been added to* ***`runs.py`***, so `evaluate.sh` had never scored it. Its exact one-variable partner is P5 — the two `fort.4`s differ in one line.

S4 − P5, 44 yr	value	threshold
SO TOA SW CRE	−1.98	±0.298
global net TOA	−0.41	±0.201
tropics net TOA	−0.10	±0.250
Siberia JJA T2m	−0.120	±0.242
SO SW RMSE	**+0.255**	±0.019

Best mean, worse pattern: it deepens opacity where cloud already exists rather than adding cloud (area +0.105 pp). Siberia is untouched, and the code says it cannot be otherwise — `cloudsc.F90:2520` scales ice nuclei by `RCL_INPSEA`+ (1 − `RCL_INPSEA`) · PLSM, which is exactly 1 over land.

43.9 The AMIP runs are not at 1850 forcing

~~All AMIP runs use 1850 GHG (`NCMIPFIXYR`=1850).~~ **False for every AMIP run in the campaign, including the control.** `NCMIPFIXYR=1850` sits in `&NAMECECMIP6`, the block 48r1 renamed and no longer reads. The runtime value is −1 and the model reads *yearly* GHG, ozone and aerosol: the log shows 1869 in the first leg and 1911 in the last.

The base runscript already documents the trap and removed the dead block on 2026–07–30, but its replacement (`oifs.mip` + `oifs.scenario`) was never added, because that path aborts with `GRID_IN MISSING FIELD` on the awi3 INIT. **The coupled runs are correctly pinned** — 11E logs `Set JYEAR=...= 1850` — so this is AMIP-only.

Z1 verified the one-line fix: `NCMIPFIXYR: 1850` in the *live* `&NAMECECMIP` block reaches the model, the run completes `CNTO` with zero aborts, and it does *not* trip the missing-field abort.

Consequences: **every lever delta stands**, because both arms of every pair share the forcing. The *absolute* energy baseline does not — mean CO_2 over the evaluation window is ≈294 ppm against a PI 284.3, worth $\approx +0.15\,\mathrm{W\,m^{-2}}$, or about a quarter of the control's +0.64. The fix is therefore deliberately *not* applied to the runs in flight: it would break comparability with P5 and all 50 prior arms to buy an offset that one dedicated control run can measure later.

The irony worth remembering: `amip_pi_base`'s `fort.4` *contains* "`NCMIPFIXYR = 1850`". Anyone grepping it sees the right number. Only `NODE.001_01` disagrees.

43.10 Coupled: 11F retired, 11E resumed

11F was cancelled at 20 of 50 years. Its lever had been superseded within hours of starting (§42.6), and the generic question it was kept for — what $\approx 2\,\mathrm{W\,m^{-2}}$ of SO brightening does with free SST — did not justify the remaining 30 years once S4 and the net-forcing re-ranking landed.

11E was cancelled by accident in the same operation and **resumed cleanly** from year 1370 (`run_number=3`), with 20 years banked and restart state intact at the end of 1369. Nothing was lost but the killed leg.

43.11 Three runs in flight, and what would disqualify them

run	configuration	why
LX1	stack + `RSNOWLIN2`	the gating run; isolates the SO penalty and the Siberian cost
LX3	+ DMS S=166 + `INPPMIN`	only arm with resolved net SO cooling
LY2	+ `ovl` 0.10	best radiative package; A1a's value, rejected in round 3

Control is P5, verified **bit-for-bit identical to T3** at year 1870 on all three radiative metrics — `ECE_CLIMR_DMS` reads the DMS field but does nothing at $S = 0$. That check saved a 9.5-hour run.

LX4 was cancelled after 12 minutes as dominated: it warms the SO (+0.377), leaves energy untouched (+0.067), and its unique content already exists as S4 − P5.

What would disqualify them, and what one year could never see. Siberia. The JJA threshold is ±0.242 K and DJF ±0.588, both 44-year numbers; every arm changes cloud, which is the boreal energy supply; and LY2 carries the value round 3 rejected *for boreal damage*. The land stack currently recovers 1.15 K of a ≈2 K period-clean bias, leaving Siberia **still ≈0.9 K cold** — so there is no headroom to spend.

43.12 The boreal forest: the coupled model destroys it, and warming brings it back

11E's Siberian result prompted the question the campaign actually exists to answer — not whether T2m improved, but whether the forest is there. It is not, and chasing that properly produced one retraction, two analysis bugs of my own, and a calibration the campaign has never had.

The claim, and the retraction. ~~The forest failure is not a growing-season problem: 11E clears the larch GDD gate on 91 % of cells and the trees still do not come, so no amount of additional summer warming will bring the forest back.~~ **Refuted by experiments already on disk.** Three 2000-year offline LPJ-GUESS spin-ups exist at TCO95 — AMIP forcing, and the same forcing warmed by +1 and +2 K:

offline 2000-yr equilibrium, Siberian box	BNS	TREE	GRASS	tree>0.5	BNS>0.1
CRUNCEP+CERES (observed forcing)	0.227	**0.415**	0.281	50.3 %	36.2 %
AMIP forcing	0.034	0.202	0.313	18.2 %	7.4 %
AMIP +1 K	0.058	0.261	0.332	24.5 %	12.5 %
AMIP +2 K	**0.099**	**0.327**	0.344	31.7 %	19.7 %

Warming restores the forest, monotonically and without saturating. BNS roughly triples from AMIP to AMIP+2 K and tree cover recovers 59 % of the gap to the observed-forcing equilibrium.

This is the campaign's founding premise, calibrated for the first time. The AMIP Siberian cold bias is ≈2 K period-clean (§38.2), and +2 K of offline warming buys back three fifths of the forest. Closing the cold bias should therefore recover most of it — previously an assumption, now a measured transfer function.

Why 11E looked like a counter-example, and was not. The coupled runs are **50-year transients**; these are **2000-year equilibria**. 11E is not showing an endpoint but the *decline* from its CRUNCEP restart toward whatever equilibrium its own climate supports, and forest composition takes centuries:

Siberian box, 473 cells	year	BNS	TREE	tree>0.5	grass>tree
offline CRUNCEP (the restart state)	3900	0.227	0.415	50.3 %	29.2 %
11E at start	1350	0.213	0.394	47.8 %	28.3 %
11E after 50 coupled years	1399	**0.017**	**0.083**	2.1 %	**69.6** %
11D after 50 coupled years	1399	0.019	0.074	0.6 %	68.1 %

Year 1350 reproduces the offline restart almost exactly, as it must. By 1399 larch has fallen **92** % and grass goes from beating trees on 28 % of cells to 70 %. **The coupled model inherits a real boreal forest and dismantles it in fifty years**, from a healthy state, while clearing the GDD gate on 91 % of cells — which is exactly what a transient toward a much poorer equilibrium looks like.

Consistent with the established mechanism, not a new one. §31.2 already separates the two claims: competition as an LPJG *parameter* error is dead (the forcing-transfer test reproduced the loss with byte-identical parameters), while competition as the *pathway* by which climate removes larch is supported — larch is climatically permitted on 84 % of cells and loses to grass 5.8 : 1. The offline warming series is the driver half of that statement, measured: `grass>tree` stays at 43–45 % across AMIP, +1 and +2 K against CRUNCEP's 29 %, so warming raises tree cover *without* dissolving the competition. Forcing is the driver; grass-versus-larch remains the mechanism.

Two of my own analysis errors, recorded because both produced plausible numbers. ~~11E Siberian FPC over 2365 cells: TREE 0.109, GRASS 0.203.~~ The glob matched all five decade directories and took each one's

own final year, so it averaged 1359–1399, not 1399. The true final-year values are *worse*: TREE 0.083. ~~Offline CRUNCEP TREE 0.617, BNS 0.617, forest on 100 % of cells.~~ That came from `tail`-ing a 13 GB `fpc.out` — but LPJ-GUESS writes *all 2000 years per gridcell* before moving to the next, so the tail sampled the last ∼30 gridcells, of which three fell in the box. Both were caught only by checking cell counts against each other; neither looked wrong on its face. **For these files, stream with `awk` filtered on year and box — 13 GB in 20 seconds — and never trust a `tail`.**

Unread, and deliberately not speculated about. Three offline experiments target the competition directly: `LPJG_COMP_EXP1_BNS_GREFF0035`, `EXP2_IBS_GREFF012`, `EXP3_C3G_PSTEMPLOW12`. They belong to the final land round.

44 Round 28 (2026–08–10→11): the L-series at full length, and the arm that was already on disk

The three 46-year runs landed, and with them the first honest scoring of the round-27 levers. Two of the three fail. The configuration that wins is S4 — one namelist number, finished on 2026–08–09, and unevaluated until now.

44.1 The three candidates, attributed

Each effect is the difference from the arm above it, 44 years, against `amip_pi_base`:

effect	SO SW	**SO net**	Siberia JJA	global net	global LW CRE
`RSNOWLIN2` (LX1−P5)	−1.13	**+0.61**	+0.14	**+0.42**	**+2.06**
DMS + `INPPMIN` (LX3−LX1)	−4.53	−3.87	+0.04	−1.96	+0.21
`ovl` 0.10 (LY2−LX1)	−4.73	−3.34	**−0.67**	−1.66	+0.85

Three things settle. **`RSNOWLIN2`'s Southern Ocean penalty is real at 44 years** — it delivers the tropical longwave gain exactly as the one-year screen predicted (+2.06 global LW CRE) and makes SO net forcing and global energy worse. **A1a's boreal damage reproduces on the new base**: LY2 costs −0.67 K of Siberian JJA, nearly 3× the ±0.246 threshold, and A1a alone on the bare base gives −0.749 — the same number. Round 3 was right. And **DMS+`INPPMIN` dominates `ovl` 0.10**: more SO cooling, no Siberian cost.

LX3 is the best boreal arm in the campaign on the seasonal table — JJA +1.331, the highest of all 54 arms, with every season neutral or positive, and it is not on the report's standing "warms JJA but cools DJF" list of eleven levers.

44.2 Two fields, not one: what the maps resolved

LX3's band mean improved while its "SW RMSE" worsened, which looked like a pattern degradation. It is not. **The two numbers are different fields**: the band mean quoted throughout this campaign is TOA SW CRE, while the RMSE is *surface* net SW over ocean (`eval_round10_A.py:190`).

~~LX3 overshoots reflection in some places and not others, so the pattern becomes patchy.~~ **Wrong.** On TOA CRE the pattern improves monotonically (RMSE 11.45 → 8.78 → 7.74). On the surface field LX3 overshoots into a *uniform* negative bias of −4.75, and since $\mathrm{RMSE}^2 = \mathrm{mean}^2 + \mathrm{variance}$ that mean is most of the degradation. The field did not become noisier; it sailed past zero — a much better problem, because it is fixable by scaling rather than by finding a different shape.

The global frame then undercut the whole framing. Extending the maps from the Southern Ocean strip to 80S–30N:

	TOA SW CRE			surface net SW	
	global	45–65S	tropics	global	tropics
P5	+0.60	+4.98	−0.71	−0.60	−0.15
LX3	**−3.17**	−0.63	**−4.76**	−4.40	−4.30

The tropics move −4.05 while the Southern Ocean moves −5.61. The change is near-uniform blue in every basin. **LX3 is a global dimming knob, not a Southern Ocean lever**, and it survives on *net* only because `RSNOWLIN2` adds +2.46 of global LW CRE in compensation. The seasonal selectivity DMS showed at one year does not become spatial selectivity at 44.

Two further things only the global frame shows. Global RMSE barely moves (TOA 11.48 → 12.40, surface 14.44 → 15.18) because the error field is dominated by the tropical Pacific dipole, the stratocumulus decks and the

land signals — none of which any lever touches, and against which **the Southern Ocean box is comparatively well-behaved**. And the surface field carries a *larger* error than TOA everywhere (14.4 against 11.5), which is an atmospheric shortwave-absorption signature and is global rather than a Southern Ocean artefact.

44.3 Scoring every arm on all three constraints at once

Southern Ocean *surface* bias against CERES, ocean points only, with the tropics and both Siberian seasons as constraints and global net TOA carried alongside:

	SO sfc	SO rmse	SO CRE	Δtrop	ΔJJA	ΔDJF	global TOA
control	+3.66	7.04	+7.69	—	—	—	+0.64
S4 `inppmin50k`	**−0.96**	5.31	+3.02	+0.26	+1.03	+0.05	**+0.25**
LX1	−0.52	5.15	+3.86	+0.75	+1.29	+0.54	+1.08
P5	+0.74	5.12	+4.98	+0.35	+1.15	−0.06	+0.66
LX3	−4.75	7.31	−0.63	−0.97	+1.33	+0.26	−0.88
LY2	−4.75	6.81	−0.87	−0.32	+0.61	+2.14	−0.58
A1a	−2.32	5.70	+1.25	−1.06	−0.75	+0.83	−1.25

S4 wins. LX1 is nominally closer to zero on the SO surface (−0.52 against −0.96) and pays 0.83 W m^{-2} of global imbalance for that last 0.4 — a bad trade. LX3 and LY2 reach the Southern Ocean by dimming the planet. A1a is the only arm that damages the tropics *and* Siberian summer, confirming its round-3 rejection on both counts. **S4 is the only arm that improves the Southern Ocean surface while landing global net TOA near the piControl target, at no cost in either Siberian season or the tropics — and it is one namelist number.**

~~LX1 fails the tropical guardrail at +0.75.~~ **My own sign error.** The test compared |Δtrop| against ±0.5 regardless of sign, but the tropics sit *below* CERES (42.61 against 45.11) and need *more* absorbed energy, so +0.75 is an improvement. The test is now one-sided. LX1 reads CLEAN; it is still disqualified, on global energy.

44.4 S4 coupled: 11G

Every coupled arm that sets `RCL_INPPMIN` — 10A, 10B, 11A, 11B, 11C, 11D, 11E, 11F — uses 70000. **The** 50000 **value has only ever run in AMIP.** So 11G is 11E with that one line changed: a strict one-variable pair against a run already complete at 50 years, needing no new control. Submitted 2026–08–11 as job 26868368.

~~The AMIP number should be a *lower* bound. The coupled amplification measured in round 27 is 1.40× … A lever that cools the SO *surface* is exactly the kind that should amplify.~~ **Falsified by 11G itself** (§44.5): the amplification is a property of the *lever*, not of coupling, and S4 does not have it. The 1.40× came from sea-ice albedo — clear-sky, not CRE — with SST falling 0.598 K and ice growing 3.2 pp, and S4 moves neither enough to engage it. 11E's remaining gap is SO CRE 5.65 short of CERES with net TOA +0.79.

11G's leg history, and the decade that is not a clean pair (2026–08–11→12). 11G reached 40 years on 2026–08–12 and leg 5 (1390–99) is running. Getting there was not clean, and one decade of it cannot be used.

Legs 3–5 were killed by a defect of mine in an unrelated branch: `couple_put` called `put_2d(GUE_Z0HV)` unconditionally, so any binary carrying the Raupach roughness code aborts with *"oasis_put is called for a variable not in namcouple"* unless the namcouple declares the field. 11G was rewound to 1370 and restarted on the production binary. Every one of the five legs is confirmed to have loaded the same LPJ-GUESS executable, md5 `8c5ab467`, so there is no binary discontinuity.

The soil map is a different matter. Staged `slt_TCO95.nc` per leg, counted by organic (type 6) cells:

years	11E type 6	11G type 6	
1350–59	547	547	matched
1360–69	547	547	matched
1370–79	**547**	**233**	**NOT a one-variable pair**
1380–89	547	547	matched
1390–99	547	547	matched (in flight)

The repaired CORE3 soil map slipped into 11G's third leg alone and was reverted after. So for 1370–79 the arm differs from 11E in *two* things, not one: `RCL_INPPMIN` and 314 cells moved off soil code 6 — a code `soil.cpp` treats on its own branch. **Score 11G against 11E on 1350–69 plus 1380–99, 40 years, and exclude 1370–79.** The campaign minimum is 30 years, so this costs nothing except the temptation to use the most recent 30, which is exactly the window that contains the contaminated decade.

A stale log file can eat a completed leg. Leg 4 ran to completion — job `COMPLETED` in 4:27:46, all ten years present *and already remapped* in the run directory — and delivered nothing, because the file-movement step hit

```
ERROR: File movement conflict
log/oasis3mct/nout.000000_13800101-13891231 already exists.  Skipping movement
```

left behind by one of the killed attempts at the same leg. esm_tools aborts the *whole* movement on the first conflict, so 10 years of model output stayed in `work/` while `outdata` showed 30 years and the date file still read 1380. The run looked like it had failed and was one resubmission away from recomputing 4.5 hours that were already on disk. Moving the dead attempts' leftovers aside and re-running `esm_runscripts -t tidy` recovered all ten years and advanced the chain. *A leg that looks unfinished should be checked in* `work/` *before it is rerun*: `ifs.stat` and the presence of `atm_remapped_*` files settle it in seconds.

44.5 S4 coupled, scored: real, resolved, and three times smaller than AMIP (2026–08–12)

11G against 11E over the **30 matched years** 1350–69 plus 1380–89; 1370–79 excluded because 11G alone carries the repaired soil map there. Thresholds are $1.96\,\mathrm{sd}\sqrt{2/n}$ from 11E's own interannual scatter over the same years.

	11E	11G	Δ	threshold
net TOA [W m^{-2}]	+0.387	+0.205	−0.182	±0.362
SO SW CRE [W m^{-2}]	−62.472	−63.946	**−1.474**	±0.479
SO net TOA [W m^{-2}]	−51.417	−52.499	**−1.082**	±0.778
SO clear-sky SW [W m^{-2}]	217.847	217.294	−0.553	±0.902
SO SST [°C]	3.561	3.297	**−0.264**	±0.098
SO sea ice [%]	9.162	9.758	+0.595	±0.806
global T2m [°C]	12.604	12.388	**−0.216**	±0.123
Siberia JJA T2m [°C]	9.670	9.777	+0.107	±0.397
Siberia DJF T2m [°C]	−32.957	−33.322	−0.365	±0.736
Siberia DJF soil [°C]	−12.606	−13.015	−0.409	±0.469

S4 transfers, and costs nothing. SO SW CRE, SO net TOA, SO SST and global T2m all resolve, all in the intended direction, and *no* Siberian term does — the AMIP promise of a free Southern Ocean gain holds coupled. The CERES gap narrows from +5.36 to +3.89, i.e. 27 % of what 11E had left.

But it is three times smaller than AMIP, not 1.40× **larger.** Against S4's own AMIP deltas (−4.704 SO CRE, −3.813 SO net TOA) the coupled ratios are **0.31**× and **0.28**×. The prediction from round 27's DMS pair was ≈ -6.5; the measurement is −1.474.

The discriminator is sea ice, and it is a property of the lever. The amplification mechanism is sea-ice albedo, which is clear-sky and invisible to CRE. DMS at $S = 166$ cooled the SO surface 0.598 K and grew 3.2 pp of ice; S4 manages 0.264 K and 0.595 pp, the latter not resolved, and its clear-sky SW response is not resolved either. The amplifier never engages. So *"treat the AMIP magnitude as a lower bound"* is not a rule about coupling — it is a rule about levers large enough to move ice, and S4 is not one.

What this does to the decision. S4 remains the best available arm: it is one namelist number, it improves the Southern Ocean, and it is free everywhere else. It is simply much smaller than advertised, and on its own it does not close the SO gap. The global energy target is untouched — −0.182 against a ±0.362 threshold is not a result. Leg 5 will take this to 50 years and 1390–99 is matched, so the clean window becomes 40 years and every threshold above tightens by ∼15 %.

44.6 Krakatoa in a pre-industrial run

The net-TOA history was regenerated with the 11 series and given the AMIP companion it had always implied (Fig. 32). The AMIP panel shows a deep *synchronised* dip in 1883–86 in every arm. That is **Krakatoa** — and a run at fixed 1850 forcing cannot contain a volcano. It is independent visual confirmation of §43.9, and stronger than the CO_2 argument: not a slow drift, but real volcanic events inside runs the logbook describes as pre-industrial.

The AMIP arms also no longer cluster near the "atmosphere-only floor" the coupled figure draws at +0.67. They straddle it: LX1 +1.08, P5 +0.66, control +0.64, S4 +0.25, LY2 −0.58, LX3 −0.88. The line is relabelled as what it is — the AMIP *control*, one arm among many.

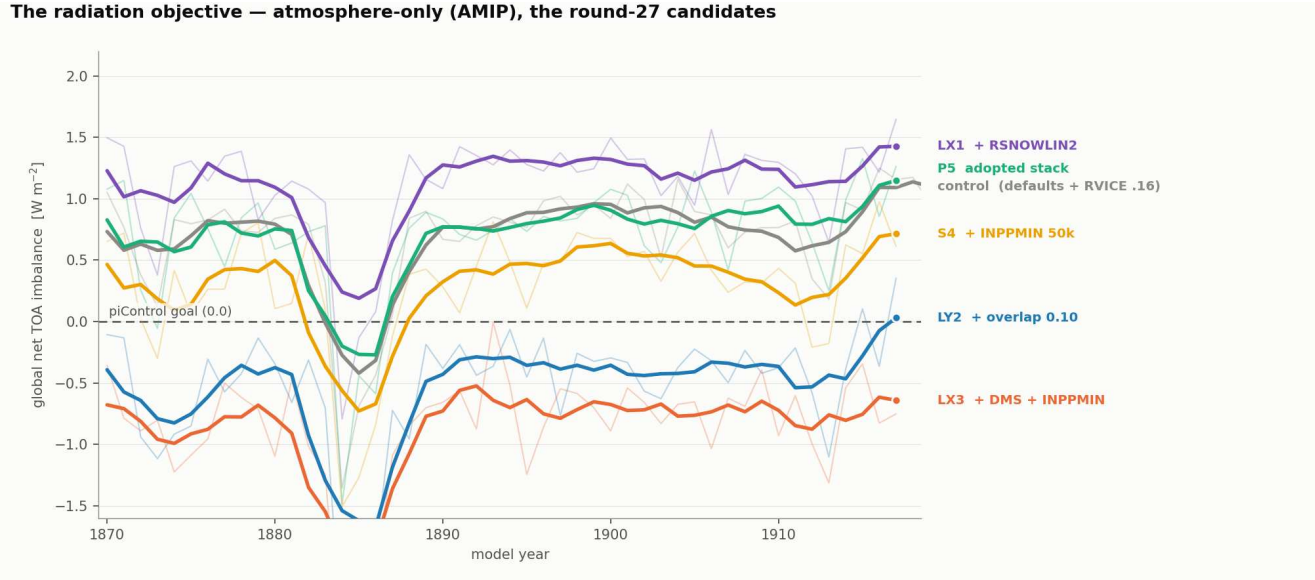


Figure 32: The AMIP radiation objective, round-27 candidates. The synchronised 1883–86 dip is Krakatoa: these "pre-industrial" runs read transient forcing, because `NCMIPFIXYR` sits in the dead `&NAMECECMIP6` block and the runtime value is −1 (§43.9). S4 lands nearest the piControl goal; LX3 and LY2 overshoot past it; `RSNOWLIN2` alone (LX1) moves it the wrong way.

44.7 Where the SO/tropics phase ends

Adopt S4 — `RCL_INPPMIN` 70000 → 50000 — pending 11G. It is the only arm that improves the Southern Ocean without spending the tropics, Siberia or the global budget.

Retire the round-26/27 lever line for tuning purposes. `RSNOWLIN2` closes the tropical longwave deficit and is the largest such response ever measured here, but it warms the Southern Ocean and worsens global energy; DMS and `ovl` 0.10 reach the Southern Ocean only by dimming the planet. None is adoptable. `RSNOWLIN2` remains interesting as *physics* — it is the one lever that moves the tropical longwave at all — and should be revisited if the atmospheric-absorption question below is resolved.

The open structural question. The surface shortwave error exceeds the TOA error everywhere (14.4 against 11.5 globally), and the TOA and surface constraints disagree about which arm is best: P5 has the surface nearly right and TOA 5 W m^{-2} short, LX3 the reverse. Closing a TOA gap of ≈5 costs the surface ≈5, and it did, almost exactly. That is an *atmospheric absorption* signature, not a cloud-amount one, and if it is the underlying error then no cloud-reflectivity lever can satisfy both — the campaign has been optimising against whichever metric happened to be quoted. Measuring it is one pass over data already on disk.

45 Round 29 (2026–08–13→16): the land-surface repair priced, and roughness falsified structurally

Three arms differing by one thing each — **11G** (adopted stack, S4), **11I** (+ land-surface repair), **11J** (+ Raupach canopy roughness) — run at 50 years to price the two remaining land changes. 11I completed. 11J reached 30 years, lost its fourth leg to a filesystem error, and was restarted. The roughness lever is dead, and not for a statistical reason.

45.1 Getting the arms to start: a fork, and two coupling defects (2026–08–13→14)

Both arms build from `run/50yr-arms` = `168a98b` (LPJ-GUESS gridlist from `TCO95-land` rather than `L096`) plus a cherry-picked `0218bbb` (cold-start a gridcell that carries no spin-up state). They *exclude* `46e7a84`; see §45.2.

Two defects had to be cleared first, both in the roughness coupling and both fatal at about 80 s:

- **`put_2d(GUE_Z0HV)` was ungated.** `ifraupachz0` gated the *computation* but not the OASIS put, so 11I — which does not request the field — sent it anyway: `oasis_put is called for a variable not in namcouple`, `mod_oasis_getput_interface.F90:685`. This is the same defect that cost 11G three legs on

2026–08–11; it had been fixed once and lost. Gating the put is what allows both arms to run from *one* binary (md5 `ff9b5507`), so 11J−11I carries no build difference at all.

- **11J's coupling fields were declared in the wrong section.** A new field needs `lpj_guess.add_veg_atm_fields` on the sending side and `oifs.add_atm_veg_fields` on the receiving side; 11J had the latter under `oasis3mct` and lacked the former. `esm_runscripts` then exits with `Coupling var not found` *without creating the experiment directory*, so there is no run to inspect — only the driver log.

45.2 `46e7a84`: 107 new `fail()` calls, classified (2026–08–14→15)

Laszlo's `46e7a84` ("Fix historical run crashes during land-use restart years", 2026–08–11, 2065 changed lines in `landcover.cpp` alone) adds **107** hard `fail()` calls and took `landcover.cpp` from 9 to 74. Three fired on our restart. Rather than patch them one at a time, the twelve failing cells — *all* between 61 and 71°N, this campaign's own region — were classified:

- **Nine are floating-point roundoff**, 3.3×10^{-9} to 4.2×10^{-7}, against `lc_bookkeeping_tol()` = `INPUT_RESOLUTION`$\times10^{-}$ 10^{-9}. That tolerance is scaled off the precision of the land-use *input*, but is applied to a sum of restart-state floats whose error floor is float epsilon; 4.2×10^{-7} is ≈3.5× float32 eps. **No restart can pass it.** A later run with the check relaxed measured 14909 metadata repairs and 1099 renormalisations across a ≈10 200-cell gridlist — essentially the whole domain fails it, not twelve cells.
- **Three are real**: stand metadata against physical stand fraction, differing by 8.0×10^{-4}, 6.1×10^{-3} and 3.3×10^{-2}. These are ours, not his; the check found something wrong in our own spin-up state.
- **One is the peat conflict** (§45.3).

The function carrying the first two is named `synchronize_carried_landcover_state` and its next statement is `gcst.frac = st_sum[s]` — **it is already a repair routine that refuses to repair**. Both CMIP6 predecessors of this configuration repair rather than assert: EC-Earth3 `lpjg_v4p1` rescales land-cover fractions by their sum with a warning and pushes the residual into `NATURAL` (it tests closure at 10^{-14}, *tighter*, but fixes); JSBACH3 `scale_cover_fract` iterates `ntiles` times toward sum=1 and calls `finish()` only if a fraction is still out of range afterwards.

45.3 Peat versus LUH3: two prescribed inputs pinned to different epochs (2026–08–16)

At 67.7917°N 19.2857°E the peat map prescribes 0.2264 and the spin-up carries a 0.1917 physical stand, while LUH3 leaves only **0.088604** NATURAL. Reproducing that number exactly required two details: our configuration is `run_barren 0`, so NATURAL also absorbs the LUH3 closure gap rather than being the raw `primf+primn+secdf+secdn` (0.043008); and cells with no usable LUH3 state fall back to NATURAL=1 instead of taking the $\approx 2 \times 10^{20}$ fill value into the arithmetic.

The conflict is **not** static-peat-versus-dynamic-land-use. Both are prescribed and both are static here: `externalinput.cpp:779` pins the LUH3 year to `fixedLUafter` unconditionally, ignoring the model calendar. Read at the run's own year the same cell gives NATURAL = 0.617646 (1350), 0.796720 (1399) — **no conflict at all**. The entire problem is an artefact of the 1850 pin, which is not negotiable because 1850 forcing *is* the piControl convention.

Neither CMIP6 predecessor could hit this. EC-Earth3 reads `PEATLAND` from the land-use file itself (`externalinput.cpp:670`), so there is no second dataset to disagree with — its `file_peat` is a dead declaration, referenced nowhere — and its coupled configuration runs `run_landcover 0` regardless. JSBACH3 has no peatland type at all.

Two policies were coded behind `peat_lu_conflict_policy`, default 0 so an unmodified `.ins` is CMIP7-compliant:

- **0, LUH3 authoritative** — required for CMIP7, where the land-use dataset is a common forcing. Peat is capped and the surplus *area and its pools* are moved by `landcover_dynamics`. Under `fixed_LU` that machinery never runs (`landcover_init` returns straight after `landcover_reconcile_restart_state`, `landcover.cpp:156-163` and `lc_changed()` is false), so this policy *refuses with an explanatory error* rather than truncating. A first attempt that capped anyway rewrote the bookkeeping while the stand kept its area and was caught one call later by the very next check (`st=2 metadata=0.0886 physical=0.1917`) — Laszlo's comment that capping "would discard state without a transfer" is not a preference but structurally true here.
- **1, physical peat authoritative** — the stand keeps its area and the non-peat categories are scaled into the remainder. Nothing is discarded, so no transfer is needed. Defensible only for fixed-forcing control runs: north of 60°N mean LUH3 rangeland at 1850 is 0.0040, while northern peat holds order 500–1000 PgC and sets the soil thermal regime this campaign's bias lives in.

Only 73 of 10 211 cells have a peat-map value above the available NATURAL, and only one aborted, because elsewhere the spin-up had already capped the physical stand lower.

45.4 11I: the repair is not free (2026–08–16)

11I−11G is not a one-variable pair, despite its runscript header saying "ONE input file". Three things differ: the `_v2` soil map, the `L096→TC095-land` gridlist, and the cold-start binary path; the last two are inseparable, because the new gridlist is what adds the 307 cells the cold-start path lets run. Read it as *the land-surface repair bundle*.

Scored over the 40 matched years (1350–69 + 1380–99; 1370–79 excluded because the repaired soil map slipped into 11G's leg 3 alone, 233 organic cells against 547):

metric	11G	11I	diff	paired t
net TOA [W m^{-2}]	0.321	0.301	−0.020	−0.33
Siberia DJF soil [°C]	−13.079	−12.771	**+0.308**	**2.49**
Siberia JJA soil [°C]	8.160	7.939	−0.221	−1.08
Siberia JJA T2m [°C]	9.664	9.434	−0.230	−1.19
SO SW CRE [W m^{-2}]	−63.964	−63.779	+0.185	1.71
global T2m [°C]	12.365	12.335	−0.030	−1.25

The statistic mattered. Every arm branches from the same 1350 state, so the difference series is *paired* and starts near zero. Scored against the control's absolute interannual scatter ($1.96\,\mathrm{sd}\sqrt{2/n}$, threshold 0.345) the DJF soil warming reads as null; paired, it is $t = 2.49$ and **resolved**, and it is still growing (first half +0.212, second +0.404).

But against the target it is a trade, not a freebie. Siberian JJA T2m bias against ERA5:

$$11\mathrm{E}\ -1.013\ \rightarrow\ 11\mathrm{G}\ -0.673\ \rightarrow\ \mathbf{11I}\ \mathbf{-1.123}\ \rightarrow\ 11\mathrm{J}\ -0.844$$

11I hands back S4's entire Siberian summer gain and lands *worse than the 11E baseline*. The increment (−0.230, $t = -1.19$) is not resolved, but the ranking is unambiguous and JJA Siberia is the bias the campaign exists for. Vegetation moves the other way: on matched decade and matched cells, TREEFPC +0.0094 and C3G +0.0276, recovering most of what S4 cost (0.0787 → 0.0649 → 0.0743).

45.5 Raupach: the lever cannot bootstrap the canopy it needs (2026–08–16)

11J−11I *is* a clean one-variable pair: same binary, one namelist flag. Over all 30 years, paired, nothing Siberian resolves — DJF soil, the *pre-registered* target, is +0.054 ($p = 0.670$); JJA soil +0.312 ($p = 0.167$); JJA T2m +0.161 ($p = 0.432$). Resolving the JJA soil point estimate at 95 % would need $(1.96\,\mathrm{sd/effect})^2 = \mathbf{57}$ years, so **even a completed 50-year 11J is underpowered for its own point estimate**: with a 1.2 K interannual noise floor this design cannot detect anything below ≈0.35 K at 50 coupled years.

That is a statement about the diagnostic. The *mechanism* was then tested at the top of the causal chain, where a roughness change acts first and the noise is far lower:

Siberian land, 30 yr paired	11I	11J	diff	p
DJF $\lvert\tau\rvert$ [N m^{-2}]	0.0955	0.0916	−0.0039	0.113
DJF 10 m wind [m s^{-1}]	2.1282	2.1911	+0.0629	0.253
DJF skt−2t [K]	−2.2719	−2.2761	−0.0042	0.787
JJA $\lvert\tau\rvert$ [N m^{-2}]	0.0753	0.0706	−0.0047	0.016
JJA 10 m wind [m s^{-1}]	1.2821	1.3637	+0.0816	0.037
JJA skt−2t [K]	0.8456	0.9497	**+0.1041**	**0.00003**

`JJA skt−2t` survives Bonferroni over all 14 tests. But the pre-registered sign test — *noise cannot know to reverse between seasons* — fails: winter was predicted rougher (stress up, wind down) and instead has the *same* signs as summer. The winter lever does not act at all.

Why. Measured in 11J's Siberian box (728 cells, 1370–79): tree height in January **2.89** m, density 0.161 indiv m^{-2} (of which BNS 0.112), implied stem diameter 0.0198 m, and total LAI collapsing Jan/Jul = 0.131. `add_woody` is called only for `lifeform == TREE`, so grass contributes nothing to λ by construction, and in winter everything rests on the stem term:

$$\lambda_{\mathrm{stem}} = \mathrm{dens} \cdot \mathrm{diam} \cdot h = 0.161 \times 0.0198 \times 2.89 = \mathbf{0.0092}$$

Fed to the code's own constants (C_S=0.003, C_R=0.3, C_{D1}=7.5, κ=0.41), that gives z_0 = **0.0132** m — the bare-soil value — against the IFS lookup's 0.058 m for the same box in DJF. **Raupach makes the winter surface** 4.4× **smoother, not** 2.4× **rougher**, which is exactly the sign the momentum budget measures.

"A leafless canopy is the roughest state" is true — for a canopy. A real boreal larch stand ($h = 18$ m, $\lambda_{\text{stem}} = 0.12$) gives $z_0 = 1.51$ m. At 2.9 m and 2 cm stems, λ is an order of magnitude below the peak of the Raupach z_0/h curve, on the rising branch where $z_0 \to 0$. The offline calculation that predicted DJF 0.058 → 0.140 m assumed a forest; the coupled model grows scrub.

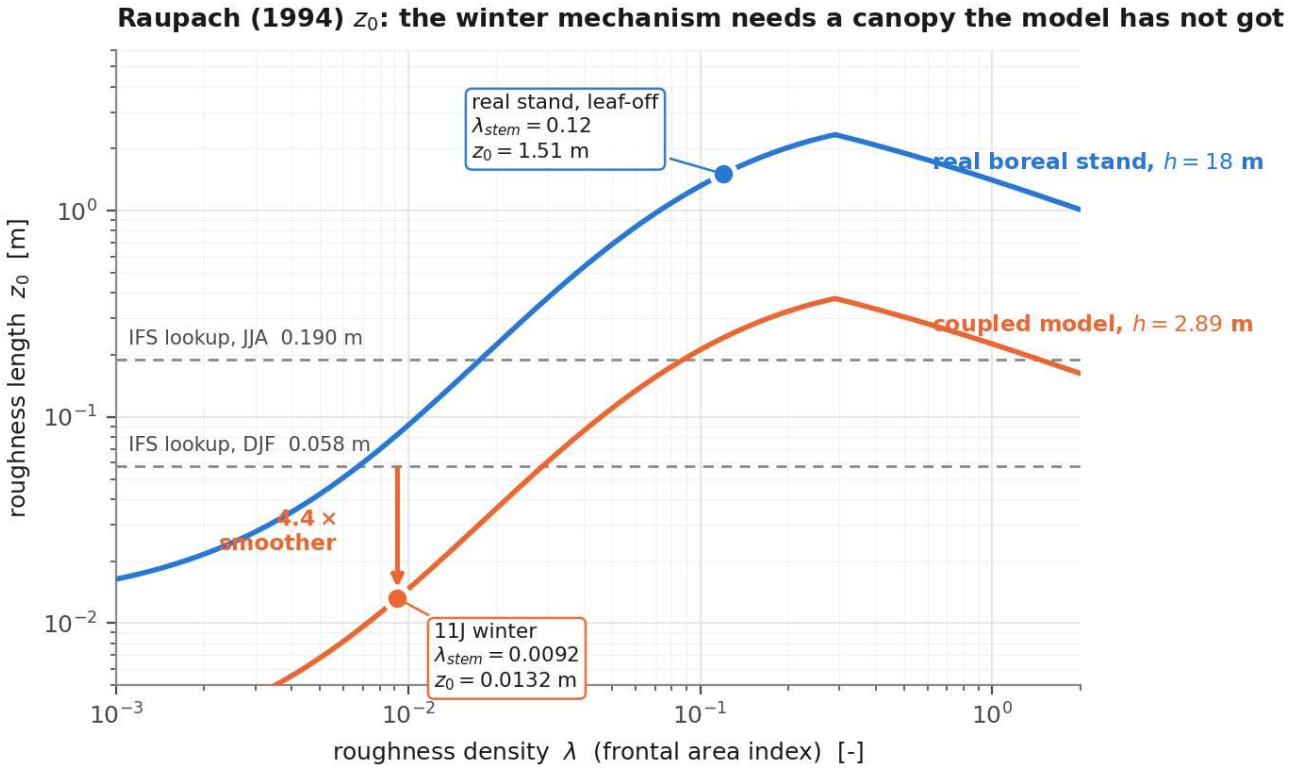


Figure 33: Why the winter lever cannot engage: the model has no canopy for it to act on. $z_0(\lambda)$ from the implementation's own constants (C_S=0.003, C_R=0.3, C_{D1}=7.5, κ=0.41), for the 2.89 m scrub the coupled model grows in Siberia (orange) and an 18 m boreal larch stand (blue). Measured in 11J's Siberian box (728 cells, 1370–79): density 0.161 indiv m^{-2}, stem diameter 0.0198 m, so $\lambda_{\text{stem}} = 0.0092$ once the leaves are gone (Jan/Jul LAI = 0.131). That lands an order of magnitude below the peak, on the rising branch, at $z_0 = 0.0132$ m — the bare-soil value — against the 0.058 m IFS lookup it replaces. **Switching Raupach on makes the winter surface** 4.4× **smoother, not** 2.4× **rougher.** A real stand at leaf-off sits at 1.51 m: "a leafless canopy is the roughest state" is true, for a canopy.

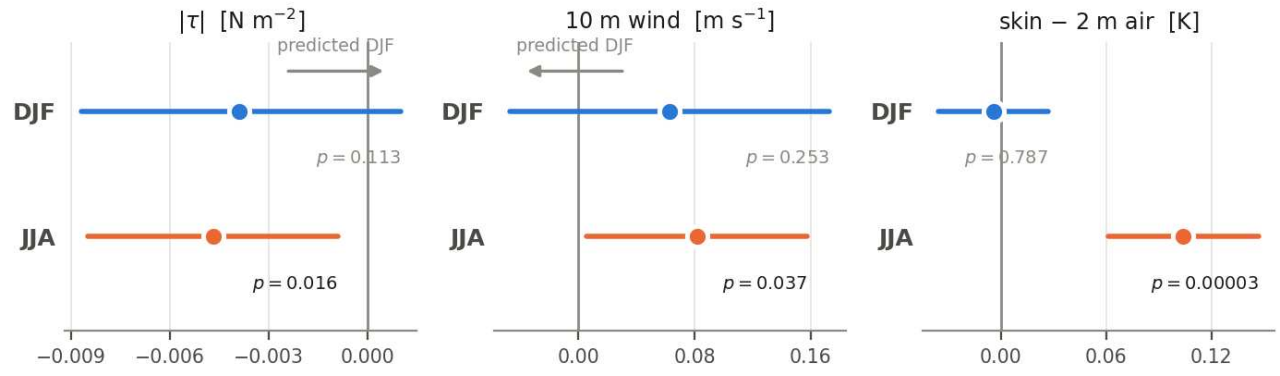


Figure 34: The momentum budget confirms it, where soil temperature could not. 11J−11I over 30 paired years, Siberian land, in native units (small multiples rather than a shared axis). Summer responds as a *smoothing*: $|\tau|$ −6.3 % ($p = 0.016$), 10 m wind +6.4 % ($p = 0.037$), and skin-to-air decoupling +0.104 K ($p = 3 \times 10^{-5}$, the only result surviving Bonferroni over all 14 tests). Winter shows the *same* signs, where the pre-registered mechanism required the opposite (grey arrows), and skin−air is flat at −0.004 K. The sign test was built as a falsification check — noise cannot know to reverse between seasons — and it fails. This is why Siberian DJF soil is +0.054 K ($p = 0.670$): not a signal buried in noise, but no winter roughness change to propagate.

The structural finding. Raupach was proposed to fix a cold bias severe enough to lose the boreal forest, and its winter mechanism *requires* that forest. The lever cannot bootstrap the canopy it needs in order to work. This is stronger than the statistical null: with 57 years we would merely be measuring the response to a roughness *decrease*. It is the same trap as §43.12 — BNS is outcompeted by C3 grass rather than climate-gate limited — seen from the other side.

Not explained: summer also got smoother, whereas a rough summer λ suggests Raupach should exceed IFS's 0.190 m. That estimate used box-total LAI, which includes grass contributing nothing, so it is untrustworthy; tree-only LAI would settle it. It does not affect the winter conclusion, which is the hypothesis under test.

45.6 A leg destroyed at finalisation, again (2026–08–15)

11J's leg 4 ran to **step** 87648 — **the full ten years** — and then died in finalisation: `error in line 261 io_netcdf_file_module.F90 NetCDF: HDF error` on FESOM rank 129. The XIOS buffer flush is destroyed by the abort, so ten completed years became nothing, the date file walked to 1390, and the run stopped. Ruled out: not our code (no `oasis_abort`, no LPJ-GUESS `fail()`), not Raupach (the failure is in FESOM's writer), not quota (2.81 P of 4.22 P), not systematic (11I never saw it in five legs). A transient Lustre/HDF fault. Recovered by resetting the date file to 1380–01–01 run 4; all three restart sets were intact, and the leg-4 `oifs` restart directory was *empty*, confirming nothing was written.

LPJ-GUESS output survived: it writes `.out` files per rank continuously, outside XIOS, so the vegetation diagnostics for the lost decade are on disk even though every netCDF field is gone. That is worth knowing for any future leg loss.

46 Round 30 (2026–08–18→19): the Southern Ocean cloud round, coupled

Six weeks of Southern Ocean work reduced to one question — *is there a lever that brightens SO cloud without wrecking Siberia?* — and it now has an answer, with the candidates named and two of them measured coupled.

46.1 The reframing: 84 % of the coupled SO error is inherited from AMIP

The same diagnostic applied to AMIP and coupled, 30 yr each, SO 45–65°S TOA SW CRE against CERES (`scripts/analysis/amip_vs_coupled_so.py`):

run	SW CRE	vs CERES	tcc
AMIP P5 control	−63.097	**+4.739**	0.8356
AMIP LY2 `ovl 0.10`	−68.983	−1.147	0.8474
AMIP LX3 DMS+INPPMIN	−68.737	−0.901	0.8435
CPL 11E	−62.186	**+5.651**	0.8365
CPL 11G +S4	−63.673	**+4.163**	0.8359

Cloud cover is 0.8356 in AMIP against 0.8365 coupled, and the CRE differs by 0.91 W m^{-2}: **the cloud field transfers**. 84 % of the coupled error is already present in AMIP over a prescribed, correct ocean, and 11G coupled (+4.163) is *closer* to CERES than the AMIP control (+4.739). The Southern Ocean bias is mostly an unclosed AMIP problem, not a coupling problem — which is why the remaining effort belongs in AMIP, where an arm costs a fraction of a coupled one.

46.2 The overlap lever taken coupled: 11L and 11M, both rejected

Both are 11G plus one namelist number, 40 yr, scored paired over the 30 clean matched years 1350–69 + 1380–89. 11G's 1370–79 carries the repaired soil map and has to be excluded; `coupled_dms_pair.py` takes an exclusion argument for exactly this.

	11L `ovl 0.35`	**11M** `ovl 0.10`	threshold
SO SW CRE	−1.565 *	−5.938 *	0.517
SO SW CRE DJF	−3.399 *	−11.433 *	1.167
vs CERES	+3.89 → **+2.33**	+3.89 → **−2.05**	
net TOA	−0.281	**−0.698** *	0.403
tropics SW CRE	−0.297 *	−1.370 *	0.209
global T2m	−0.242 *	−0.604 *	0.162
Siberia JJA T2m	**−1.000** *	**−2.321** *	0.406
Siberia JJA soil	−1.014 *	−2.377 *	0.402

It transfers — the first SO lever that does. AMIP predicted 11L at +1.992 against CERES; coupled lands at +2.33. The transfer ratios are 0.57× and 1.01× on annual SO CRE, against DMS's 0.19–0.44× and S4's 0.28–0.31×. The discriminator, consistent with §76.5, is how much sea ice the lever grows.

Both are rejected anyway. 11M fails net TOA outright. 11L passes net TOA but costs 1.000 K of Siberian JJA — it spends the entire budget of target 2 to buy 40 % of target 1. The vegetation cost is proportionate: against 11G, 11L loses 39.5 % of boreal needleleaf and 11M loses 89.2 %, all but eliminating larch.

46.3 Ocean drift, and a correction that inverts its own conclusion

Whole-ocean potential temperature (`thetaoga`), steady rate after the first decade:

arm	K/century	per millennium
11E	+0.22	+2.2 K
11G	**+0.19**	+1.9 K
11L	+0.12	+1.2 K
11M	−0.02	∼ 0

Not decelerating, so a multi-millennial spin-up on 11G is untenable.

Corrected the same day. An earlier reading of this table reported a "−1.0 W m^{-2} structural leak, invariant across arms" between TOA and surface, and concluded the drift could not be fixed radiatively. **That −1.0 is snow enthalpy.** $sf \cdot L_f$ measures 0.962/0.976/1.018/1.073 against residuals of −1.006/ − 0.996/ − 1.022/ − 1.088. The convention was already on record in §11.1: surface = $ssr + str + sshf + slhf - sf\rho L_f$. With the term included the column conserves to 0.02–0.04 W m^{-2} and the ocean warming matches what TOA implies, so **tuning net TOA to zero does stop the drift** — 11M, at TOA −0.016, drifts −0.02 K/century. The conclusion inverts.

A second error in the same reading was a window mismatch: 11G's net TOA is +0.205 over 1350–69+1380–89 but +0.732 over 1380–99. Net TOA is itself drifting upward through these runs, which matters separately for spin-up planning.

47 The 11-series coupled campaign (2026–08–16→21): LX4 taken coupled

47.1 The roster

arm	content	role
11E	campaign base	the reference the others are read against
11G	11E + S4 (`RCL_INPPMIN` 50000)	the adopted base for rounds 29–30
11I	11G + v2soil (LPJG repair)	prices the land-surface repair
11J	11I + `ifraupachz0`	the roughness pair, one binary, run-time gate
11L	11G + `ovl 0.35`	SO overlap lever, rejected (§46.2)
11M	11G + `ovl 0.10`	SO overlap lever, rejected (§46.2)
11N	LX4 stack, 1850 forcing	the best AMIP tuning, taken coupled
11P	LX4 stack, constant 1990 forcing	the same stack, period-matched to ERA5

47.2 Is the 1990 forcing actually 1990? Everything except LPJ-GUESS

11P exists to remove a confound that had been quietly distorting every comparison against ERA5: a PI run read against a present-day reanalysis is cold *by construction*, so part of the reported land bias was never a model error at all. `NCMIPFIXYR` pins GHG, ozone, composition, stratospheric aerosol, SO_4, solar TSI *and* MACv2SP. It does **not** pin LPJ-GUESS, which has its own `fixed_CO2`, `fixed_ndep` and `fixed_LU`.

Two traps were hit while verifying this and are worth recording:

- The `&NAMECECMIP6` block accepts `NCMIPFIXYR` and is **never read**. Liveness must be checked as `Set JYEAR` in `NODE.001_01`, not by finding the name in `fort.4`.
- **A correct run was cancelled on a misreading.** The log line `SP_SETUP: IYR` reports the MACv2SP *file* that was opened, not the year the profile is evaluated at. The evaluation year is `YEAR_FR`, which honours `NCMIPFIXYR` in both source trees (`radiation_scheme.F90:387`). 11P was already right; the patch written to "fix" it was reverted and the arm resubmitted unchanged.

47.3 What the stack bought: the land bias by band and season

Land T2m bias against ERA5, 1380–89, from `scripts/analysis/land_bias_season_band.py`:

	11G					**11N** (1850)					**11P** (1990)				
band	DJF	MAM	JJA	SON	ANN	DJF	MAM	JJA	SON	ANN	DJF	MAM	JJA	SON	A
60–90N	−6.35	−4.60	−1.96	−3.65	−4.14	−4.49	−2.99	−1.52	−2.26	−2.82	−3.63	−1.64	−0.58	−1.29	−
30–60N	−3.83	−3.56	−0.80	−1.84	−2.51	−3.18	−3.37	−0.17	−0.93	−1.91	−2.90	−2.49	+0.72	−0.22	−
0–30N	−2.96	−2.21	−2.46	−2.33	−2.49	−2.59	−1.98	−1.99	−1.88	−2.11	−1.90	−0.98	−1.16	−1.19	−
30S–0	−2.75	−2.83	−2.21	−1.68	−2.37	−2.30	−2.20	−1.61	−1.00	−1.78	−1.76	−1.76	−0.90	−0.63	−
60–30S	−0.89	−1.46	−2.29	−1.52	−1.54	−0.32	−0.90	−1.73	−1.00	−0.99	+0.07	−0.66	−1.13	−0.86	−
ALL LAND	−3.47	−3.27	−2.14	−2.29	−2.79	−2.76	−2.66	−1.48	−1.54	−2.11	−2.23	−1.85	−0.73	−0.83	−
ocean	−1.86	−1.64	−1.53	−1.49	−1.63	−1.44	−1.24	−1.15	−1.10	−1.23	−0.86	−0.71	−0.64	−0.56	−

Read the two steps differently. 11G → 11N is a genuine tuning gain at a fixed forcing basis: all-land annual −2.79 → −2.11, and 60–90N DJF −6.35 → −4.49, the largest single winter improvement in the campaign. 11N → 11P is *not* a tuning gain — it is a change of comparison basis, from a 1850-forced run read against a present-day reanalysis to a period-matched one. Only 11P should be quoted as a bias against ERA5.

And the shape of what remains is the finding. In 11P the land–ocean gap is −1.37 K in DJF against −0.09 K in JJA. Siberian JJA is essentially spot on (10.649 against ERA5's 10.666). The campaign was built around a boreal *summer* cold bias; what is left is a *winter* one, and §48 resolves where in the column it sits.

48 Round 31 (2026–08–21): the winter land bias is above the screen

Three surface-side levers are scored null in this round, and the reason turns out to be the same for all three.

48.1 Round 16 scored at last: skin conductivity is not the route

Round 16 (`ECE_LAMSK_SN`, the exposed-snow skin conductivity Z_{SNOW} $7 \to 15 \to 25$) ran 48 years on 2026–08–04 and then sat unscored for 17 days. Scored paired against I1 over 1872–1915:

arm	60–90N DJF	threshold	60–90N ANN	land−ocean gap DJF
J1 `LAMSK_SN` 15	+0.125	0.41	+0.062	+0.009
J2 `LAMSK_SN` 25	−0.039	0.45	−0.015	−0.073

The round predicted +1 to +3 K. It measured nothing, at 3.6× the default, with power to detect 0.4 K.

This is **not** a seventh silent no-op, and that was checked before the null was believed: the symbol lives in `libsurf.SP.so`, *not* `libarpifs.SP.so` — grepping only arpifs returns zero and looks exactly like the W1 failure — `fort.4` carries 15/25, and the arms differ from I1 by up to 21 K locally in year 1 across 53 % of cells. The lever is live and its mean response is genuinely zero. Round 16's own pre-registered falsifier therefore fires: the winter bias lies in boundary-layer mixing instead.

48.2 Raupach roughness, measured: null, as §45.5 predicted

11J = 11I + `ifraupachz0`, one binary with a run-time gate, both branched from 1350. Verified before scoring: 11J's `guess.ins` carries `ifraupachz0 1` and `fort.4 ECE_CPL_LPJG_Z0 = .true.`; 11I has neither.

window	60–90N DJF	30–60N DJF	land−ocean gap DJF
1380–1389	+0.302 (0.89)	−0.234 (0.41)	−0.017 (0.20)
1360–1379	−0.172 (0.58)	−0.161 (0.43)	−0.003 (0.19)

The sign flips between windows and the DJF gap is zero in both. This is the T2m confirmation of the mechanism argument already made in §45.5: Raupach derives z_0 from FPC and *canopy height*, so wherever LPJ-GUESS has lost the forest it too returns a low z_0. It changes the functional form, not the missing canopy.

48.3 The decomposition: the skin is already right

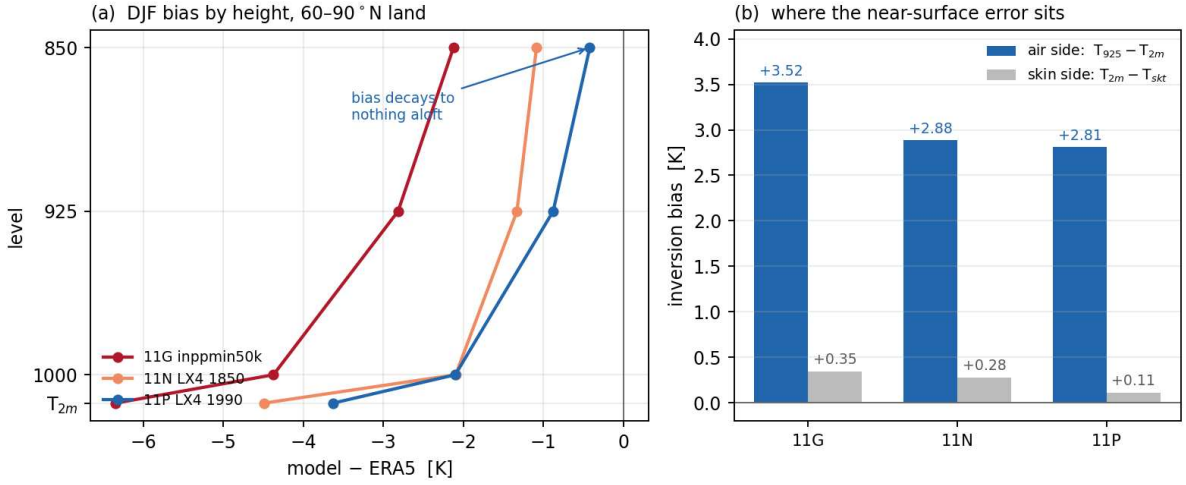


Figure 35: The DJF land cold bias is surface-confined, but the surface is not what is broken. **(a)** Model−ERA5 DJF bias over 60–90°N land against height. In 11P the bias runs −3.63 K at the screen to −0.42 K at 850 hPa, so it decays to nothing aloft and a *redistribution* lever is the right class. **(b)** Splitting the near-surface error: the air-side inversion bias ($T_{925} - T_{2m}$) is +2.81 K in 11P while the skin-side one ($T_{2m} - T_{skt}$) is +0.11 K. The skin is already right, so the entire excess inversion sits in the layer between the screen and 925 hPa. That is why skin conductivity (§48.1), Raupach z_0 (§48.2) and F1 `RVZOH` all scored null: each adjusts a part of the column carrying no error. Note also that the campaign's tuning improved both terms, 11G→11N→11P.

Measured against ERA5 monthly pressure-level T (`/pool/data/ERA5/E5/pl/an/1M/130`, DJF 1990–2014, remapped to the model grid), with land cells excluded *per level* where the level lies below ground. That mask is not cosmetic: without it the "bias" at 1000 hPa reads +28 K, which is mountains being extrapolated rather than error.

11P, DJF	T_{2m}	T_{1000}	T_{925}	T_{850}	lowinv bias	sfcinv bias
60–90N	−3.63	−2.10	−0.88	−0.42	**+2.81**	**+0.11**
30–60N	−2.90	−1.86	−1.31	−1.31	+1.07	−0.27
ALL LAND	−2.23	−1.43	−1.25	−1.22	+0.86	−0.36

Both inversion columns are differences of differences, so any systematic model−ERA5 offset cancels out of them.

Splitting the all-land number: of the −2.23 K DJF bias, roughly −1.0 K is excess inversion and roughly −1.2 K is a **deep tropospheric cold bias** still present at 850 hPa, which no mixing lever can remove and which tracks the ocean bias (−0.69 K) and the TOA budget. **Only about half of the remaining land bias is addressable by a surface or boundary-layer lever at all**, and that half is concentrated at high latitude.

48.4 The lever: `&NAMVDFS`, and why it had to be created

The 2 m–925 hPa layer is mixed by the stable branch of `VDFEXCU` (Sandu & Beljaars 2013, cited in the routine's own header at `vdfexcu.F90:81`), and every constant governing it was a hardcoded literal. `scripts/patches/expose_sbl_mixing.p` moves five into `YOEVDFS`, set in `SUVDFS` from a new namelist:

name	default	what it does
`RSBLB`	5.0	coefficient b of the LTG stable stability functions
`RSBLD`	1.0	coefficient d of the same
`RSBLLMIN`	30.0	asymptotic mixing length above the stable BL, and floor within it [m]
`RSBLLMAX`	300.0	cap on the stable mixing length [m]
`RSBLPBLF`	0.1	fraction of stable BL depth taken as the mixing length

Defaults equal the removed literals, so a `fort.4` without the group is bit-identical to the as-released model. `ZCB`, `ZCD` and `ZLMIN` were checked to appear only inside `IF (ZRI > 0)` before patching, so the knobs cannot reach unstable or convective mixing.

With $F_H = 1/(1 + 2b\,Ri\sqrt{1 + d\,Ri})$, **lowering b lengthens the stable tail** and the gain grows with Ri, concentrating the change in exactly the deep-inversion columns that carry the error:

Ri	$b = 5$	$b = 3$	$b = 2$
0.5	0.140	0.214 (×1.5)	0.290 (×2.1)
2.0	0.028	0.046 (×1.6)	0.067 (×2.4)

Which knob bites where. Inside the stable BL the mixing length is clamp($0.1h$, 30, 300), so the $0.1h$ term only takes over once $h > 300$ m and the cap needs $h > 3000$ m. An Arctic winter SBL is typically 50–300 m deep, so in the regime being targeted λ is pinned at the 30 m floor and `RSBLPBLF` never fires. The operative pair is therefore `RSBLB` (always evaluated) and `RSBLLMIN`; `RSBLLMAX` is effectively inert.

`POSNAM` is deliberately not used: it aborts when a group is absent, and no existing `fort.4` carries `&NAMVDFS`. A plain `READ` with `IOSTAT` keeps the defaults when the group is missing (< 0) and **aborts** when it is malformed (> 0), rather than swallowing a typo into an arm identical to its control. `SUVDFS` echoes the values to `NULOUT` unconditionally so `NODE.001_01` proves what ran.

48.5 The SB arms, and the premise check they rest on

run	setting	prediction
SB0	LX4, no `&NAMVDFS`	rebuilt control (see below)
SB1	LX4 + `RSBLB: 3.0`	60–90N DJF warms; JJA and tropics barely move
SB2	LX4 + `RSBLB: 2.0`	the same, roughly twice the size

SB0 exists because the AMIP library was last built 2026–08–09 and commit `75a4e45` has since completed the DMS ICMCL ingest chain — not an inert namelist exposure but a change to which surface field is ingested and under which GRIB code. LX4 reads DMS, so SB1 minus the old LX4 output would mix the `RSBLB` effect with an ingest change. The Raupach pair was deliberately run from one binary for this reason; the same standard applies here, and since a rebuild cannot be a run-time gate, the control is re-run instead.

Does the AMIP base share the defect? It has to, or the screen is meaningless, and the bias being screened was measured on *coupled* 11P. Checked before the compute was spent (`scripts/analysis/amip_vs_coupled_inversion.py`), low-level inversion bias, DJF:

band	AMIP LX4	coupled 11P
60–90N	+1.54	+2.81
30–60N	+0.37	+1.07
ALL LAND	+0.43	+0.86

Same sign and same vertical shape, so the screen tests the right defect, and +1.54 K against a 44-year paired threshold of ~ 0.4 K resolves a lever that removes a quarter of the bias. But T_{925} is *identical* in the two (−0.88), so the whole AMIP–coupled difference sits at the screen: about 1.3 K of the coupled excess at 60–90N has no AMIP counterpart. AMIP prescribes sea ice, which is the obvious home for it. **A clean AMIP result bounds the direct land effect and cannot close the coupled gap**, and the sea-ice guardrail cannot be tested in AMIP even in principle.

Scored in §49. The falsifier below did not fire: the inversion responded, and coupled the screen followed.

Falsifier on record. If 60–90N DJF does not respond, the excess inversion is not set by the stable stability function, and what remains is the mixing length (`RSBLLMIN`) or a radiative rather than turbulent cause. **Guardrails:** 60–90N DJF must not overshoot warm, the inversion bias must not cross zero, global net TOA must stay within ±0.3 at 44 years, Siberian JJA must not go backwards, and the Southern Ocean must be re-checked — $Ri > 0$ does not know about land, and the stable regime is rare over open ocean but common over ice.

48.6 Infrastructure fixed on the way

- `esm_master recomp-oifs-*` died in the config parser with `KeyError: 'Key general.with_co2_oce_coupling was not defined'`. The `feat/awiesm3-v3.4-co2` merge added a `choose_general.` block to the *component* yaml, and a standalone component build has no setup to define the key. Fixed with esm_parser's documented `"*"` wildcard, which drops the block when the key is undefined; every setup defines it, so the wildcard never fires for them.
- The build then failed three times with `ifort: error #10103: can't fork process`. Not a code or HPC change: `ulimit -u` counts *threads* on Linux, and the IDE/agent session alone held ∼1310 of the 2048 (devin 399, language_server 300, github-mcp 218, claude 206), leaving ∼700 for a build that spawns `make -j32` with `fcm make -j 256` inside it. The first failure also orphaned ∼250 fcm workers, which then broke the retries. Resolved by building on a compute node (`scripts/build/build_oifs_compute.sh`); the as-released `FCM_PARALLEL` is left untouched. Completed in 8:16 with zero fork failures.

49 Round 32 (2026–08–21→23): the stable-BL lever, scored

Round 31 ended with a diagnosis and an untested knob. This round scores it: four AMIP arms, one falsification, a coupled pair, and the first coupled state in this campaign that improves the winter land bias without paying for it somewhere else.

49.1 The AMIP screen, and the shortwave bill

SB0/SB1/SB2 ran 46 years each on one binary, scored paired over 1872–1915. SB0 exists because the AMIP library predated commit `75a4e45`, which changed the DMS ICMCL ingest — LX4's own output was therefore not a valid control (§48.5).

DJF, 44 yr	**SB1** b=3	**SB2** b=2
inversion 60–90N	−0.424 *	−0.713 *
inversion all land	−0.148 *	−0.282 *
T2m 60–90N	+0.029	+0.285
T2m 30–60N	+0.060	+0.492 *
T2m all land	+0.077	+0.233 *
T925 60–90N	−0.396 *	−0.428 *
Siberia JJA	−0.002	+0.000
global net TOA	+0.090 *	+0.175 *

Three things to read here. The inversion responds strongly and **dose-consistently**. The Arctic *screen* does not — +0.285 against a 0.345 threshold — and the inversion weakens roughly 40 % by warming the screen and 60 % by cooling the air above it. And **Siberian JJA is exactly null**, which makes this the first lever in the campaign to move the winter bias without spending boreal summer.

The TOA bill is **shortwave, and cloud-mediated**: of SB2's +0.175, SW carries +0.319 against LW −0.144, and SW CRE carries +0.284 of the SW with total and low cloud both falling (−0.007, −0.005). Enhanced stable mixing dries and erodes low cloud. Regionally the tropics carry all of it (SW CRE +0.515, net +0.235) because *subtropical stratocumulus decks are stable boundary layers*; the Southern Ocean mildly improves (−0.182); and the Arctic's +0.620 is almost all LW, a symptom of the cooling aloft reducing outgoing longwave rather than a gain.

49.2 SB3: the mixing-length route, falsified

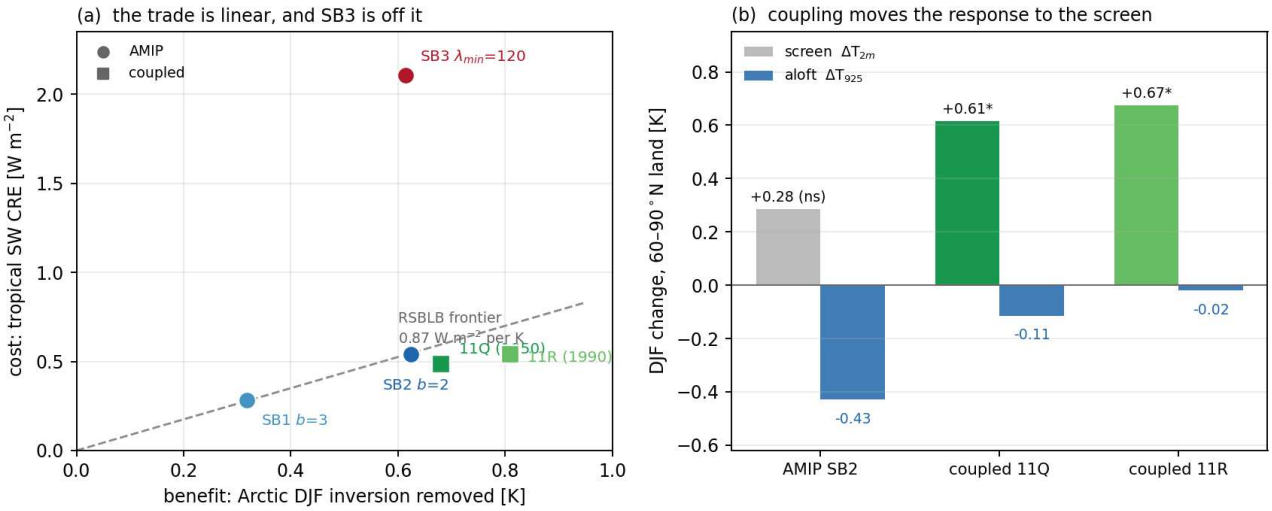


Figure 36: (a) What each arm pays (tropical SW CRE) against what it buys (Arctic DJF inversion removed). SB1 and SB2 lie on one line through the origin at 0.87 W m^{-2} per K, so the trade is *linear in dose* and cannot be dodged by tuning gentler — halving `RSBLB` halves both terms. SB3, the mixing-length route, sits far off that line at 3.43. The coupled arms land on the same frontier. **(b)** Why the coupled pair was necessary: in AMIP the inversion weakens mostly by cooling the air aloft and the screen response is not significant, whereas coupled — with sea ice free to respond — it is essentially all screen.

`RSBLLMIN` 30 → 120 m was a regime-discriminator hypothesis: inside the stable BL the length is clamp$(0.1h, 30, 300)$, so the 30 m floor binds only where the stable layer is shallower than 300 m, which a shallow Arctic winter SBL is and a deep subtropical Sc boundary layer is not. Crucially this is a *regime* discriminator, not a geographic one; no latitude-dependent parameter is invented anywhere.

The runscript recorded, before the run, the reason it might invert: `RSBLLMIN` also sets the length *above* the stable BL, because the `ELSE` branch of `vdfexcu` assigns `ZKLENT = ZLMIN` in free-shear layers. Over a Sc deck that is enhanced entrainment across the capping inversion, a known way to destroy stratocumulus. That is what happened.

1874–1901, 28 yr	SB1 b=3	SB2 b=2	**SB3** λ_{min}=120
Arctic DJF inversion	−0.318 *	−0.624 *	−0.614 *
tropics SW CRE	+0.281 *	+0.541 *	**+2.106** *
global net TOA	+0.094 *	+0.187 *	**+1.673** *
SO SW CRE	−0.170	−0.182	+1.395 *
cost per K of inversion	0.88	0.87	**3.43**

SB3 buys SB2's Arctic benefit at four times the tropical cost and five times outside the ±0.3 TOA guardrail, and it damages the Southern Ocean where SB2 mildly helped. **The exchange rate is the durable result**: constant across a factor-two `RSBLB` dose change, so the tropical shortwave cost is intrinsic to enhancing stable mixing in this scheme rather than an artefact of which knob is turned. `RSBLB` is the efficient frontier.

49.3 Coupled: 11Q and 11R

11Q = 11N + `RSBLB 2.0` (1850) and 11R = 11P + `RSBLB 2.0` (1990), 40 yr, paired against their controls. No re-run controls were needed here, unlike AMIP: the coupled source has no commits since the 08–12 build the controls used, all four pre-existing working-tree modifications predate it, and the patch is bit-identical with `&NAMVDFS` absent.

1360–1389	**11Q** − 11N	**11R** − 11P	AMIP SB2
DJF inversion 60–90N	−0.680 *	−0.809 *	−0.713 *
DJF T2m 60–90N	**+0.614** *	**+0.674** *	+0.285
DJF T2m all land	+0.566 *	+0.535 *	+0.233 *
DJF T925 60–90N	−0.114	−0.020	−0.428 *
ANN T2m all land	+0.516 *	+0.494 *	—
global net TOA	**−0.001**	**−0.027**	+0.175 *
tropics SW CRE	+0.486 *	+0.540 *	+0.541 *
tropics net TOA	−0.059	−0.048	+0.247 *
SO SW CRE	−0.367 *	−0.713 *	−0.182
Siberia JJA	+0.767 *	+0.045	+0.000

All three questions AMIP could not settle come out favourably.

The Arctic screen follows, 2.2–2.4× the AMIP response and significant in both arms. That is the sea-ice amplification the pair was run to test: AMIP prescribes ice and suppresses it, which is also why 45 % of the coupled Arctic inversion bias has no AMIP counterpart (§48.5).

The cooling aloft disappears. T925 and T850 are null in both arms. Coupled, the inversion weakens by warming the screen rather than by chilling the layer above — the mechanism, not merely the metric.

The TOA cost vanishes. Tropical SW CRE is unchanged at +0.5, so the cloud is still lost, but tropical *net* is null and global net TOA is zero to 0.03 W m^{-2}: the ocean warms 0.13–0.15 K and the longwave response cancels the shortwave gain. The cost was paid as warming rather than as imbalance, and in a model this cold that is where it was wanted.

49.4 `cmpitool`: the best-scoring run in the campaign

run	CMPI
11E base	0.8237
11G	0.8446
11N LX4 1850	0.8198
11P LX4 1990	0.7709
11R +RSBLB2 1990	**0.7577**

Against 11P, the largest gains are `tas` −0.109 and `zg` −0.088 — *precisely the two fields where 11G regressed against the base* (+0.160, +0.149, §44.4). Sea ice also improves (−0.024, with Arctic SON 0.891 → 0.764).

A prediction of this report, falsified by its own test. §49.1 argued the tropical low-cloud loss would land on `nino34`, where 11G had already regressed. It does not: `nino34` is the *most improved* region at −0.030 (`tas` −0.258, `zg` −0.124, `zos` −0.094), and the tropics as a whole come out neutral at −0.0002. The cost is real but lands elsewhere — on `thetao`, +0.018 globally and +0.051 in the tropics, i.e. the ocean absorbing the shortwave. That is a spin-up concern rather than a present-day skill one, and it belongs with the standing imbalance rather than with this lever.

49.5 Base → best → this round, at 1850

All three arms at 1850 forcing, so the differences are clean even though the absolute biases against a present-day ERA5 carry a forcing offset of order a kelvin that is *not* model error.

1360–1389	11E base	11N LX4	**11Q +RSBLB2**	11Q−11N	11Q−11E
DJF T2m 60–90N	−6.149	−5.528	**−4.913**	+0.614	+1.236
DJF T2m 30–60N	−3.854	−3.757	**−2.875**	+0.882	+0.979
DJF T2m all land	−3.351	−3.213	**−2.647**	+0.566	+0.704
ANN T2m all land	−2.509	−2.370	**−1.854**	+0.516	+0.655
ANN T2m ocean	−1.425	−1.420	−1.266	+0.154	+0.159
JJA T2m Siberia	−2.532	−2.299	**−1.532**	+0.767	+1.000
DJF inversion 60–90N	+3.271	+3.109	**+2.428**	−0.681	−0.843
net TOA	+0.778	+0.878	+0.877	−0.001	+0.099
SW CRE SO 45–65S	−62.285	−65.462	−65.828	−0.367	−3.543
NH ice Mar [10^6 km^2]	14.142	13.862	13.697	−0.165	−0.445

11Q gains more over 11N than 11N gained over the base, on every land metric, and the annual land bias falls 26 % from base. The result that was not predicted is **Siberian JJA improving by** 1.000 **K over base**,

0.767 of it from `RSBLB` itself: this campaign's history is levers that buy winter by spending summer or spend energy to buy Siberia, and this is the first time both targets move together. It is free in energy terms, net TOA changing by $0.001\,\mathrm{W\,m^{-2}}$.

What it does not fix. Net TOA is +0.88 in all three arms. That is the campaign's standing imbalance, far outside the ±0.3 target and the other half of this report's title, and `RSBLB` neither helps nor hurts it.

49.6 Two caveats to carry forward

- **The winter Arctic gain may not be stable.** Restricted to 1380–89 the DJF Arctic screen reads +0.104 (11Q) and +0.639 (11R), neither significant — but n=10 gives a 0.685 K threshold, so that window cannot resolve a 0.6 K signal either way. What is informative is the fitted trend across the 30 years: −0.456 K/decade for 11Q and −0.210 for 11R. The *annual* land and global warming does not decay (+0.409 */+0.402 * all land in the last decade alone).
- **The ocean is absorbing the shortwave.** `thetao` degrades in `cmpitool`, global T2m rises 0.26–0.28 K, and net TOA remains +0.88. Ocean heat content should be priced before this state is committed to a long PI control.

49.7 One infrastructure note

11Q lost its 1380–89 leg once to LPJ-GUESS reporting `failed to find element to deserialize`. The state was *not* corrupt: all 65 files re-read cleanly at 446 MB/s afterwards, the failing `32.state` is 55.14 MB against 55.04 at 1370, and 11R passed the identical leg minutes later. It was a transient read, consistent with the DKRZ Lustre/Infiniband incident open since 2026–08–17. The leg was re-run and the failed attempt preserved at `run_13800101-13891231_dead_lustre_20260822`. Recorded because the obvious misreading — a genuine LPJ-GUESS restart bug — would have argued for switching `guess` binaries mid-experiment and destroyed the pairing against 11N.

50 Round 33 (2026–08–23): the Southern Ocean has two problems, not one

Round 32 left the model with a −0.74 K global cold bias and a $+0.88\,\mathrm{W\,m^{-2}}$ imbalance, i.e. still warming toward equilibrium. The question this round answers is where that warming lands, and it turns out the Southern Ocean is not a passive recipient of a global imbalance — it is where the imbalance is generated.

50.1 The drift is in every arm, including the 1850 controls

Drift from 1350–59 to 1380–89, SH ocean south of 60S:

arm	forcing	ΔT2m	ΔSep ice [10^6 km^2]
11N	1850	+0.70	−2.21
11Q	1850	+0.81	−1.94
11P	1990	+0.90	−2.61
11R	1990	+0.97	−2.50

The 1850 arms drift almost as hard as the 1990 arms, so this is **not** the response to present-day forcing — that adds only about +0.2 K and -0.5×10^6 km^2 on top. The PI controls are warming at roughly +0.8 K and shedding 2×10^6 km^2 of September ice per 40 years.

Two consequences. First, at 1350–59 the T2m bias south of 60S was −0.06 to −0.65 K, i.e. neutral to slightly cold: **the air was not too warm at the start**, so the ocean cannot have been warmed by sitting under warm air. Both warm together. Second, September ice goes 20 → 17.6 against 16.27 observed, so the model is passing *through* the right answer on its way past it — anything tuned to match the ice now is tuned to a transient.

50.2 The imbalance is generated in the Southern Ocean

Net TOA bias against CERES, 11R, with each band's share of the global mean:

band	area %	net bias	contribution	SW CRE bias
90–60S	6.7	**+5.76**	**+0.383**	+3.65
60–45S	7.9	**+4.46**	**+0.353**	−0.53
45–30S	10.3	−3.59	−0.370	−6.82
30S–30N	49.7	−1.75	−0.872	−3.35
30–45N	10.3	+2.41	+0.248	+3.40
45–60N	7.9	+3.75	+0.296	+3.47
60–90N	6.7	+2.05	+0.137	−5.83

Superseded by §50.7, which redoes this over ocean. The two Southern Ocean bands are the largest positive contributors, $+0.74\,\mathrm{W\,m^{-2}}$ combined, against a tropical −0.87. The global figure is a small residual of large regional cancellations, so "close the TOA imbalance and the SO follows" is circular: the SO ice loss absorbs the shortwave that *is* the imbalance.

50.3 What the campaign has actually done to those bands

SW CRE bias against CERES; positive means the model cloud reflects too little.

arm	90–60S ANN	90–60S DJF	60–45S ANN	60–45S DJF
11E base	+6.25	+13.63	+4.67	+4.84
11G +S4	+5.99	+12.09	+2.61	+2.39
11I v2soil	+6.33	+13.49	+2.61	+1.55
11N LX4	+4.39	+8.91	+1.25	+0.34
11P LX4 1990	+4.37	+9.05	+0.08	−2.20
11Q +RSBLB	+4.70	+9.85	+0.35	−1.46
11R +RSBLB 1990	**+3.65**	**+6.98**	**−0.53**	**−2.97**

The polar band has been more than halved, +13.63 → +6.98 in DJF, so it is *not* an untouched region. What the table actually shows is a **meridional gradient in lever effectiveness**: the same levers drove 60–45S from +4.84 through zero to −2.97, overcorrected, while 90–60S still reads +6.98. Pushing any of them harder buys the last of the polar band at the cost of driving the sub-Antarctic further negative — and 60–45S is inside the campaign's own 45–65S SO metric, which is part of why this looked solved.

Only 11P→11R is a clean like-for-like step here; the 1850 arms carry a forcing mismatch against a present-day CERES. The 11N→11Q reading of +0.31 has the wrong sign against the −0.367 * measured for 45–65S over 30 years, and is a 10-year window on a noisy band.

50.4 Amount versus opacity: the two bands need opposite corrections

Superseded by §50.7: over ocean the split is half amount and half opacity, not predominantly opacity.

DJF, SW CRE per unit cloud cover, which isolates opacity from amount:

band		CRE	cover	CRE / cover
90–60S	CERES	−65.99	0.748	−88.2
90–60S	11R	−59.01	0.726	**−81.3**
60–45S	CERES	−121.01	0.879	−137.7
60–45S	11R	−123.98	0.827	**−149.9**

At **60–45S** the model has 5 pp too *few* clouds, each 9 % too *bright*; they compensate, which is how the net CRE ended up slightly overcorrected. At **90–60S** the amount is nearly right (−2.2 pp) but each cloud is 8 % too *dim*. Splitting the $+6.98\,\mathrm{W\,m^{-2}}$: about 5.2 is opacity and 2.0 amount. So the polar residual is an **opacity** problem — the opposite of the 45–65S diagnosis that closed the SO cloud question in round 30.

That also explains the gradient above: every campaign lever works by raising opacity, and at 45–60S opacity was already too high.

The phase measurement agrees, and it flips sign at 60S. Low-level liquid fraction against CALIPSO-GOCCP: 45–65S model 0.873 vs GOCCP 0.858 (too liquid, +0.015); **60–90S model** 0.731 **vs** 0.759 **(too icy,** −0.028); Siberia 0.813 vs 0.818. Icier cloud precipitates out and carries less liquid water, which is how a cloud can be present and optically thin. **The INP family was retired on a phase measurement taken in the one band where the model is too liquid, and that conclusion does not extend poleward.**

50.5 The sea ice itself

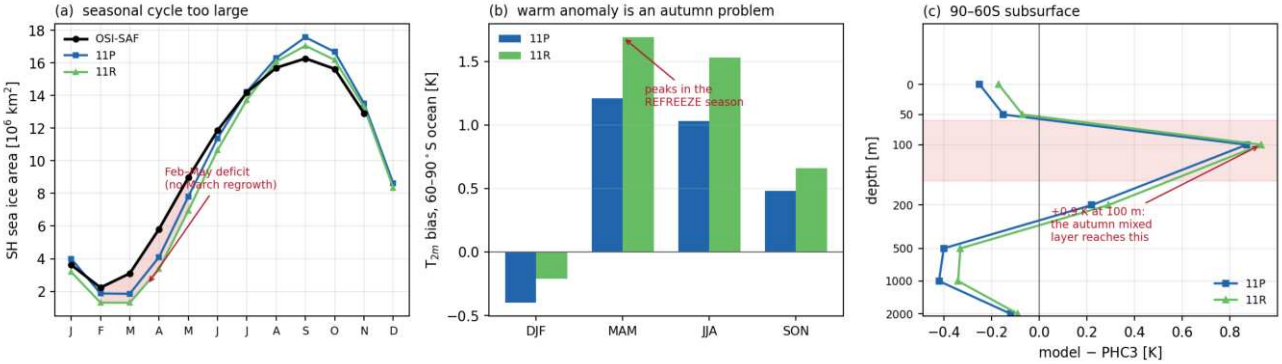


Figure 37: Two Southern Ocean problems in two different seasons. **(a)** The SH ice seasonal cycle is too large: September is +0.78 (10^6 km^2 area, +2.01 extent) against OSI-SAF, while February–May is far too low and the March regrowth is absent entirely (model 1.30 against 3.10 observed). **(b)** The T2m warm anomaly over the ice zone peaks in *MAM*, the refreeze season, which no shortwave mechanism explains. **(c)** A +0.9 K anomaly at exactly 100 m, directly under a neutral surface layer and above *cold* water at 500–1000 m — the depth the autumn mixed layer reaches.

The ice is **not too dark**: September albedo over compact ice is 0.760 at $ci \geq 0.9$ and 0.778 at $ci \geq 0.95$, with snow albedo 0.850, inside the observed range. But in January–March there is **no compact ice anywhere** — every SH cell is below 90 % concentration — so the area-mean albedo the sun sees falls to 0.49–0.51. The pack is also too diffuse: mean concentration inside the ice edge is 0.767 in September against OSI-SAF's 0.804, spread over 2×10^6 km^2 more extent. Diffuse ice has less volume per unit area to lose and melts out rather than retreating to a compact core.

50.6 The subsurface, measured at last

Temperature against PHC3, area-weighted zonal means, 90–60S:

depth [m]	PHC3	11P	11R	11N	11Q
0	−0.52	−0.25	−0.17	−0.17	−0.25
50	−0.69	−0.15	−0.07	−0.07	−0.15
100	−0.78	**+0.87**	**+0.93**	**+0.76**	**+0.79**
200	+0.50	+0.22	+0.29	+0.09	+0.19
500	+1.14	−0.40	−0.33	−0.54	−0.35
1000	+1.03	−0.42	−0.34	−0.52	−0.37

A sharp +0.9 K anomaly at 100 m, beneath a near-neutral surface and *above* cold water at depth. Not a deep warm drift — a shallow one, at the depth the winter mixed layer reaches. It is present in all four arms, so it predates `RSBLB`, which worsens it by only 0.03–0.06 K.

This closes the melt-from-above versus melt-from-below question: it is both, in different seasons. In DJF, +6.98 W m^{-2} of cloud shortwave melts from above. In MAM, autumn cooling deepens the mixed layer into the 100 m warm layer and entrains it, which is why the MAM anomaly is +4.48 W m^{-2} of *longwave* with almost no shortwave, why the surface cannot cool, and why March ice never regrows. Each mechanism explains the season the other cannot.

A cloud-opacity lever therefore addresses the summer half only. Even a perfect fix to the +6.98 leaves the autumn refreeze failure intact, because that one is fed from 100 m down.

50.7 Corrected: recomputed over ocean, and the numbers above understate it

Everything in §50.2–§50.4 was area-averaged over whole latitude bands. That folds the **Antarctic ice sheet** into the 90–60S band. The ice sheet is brilliant white and radiatively unlike the sea-ice zone, and it cancels part of the signal: over the whole cap the surface net SW bias reads −0.57 W m^{-2}, over ocean alone +7.99. The sea-ice question is about the ocean, so the band averages are the wrong domain and the committed numbers are too small.

Two further points of method, both checked rather than assumed. A model's `tsrc` is *total-region* clear sky, so the comparable CERES field is `toa_sw_clr_t`, not `clr_c` which is clear sky over cloud-free scenes only; the choice moves the polar numbers by 1.7 W m^{-2}, not enough to change any conclusion here. And the surface flux is now compared alongside the TOA, because agreement at two semi-independent levels is what distinguishes a real radiative error from a cloud-mask artefact.

SW CRE bias against CERES `clr_t`, OCEAN ONLY:

arm	90–60S DJF	90–60S ANN	60–45S DJF	60–45S ANN
11E base	+24.24	+11.46	+4.20	+4.31
11G +S4	+21.66	+11.01	+1.75	+2.21
11N LX4	+16.50	+8.44	−0.36	+0.85
11P LX4 1990	+16.91	+8.48	−2.91	−0.29
11Q +RSBLB	+18.07	+8.96	−2.17	−0.04
11R +RSBLB 1990	**+13.36**	**+7.26**	**−3.70**	**−0.93**

The polar ocean deficit is **+13.36** W m^{-2} in DJF, not the +6.98 reported above — close to double. The campaign's progress is also larger than it looked, 24.24 → 13.36, and the sub-Antarctic overcorrection is deeper, −3.70 rather than −2.97. The qualitative conclusion — a meridional gradient in lever effectiveness, with 60–45S driven through zero while 90–60S still leaks — is unchanged and in fact sharper.

The surface confirms it, which §50.4 could not show:

ocean band, summer	CRE	cover	CRE/cover	sfc net SW	CERES	bias
60–90S DJF, model	−92.62	0.858	−108.0	139.4	131.4	**+7.99**
60–90S DJF, CERES	−105.98	0.916	−115.7			
50–70S DJF, model	−116.34	0.851	−136.7	163.0	163.4	−0.40
50–70S DJF, CERES	−120.75	0.912	−132.5			
60–90N JJA, model	−90.00	0.884	−101.8	114.2	127.6	**−13.34**
60–90N JJA, CERES	−71.78	0.826	−86.9			

The Southern Ocean sea-ice zone absorbs +7.99 W m^{-2} too much shortwave at the surface in summer. That is the direct measurement the melt mechanism needed and did not have: §50.6 inferred the excess from the ice and the 100 m heat, and it is now confirmed at the surface. Note it is confined to the ice zone — 50–70S reads −0.40, i.e. nothing.

But the amount/opacity split changes. Over ocean the polar deficit is roughly *half amount and half opacity*: cover is 5.8 pp too low (0.858 against 0.916) and each cloud is 6.7 % too dim (−108.0 against −115.7), each accounting for about 6.5 of the 13.36. The band-averaged version reported ≈ 72 % opacity, and that was an artefact of including the ice sheet. A pure opacity lever therefore addresses at most half of this, which is consistent with SC2/SC3 underperforming.

The Arctic, and why it is not yet a target. Over ocean in JJA the model has 5.8 pp *more* cloud than CERES and each is 17 % brighter, giving TOA CRE −18.22 and a surface shortwave deficit of −13.34 W m^{-2}. TOA and surface agree, so the radiative error is real rather than a cloud-mask artefact. *But the temperature does not corroborate it*: 60–90N ocean T2m is +0.84 K **warm** in JJA. A 13 W m^{-2} surface shortwave deficit that leaves the surface warm means something else is compensating, and until that is explained the Arctic cloud error should not be treated as a lever target — even though it is three times the Antarctic one and points toward the campaign's founding cold bias.

50.8 Method notes from this round

- **reval's PHC3 ocean plots are empty for every run in this campaign** — grey maps, `rmsd=nan`, `bias=nan`, byte-identical 153 687-byte files for 11E, 11G, 11N, 11P and 11R. The ocean interior had never been validated. The comparison above was done by hand because of it.
- PHC3 as staged is itself in FESOM format but on a *different* mesh (126 858 nodes against CORE3's 211 567), so it cannot be differenced node-by-node. Both share an identical 47-level vertical grid, so binning onto a common (latitude, depth) grid avoids interpolation entirely. Node coordinates for the PHC3 mesh are in its `fesom.mesh.diag.nc` as `nodes` (2, nod2) in *radians*; the CORE3 diag instead carries explicit `lat/lon` in degrees.
- Sea ice AREA and EXTENT are different numbers and mixing them inverts the sign of the answer. September: area +0.78 but extent +2.01.

- **SST is the wrong variable for this question.** Under sea ice it is pinned near freezing and reads −0.9 to +0.0 K in the ice zone, which says nothing. T2m is the free variable and reads +1.5 to +1.9 K at 70–80S. An early reading of this round used SST and concluded the Southern Ocean was too cold and that the ice-albedo feedback would work in the model's favour as it warms. Both were wrong.

51 Round 34 (2026–09–06→09): four defects in the coupled ice skin

A separate thread from the radiative campaign, recorded here because it changes what the coupled arms mean. The coupled-slab ice surface temperature diverged to −36,470 K and overflowed single-precision OpenIFS at `ece_updclie_cpl.F90`:104 after ∼43 model days. Four independent defects were found; the first is the one that mattered.

51.1 The anchor that was never delivered

`oasis_put` returns `OASIS_Sent` (4) only to the call that completes a coupling group. Every earlier field in the group gets `OASIS_Waitgroup` (14) — the data is accepted and buffered, and transmission happens when the last field puts. FESOM's `cpl_driver.F90` tested only for 4 and 8, read 14 as "not sent", and never wrote the received field.

The consequence was silent and specific. `ist_ref`, the reference skin temperature the implicit solve is expanded about, was 0.0 in all 63,923 logged events. `tref` then fell back to the model's own previous `t`, at which point the ζ_{lam} terms cancel identically:

$$t = \frac{c_p \Delta t^{-1} t + a_{2i} h f + \zeta_{\mathrm{lam}} t_{\mathrm{ref}} + \zeta_{ice}}{c_p \Delta t^{-1} + \kappa/h + \zeta_{\mathrm{lam}}} \xrightarrow{t_{\mathrm{ref}}=t} t^* = T_{\mathrm{frz}} + \frac{a_{2i} h f}{\kappa/h},$$

an unbounded steady state with no restoring term at all. The fix is one line — `action=(info==OASIS_Sent .OR. info==OASIS_SentOut .OR. info==OASIS_Waitgroup)`.

51.2 The other three, and what a year of running says

- Recovering the advected ice temperature divided by `max(a_ice, 1e-6)`, which injects ≈0 K wherever concentration collapses rather than leaving the node alone.
- The fallback anchor had no sanity bound; it now clamps outside 173.15–400 K to 271.35 K.
- The skin is now taken from OpenIFS (`A_Ice_tskin`, new coupling field) instead of FESOM solving a second, independent one. Two models solving the same skin and exchanging only part of the state is the structural version of the same bug.

Verified over a full coupled year: `ist` minimum −36,470 K → 224.39 K, p1 233.57 K, **zero** points below 200 K, no overflow.

51.3 GSMART is not smart enough for a field that changes sign

`A_Q_ice` carried `CONSERV` method `GSMART`, which is `GLBPOS` (multiplicative, $\Sigma_{src}/\Sigma_{dst}$) with a fallback used only when the denominator is *exactly* zero — a condition floating point never meets. The field changes sign between the hemispheres, so the denominator passes through zero and the factor blows up near it. Measured over a coupled year, GSMART delivered **9105 excursions above 1 kW m^{-2} onto real ice, maximum** 22,550 **W m^{-2}**; `GLOBAL` and the new `GSSMAR` deliver **none**. `GSSMAR` (OASIS `ip_cgssmar`, method 9) scores the additive, whole-field multiplicative and per-sign multiplicative corrections at each step and applies whichever distorts the field least, with no extra MPI reductions. Note the keyword is six characters because the `namcouple` parser tokenises on that width; a seven-character name shifts the line and aborts with a garbage model name.

51.4 A restart that was never collected

11Y ran its first ten years cleanly and then lost legs 2 and 3. FESOM 2.8 renamed the NetCDF restarts — one directory per restart *instant*, stamped with its date (`fesom.1360-01-01.oce.restart`), where pre-2.8 wrote one directory per year named by that year (`fesom.1359.oce.restart`). The reader still falls back to the old name, so a leg branched off a pre-2.8 restart works and the change is invisible for exactly one leg; the writer only ever

emits the new name. esm-tools' unchanged patterns therefore collected nothing, `restart/fesom` stayed empty, and the next leg died reading a file that had never been staged — while the date file advanced, so the wreckage looked like progress. Fixed version-bound in the awiesm3 `v3.5` arm.

52 Round 35 (2026–09–09): the heat does not go down in the tropics

The campaign's standing account of the imbalance is that the model removes heat from the surface faster than it should, so the surface never warms enough to radiate the excess away. That account had never been located: "too much mixing" was an inference from a cold surface and a warm interior, not a measurement. This round measures it, on 11X — the 40-year 1850 control on the corrected CORE3 mesh (220509 nodes). The conclusion is that the mechanism is real and the region was wrong.

52.1 Definitions first, because they are not the same quantity

"Mixed layer depth" names four different things here, and a model-minus-obs map is worthless if the two numbers are not the same one.

source	criterion	reference
WOA18 `M_an`	σ_θ exceeds reference by $0.125\,\mathrm{kg\,m^{-3}}$	**10 m**
FESOM `MLD2`	the same $0.125\,\mathrm{kg\,m^{-3}}$	**top level, 2.5 m**
FESOM `MLD3`	CMOR $0.03\,\mathrm{kg\,m^{-3}}$	top level
FESOM `MLD1`	Large et al. (1997) buoyancy gradient	—

WOA's criterion is stated verbatim in the file's `summary` attribute rather than inferred; FESOM's is `oce_ale_pressure_bv.F90`
with the threshold set at line 219. Comparing `MLD2` against `M_an` compares a surface-referenced depth with a 10 m-referenced one, and where a warm or fresh skin sits above 10 m the surface-referenced version trips its threshold earlier and reads *shallower* — it flatters the model on precisely the hypothesis under test. So the model MLD is recomputed offline from the model's own monthly T and S under WOA's criterion, with EOS-80 σ_θ (the equation of state WOA is built on).

That recompute was validated before it was used. Reproducing FESOM's *own* convention from the same fields gives $r = 0.99763$ against the written `MLD2`, a global area-weighted 91.08 against 93.33 m, and a median per-node difference of 1.42 m — the residual being FESOM's Jackett–McDougall EOS against EOS-80. The definitional effect itself is small: switching the reference from 2.5 m to 10 m deepens the global mean by +1.9 m, at most +3.6 m in any band. **So the definition matters at the few-metre level and cannot explain anything below.** That is a measurement, where before it was a worry.

52.2 The mixed layer: right in the tropics, catastrophic in the Southern Ocean

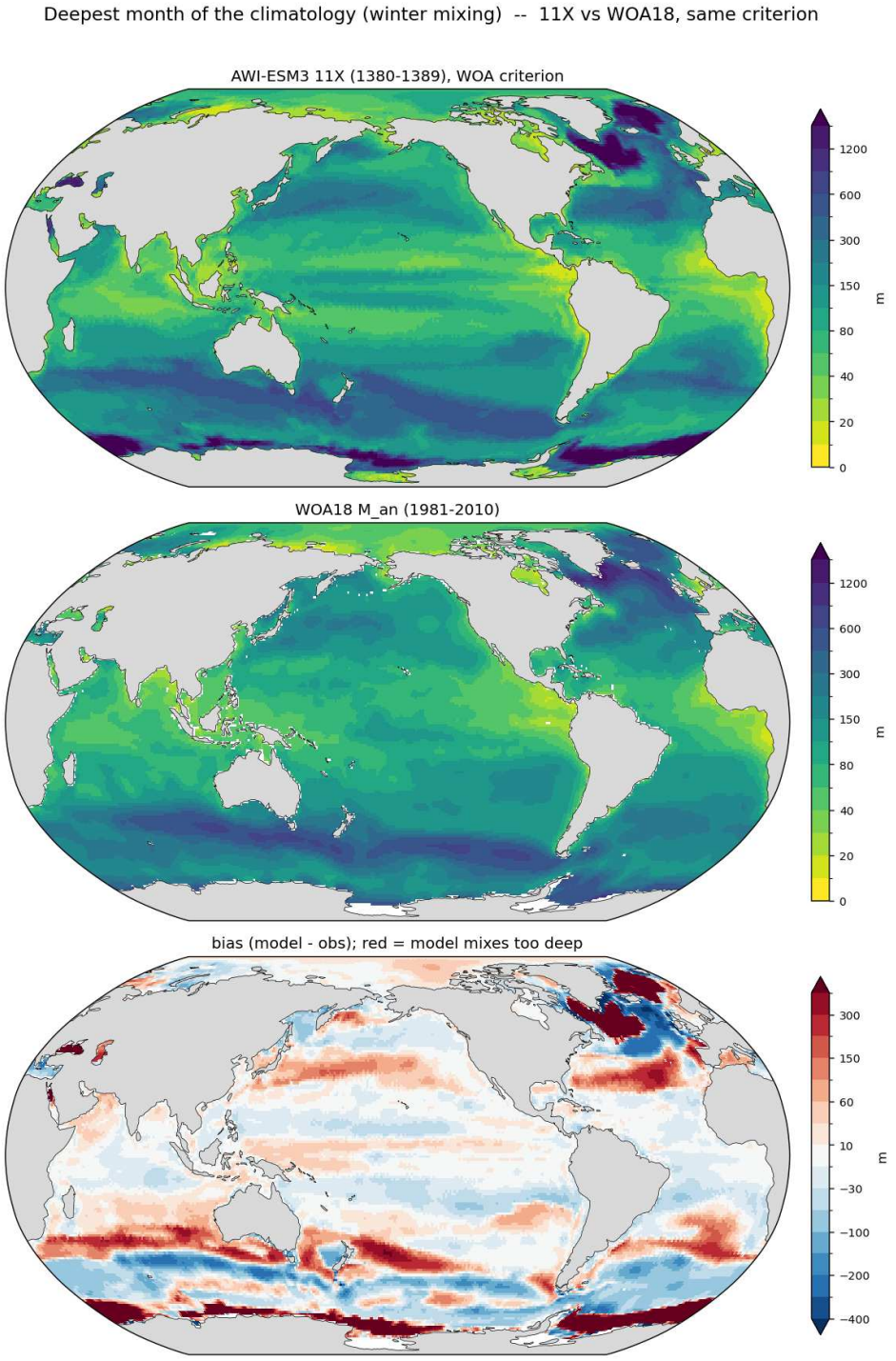


Figure 38: Deepest month of the climatology, 11X (1380–89) against WOA18, **both on WOA's** $0.125\,\mathrm{kg\,m^{-3}}$ **/ 10 m criterion**. The model's over-deep mixing is confined to the Southern Ocean, the subpolar North Atlantic and the western boundary currents; the tropical band is unbiased. Red is the model mixing too deep.

band	ann model	ann obs	**ann bias**	max model	max obs	**max bias**
90–60S	371.4	123.2	**+248.2**	840.1	232.7	**+607.4**
60–45S	140.0	169.8	−29.8	264.1	301.2	−37.1
45–30S	105.0	97.6	+7.5	251.1	202.2	+48.9
30S–30N	43.6	51.6	**−8.1**	81.0	80.8	**+0.3**
30–45N	78.5	70.8	+7.6	206.4	157.5	+48.9
45–60N	141.5	104.3	+37.2	389.8	234.0	+155.7
60–90N	118.9	80.2	+38.7	272.5	176.9	+95.5

The tropics are the one band that comes out clean, and the sign is wrong for the mixing story: 8.1 m too *shallow* in the annual mean, and a dead heat at the seasonal maximum (81.0 against 80.8). The seasonal cycle (Fig. 39) shows the model shallower than WOA in all twelve months, not only in the annual mean. The convention also works against the model here: MLD computed from a monthly-*mean* profile is biased shallow relative to WOA's per-profile-then-analyse, so correcting it moves the tropical model toward the observations, not away.

Where the model does over-mix is entirely extratropical and entirely winter, and 90–60S is not a mixing-coefficient error at all: **18.7 % of model month-nodes there mix to the sea floor** (60–90N 12.5 %, 45–60N 7.9 %, tropics 1.4 %). That is open-ocean deep convection, and no diffusivity knob addresses it.

Two caveats, stated rather than buried. WOA's Southern Ocean coverage is 13.9 % of analysed cells annually and 6.6 % **in September**, so the +607 leans on a sparsely constrained analysis — though the "reaches the bottom" statistic is a model-side number needing no observations. The Arctic is 18.7 %; every other band is 50–73 %. WOA18 is also 1981–2010 against an 1850 control.

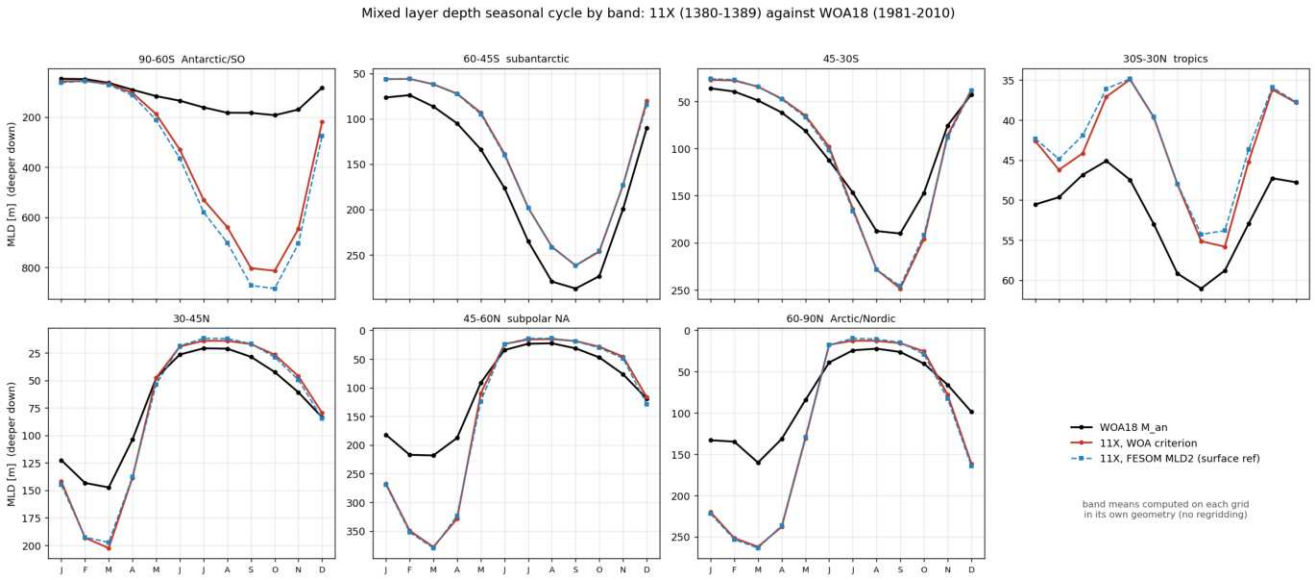


Figure 39: Seasonal cycle by band. The Southern Ocean panel is the failure: model and observations agree from January to March, then the model runs away to 840 m in September–October while WOA stays near 210 m. The tropical panel shows the model shallower than WOA in every month. Band means are computed on each grid in its own geometry, so no regridding enters.

52.3 Where the heat actually lands

The mixed layer is not heat uptake, so the next measurement is the heat itself: the drift of 11X's own temperature between its first and last decade (1350–59 to 1380–89), which is unambiguously what *this run* put there rather than what it inherited.

The ocean gains +1.502 **W m^{-2} per unit ocean area** (≈ +1.07 globally), which is the right size to be the TOA imbalance — so this is the missing heat, not a residual. Its depth distribution is the finding:

band (share of global budget)	0–100 m	100–300 m	300–700 m	700–2000 m	>2000 m	total
90–60S	+0.001	−0.004	−0.020	−0.063	+0.009	**−0.076**
60–45S	+0.016	+0.022	+0.060	+0.007	+0.038	+0.144
45–30S	+0.005	−0.018	+0.157	+0.245	+0.072	+0.462
30S–30N	−0.043	−0.217	+0.373	+0.411	+0.082	**+0.607**
30–45N	+0.013	+0.035	+0.084	+0.029	+0.031	+0.192
45–60N	+0.015	+0.005	+0.014	+0.039	+0.024	+0.098
60–90N	+0.006	+0.006	+0.010	+0.040	+0.013	+0.075
GLOBAL	**+0.013**	**−0.170**	**+0.678**	**+0.710**	+0.270	**+1.502**

The top 100 m gains essentially nothing and the 100–300 m layer cools. Almost all of it lands between 300 and 2000 m, peaking at +0.50 K near 580–680 m with the zero crossing at ∼340 m. And 90–60S is the one band with a *net heat loss*, cooling at every depth from 70 m to 2400 m and peaking −0.26 K near 1000 m — the

runaway convection ventilates the abyss with cold surface water. **The Southern Ocean releases heat; it does not bury it.**

52.4 Heave versus water mass: the discriminator

A cooling-above / warming-below dipole at fixed depth has two possible causes, and they call for opposite fixes: heat crossing density surfaces (diapycnal mixing or a changed source water), or the surfaces themselves moving (circulation). Temperature on isopycnals separates them.

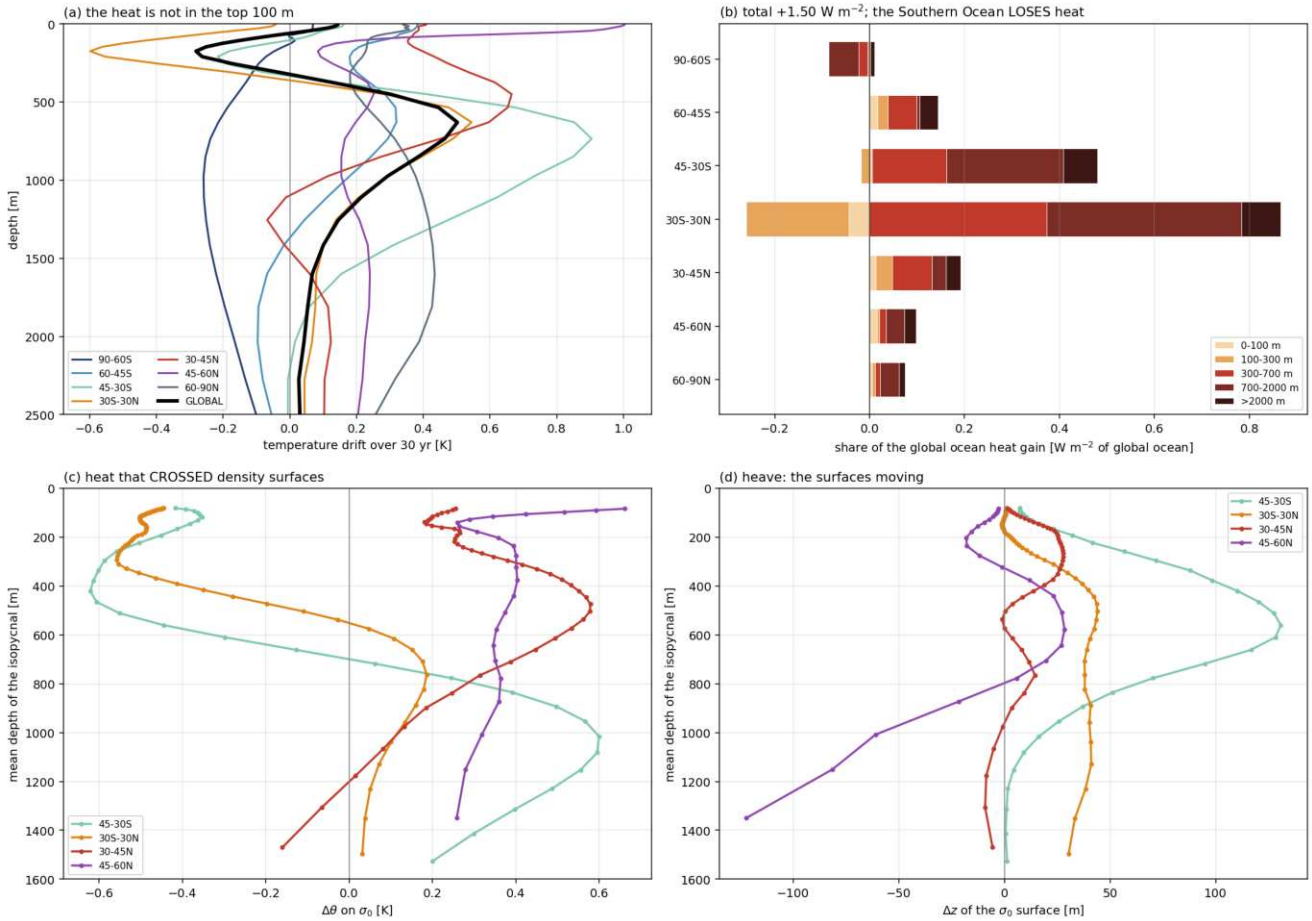


Figure 40: (a) The heat is not in the top 100 m: the global drift is negative from ∼60 to ∼340 m and peaks +0.50 K at 580–680 m. (b) The tropics take the largest share of the global heat gain and the Southern Ocean is the only band that *loses* heat. (c) Heat genuinely crossing density surfaces is a mid- and high-latitude phenomenon (30–45N +0.58 K at 500 m, 45–30S +0.57 K at 950 m); the tropical upper thermocline goes *colder* on isopycnals. (d) The tropical deep warming is instead heave — the $\sigma_0 = 27.0$ surface sinks +42 m. Only isopycnals present in $\geq 25\,\%$ of a band's nodes are drawn; sparser ones give hundreds of metres of spurious heave.

In the tropics at $\sigma_0 = 27.0$ (mean depth 575 m) the water on that surface warmed by +0.047 **K** while the surface itself sank +42.4 **m**. The fixed-depth warming there is +0.477 K, and 42 m against the local $\sim 0.01\,\mathrm{K\,m^{-1}}$ gradient accounts for 0.42 of it. **Roughly 90 % of the tropical deep warming is heave.** Above it, the tropical upper thermocline went *colder* on isopycnals, −0.45 to −0.56 K from 83 to 350 m, with salinity down −0.153 psu over 0–100 m — colder and fresher at fixed density is a water-mass change running the opposite way from heat burial.

Heat that genuinely crossed density surfaces is mid- and high-latitude: 30–45N +0.53 to +0.58 K at 400–500 m, 45–60N +0.35 to +0.52 K over 100–700 m, 45–30S +0.57 K at 950 m, 60–90N +0.43 to +0.72 K at 280–750 m. **Those are the winter outcrop regions, and they are exactly where the mixed layer is too deep** — 30–45N +48.9 m, 45–30S +48.9 m, 45–60N +155.7 m at the seasonal maximum.

52.5 What this rules out, and what it does not

- **Ruled out: a tropical interior mixing lever.** The tropical mixed layer is not too deep, and 90 % of the tropical deep warming is heave rather than diapycnal flux. A diffusivity knob there has almost nothing to act on.

- **Ruled out: the Southern Ocean as the burial route.** It loses heat at every depth. Its convection is a separate and larger pathology, and it works against the interior warming rather than causing it.
- **Not established: causality at the outcrops.** The subduction warming sits in the same regions as the too-deep winter mixed layer. That is a coincidence of location, not a demonstration; only a lever test settles it.
- **Method limits.** σ_0 is a poor coordinate below ∼2000 m, so the deepest rows are weak. Heave is a local decomposition and can be driven remotely. And band drift cannot separate local vertical transfer from lateral import — which is precisely why the isopycnal split was needed.

52.6 The free checks, and the two hypotheses they killed

Before pricing any lever, four checks that cost nothing and could each have invalidated the whole framing. Two of them did.

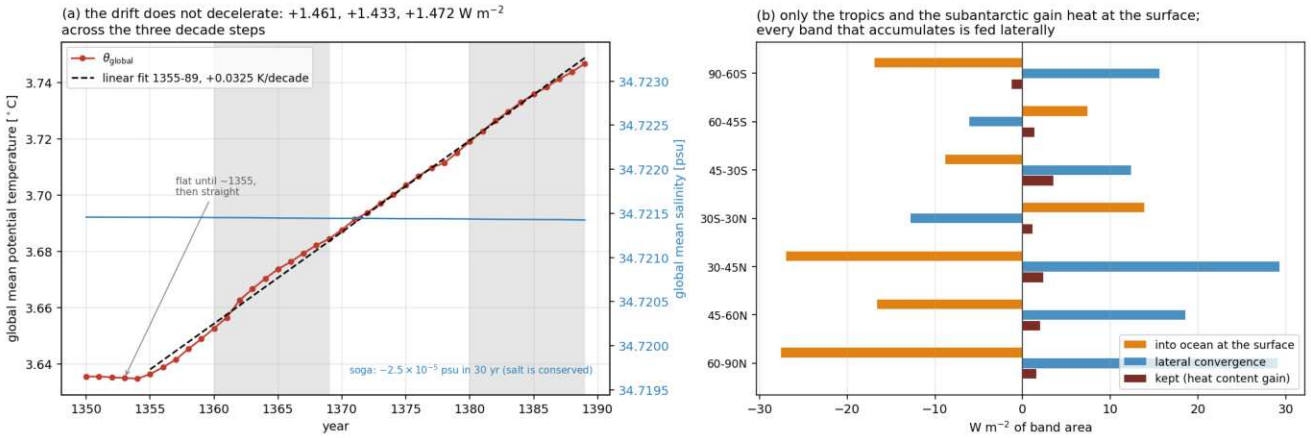


Figure 41: (a) The drift is flat for about five years after the branch and then perfectly straight: the three decade-to-decade steps are +1.461, +1.433 and +1.472 W m^{-2}, so this is a standing bias and not an equilibrating transient. Global mean salinity (blue, right axis) drifts -2.5×10^{-5} psu in 30 years — salt is conserved, which kills the virtual-salt-flux hypothesis. (b) Only the tropics and the subantarctic take heat in through the surface; every band that accumulates strongly (45–30S, 30–45N, 45–60N) *loses* heat at the surface and is fed by lateral convergence.

Check 1: the drift does not decelerate. `thetaoga` over all forty years gives decade-to-decade steps of +1.461, +1.433, +1.472 W m^{-2} — flat to 3 %. There is a short adjustment over roughly the first five years after the branch and then a straight ramp. **So the imbalance is a standing bias, and tuning against it is legitimate** rather than tuning against a spin-up that would have equilibrated anyway.

Check 2: salt is conserved, and the virtual-salt-flux hypothesis is dead. This was the leading candidate: `which_ale = 'linfs'` applies freshwater as a virtual salt flux, and the surface salinity trends at the outcrops (+0.30 psu at 45–60N, +0.37 psu at 60–90N over 30 years) looked like the numerical signature of it. If salt were accumulating at the outcrops it would deepen the winter mixed layer there, and every mixing coefficient would be the wrong end of the problem.

It is not. `soga` drifts −0.000025 psu over the same 30 years, five orders below the regional trends; `balance_salt_water` closes the budget. And the sign kills it a second time: at 45–60N the virtual salt flux is *freshening* (-8.3×10^{-7} m s^{-1} psu, equivalent to −2.6 psu per decade if it stayed in the top 100 m) while the layer became 0.30 psu *saltier*. **The salinification is advective and convective — a consequence of the too-deep mixing, not its cause.**

Check 3: the winds did not drift. Zonal surface stress changes by at most 4.3 % in any band carrying meaningful stress (60–45S −2.4 %, 45–30S +4.3 %, tropics +2.2 %; the two larger percentages are on stresses of 0.01 N m^{-2} and are noise). So the 42 m of tropical thermocline heave in §52.4 is **adjustment to a standing bias, not a trend inside the run** — and with check 1 showing no deceleration, that adjustment is still running after forty years.

Check 4: the surface budget closes, and the accumulating bands are fed laterally. FESOM's `fh` is positive *upward*; its 1380–89 global mean is −1.535, against a storage rate of +1.455 W m^{-2} from `thetaoga`, so the budget closes to 0.08 W m^{-2} only with that convention. An earlier pass read it as positive-downward and produced lateral terms with the wrong sign throughout. With the sign right, `fh` and the CMOR pair `opottempdiff`/`opottemprmadvect` agree band by band to about 1 W m^{-2}, which is what licenses the table:

band	into ocean at surface	lateral convergence	kept
90–60S	−16.88	+15.62	−1.260
60–45S	**+7.42**	−6.09	+1.331
45–30S	−8.83	+12.36	**+3.524**
30S–30N	**+13.91**	−12.74	+1.164
30–45N	−26.98	+29.33	+2.356
45–60N	−16.58	+18.57	+1.994
60–90N	−27.54	+29.10	+1.559
GLOBAL	**+1.53**	—	+1.455

Only the tropics and the subantarctic are net surface heat gainers. Every band that accumulates the most — 45–30S, 30–45N, 45–60N — loses heat at the surface and imports more than it loses. So the burial is not the mid-latitude surface pushing heat down. It is heat entering at the tropical surface, being exported poleward, and being placed into the 300–2000 m layer at the mid-latitudes.

One limitation, and a fix for the next run. `opottempdiff` is written column-integrated, so it cannot separate interior dianeutral mixing from the surface boundary flux — integrated over a column the interior term nearly cancels and what remains is essentially the surface flux, which is why it reproduces `fh`. The 3D field would measure the interior diapycnal heat flux directly and settle §52.4 without the isopycnal inference. **Add `opottempdiff` and `opottemprmadvect` in 3D to the FESOM `io_list` for the next arm.**

What survives. The subduction account survives but its *source* moves: the heat enters in the tropics. That raises the question none of these checks asked — whether $+13.9\,\mathrm{W\,m^{-2}}$ of net tropical surface uptake is itself too much, which ERA5 and NCEP2 on disk can answer. And the structural candidate is untouched by any of this: **FESOM 2.7 carries no Fox–Kemper mixed-layer-eddy restratification** (nothing in `src/` matches), whose absence is the standard reason coarse models over-deepen winter mixed layers at exactly these outcrops.

52.7 Method notes from this round

- **Read the observational product's own definition out of the file.** WOA18's criterion is in the `summary` global attribute. Two products can share a threshold ($0.125\,\mathrm{kg\,m^{-3}}$) and still differ in the reference depth, which is the part nobody writes on the axis label.
- **Reproduce the model's own diagnostic before trusting a recompute of it.** $r = 0.99763$ against `MLD2` is what licenses the WOA-convention number; without that check a difference of the size found here would be indistinguishable from a coding error.
- **Do not mask an objectively analysed field on raw observation counts.** A first pass masked WOA `M_an` on `M_dd` ≥ 3 and deleted most of the ocean — `M_an` is complete over the ocean (41088 of 64800 cells) while `M_dd` is sparse. Coverage belongs in a caveat, not in a mask.
- **Compute nonlinear diagnostics on the native mesh, regrid afterwards.** MLD is a nonlinear functional of the profile; regridding T and S first and computing after gives a different quantity. Band means were additionally computed with no regridding at all as a cross-check.
- **Check a flux variable's sign convention against a budget, not against its name.** FESOM's `fh` is positive upward. Taking it as positive-downward inverted every lateral-transport term in §52.6; the error was caught only by requiring the global surface flux to match the `thetaoga` storage rate.
- **Require coverage on isopycnal averages.** Density surfaces that outcrop in a handful of nodes produced −1134 m of "heave" in 45–60N. A 25 % band-coverage floor removes it.

53 Round 36 (2026–09–10→12): mid-depth heat storage sets the TOA plateau

Two things changed underneath the coupled work before this round, and both are documented in `notes/ICE_SKIN_OPTIONS_2026-09-10_14series.md`. The single-precision FESOM lost about $0.5\,\mathrm{W\,m^{-2}}$ of ocean heat: the FCT low-order tendency was rebuilt as $LO \cdot h_{new} - T \cdot h$ from a float32 provisional solution, losing every per-step increment below half an ulp. It is fixed upstream (FESOM PR #1054). **15F** is the ice-physics configuration of 15C (OpenIFS column-heat flux `ECE_CPL_FESOM_ICE_QOUT`, ungated snow melt, 1 K albedo ramp, ECHAM6 ice albedos) on the fixed ocean. Its ocean heat budget closes at $-0.069\,\mathrm{W\,m^{-2}}$ over 10 years (-0.035 over 20), against -0.073 for the double-precision 13A and -0.544 for the unfixed 15C, and Southern Ocean convection returns ($2.39\,\mathrm{M\,km^2}$ deeper than 1000 m in years 1–10, 5.66 in years 11–20). 15F is the new coupled baseline.

This round asks what sets the TOA imbalance a coupled run settles at, using every coupled run since the 06 series. Annual global net TOA and annual ocean heat content in four depth ranges are kept for 76 runs in one store, `data/coupled_annual_diag.nc`.

53.1 Every run ramps, then settles at a configuration-dependent plateau

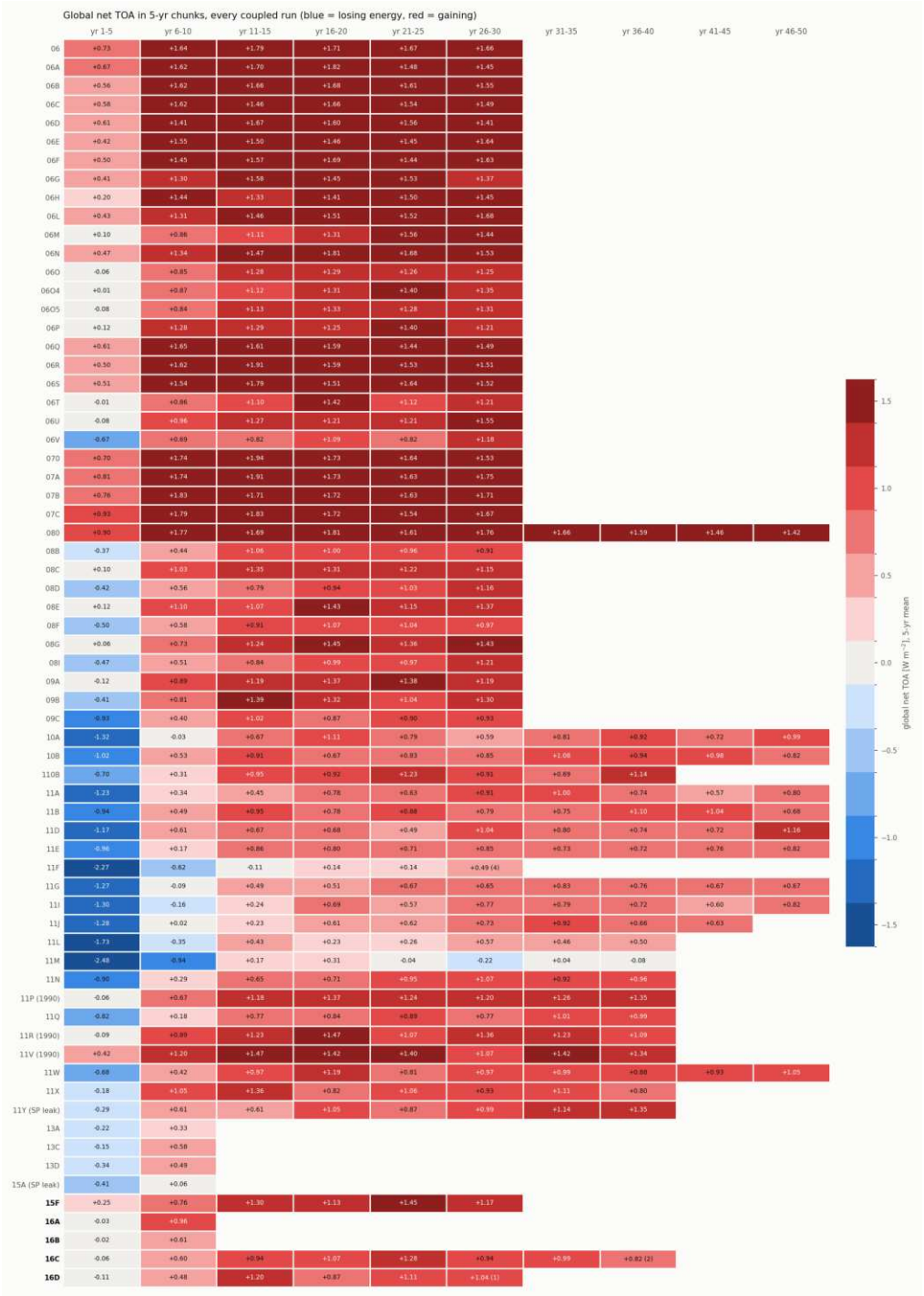


Figure 42: Global net TOA in 5-year chunks counted from each run's start, every coupled run with at least 10 complete years (the CMIP7 spin-ups are excluded: their TOA comes out near $+10\,\mathrm{W\,m^{-2}}$, a different output convention). The first chunk is the lowest in almost every run; TOA rises through years 6–15 and is flat afterwards. The level it settles at is the configuration's: +1.4 to +1.8 for the 06/07/08 series, +0.7 to +1.0 for the new-sea-ice chain 09C–11J, 0.9–1.0 for 11W and 11X, 1.2–1.4 for the 1990-forced 11P/11R/11V. 15F reads 0.25, 0.76, 1.30, 1.13, 1.45, 1.17, the highest 1850 plateau since the 07/08 series. The darker ice of 15F absorbs about $0.9\,\mathrm{W\,m^{-2}}$ more shortwave than 13A. Of the 40-year 16 arms, 16C and 16D settle at 0.9–1.1 and **16E at** 0.66–0.83, the lowest plateau of any run the campaign has kept.

A single year scatters by about $\pm 0.3\,\mathrm{W\,m^{-2}}$, so a 5-year mean carries about ± 0.13 and differences below ~ 0.35 between two chunks are not resolved. 11E is flat at 0.71–0.86 for 40 years once past year 10; 11X oscillates around 1.0; 11Y, the single-precision run with the heat leak, is the only one that keeps climbing ($0.61 \to 1.35$).

53.2 Where the heat is stored



Figure 43: Ocean heating rate in 5-year chunks at 700–2000 m (left) and below 2000 m (right), in $\mathrm{W\,m^{-2}}$ of Earth area so it compares directly with TOA. From annual-mean temperature, $\rho c_p T\, A\, \Delta z$ with below-bottom points masked, on each run's own mesh. $*$ marks a chunk shorter than 5 years. Every run starts at 0.7–1.0 at mid-depth; only the new-sea-ice chain lets it decay.

- **The first decade is universal.** Every run puts 0.7–1.0 $\mathrm{W\,m^{-2}}$ into 700–2000 m and 0.3–0.5 below 2000 m.
- **Only 09C–11Q let the mid-depth uptake decay.** By years 25–45, 11G reads 0.22–0.41 and 11I 0.21–0.44 at mid-depth, both 0.10–0.17 below 2000 m. The 06–08 series never decay (0.55–0.8 and a constant 0.3–0.4 at depth).

- **11W and 11X decay only partly**, to 0.50–0.62 at mid-depth and 0.15–0.23 below. 15F tracks 11X after its first decade (0.65, 0.72, 0.61 against 0.68, 0.65, 0.51), and its first decade is the highest of the recent runs (0.92, 0.96).
- **Stronger GM moves the whole column down from the first chunk.** 16C and 16D decay to 0.29–0.37 at mid-depth by years 31–40. **16E starts at** 0.37 **and stays at** 0.24–0.45, and below 2000 m it is nearly shut: 0.04–0.09 throughout, against 0.16–0.36 in 15F. The heat 16E does take goes into 100–700 m instead (0.57 in years 31–35).
- 11Y's 0.84–1.30 $\mathrm{W\,m^{-2}}$ below 2000 m is the heat-leak artefact, not physics.

53.3 The relationship: lower mid-depth storage, lower decade-4 TOA

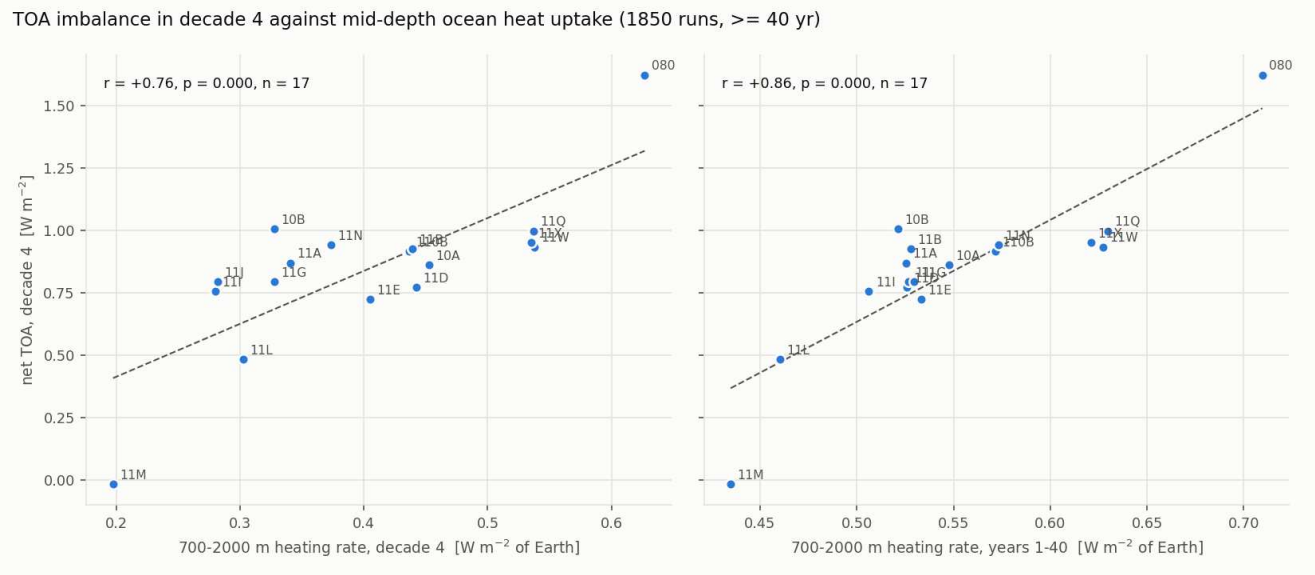


Figure 44: Net TOA in decade 4 (years 31–40) against the 700–2000 m heating rate in decade 4 (a) and averaged over years 1–40 (b), for the 20 runs with 1850 forcing and at least 40 years, now including the 40-year arms 16C, 16D and 16E. The 1990-forced runs and the single-precision heat-leak runs are excluded. Dashed: least-squares fit. 16E has the lowest 40-year mid-depth uptake of any run and sits well above the line.

decade-4 TOA against	all 20	without 080	without 080, 11M	without 080, 11M, 11L
700–2000 m uptake, years 1–40	+**0.72**	+0.59 ($p = 0.008$)	+0.59 ($p = 0.01$)	+0.58 ($p = 0.015$)
700–2000 m uptake, decade 4	+0.73	+0.61 ($p = 0.006$)	+0.49 ($p = 0.04$)	+0.47 ($p = 0.06$)
total ocean uptake, decade 4	+0.96	+0.94	+0.82	+0.72

The leave-out columns were computed from the tabulated values to two decimals. 080 is the old configuration at the top of both axes, and 11M and 11L are the rejected overlap levers at the bottom. Without them the 40-year relation stays significant (+0.58, $p = 0.015$, 17 runs), and cumulative mid-depth storage still predicts decade-4 TOA better than the decade-4 rate does (+0.58 against +0.47): what the run has already buried sets how far it still is from balance. With the extremes included the two are equal.

The first version of this table (17 runs, $r = +0.86$) overstated the slope. Its fit, 4.1 $\mathrm{W\,m^{-2}}$ of decade-4 TOA per $\mathrm{W\,m^{-2}}$ of 40-year mid-depth uptake, was set by 080 and 11M. It predicted the three GM arms far too low: 16C +0.59 (measured +0.96), 16D +0.67 (+0.91), **16E** −0.03 **(+0.70, 4.4 residual sd off)**. The fit without the extremes (slope 1.3) predicts 16E at +0.59, close to what it did. So the relation holds, but cutting mid-depth uptake buys about a third of what the headline correlation implied: stronger GM lowers the mid-depth heating rate by more than it lowers TOA, and part of the heat it keeps out of 700–2000 m is stored higher up instead (§53.2). The last row is largely the energy budget itself (slope 0.91) and confirms the bookkeeping rather than adding evidence. From 11X to 11G, about 0.1 $\mathrm{W\,m^{-2}}$ less 40-year mid-depth uptake goes with 0.15–0.2 lower decade-4 TOA (0.95 against 0.80).

53.4 The 16 series: three ocean levers against the mid-depth uptake

Each arm is 15F plus one change, 10 years from the same 1350 restart and libraries (FESOM `540e1029`, OpenIFS `3726203165e7`; 16B uses the MLE build `79dfae3f`). Rates are 5-year chunks, the second ending at 1359 and

therefore 4 years long.

arm	change	700–2000 m [W m^{-2}]	> 2000 m [W m^{-2}]	NH Mar ext. [M km^2]	global T2m bias [K]
15F	baseline	0.92 / 0.96	0.35 / 0.36	18.8	−1.52
16A	`Redi_Kmax` 300	0.98 / 1.02	0.35 / 0.33	19.9	−1.79
16B	Fox–Kemper MLE	0.89 / 0.91	0.37 / 0.35	19.5	−1.37
16C	`K_GM_max` 1500, Redi 1000	**0.67** / **0.62**	**0.25** / **0.22**	20.2	−1.41

Ice and T2m are 1355–59 means; the 4-year thresholds from 15F's own detrended scatter are ±0.64 M km^2 for NH March extent and ±0.16 K for global T2m.

- **16C (stronger GM) cuts first-decade mid-depth uptake by about 30 %** and deep uptake by 0.1 W m^{-2}. Nothing else degrades beyond noise. At 60–45S the upper ocean is the closest of any run to PHC3 (−0.37/−0.16 K at 0/100 m against 15F's −0.62/−0.44), and the Antarctic convection excess shrinks (+491 against +594 m at the seasonal maximum). **The one cost is NH winter ice**: March extent +1.4 M km^2, beyond threshold, and trending up at +1.8 M km^2 per decade where 15F is flat.
- **16A (weaker Redi) goes the wrong way on everything**: more mid-depth uptake, 0.27 K colder, more Southern Ocean convection (3.35 against 2.39 M km^2), more ice in both hemispheres. Rejected.
- **16B (MLE) leaves mid-depth uptake unchanged.** It cuts 100–700 m instead, and it fixes the subtropical winter outcrops (45–30S +2 m against +35, 30–45N +28 against +50) and SH summer ice (3.28 against 4.65 M km^2, obs 3.0). But it over-shoals the subantarctic mixed layer (−103 m at the seasonal maximum).
- **None of the three changes TOA within the first decade beyond noise** (0.60 to 0.96 against 15F's 0.76 in 1355–59). The first decade is the adjustment in every run (§53.1); the lever's effect on the plateau shows in decades 2–4.

53.5 The 40-year arms: 16E settles lowest, and pays in NH ice and AMOC

Three arms ran 40 years from the 1350 restart: 16C (`K_GM_max` 1500, Redi 1000), 16D (16C plus Fox–Kemper MLE) and 16E (`K_GM_max` 2500, Redi 1000). The 15F control stopped at 1380, so it enters with its third decade. Heating rates are over the decade from annual heat content, in W m^{-2} of Earth.

run	decade	net TOA	100–700 m	700–2000 m	> 2000 m	TOA decade 3
15F	1370–79	1.31	0.36	0.57	0.19	1.31
16C	1380–89	0.96	0.37	0.36	0.15	1.11
16D	1380–89	0.91	0.34	0.31	0.16	0.99
16E	1380–89	**0.70**	0.40	**0.28**	**0.07**	**0.77**

Clean pairs from the same restart, mean over the common years with the standard error from the scatter of the annual difference:

pair	net TOA	700–2000 m	> 2000 m	total ocean
16C − 15F (1350–79)	-0.21 ± 0.06	-0.19 ± 0.03	-0.09 ± 0.01	-0.24 ± 0.04
16E − 15F (1350–79)	$\mathbf{-0.45 \pm 0.08}$	-0.37 ± 0.03	-0.19 ± 0.01	-0.48 ± 0.05
16E − 16C (1350–89)	-0.24 ± 0.06	-0.15 ± 0.02	-0.09 ± 0.01	-0.25 ± 0.05

- **The GM dose response continues to 2500.** 16E takes 0.45 W m^{-2} off 15F's TOA and another 0.24 off 16C's, and settles at +0.66 to +0.83 in every 5-year chunk after year 10 with no upward drift. It is the lowest plateau of any run the campaign has kept.
- **The pairs measure the slope directly**: TOA falls 1.2 ($16E - 15F$) to 1.6 ($16E - 16C$) times the mid-depth cut and about one-for-one with the total ocean uptake. That is the shallow slope of the extremes-free fit in §53.3, not the steep one of the first 17-run fit.
- **The deep sink is nearly shut**: below 2000 m 16E takes 0.04–0.09 in every chunk. What 16E still stores goes to 100–700 m (0.40 in decade 4, the most of the four).
- **MLE on top of GM 1500 adds nothing resolvable at 40 years**: 16D minus 16C is −0.05 in decade-4 TOA and −0.05 at mid-depth.

The climate, decade means (15F: 1370–79; others 1380–89):

run	AMOC 26.5N [Sv]	AMOC 40–60N [Sv]	NH Mar ext. [M km^2]	NH Sep ext. [M km^2]	SH Sep ext. [M km^2]	global T2m [K]
15F	12.5	14.9	16.4	7.1	20.9	286.28
16C	14.9	17.3	15.7	7.4	20.6	286.60
16D	15.1	18.1	16.1	7.2	20.3	286.62
16E	13.3	14.5	17.2	6.8	20.2	286.56

Differences below are 16E minus 15F over 1350–79 unless stated. The standard error of a 30-year mean difference is about sd$\sqrt{2/30}$ from 16E's detrended interannual scatter: 0.24 M km^2 for NH March extent, 0.15 for SH September, 0.04 K for global T2m, 0.23 K for Arctic winter land.

- **NH winter ice is the main cost**: March extent +1.3 M km^2 against 15F and +0.7 against 16C over 40 years; in decade 4 16E has 17.2 against 16C's 15.7 and the observed 15.0. It shrinks at −0.8 M km^2 per decade, as 16C does, but from a higher start. September extent is 0.2 lower.
- **SH winter ice shrinks, as expected with less heat stored**: September extent −0.4 against 15F, and 20.2 in decade 4. The direction holds; the size is small.
- **The surface warms**: global T2m +0.21 K, 90–60S +0.39 K. Arctic winter land +0.20 K is not resolved. Southern Ocean SW cloud radiative effect (45–65S) is 0.76 W m^{-2} more negative.
- **The AMOC does not keep the GM 1500 gain** (§53.7): 16E is 1.0 ± 0.2 Sv weaker than 16C at 26.5N and 1.7 ± 0.2 at 40–60N over 40 years. Against 15F it is level at 26.5N ($+0.04 \pm 0.24$) and 0.67 ± 0.23 weaker at 40–60N. 16E's first decade already showed a subpolar North Atlantic winter mixed layer 144 m shallower than 15F's.

53.6 The stable-BL lever carries part of the late mid-depth uptake, but not the TOA

Within the 11 chain the late mid-depth uptake splits along `RSBLB = 2` (the stable boundary-layer stability coefficient, §49.3): 11E/11G/11I read 0.21–0.48 W m^{-2} at 700–2000 m in years 26–45, 11Q/11W/11X 0.49–0.62. Two pairs differ only in `RSBLB` (5 → 2), 40 years each on core3_beta. RSBLB arm minus control, 700–2000 m heating rate [W m^{-2} of Earth], with the standard error of the 40-year mean from the year-to-year scatter:

pair	yrs 1–10	11–20	21–30	31–40	40-yr mean
11Q − 11N (1850)	+0.02	−0.05	+0.10	+**0.17**	$+0.057 \pm 0.037$
11R − 11P (1990)	−0.02	+0.05	+0.01	+**0.19**	$+0.054 \pm 0.033$

- **Both pairs agree**: the same sign and nearly the same size, together about $+0.055 \pm 0.025$ W m^{-2}. The effect grows with time and in decade 4 is most of the gap between 11N (0.37) and 11Q (0.53).
- Below 2000 m nothing changes (+0.01, +0.03); total ocean uptake rises by $+0.09 \pm 0.05$ and $+0.13 \pm 0.03$.
- **Net TOA does not move**: -0.003 ± 0.066 and $+0.002 \pm 0.053$ W m^{-2} over 40 years, and decade 4 disagrees in sign between the pairs. The extra ocean heat is drawn from another reservoir, not added at the top of the atmosphere; stronger stable mixing pushes heat from the air into the surface over ice and land. Which reservoir gives it up is not checked.
- **Consequence for §53.3**: part of what separates 11Q/11W/11X from 11G/11I at mid-depth is `RSBLB`, and that part does not raise TOA. The cross-run correlation is therefore partly carried by other differences between those runs, and `RSBLB = 2` stays: switching it off would give up its Arctic winter (+0.6–0.7 K) and Siberian summer (+0.77 K) gains for no measured TOA benefit.

53.7 The AMOC: GM 1500 strengthens it, GM 2500 gives it back

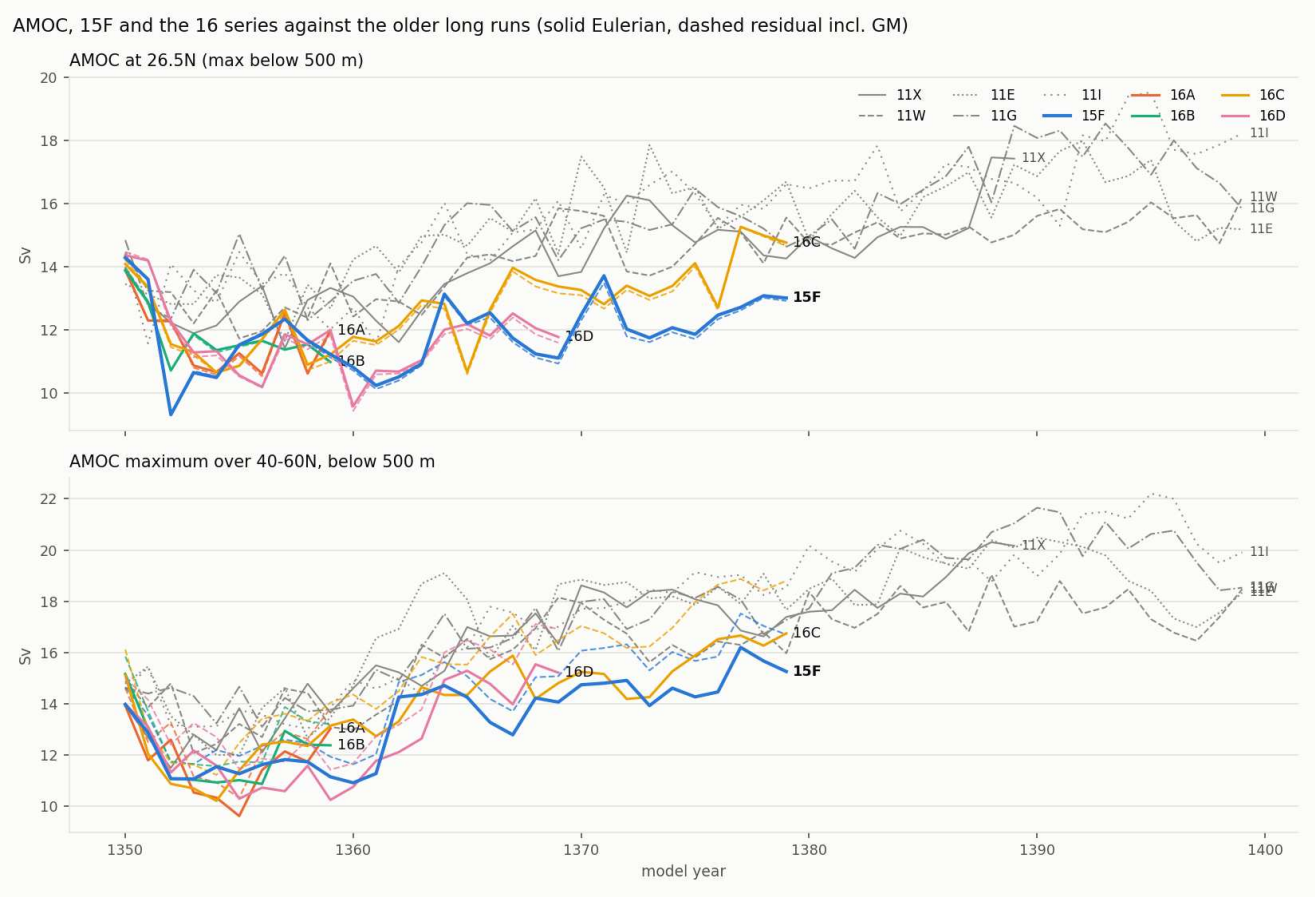


Figure 45: Atlantic overturning maximum below 500 m at 26.5N (a) and over 40–60N (b), annual means. Solid: Eulerian, from w; dashed: residual, $w + w^*$ with the GM bolus velocity. Older runs in grey. Every run strengthens as it spins up; 15F is the weakest of the older configurations, 16C (GM 1500) and 16D sit above it, and 16E (GM 2500) falls back to 15F's level at 26.5N and below it at 40–60N.

AMOC at 26.5N [Sv]	yrs 1–10	11–20	21–30	31–40	41–50
11X	12.7	13.4	15.0	15.4	
11W	12.9	13.7	14.8	15.0	15.5
11E	13.1	14.9	16.3	16.0	16.4
11G	13.5	14.6	15.4	16.3	17.5
11I	13.2	14.4	16.0	16.8	17.8
15F	11.7	11.4	12.5		
16C	11.8	12.6	13.8	**14.9**	
16D	12.0	11.4	13.5	15.1	
16E	12.0	11.7	12.0	**13.3**	

- **15F has the weakest AMOC of all runs**, 11.4–12.5 Sv and 2–3 Sv below 11X over the same years; RAPID observes about 17 Sv today. The likeliest cause is the new ice physics (a warmer, less icy North Atlantic and Nordic Seas forming less dense water); this is not tested.
- **The response depends on the dose.** GM 1500 strengthens it: 16C minus 15F is $+0.83 \pm 0.21$ Sv at 26.5N and $+0.63 \pm 0.19$ at 40–60N over 1350–79. GM 2500 gives that back: 16E minus 16C is -0.99 ± 0.21 and -1.68 ± 0.21 over 40 years, and in decade 4 16E reads 13.3 and 14.5 Sv against 16C's 14.9 and 17.3.
- **MLE does not take the GM 1500 gain back**: 16D dipped in decade 2 (11.4 Sv) and caught up by decade 4 (15.1); 16D minus 16C is -0.25 ± 0.18 Sv over 40 years.
- The GM eddy-induced part is negligible at 26.5N (Eulerian and residual differ by at most 0.2 Sv) and adds 1–2 Sv at 40–60N in every run.
- ~~So the AMOC is no obstacle to a stronger GM dose.~~ 16E killed this: a larger dose than 1500 weakens it again (§53.5). 16C's early NH ice excess ($+1.4$ M km^2 in 1355–59, gone by 1375–79) is still not an overturning effect.

53.8 Method notes from this round

- `scripts/analysis/coupled_annual_store.py` keeps `data/coupled_annual_diag.nc` (run × year: `toa`, `ohc_0_100`, `ohc_100_700`, `ohc_700_2000`, `ohc_gt2000`) and computes only missing entries on each call, checkpointing every 200. New runs and years are added by running it again.
- Only years with all 12 months count. 15B and 15C wrote to `work` only and are read from there. The mesh is chosen by node count (core3 220509, core3_beta 211567).
- Figures: `scripts/figures/toa_decadal_heatmap_all_runs.py` (`CHUNK=5`), `scripts/figures/ohc_rate_heatmap_all_runs.py`, `scripts/analysis/toa_vs_middepth_uptake.py`.
- AMOC: `scripts/analysis/amoc_annual_store.py` keeps `data/amoc_annual_diag.nc` with pyfesom2 `xmoc_data` (as reval part 17), `Atlantic_MOC` mask, and grows incrementally. It needs `fesom.mesh.diag.nc`, which FESOM writes only on a cold start (`fesom_module.F90:698`); none of the restarted coupled runs had one, so it was generated for the corrected CORE3 mesh with FESOM's `fesom_meshdiag` utility (`scripts/model/comp_fesom_meshdiag.sh`, `scripts/model/run_fesom_meshdiag.sh`) and placed in the mesh directory. The RSBLB pairs: `scripts/analysis/rsblb_pair_ohc.py`.
- Clean-pair differences (heat by depth, TOA, AMOC) with standard errors: `scripts/analysis/pair_diff_store.py` (`PAIRS=15F:16E,16C:16E`). Per-year T2m, TOA, SW CRE and ice extent for any arm: `scripts/analysis/detection_threshold_4yr.py` (`ARM=16E Y1=1389 SERIES=1`).

54 Round 37 (2026–09–13→17): five levers at the Southern Ocean, none of them moves it

Round 33 split the Southern Ocean problem in two: a winter over-ice bias and a warm, shallow CDW. This round spends five 20-year branches of **PI200** on the second, and the result is uniformly negative. The value of the round is the exclusion, so the arms are recorded with what each was supposed to do.

Four arms branch at 1560, control **PI200** 1560–1579, everything else identical (v3.5 defaults, `INPSEA` 0.1 / `INPPMIN` 50000, FESOM and LPJ-GUESS from PI200's 1560 restarts, atmosphere cold start):

- **PI200_spp** — salt-plume parameterisation on. Brine rejection reaches deeper than the mixed-layer scheme allows, which is the textbook route to dense shelf water.
- **PI200_h0** — lead closing `h0min` 0.5 → 0.25 m. The pack is dispersive because FESOM closes leads at the thickness of the ice already there.
- **PI200_mle** — Fox-Kemper mixed-layer eddies, settings as 16B.
- **PI200_all3** — all three together, to catch a combination effect that the single levers miss.

A fifth arm attacks the same loop from the ice shelves. One-year screens **PI200_cav05/06/07** gave cavity basal melt of 1038 / 1190 / 1402 Gt yr^{-1} against 1893 at `cavity_gamma_scale` 1.0 and 1100–1325 observed, so 0.6 was taken forward as **PI200_cavg06** (1600–1619, control PI200 1600–1619). The hypothesis was a loop: warm CDW drives melt, meltwater freshens the shelf, no dense water forms, the CDW stays warm. Less meltwater should break it.

54.1 The result, and what it costs to state it

None of the five fixes the Southern Ocean in 20 years. The winter over-ice bias is a *freshwater regime*, not a mixing deficiency: the model sits in one of two states, salty and venting or fresh with warm CDW, and none of these levers moves it between them. The basal-melt calibration is worth keeping for its own sake — 1190 Gt yr^{-1} against 1893 is a defensible number where the uncalibrated one was not — and it became the v3.5 CORE3 default `cavity_gamma_scale: 0.6` (FESOM PR #1064, merged upstream 2026–09–17). But it is not a Southern Ocean fix.

Recorded as an exclusion list, each with the observation that killed it, in §77.2. **The SSS restoring lever stays excluded**: `surf_relax_s` is 0 and remains 0, even for Weddell convection.

55 Round 38 (2026–09–18→22): the PICAL production line, and the albedo that had to be computed

With the lever search exhausted the campaign switches from screening to producing. The v3.5 TCO95/CORE3 defaults go into `awiesm3.yaml` rather than living in runscripts (2026–09–21), and a single pre-industrial line is carried forward: **PICAL** (1850–1929) → **PICAL_momixoff** (branched 1920, to 2049) → **PICAL_ccnice** (branched 1940).

PICAL_v35def and **PICAL_v35def_darkice** are the paired control for the two namelist values the campaign had retired but the production line still carried (`RCL_INPPMIN` 50000 vs 70000) and for the ice albedos (`albi` 0.70 → 0.66, `albim` 0.68 → 0.64). Both are drawn in the campaign TOA figure as context: they are the counter-example, an Arctic volume runaway that fifty years of correction did not arrest.

55.1 Sea-ice albedo, computed rather than guessed

The ice was too dark and the first instinct — darken the ponds — was wrong: the four `ice_therm` albedos are *pond-free*, `ice_thermo_cpl.F90` builds `alb_noponds` from them and only then applies `albpnd` over the pond fraction, so darkening them to stand in for ponds double-counts. Ponds stay dark; the ice and snow get brighter.

The two snow albedos do different jobs, and the split was *measured* on the model's own melt seasons rather than assumed:

	$\partial\alpha/\partial$ `albsn`	$\partial\alpha/\partial$ `albsnm`
NH JJA	0.149	0.504
SH DJF	0.908	0.051

The Antarctic pack surface rarely reaches melting point, so `fmelt` ≈ 0 there and `albsn` carries the southern response; `albsnm` is the Arctic knob. Both stay global: the split is measured surface state, not a hemispheric parameter. `albsn` was raised to 0.80 (2026–09–22).

The sensitivities are state-dependent, which is the part that generalises. As the Arctic cooled, NH $\partial\alpha/\partial$ `albsn` moved 0.149 → 0.504, a factor of three. A number measured in one state is not reusable in another, and albedo changes have to be priced in W m^{-2} before they are made: the Arctic runs away above about +5.5 W m^{-2} while the Antarctic is stable to at least +10.4.

55.2 Lead closing: the right flux response, the wrong state response

`h0min` was tried once more on the production line, 0.5 → 1.0 from 2000. The flux response was exactly as designed — the area-to-volume ratio of freezing fell 1.71 → 1.10 — and the state did not follow: mean thickness 0.61 → 0.62 m, and the 60–90 S bias went +0.82 to +1.45 K. Reverted. The namelist exposure survives as FESOM branch `feat/leadclose-h0min` and is *not* in `main`, which matters later (§57.1).

56 Round 39 (2026–09–22→24): vertical mixing, and two real bugs in the CVMix path

Four vertical mixing schemes, 20 years each (2020–2039) from one restart, differing only in `mix_scheme`:

arm	scheme	SH Feb volume	SH Feb extent	cavity melt
PICAL_ccnice	KPP (control)	—	—	1255.6
PICAL_cvKPP	cvmix_KPP	—	—	1283.5
PICAL_cvTKE	cvmix_TKE	1.31	1.95	1174.5
PICAL_cvTKEIDEMIX	cvmix_TKE+cvmix_IDEMIX	1.62	2.26	1062.1

Volumes in 10^3 km^3, melt in Gt yr^{-1}; observed SH Feb volume about 2.07, extent about 3.0. Both KPP variants lose Antarctic summer ice, both TKE variants grow it, and `cvmix_KPP` reproduces the KPP control closely enough to serve as the null. TKE+IDEMIX comes closest to the observed summer pack and brings cavity melt from 1255.6 to 1062.1 Gt yr^{-1}, towards Adusumilli (2020) ∼1100 from Rignot (2013) ∼1500.

Adopted: `cvmix_TKE` from 2060, `cvmix_TKE+cvmix_IDEMIX` from 2080, and the pair became the v3.5 CORE3 default (esm_tools `35a0524f7`). The cost was stated at adoption and is northern: NH April volume 38.04 rising +2.46/decade against about 31.2 observed, where TKE alone sat at 35.17 and was flat.

56.1 Bugs VII and VIII: IDEMIX read garbage, and one bug hid the other

IDEMIX would not initialise. Four hypotheses were wrong before the right one (§78); the two real defects are in the CVMix forcing path and are now FESOM PR #1087.

Bug VII, the implicit netCDF fill value. `read_other_NetCDF` reads `missing_value` into `miss` and fills those points before interpolating. A variable with neither `missing_value` nor `_FillValue` leaves `miss` at whatever it held, in practice 0.0. Every genuine zero is then treated as missing while the real sentinel, netCDF's implicit float fill 9.96921×10^{36}, is left in; the nearby-point average pulls it in and the field is `Infinity` before it reaches `interp_2d_field`, which documents that it assumes no dummy values exist. On the IDEMIX near-inertial surface forcing this masked 146694 real zeros and gave min/max of -7.6×10^{-4} to Infinity.

Bug VIII, an MPI type mismatch. `loc_Etot` and `glb_Etot` are `real(kind=WP)`, but five `MPI_AllREDUCE` calls declared the buffer `MPI_DOUBLE_PRECISION`. In a single-precision build that reads 8 bytes from a 4-byte variable, so half of every energy total is whatever sits next to it in memory. It is not only a diagnostic: `idemix_botforc_Etot` rescales the field by `Etot/glb_Etot`, so the corrupt total goes back into the physics.

With both fixed, `Etot_bot` reads 0.9135067 TW against the forcing file's own `Etot_in_TW_bin` attribute of 0.91798, and `Etot_srf` 0.3211645 TW. That pair is now the standing gate on any IDEMIX leg: the signature of an unfixed build is `Etot_srf` 6.05×10^{-4} with `Etot_bot` -1.54×10^{16}.

57 Round 40 (2026–09–24→25): PICAL_ccnice reaches 2100, and it is drifting cold

PICAL_ccnice completed 1940–2099 and stopped at its configured `final_date`. Throughput was about 77 SYPD on 47 nodes under TKE+IDEMIX (3:18:32 for 2080–89, 2:58:12 for 2090–99), against about 86 for `cvmix_TKE` alone and 75 for KPP.

57.1 Bug IX: the staged library was a symlink, and `md5sum` follows it

The 2080 leg — the first with IDEMIX on — aborted 100 s in with `Etot_bot` $= -1.54 \times 10^{16}$ TW, the exact unfixed-build signature, and NaN'd the ocean. `bin_sources` named the right file and the staged `fesom` executable matched the source md5. The defect was one level down: `<exp>/bin/fesom/lib/fesom/libfesom.so` is a *symlink* into esm_tools' own per-leg snapshots `libfesom.so_<leg>`, and it pointed at a pre-fix one. `md5sum` follows symlinks, so every check passed while the wrong library loaded. The FESOM executable is a driver stub, byte-identical across all these builds, so a matching binary checksum proves nothing: **only the `.so` distinguishes them, and `ls -l` must come before `md5sum`**.

57.2 The drift: cooling at the surface, flat in the whole ocean

Over 2060–2099, the whole cvmix era, every headline metric is outside its confidence interval:

metric	2060–64	2095–99	trend/decade	95 % CI
T2m global	14.439	13.645	−0.214	0.025
T2m 60–90 N	−8.143	−9.842	−0.511	0.118
NH extent, March	16.262	18.188	+0.522	0.103
net TOA	−0.775	+0.236	+0.272	0.079

The run cools 0.21 K/decade globally while the Arctic grows ice, and net TOA over the last two decades is *positive* at $+0.167\,\mathrm{W\,m^{-2}}$. That is not a contradiction and it is not an energy leak. Total ocean heat content is flat, +0.04 ZJ/decade, and the cooling is a redistribution inside the ocean:

band	ZJ/decade	$\mathrm{W\,m^{-2}}$ equivalent
0–100 m	−17.8	−0.110
100–700 m	−34.4	−0.214
700–2000 m	+19.4	+0.121
>2000 m	+32.7	+0.203
total	+0.04	0.000

The upper 700 m loses heat to below 700 m at about $0.32\,\mathrm{W\,m^{-2}}$ equivalent, almost exactly balanced. Split by mixing era the mechanism follows the scheme changes: under KPP (2040–59) the upper ocean was *gaining*

(+0.061); `cvmix_TKE` flipped it hard negative (−0.488) and the whole system lost energy (net TOA −0.468); adding IDEMIX halved the upper-ocean loss (−0.201) and drove deep uptake up (+0.176 → +0.452), putting the total back positive. **IDEMIX did what internal-wave-driven diapycnal mixing should do — it moved the burial deeper and stopped the net drain — but the upper ocean is still losing heat, and that is what keeps the surface cooling.**

57.3 Where the state stands against observations

Against GIOMAS 1989–1999 (the early record is the fairer anchor for a PI run; the Arctic thinned sharply across 1989–2014 while the Antarctic has no significant trend):

10^3 km^3	model 2090–99	GIOMAS 1989–99	bias
NH April volume	34.56	31.19	+3.37
NH September volume	13.97	11.96	+2.01
SH September volume	12.33	19.89	**−7.56**
SH February volume	0.73	2.07	**−1.34**

The Antarctic sits at about 62 % of observed winter volume. NH April volume is still rising at $+1.145 \pm 0.968$ per decade over 2080–2099, which is roughly half the +2.46 priced at adoption but not flat.

CMPI went 0.844 (1980–89) → 0.879 (2000–09) → 0.902 (2080–89) → 0.903 (2090–99). **The degradation is tropical, not polar**: nino34 +0.155 and tropics +0.134 against arctic +0.003 and antarctic +0.008; by variable `tas` +0.149, `thetao` +0.127, `zg` +0.111.

57.4 Drake Passage, and the price of the GM tuning

Computed with tripyview's model-accurate transect path, which resolves the 105 mesh edges the 66 W line crosses and integrates the flux FESOM's own discretisation carries. At 2090–99 the net transport is **98.9 Sv** (128.1 eastward, −29.2 westward), against Cunningham (2003) 134 ± 11.2 and Donohue (2016) 173.3 ± 10.7. It has been declining slowly: 105.1 (2040–49) → 98.9 (2090–99), with no step at either mixing-scheme switch, so the ACC weakness is structural rather than a cvmix artefact.

The GM tuning of Round 36 is a different matter. On identical years 1580–1599 from identical restarts:

run	GM	resolved	**net**
PI200	resolution scaling (before)	110.0	**110.0**
PI200_gmR1500	Rossby cutoff, K_{GM}^{max} 1500	103.2	**103.2**
PI200_gmR2500	Rossby cutoff, K_{GM}^{max} 2500	93.3	**93.3**

The GM tuning cost 16.7 Sv of ACC transport, 15 %, monotonic in K_{GM}^{max}, far outside the ∼1.5 Sv decadal scatter. The bolus term responds cleanly and monotonically ($\pm 1.8 \rightarrow \pm 2.7 \rightarrow \pm 3.6$ Sv) but its *net* is zero in every arm, because the eddy-induced circulation across the passage is a closed overturning cell. GM 2500 was adopted for the CDW improvement; Drake Passage was not scored at the time, and it is a real cost that belongs on the ledger next to the benefit.

57.5 The boreal forest is gone, and the canopy work needs one

In the Siberian box 55–75 N / 60–180 E at 2099: BNS 0.001, BNE 0.024, TREEFPC 0.050, GRASSFPC 0.216, and grass beats tree on 84.9 % of cells. The reference points are the offline CRUNCEP state it descends from (BNS 0.227, TREE 0.415, grass>tree 29.2 %) and 11E after 50 coupled years (0.017 / 0.083 / 69.6 %). The state is stable between 2090 and 2099, so this is settled, not transient. **The mechanism is competition with C3 grass, not a climate gate**: every BNS establishment gate stays open while the forest disappears, the only temperature gate is a warm one 34 K away, and GDD5 clears on 91 % of cells.

This matters for the LPJ-GUESS canopy work merged on 2026–09–25 (stem area index via `ltos`, shrubs routed to low IFS vegetation, `iftreefracca 1`). Laszlo Hajdu measured +0.46 K global T2m from it, concentrated in the boreal zone, in a run that has a boreal forest — and it improved his ERA5 comparison, bias −0.46 against −0.92 K and RMSD 1.38 against 1.77. **That result is not transferable here**: the change acts on the tree side of the tile and this configuration has 5 % tree cover. **PICAL_crunveg** was built to answer it — identical in every respect except that LPJ-GUESS starts from the CRUNCEP spin-up end state — and was cancelled before starting when the campaign moved to albedo.

57.6 Bug X: the instruction file nothing reads

The canopy parameters were first put in `namelists/lpj_guess/global.ins`, on the evidence that `run_coupled_4_1_2.ins` imports it and that the staged copy matched. That was wrong, and the correct chain has to be traced through the source:

```
framework.cpp:1063   xtring insfilename="guess.ins";        <- hard-wired
awiesm3.yaml:1331    ecearth.ins.j2 -> runPROC_NUM/ecearth.ins   (one per rank)
runN/guess.ins       import "ecearth.ins"
```

`global.ins` and `run_coupled_4_1_2.ins` are staged into the top of the work directory and never reached; `run_coupled_4_1_2.ins` even imports a path that does not exist, which nothing noticed because nothing reads it. The two copies had drifted apart in exactly the way that catches you out: `global.ins` carried `iftreefracca 1` while the file actually read carried `0`. Corrected in esm_tools `cf0a27943`.

Both 50-year continuations queued on 2026–09–25 were cancelled before starting, partly for this: they would have run with the canopy work compiled and inert.

58 Round 41 (2026–09–26→30): albedo, the crunveg line, and what a forest costs in ice

The campaign moved to albedo for queue time. Runs live under `/albedo/work/projects/p_awiesm3_cmip7/jstreffi/runtime/awiesm3-v3.4/`; year 2100 of every continuation is a cold-started OpenIFS and is discarded. The production layout at the 1800 s ocean step is 768 FESOM ranks at 96 per node on **25 nodes**; the 1200 s step needs **33 nodes** for comparable throughput, which is why 1800 s is the production step and 1200 s is not.

PICAL_crunveg ran 2100–2119: PICAL_ccnice's ocean and atmosphere at 2100, LPJ-GUESS restarted from the CRUNCEP spin-up end state, with the canopy work merged on 2026–09–25 (§57.5). **PICAL_ccnice** was continued alongside to 2129 as the unforested control, so 2101–2119 is a pair in which only the land state differs.

question	measured	on
does the forest regrow on ccnice?	Siberian tree FPC 0.062–0.067 per decade after 2100 against 0.061–0.065 before; BNS ≈0.0015. **No.**	ccnice 2100–2129
does it survive in crunveg?	BNS retained **81 %** after 19 years against 34 % (080a) and 20 % (11G), same CRUNCEP start	crunveg 2100–2119 pre-merge pair
what does the forest do?	global T2m +0.24 K, 60–90 N +0.91 K NH April / September volume −2.0 / -2.35×10^3 km^3	crunveg − ccnice 2101–2119

Retention is higher in crunveg also at matched GDD5 and at matched grass PAR, so the merged canopy code is not only surviving in a warmer climate. It is *not* yet separated from state migration or from non-GDD climate; a clean attribution needs an offline pair of old and new LPJ-GUESS binaries on identical forcing (`scripts/analysis/bns_retention_vs_climate.py`, `data/veg_traj/`). **A boreal forest costs about 2×10^3 km^3 of NH ice**, and the forested NH sits at or below GIOMAS 1989–99, so carrying a forest in the production line means the sea ice no longer has to be tuned against an unforested Arctic.

From crunveg's 2120 restart a family of ten-year branches followed: **ob1200** and **ob1800** (the old FESOM build at 1200 and 1800 s), **ts1200** (the new build at 1200 s) and **ihf0**/**ihf1** (the new build at 1800 s, McPhee ocean-to-ice heat flux off/on, §62.2). Both new-build 1800 s arms lost Labrador Sea convection, which opened Round 42.

59 Round 42 (2026–10–01): the Labrador Sea — a time-step artefact, a noise floor, and the GM regime

59.1 IDEMIX smooths once per step, so 1800 s mixes less

Two-month screens (`TEST_dt*`) showed a Kv/Av deficit at 1800 s against 1200 s below 100 m of −0.017 to −0.025 in $\log_{10}$. It **disappears with `idemix_n_hor_iwe_prop_iter 0`** and is unchanged with 8 sub-iterations. The cause is numerical: with horizontal propagation on, the IDEMIX internal-wave energy goes element → node → element once per step, an un-Δt-scaled grid-scale smoother applied 72 times a day at 1200 s and 48 times at 1800 s. Its strength per day scales as $1/\Delta t$.

The fix is a namelist reference step, `idemix_hor_smooth_dtref` (s), that applies the smoothing per unit time rather than per step. At 1800 s with `dtref` 1200 it removes about 85 % of the deficit at no cost. Removing the round trip altogether cuts Kv by 12–19 % ($\log_{10}$ −0.055 to −0.090, `data/clim/step_screens_xd.txt`), so **the tuned state leans on this numerical smoothing**. Upstream as FESOM PR #1103 (open, not merged).

It does not bring the Labrador Sea back. PICAL_crunveg_nx1800 (ihf0 plus the fix, `dtref` 1200) restored the deep Kv and reached a March mixed layer of 414 m over its first eight winters against ihf0's 370 (Table below). The second leg was cancelled on 2026–10–01; the 2130 restarts are kept.

59.2 The noise floor: the 2120s attribution rests on one run

March MLD in the convection box 56–62 N, 60–50 W along the whole lineage, as nine-winter windows (`scripts/analysis/labrador_mld_noise_floor.py`, `data/clim/labrador_mld_noise_floor.txt`):

null / arm	mean MLD (m)	percentile	collapse winters
KPP 1900–2059	1331	—	4 of 160 (2 %)
TKE 2060–2079	1429	—	0 of 20
TKE+IDEMIX, ccnice 2080–2129	1053	—	7 of 50 (14 %)
TKE+IDEMIX, crunveg 2101–2119	1164	—	0 of 19
ccnice continuation, old 1200 s	556	1.9	3
ob1200, old build 1200 s	**979**	**28.3**	0
ts1200, new build 1200 s	520	0.0	1
ob1800, old build 1800 s	344	0.0	5
ihf0, new build 1800 s	370	0.0	2
ihf1, new build 1800 s + McPhee	308	0.0	4
nx1800, new 1800 s + IDEMIX fix (8 yr)	414	0.0	3

Arms are 2121–2129; the percentile is the share of the 53 pooled IDEMIX-era null windows at or below the arm mean. **Every arm except ob1200 sits at the bottom of the null, and so does the unbranched ccnice continuation.** The build and step effects inferred earlier were differences against ob1200, a single realisation. The phenomenon is the slow weakening of ccnice in the IDEMIX era — 1469, 1053, 1207, 870, 668 m for the 2080s to the 2120s, with 3 collapses in the 2120s (`data/clim/labrador_mld_decades.txt`) — not a step change at the branch.

59.3 The Labrador ice is a GM 2500 regime

Sea ice, not mixing, was then taken as the primary symptom and controlled for global temperature. Every coupled run with at least ten years on levante and albedo was searched: **61 runs, 468 run-decades** (`scripts/analysis/labrador_any_mesh.py`, `scripts/analysis/labrador_ice_vs_global_sst.py`, `data/clim/labrador_ice_vs_global_sst.txt`). The metric is March ice fraction in the convection box against decade-mean global SST; HadISST2 March 1979–2008 is 0.16 in the box and 0.37 over the basin.

- AWI-CM3/CORE2, AWIESM700/720, HR and Tuning_test_06 to 11V (`K_GM_max` 1000, Redi 0) stay at 0.05–0.3 at every global SST from 17 to 20 °C, with only a short first-decade shock.
- The shock grows with GM at matched global SST: 15F (1000/0) 0.37, 16C/16D (1500/1000) 0.54–0.59, 16E (2500/1000) 0.70–0.79 for about three decades.
- One clean pair: PI200_gmR2500 **0.684** against PI200_gmR1500 **0.280** on identical years 1580–1599 from one branch, at global SST 18.41 and 18.30 °C.
- The persistent capped state exists *only* under 2500/1000: the momixoff continuation sits near 0.9 for 90 years, v35def and v35def_darkice at 0.885 and 0.922. **Every PICAL run uses 2500/1000.**

The capped Labrador regime appeared with Round 36's `K_GM_max` 2500 / `Redi_Kmax` 1000 (§53), which was adopted for the lowest TOA plateau. Its costs already on the ledger were the AMOC at 40–60 N, 1.7 Sv weaker than GM 1500, and 16.7 Sv of Drake Passage (§57.4); the Labrador Sea is a third.

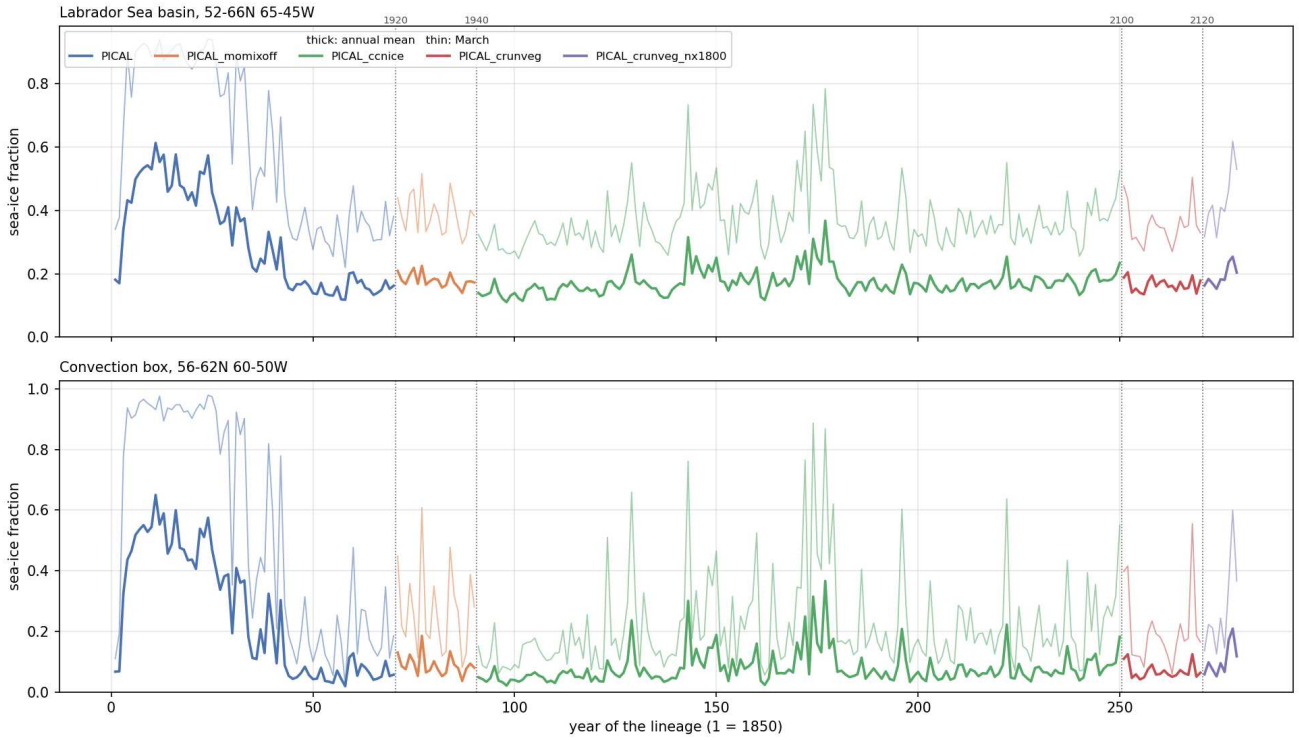


Figure 46: Labrador Sea ice along the PICAL lineage (1 = 1850), basin 52–66 N 65–45 W (top) and convection box 56–62 N 60–50 W (bottom); thick annual mean, thin March. The 2120 segment is nx1800.

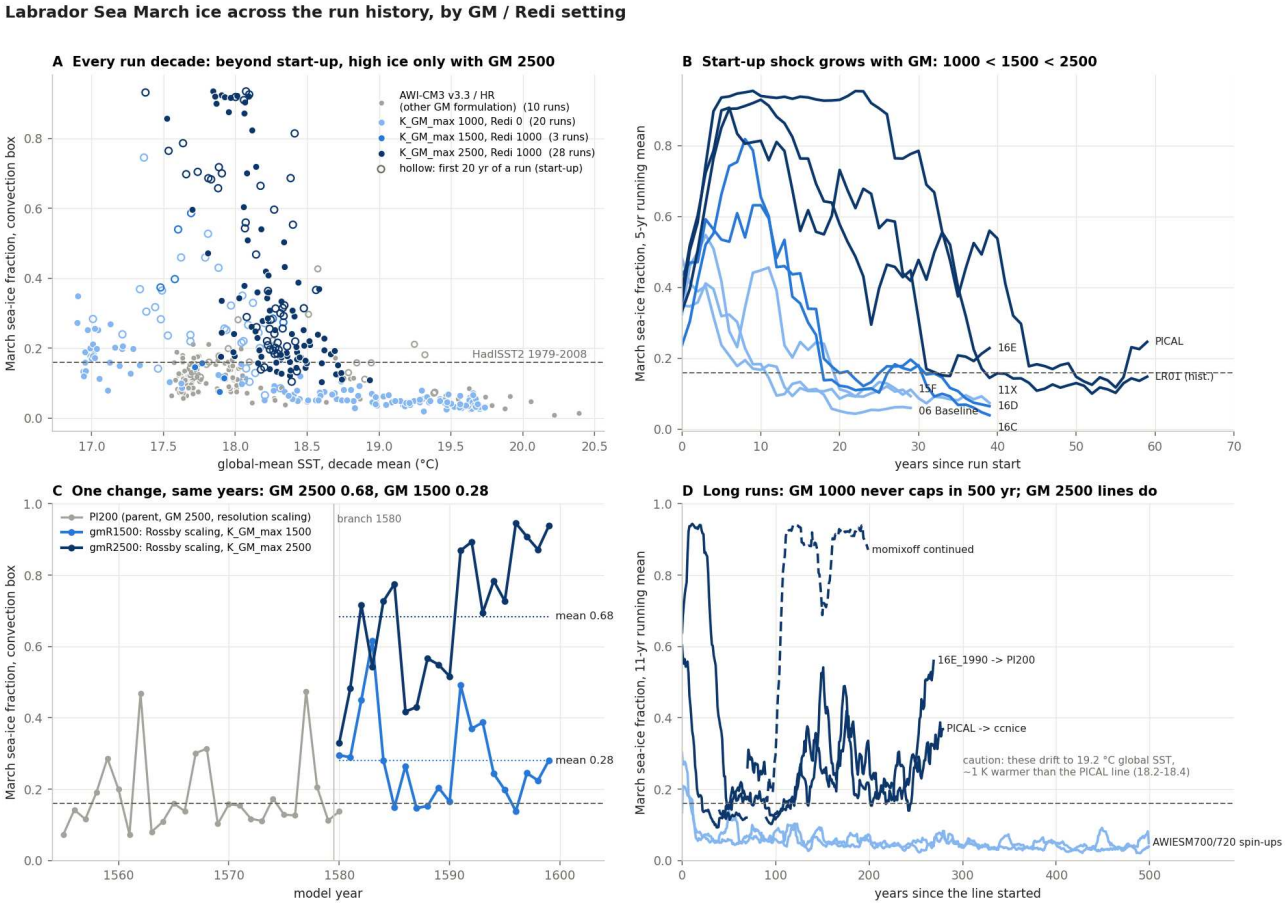


Figure 47: Labrador March ice across the run history, by GM/Redi setting. (A) every run-decade against global SST: beyond start-up, high ice only with GM 2500. (B) the start-up shock grows 1000 < 1500 < 2500. (C) one change on identical years: gmR2500 0.68 against gmR1500 0.28. (D) long runs: GM 1000 never caps in 500 years; the GM 2500 lines do.

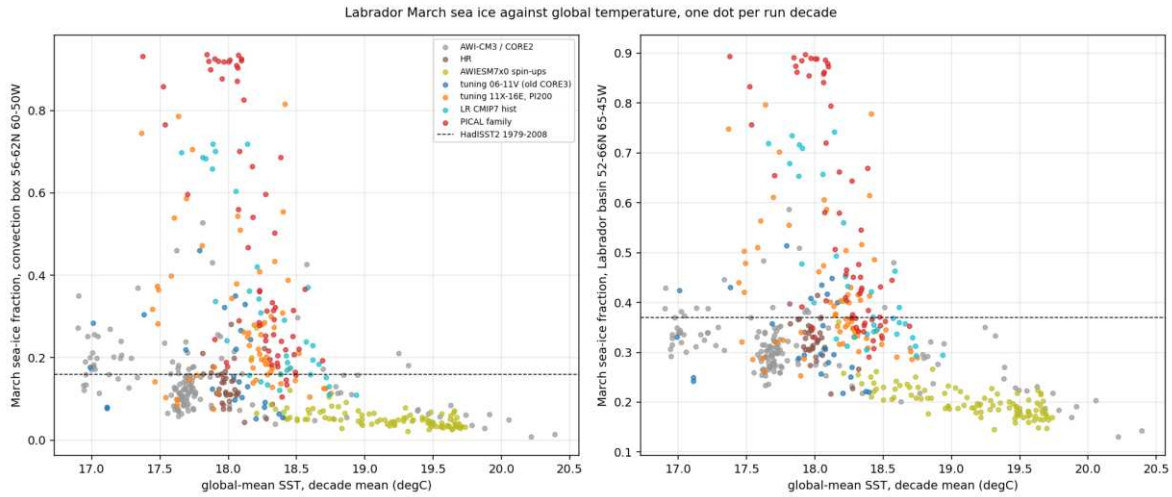


Figure 48: Labrador March ice against decade-mean global SST, one dot per run-decade, coloured by model family; convection box (left) and basin (right). The PICAL family is the cluster near 0.9 at 17.8–18.1 °C.

60 Round 43 (2026–10–01→02): where the GM benefit comes from, and four ways to split it

60.1 Why CORE2 never capped

AWI-CM3 v3.3 on CORE2 ran `K_GM_max` 3000 with linear resolution scaling and a ramp that switches GM off below 30 km and fully on above 40 km. On CORE2 that gave a top-level K of about 113 $m^2 s^{-1}$ in the Labrador Sea and about 2000 at 55–65 S (2563 at 40–55 S), because CORE2's Southern Ocean was coarse and its North Atlantic fine: resolution-based GM separated the two for free. CORE3 refined the Southern Ocean margin to the Labrador Sea's spacing, and production GM gives **983 in the Labrador Sea and 981 at 55–65 S** (Fig. 49, `scripts/figures/gm_k_maps.py`). On CORE3 the v3.3 ramp would switch GM off in both.

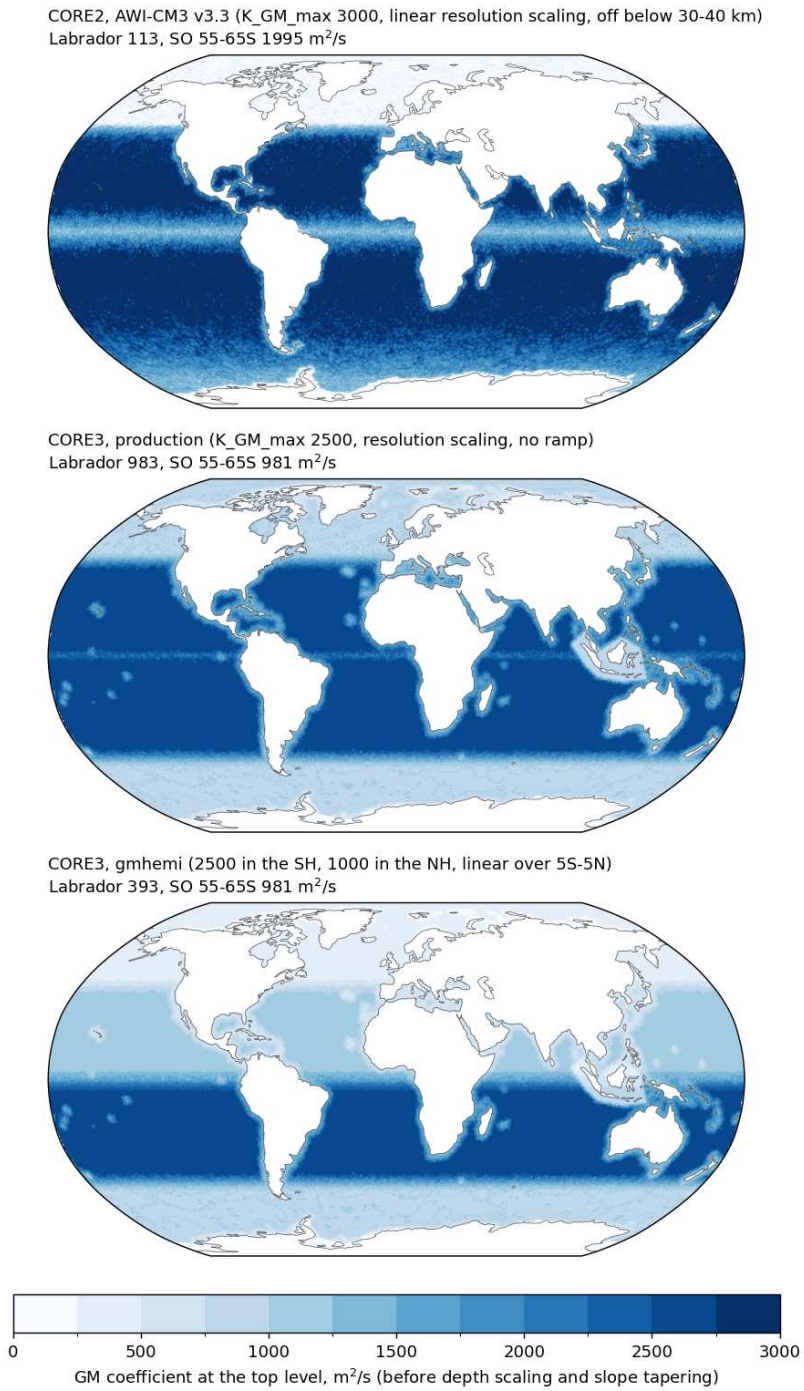


Figure 49: Top-level GM coefficient before depth scaling and slope tapering. CORE2/AWI-CM3 v3.3 (top): Labrador 113, SO 55–65 S 1995 $\mathrm{m^2\,s^{-1}}$. CORE3 production (middle): 983 and 981. CORE3 gmhemi (bottom, Round 43 arms): 393 and 981.

60.2 The 1200 s test: both arms open the Labrador Sea and give the benefit back

Two 1200 s branches from crunveg 2119, paired against **gm2500** (run directory `PICAL_crunveg_ob1200`; same branch, build, step and years, only GM differs): **gm1500** (`K_GM_max` 1500) and **gmramp** (2500 with the v3.3 ramp, off below 30 km, full above 40 km). 2121–2129, differences against gm2500, "$*$" beyond the paired $2\sigma/\sqrt{N}$ (`scripts/analysis/gm_arms_global.py`, `data/clim/gm_arms_global_2121-2129.txt`):

	gm2500	gm1500 − gm2500	gmramp − gm2500
net TOA ($\mathrm{W\,m^{-2}}$)	0.487	+0.384*	+0.442*
heat 700–2000 m ($\mathrm{W\,m^{-2}}$)	0.240	+0.250*	+0.286*
heat >2000 m ($\mathrm{W\,m^{-2}}$)	0.214	+0.137*	+0.213*
T2m global (K)	14.096	−0.197*	−0.077
T2m 60–90 S (K)	−16.557	+0.040	+0.881*
SH September volume ($10^3\,\mathrm{km^3}$)	11.842	−1.673*	−4.721*
SH September extent ($\mathrm{M\,km^2}$)	18.141	−1.129*	−4.202*
NH September volume ($10^3\,\mathrm{km^3}$)	10.726	+0.414	+1.238*

Both arms open the Labrador Sea (March MLD 2121–2128: gm1500 1489 m, gmramp 1968 m with ice 0.06, gm2500 979 m) and **both give back the heat-uptake benefit**, all of it into the ocean below 700 m. Switching GM off on the fine mesh costs the Antarctic most: gmramp loses 40 % of SH September volume. The second decades were cancelled; both keep their 2120–2129 output and 2130 restarts. This revised the storage-based conclusion of 2026–10–01 that the uptake benefit was separable from the Labrador Sea (§.4).

60.3 Heat pathways: the benefit is Southern Ocean ventilation

Three GM 1500 versus 2500 pairs, region by region (`scripts/analysis/heat_pathways.py`, `data/clim/heat_pathways_compare.txt`, `data/clim/so_sfc_entry_bands_compare.txt`), GM 1500 minus GM 2500 in $\mathrm{W\,m^{-2}}$ of Earth surface:

	16C − 16E 1351–1389	gmR1500 − gmR2500 1581–1599	gm1500 − gm2500 2121–2129
total surface entry	+0.246	+0.193	+0.387
SO surface entry (<34 S)	+0.188*	+0.190*	+0.217*
of which 34–55 S	+0.213	+0.189	+0.262
of which south of 65 S	−0.039*	−0.024*	−0.048*
SO lateral (export north)	−0.252*	−0.369*	−0.324*
Indo-Pacific storage, all depths	+0.248*	+0.237*	+0.276*
of which 700–2000 m	+0.119*	+0.169*	+0.128*
subpolar N Atlantic surface entry	−0.163*	−0.082*	−0.079
Atlantic heat transport at 55 N (PW)	+0.306*	+0.348*	+0.225*

All three agree. **The extra uptake under GM 1500 enters through the Southern Ocean surface at 34–55 S and is stored in the Indo-Pacific at 700–2000 m**: the AAIW/SAMW route. South of 60 S the sign reverses. The subpolar North Atlantic goes the other way, releasing more heat to the atmosphere and importing more from the south. So the benefit GM 2500 buys is a Southern Ocean ventilation effect, and the Labrador capping is a separate North Atlantic side effect of the same parameter.

60.4 The Rossby radius cannot separate them either

From gm2500's own stratification (`scripts/analysis/gm_k_rossby.py`, `data/clim/gm_k_rossby.txt`) the Rossby radius R_d is 8.0 km in the Labrador Sea, 9.3 km at 55–65 S, 18.2 km at 40–55 S and 35.4 km in the subtropics. $K = 2500\,\mathrm{m}\min\left(1, (R_d/20\,\mathrm{km})^2\right)$ gives Labrador 405, 40–55 S 1913 (production 1661), 55–65 S 606, south of 65 S $191\,\mathrm{m^2\,s^{-1}}$. It would keep the 34–55 S uptake benefit, but R_d **does not distinguish the Labrador Sea from 55–65 S**, and its risk is low K at the Antarctic margin — gmramp's ice loss in another form.

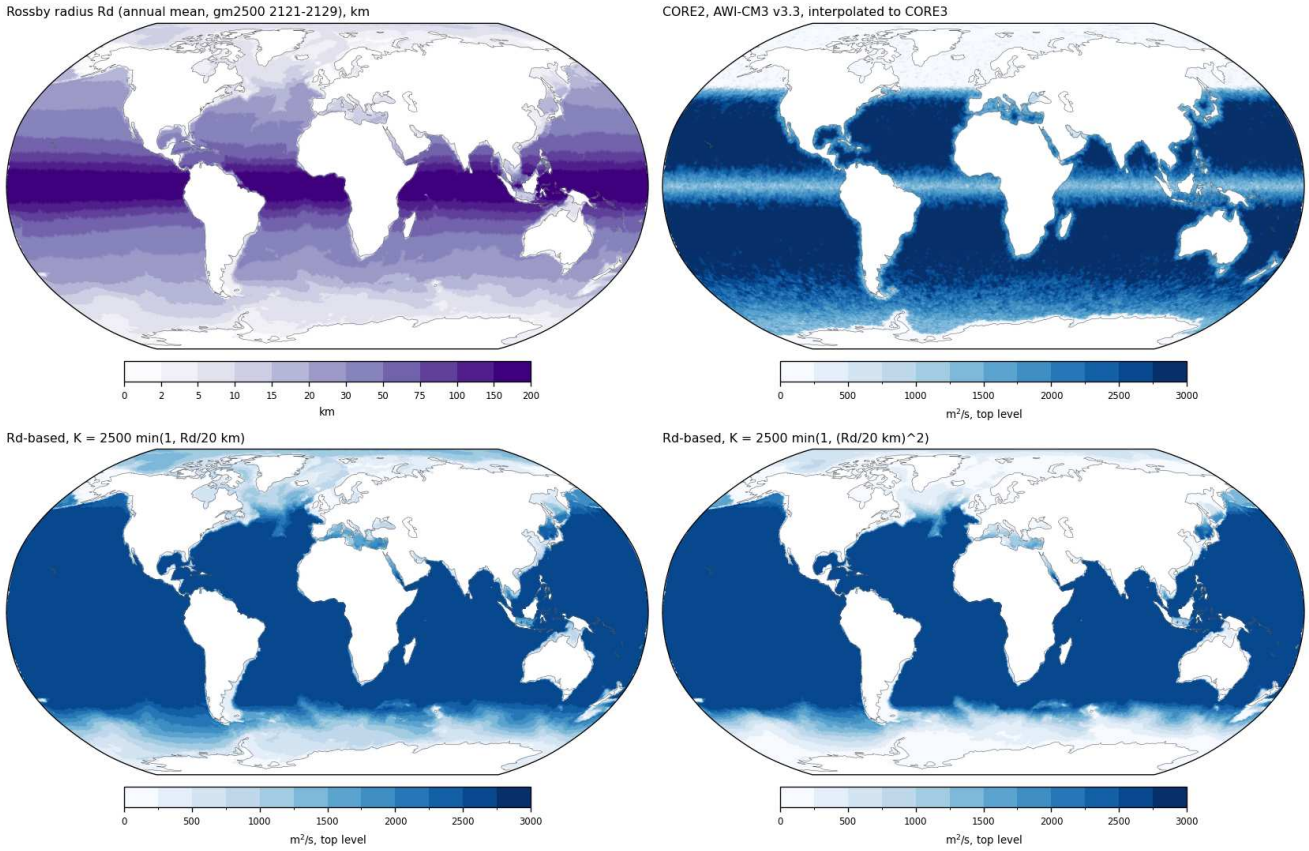


Figure 50: Rossby radius from gm2500, 2121–2129 (top left); CORE2/AWI-CM3 v3.3 GM interpolated to CORE3 (top right); two R_d-based K forms with a 20 km reference, linear (bottom left) and quadratic (bottom right).

60.5 The rule, and the one exception

The campaign rule recorded on 2026–10–02 is that **tuning must be physical**: no arbitrary regional gates chosen to make the present or PI state look better. A parameter may differ by region only through a process that genuinely differs there. A hemisphere-gated INP lever was rejected under it the same day. The hemispheric GM gate below is the one exception: it was accepted explicitly, after consulting the FESOM core developers, and it is recorded as a gate, not as eddy physics.

60.6 Four arms at the production step

All four branch from the same restart, run 2120–2139 at 1800 s on the old build, and differ only in GM:

- **ob1800** — the GM 2500 reference (called gm2500 in discussion at this step).
- **gmferr1800** — Ferreira vertical scaling ($K \propto N^2$ below the mixed layer, `bvref` 1) in place of the depth exponent, `K_GM_max` 2500.
- **gmhemi1800** — `K_GM_max` 2500 in the SH and 1000 in the NH, linear over 10° centred on the equator. FESOM branch `feat/gm-hemispheric-kmax`, namelist `K_GM_max_NH` and `K_GM_hemi_trans` in `oce_dyn`.
- **gmrd1800** — K from the local Rossby radius, `K_GM_rd_ref` 20 km, `K_GM_rd_pow` 2, resolution scaling off. `init_Redi_GM` runs every ocean step, so K follows N^2 online.

2121–2139 means, differences against ob1800, "$*$" beyond the paired $2\sigma/\sqrt{N}$ of the two runs (`data/clim/gm_arms1800_global_2121-2139.txt`; volumes in $10^3\,\mathrm{km}^3$, extents in $\mathrm{M\,km}^2$, heat in $\mathrm{W\,m}^{-2}$ of Earth surface):

	ob1800	gmferr1800	**gmhemi1800**	gmrd1800
T2m global	13.927	−0.447*	+0.002	+0.105*
T2m 60–90 N	−8.693	−1.002*	−0.153	−0.265*
T2m 60–90 S	−16.846	−0.553*	+0.080	+0.334*
T2m Siberia	−8.995	−1.000*	−0.037	−0.109
net TOA	0.526	+0.372*	+0.032	−0.034
ocean heat uptake	0.667	+0.379*	+0.049	−0.033
heat 0–700 m	0.080	−0.223*	+0.016	+0.051
heat 700–2000 m	0.267	+0.380*	+0.055*	−0.088*
heat >2000 m	0.220	+0.197*	−0.032*	+0.017
NH March volume	29.848	+2.459*	+0.000	+0.150
NH September volume	11.335	+2.173*	+0.572	−0.770*
SH February volume	1.209	+0.011	−0.064	−0.253
SH September volume	13.045	−0.023	−0.066	−1.669*
NH March extent	16.853	+0.325*	−0.657*	+0.300
NH September extent	6.810	+0.683*	+0.043	−0.066
SH September extent	18.642	+1.380*	+0.114	−0.725*

The second decade alone sharpens the verdicts (`data/clim/gm_arms1800_global_2130-2139.txt`): gmferr1800 net TOA +0.456* and global T2m −0.499*; gmhemi1800 net TOA −0.032, NH March extent −0.460*; gmrd1800 SH September volume −2.056* and extent −1.070*.

Labrador, March	ob1800	gmferr1800	gmhemi1800	gmrd1800
MLD 2121–29 (m), ice	344, 0.32	1327, 0.17	2069, 0.13	1980, 0.10
MLD 2130–39 (m), ice	111, 0.43	1723, 0.18	2026, 0.10	2487, 0.07
collapse winters	5 of 9, 9 of 10	0 of 19	0 of 19	0 of 19

HadISST2 1979–2008: 0.16 in the box (`data/clim/gm_arms1800_labrador.txt`; the ob1800 2130–39 values are from the campaign notes of 2026–10–02 and are not in that file).

Verdicts.

- **gmhemi1800: adopted.** Labrador mixed layer near 2050 m with no collapses, and every global, heat and ice-volume difference is null except NH March extent and the small 700–2000 m / >2000 m shift. Global T2m, TOA, Siberia and ice volume are unchanged. It **overshoots**: March Labrador ice 0.10–0.13 against 0.16 observed. Upstream as FESOM PR #1112 (open).
- **gmferr1800: rejected.** The most realistic Labrador ice (0.17–0.18), but net TOA +0.46* in decade 2 with the heat carried below 700 m, global −0.5 K, and NH ice volume +2.2 to $+2.5 \times 10^3$ km^3.
- **gmrd1800: rejected.** Small R_d at high latitude means low K there: SH September volume -1.67×10^3 km^3 and extent −0.73 M km^2 over 2121–2139, 60–90 S +0.33 K.

Suggested follow-up: gmhemi with an NH value of 1500 to reduce the Labrador overshoot.

61 Round 44 (2026–10–02→03): gmhemi1800 against observations, and against a pre-industrial reference

61.1 Release evaluation

`reval` on gmhemi1800 2125–2139 (`plots/reval_gmhemi1800_2125-2139/`). CMPI is the model error over the CMIP6 ensemble-mean error (cmpitool default, 30 CMIP6 models; below 1 is better than CMIP6); a PI control scored against present-day observations carries the epoch offset in every number (`plots/reval_gmhemi1800_2125-2139/cmpi_vs_cmip6.txt`).

	ccnice 2101–2129	crunveg 2101–2119	**gmhemi1800 2125–2139**
CMPI overall	0.865	0.870	**0.861**
entries better than CMIP6	286/400	286/400	**296/400**
nino34	1.109	1.119	1.084
tropics	0.965	0.981	0.954
northmid	0.821	0.819	0.801
southmid	0.812	0.819	0.831
arctic	0.830	0.811	0.839
antarctic	0.675	0.692	0.678

Largest gains against crunveg: Niño3.4 10 m temperature and T2m (−0.230, −0.228), NH mid-latitude sea ice (−0.176), tropical 300 hPa wind (−0.118), SH mid-latitude MLD (−0.080), Antarctic sea ice (−0.071). Largest losses: NH mid-latitude MLD (+0.179, the Labrador overshoot), SH mid-latitude salinity at 10 and 100 m (+0.142, +0.155), Arctic surface salinity (+0.101) and Arctic winds (+0.114 at 300 hPa), NH mid-latitude 1000 m temperature (+0.071). Note that crunveg and gmhemi1800 are not on the same years, so drift between the periods is included.

61.2 Bug XI: the Gregory plot dropped the snow enthalpy

The `reval` part20 Gregory plot read a surface imbalance of 1.51 W m^{-2} for this window. It used $L_f = 334000$ (J kg^{-1}) to convert snowfall that is in metres of water equivalent, where $\rho_w L_f = 3.3355 \times 10^8$ J m^{-3} is needed, so the snow-enthalpy term came out a thousand times too small, effectively zero. With the term restored the snow enthalpy is 0.81–0.86 W m^{-2} and the surface imbalance **0.67** W m^{-2} (2130–2139: 0.673), the same omission Round 30 found behind the "−1.0 W m^{-2} structural leak" (§46). The "net surface flux 1.47" quoted for gm2500 in the GM discussion had the same omission; corrected, 0.641 (`data/clim/gm_arms_global_2121-2128.txt`). Fixed together with seven other albedo-port fixes in `release_evaluation_tool2` PR #7 (`https://github.com/AWI-ESM/release_evaluation_tool2/pull/7`, open).

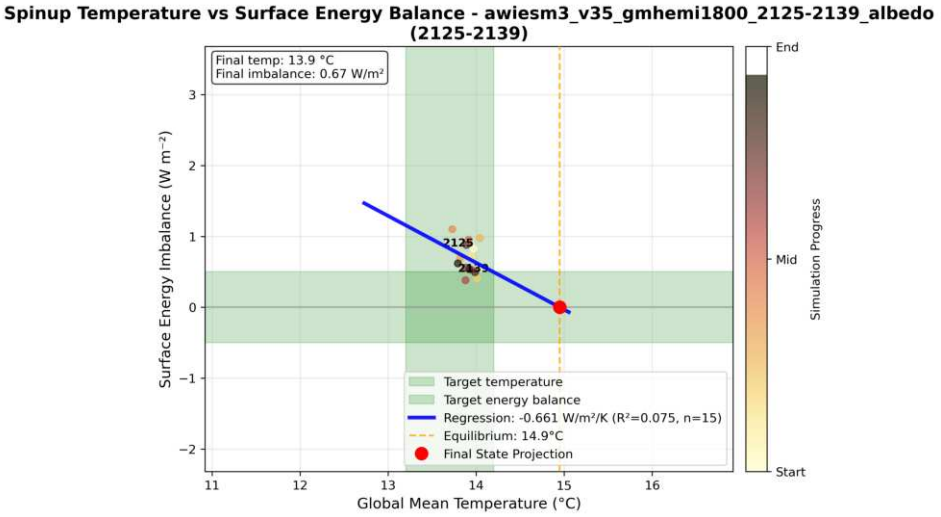


Figure 51: Corrected Gregory plot, gmhemi1800 2125–2139: global T2m against surface energy imbalance, final 0.67 W m^{-2}. The regression explains almost nothing ($R^2 = 0.075$, $n = 15$), so its extrapolated equilibrium of 14.9 °C carries no weight.

61.3 The bias against a pre-industrial reference

ERA5 1990–2014 is not a PI target. The PI bias is built per 5° box as (model − ERA5 1990–2014) + the HadCRUT5 warming from 1850–1900 to 1990–2014 (`scripts/analysis/pi_t2m_bias_epoch_corrected.py`, `scripts/analysis/pi_t2m_bias_land_ocean.py`, `scripts/figures/pi_t2m_bias_maps.py`). Globally, the model is −0.36 K against ERA5 and the HadCRUT5 global series gives 0.817 K of warming: **the model is +0.46 K too warm for the pre-industrial**. The box-based estimates agree (+0.48 with uncovered boxes filled by the latitude-row mean, +0.52 over covered boxes only, 39 % of the area). ob1800 is within 0.1 K of gmhemi1800 globally (+0.49 against +0.52 over covered boxes) but not in the Arctic winter, where gmhemi1800 is colder by 0.48 K at 60–90 N and 0.69 K at 70–90 N (`data/clim/pi_t2m_bias_ob1800_2125-2139.txt`).

PI bias (K)	ANN	DJF	JJA	cover	by surface	land	ocean
global	+0.52	+0.37	+0.49	39 %	JJA 30–45 N	+2.21	+1.42
90–60 S	+0.66	+0.23	−0.85	2 %	JJA 45–60 N	+2.08	+1.37
60–30 S	+0.97	+1.28	+0.55	53 %	DJF 60–30 S	+1.94	+1.38
30 S–30 N	+0.03	+0.09	+0.03	38 %	DJF 60–75 N	−0.81	+0.43
30–60 N	+1.01	+0.18	+1.66	70 %	DJF 75–90 N	−2.08	−0.61
60–90 N	+0.60	−1.01	+0.53	31 %	ANN 60–75 N	+0.17	+1.36

gmhemi1800 2125–2139 (`data/clim/pi_t2m_bias_gmhemi1800_2125-2139.txt`, `data/clim/pi_t2m_bias_land_ocean_gmhemi1800_2125-2139.txt`).

The pattern: **the tropics are right**; the excess sits in **NH summer mid-latitudes** (land +2.1/+2.2, ocean +1.4) and in **SH summer mid-latitude ocean** (+1.4, the known Southern Ocean cloud deficit). **The Arctic winter is cold** (land 60–75 N −0.8, 75–90 N −2.1). The maps add an Antarctic winter warm ring at 60–70 S, warm eastern-boundary upwelling coasts and a cold blob south of Greenland (Fig. 52).

Two caveats bound this. HadCRUT5 1850–1900 coverage is thin at the poles (2 % of 90–60 S, 31 % of 60–90 N), so the polar numbers rest on few boxes. And the AMIP pair showed that the model under-warms Siberia between epochs (§17.2), so an observation-based PI target is not the model's own PI.

The line still cools. Over crunveg 2101–2119 followed by ob1800 2120–2139, global T2m trends -0.059 ± 0.025 K/decade and 60–90 N -0.141 ± 0.115 (2σ; `data/clim/crunveg.csv`, `data/clim/series_ob1800.csv`); with gmhemi1800 as the continuation, -0.053 ± 0.025 and -0.203 ± 0.099 (`data/clim/series_gmhemi1800.csv`). Over 2130–2139 gmhemi1800 has net TOA 0.52, net surface 0.67 and ocean uptake 0.69 W m^{-2}, most of it below 700 m (0–700 m 0.084, 700–2000 m 0.332, >2000 m 0.167). **This reframes the campaign's founding trade-off**: against ERA5 the model was too cold with a positive imbalance, two faults pulling against each other; against a PI reference it is too warm with a positive imbalance, and the two point the same way.

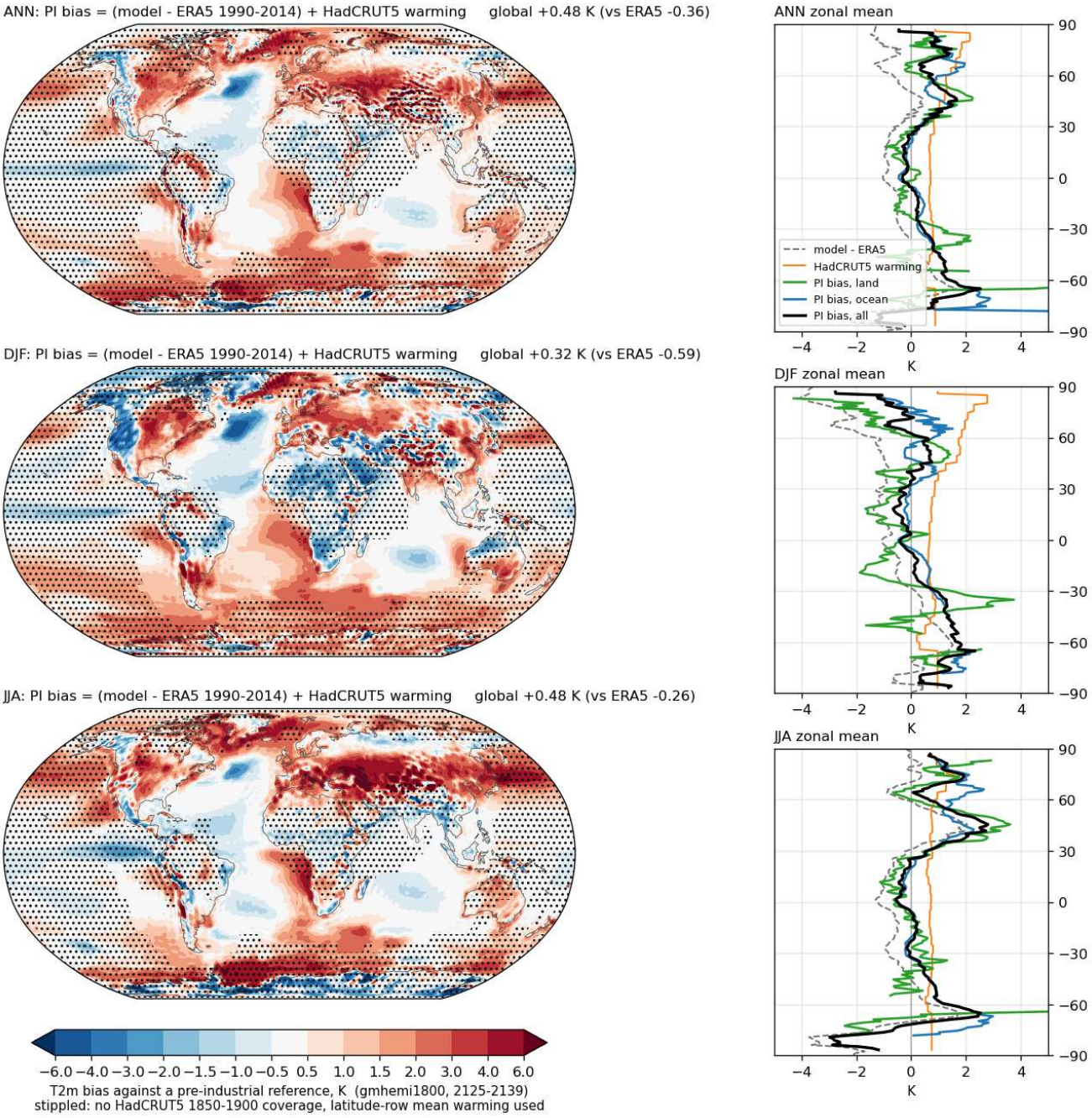


Figure 52: T2m bias of gmhemi1800 2125–2139 against a PI reference, (model − ERA5 1990–2014) + HadCRUT5 warming 1850–1900 → 1990–2014. Stippled: no HadCRUT5 1850–1900 coverage, latitude-row mean warming used. Right: zonal means for land, ocean and all, with the uncorrected model − ERA5 dashed and the HadCRUT5 warming in orange.

61.4 The warm NH summer land is dry and sunny

JJA, gmhemi1800 2125–2139 against ERA5 T2m and GPCP precipitation 1990–2014 and CERES surface SW 2000–2021 (`scripts/analysis/warm_regions_diag.py`, `data/clim/warm_regions_diag_gmhemi1800_2125-2139.txt`):

box	T2m − ERA5 (K)	precip / GPCP (mm d^{-1})	SW↓ − CERES (W m^{-2})	evap. fraction	soil water top layer
C Asia 40–55 N 50–110 E	**+2.9**	0.7 / 1.3	+20.1	**0.25**	**0.09**
Europe 45–60 N 0–40 E	+1.3	2.1 / 2.7	+27.8	0.65	0.21
N America 40–55 N 115–85 W	+1.6	2.2 / 2.5	+10.8	0.63	0.23
Siberia 55–75 N 60–180 E	−1.1	2.3 / 2.0	−7.9	0.64	0.34

Central Asia is the clearest case: half the observed rain, 20 W m^{-2} too much sun, a quarter of the available energy

going into evaporation, and a dry top soil layer.

Vegetation cover. LPJ-GUESS sends daily foliage projective cover, FPC $= 1 - e^{-k(\mathrm{LAI+SAI})}$, as `cvl`/`cvh` (`lpj_guess/framework/framework.cpp`, lines ∼1331–1477), and when LPJ-GUESS is coupled OpenIFS uses those fields directly (`ifs-source/surf/module/surfbc_ctl_mod.F90`, ∼285–320); uncoupled OpenIFS multiplies climatological cover by `RVCOV` per type (`susveg_mod.F90`). ~~The coupled cover is double-counted: LPJ-GUESS already applies a Lambert–Beer cover and OpenIFS multiplies it by `RVCOV` again.~~ The `RVCOV` branch is taken only when LPJ-GUESS is *not* coupled; raised and falsified the same day. Like-for-like effective JJA cover, IFS default against LPJ-GUESS: central Asia 0.57 against 0.31, Europe 0.87 against 0.69, N America 0.79 against 0.65. **LPJ-GUESS vegetation is sparser, most of all over central Asia.** Whether the sparse vegetation causes the dry, warm summer or follows from it cannot be separated from this run; a dry-feedback loop is possible. ~~The steppe LAI of 4.9 is too high.~~ Those are per-tile LAI values; the cover-weighted LAI over the central Asian box is 0.82.

61.5 Marine biogenic CCN, reconsidered and declined

With the global PI bias now warm, the DMS lever of Rounds 24–26 was reconsidered as a global cooling lever. Its record: $S = 166$ gives global SW CRE $-1.72\,\mathrm{W\,m^{-2}}$ in AMIP and SO DJF -7.25 (§40); the tropical cost binds below the published coefficient floor (§42.4); the coupled probe 11F was cancelled at 20 of 50 years and never scored (§43.10). **Declined for the last round.**

62 Round 45 (2026–10–03): the Antarctic pack is diffuse, not missing, and drives too freely

62.1 Where the warm ring sits

T2m against ERA5 by ice class, ice threshold 0.15, observed ice HadISST 2.2 1990–2014 (`data/clim/warm_regions_diag_gmhemi1800_2125-2139.txt`):

case	class	area	model − ERA5	conc. model	conc. obs
SH JJA	both ice (pack)	37.7 %	**+1.77**	**0.74**	**0.84**
	obs ice, model open	2.4 %	+1.35	0.10	0.28
	model ice, obs open	1.6 %	−0.91	0.27	0.07
	both open	58.3 %	+0.34	0.01	0.01
NH DJF	both ice (pack)	43.2 %	−2.30	0.83	0.89

The Antarctic winter warm ring sits over a too-diffuse pack, not over a missing edge: the pack where both agree there is ice is 1.8 K too warm with concentration 0.74 against 0.84, and the missing-ice class is only 2.4 % of the area. The NH winter pack is cold.

62.2 McPhee ocean-to-ice heat flux: wrong in both hemispheres

In the coupled build FESOM grows ice from $A(Q_{atm,ice} - Q_{oc,ice})$ with $Q_{oc,ice} = (T - T_f)\,\gamma_t\,\rho c_p$ and γ_t a constant $10\,\mathrm{m\,d^{-1}}$. The McPhee form replaces it with $St^* u^*$ ($St^* = 0.0057$, u^* from the ice–ocean velocity difference, $u^*_{min} = 0.005\,\mathrm{m\,s^{-1}}$), which exchanges at 0.31–0.56 of the old constant over the pack in 2127–2129 (`data/clim/ice_ocean_ustar_ihf1.txt`). FESOM branch `feat/ice-ocean-heat-ustar` (`1e6cb53a`); **PICAL__crunveg__ihf1** (on) against **ihf0** (off), new build, 1800 s, 2120–2129. ihf1 − ihf0 (`data/clim/ihf0.csv`, `data/clim/ihf1.csv`):

$10^3\,\mathrm{km^3}$	ihf1 − ihf0	paired $2\sigma/\sqrt{N}$
SH September volume	-1.25^*	1.22
NH September volume	$+1.04^*$	0.49
NH April volume	$+0.62^*$	0.54

Wrong direction in both hemispheres: the Antarctic needs more ice and loses it, the Arctic has too much and gains it. Rejected (`data/clim/ice_ocean_heat_ihf0.txt`, `data/clim/ice_ocean_heat_ihf1.txt`, `data/clim/marginal_ice_budget.txt`).

62.3 Bug XII: lead closing is hard-coded in the coupled build

In the OASIS-coupled build FESOM uses `ice_thermo_cpl.F90`, which declares `h0min = 0.5, h0max = 1.5` as local constants (line 62 on `main 82c67a4a`). The namelist `h0` and `h0_s` are read only by `ice_thermo_oce.F90`, which is compiled out when coupled: **they are inert in AWI-ESM3**. Separately, the `/ice_therm/` namelist read (`ice_init.F90:123`) passes `iostat` and never checks it, so an unknown name silently stops the read and every later name in the group falls back to its default. The hemispheric lead closing (SH `h0min` 1.0) was removed from the coupled code in 2022 (`a06d3ff0`, `d0556ef7`). Round 38's `h0min` experiment ran on the separate `feat/leadclose-h0min` branch that exposes it (§55.2). **`h0min` is ruled out as a lever.**

62.4 The winter deficit opens in autumn

SH 50–90 S monthly area budget, gmhemi1800 2130–2139, M km^2 per month (`scripts/analysis/ice_winter_budget_hemispheres.py`, made machine-portable through the environment variable `SIC`; `data/clim/ice_winter_budget_hemispheres_gmhemi1800_2130-2139.txt`):

month	3	4	5	6	7	8	9	10
thermodynamic	+1.44	+3.60	+5.24	+5.07	+4.32	+3.32	+2.21	+0.24
dynamic	−0.40	−1.17	−1.69	−2.09	−2.45	−2.54	−2.41	−1.58
area model	1.21	3.66	7.64	11.43	14.24	15.99	16.38	15.51
area HadISST	2.20	6.23	10.57	13.92	16.89	18.43	19.23	18.84

The area deficit opens in April–May (−2.6 to −2.9 M km^2) and then stays flat through June–September while the dynamic loss peaks. That is consistent with Round 33's autumn entrainment of the +0.9 K layer at 100 m (§50.6), not with a winter dynamics deficit.

62.5 Ice drift against observations

OSI-455 v1.0 daily 24-hour drift, 2005–2014, satellite vectors only (`/albedo/work/user/jstreffi/obs/osisaf_drift`); NCEP-2 daily 10 m winds over the same years (`/albedo/work/user/jstreffi/obs/NCEP2/drift_compare`). Model gmhemi1800 2130–2139, JJA in the SH (`scripts/analysis/ice_drift_vs_osisaf.py`, `scripts/analysis/ice_wind_response.py`, `data/clim/ice_drift_vs_osisaf_gmhemi1800_sh.txt`, `data/clim/ice_wind_response_gmhemi1800.txt`):

	speed (cm s^{-1})		divergence (% d^{-1})		|div|	wind factor (%)	turning (°)
sector	obs	model	obs	model	ratio	obs / model	obs / model
all	15.17	19.64	+0.265	+0.306	0.95	1.43 / 2.43	+23.6 / +22.7
Weddell	14.82	18.30	+0.250	+0.042	**0.61**	1.55 / 2.65	+23.4 / +11.8
Indian	15.78	20.64	+0.358	−1.371	1.09	1.35 / 2.88	+22.5 / −13.0
Pacific	12.31	22.75	+0.637	−0.828	1.00	0.86 / 2.82	+23.7 / +5.4
Ross	16.19	19.76	+0.262	**+1.678**	1.27	1.57 / 2.76	+24.1 / **+62.7**
Amundsen-Bell.	15.23	19.93	−0.074	**+1.255**	0.90	1.47 / 2.75	+25.0 / **+45.4**

The SH ice moves 1.29 times too fast (19.6 against 15.2 cm s^{-1}); the NH winter ice 1.19 times (11.4 against 9.6, `data/clim/ice_drift_vs_osisaf_gmhemi1800_nh.txt`). **Antarctic-wide the deformation is about right** — net divergence +0.31 against +0.27 % d^{-1}, total deformation ratio 0.95 — **but by sector it is wrong**: Ross and Amundsen–Bellingshausen diverge where observations are near neutral, the Indian and Pacific sectors converge where observations diverge, and the Weddell deforms at 0.61 of observed. The concentration deficit is near-uniform at 0.03–0.06 by sector except Amundsen–Bellingshausen, 0.63 against 0.79, which holds 0.69 of the 2.45 M km^2 of loose ice.

The winds are not the cause. Model 10 m wind over the ice is 7.2 against NCEP-2's 8.0 m s^{-1}. The JJA Amundsen Sea Low is 977.4 hPa at 138 W against 980.5 at 152 W, with the same relative depth (−6.1 against −6.2 hPa below the sector mean). The model wind field over the ice is less convergent than observed (−19.0 against −32.1 % d^{-1}). **The ice over-responds**: wind factor 2.4 % (2.7–2.9 by sector) against an observed 1.4 % (0.9–1.6); the observed turning angle is about +24° everywhere, the model's +23° overall but +63° in the Ross, +45° in Amundsen–Bellingshausen, −13° in the Indian and +12° in the Weddell sector. The linear wind-to-drift fit has $R^2 = 0.70$ overall and 0.91–0.94 in four of five sectors, close to free drift, against 0.53 observed (0.31–0.59 by sector).

The effective air–ice drag, from FESOM's `atmice` stress and OpenIFS daily 10 m wind over the pack ($a \geq 0.8$, 55–80 S; `scripts/analysis/ice_drag_effective.py`, `data/clim/ice_drag_effective_gmhemi1800.txt`),

is $C_{d,ai} = 2.0\text{–}2.2 \times 10^{-3}$; with `Cd_oce_ice` 0.0055 the implied free-drift factor is 2.2 %, and the ice–ocean turning angle θ_{io} is 0.

Conclusion: the winds are fine; the ice over-responds and turns the wrong way in the Ross and Amundsen–Bellingshausen sectors. Slowing the ice would tighten the A–B/Ross pack, but would likely reduce SH extent and add Arctic volume, so this is **documented as a known bias and not tuned**. Caveats: the observation years 2005–2014 had a positive SAM, and satellite drift misses fast marginal-ice-zone ice.

62.6 The stack, the pull requests, and the last round

Pull requests. FESOM #1060 (ice skin from the atmosphere) merged as `57c433e5`, #1087 (the two CVMix bugs) as `18926b72`, #1064 (`cavity_gamma_scale`) as `f3c3d3fa`. Open: #1103 (IDEMIX smoothing per unit time) and #1112 (hemispheric GM), and `release_evaluation_tool2` #7.

A fresh install. `awiesm3-develop-cc` was installed on 2026–10–02/03 at `/albedo/work/user/jstreffi/model_codes/awiesm3-develop-cc` with `esm_master comp-awiesm3-develop-cc`:

component	branch	commit
fesom-2.8	`main`	`82c67a4a` (contains #1060, #1064, #1087)
lpj_guess	`lpj_guess_awiesm3`	`179fb0d`
oasis	`local_combined_fixes`	`898c8b2`
oifs-48r1	`awiesm3-develop` (fork)	`f6d1860`, 5 commits past `872c4d2`: 2 m T to the ice sheet, SP fixer
rnfmap	`enthalpy_on_ocean_side`	`a635cc6`
xios	`main`	`9c33506d` (re-cloned)

FESOM is built coupled, in double precision, with `RECOM_COUPLED=ON` and XIOS on.

The last round, as agreed.

1. gmhemi with NH `K_GM_max` 1500, against the Labrador overshoot.
2. Continue the current tuning on the `-cc` configuration: FESOM and LPJ-GUESS warm start, REcoM on in concentration-driven PI mode, OpenIFS cold start.
3. Document as structural biases of v3.5: the Southern Ocean cloud deficit, the diffuse Antarctic pack, the stratocumulus coasts, and the warm NH summer land.

63 Round 46 (2026–10–03): the critical path as one record, and vertical mixing re-scored on heat

Two things happened on the same day and belong together: the adopted tuning was taken to the carbon-cycle configuration, and the whole production line was put on one axis for the first time. The second changed how Round 39 reads.

63.1 The line as one record, 1850–2139

The critical path is five experiments: **PICAL** 1850–1919, **PICAL__momixoff** 1920–1939, **PICAL__ccnice** 1940–2099, **PICAL__crunveg** 2100–2119 and **PICAL__crunveg__gmhemi1800** 2120–2139. `release_evaluation_tool2` branch `feat/critical-path-spinup` stitches them: a per-year cache for the diagnostics that read 3D ocean fields (so the 250 years on levante were reduced there and only the cache shipped), a new AMOC time series (`part27`), and annotations for the eleven changes along the line. The evaluation window is gmhemi1800 2125–2139. Output: `plots/reval_critical_path_1850-2139/`. OpenIFS was cold-started at 1920, 1940, 2100 and 2120, so the first year of each segment is not a response.

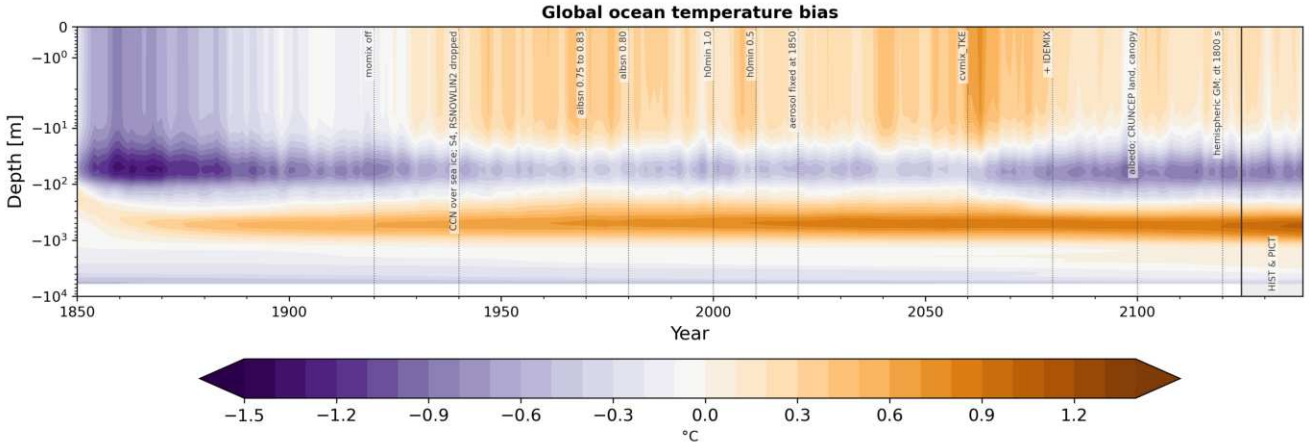


Figure 53: Global-mean ocean temperature against PHC3 along the critical path. The warm layer at 300–1000 m builds through all 290 years and no change along the line reverses it; the cold layer at 50–100 m weakens under KPP and returns with `cvmix_TKE` at 2060.

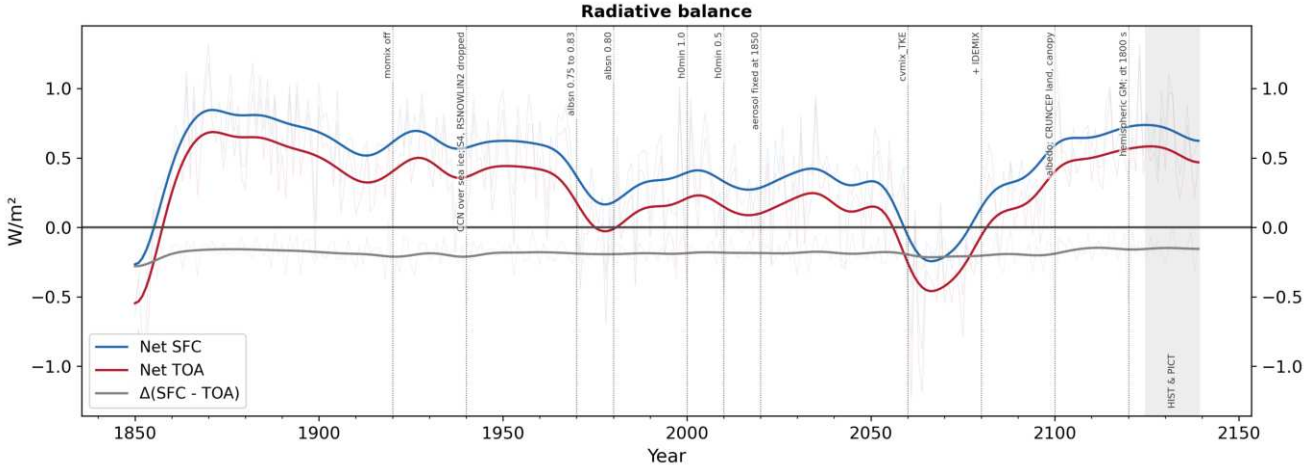


Figure 54: Net TOA and net surface flux along the critical path (smoothed; thin lines annual). TOA starts near $+0.7\,\mathrm{W\,m^{-2}}$, reaches zero in the `albsn` 0.83 decade, goes negative under `cvmix_TKE` alone and returns to +0.5 to +0.6 with IDEMIX and the forested land. The surface-minus-TOA offset stays at −0.15 to −0.2 throughout: the atmosphere's own closure error does not drift.

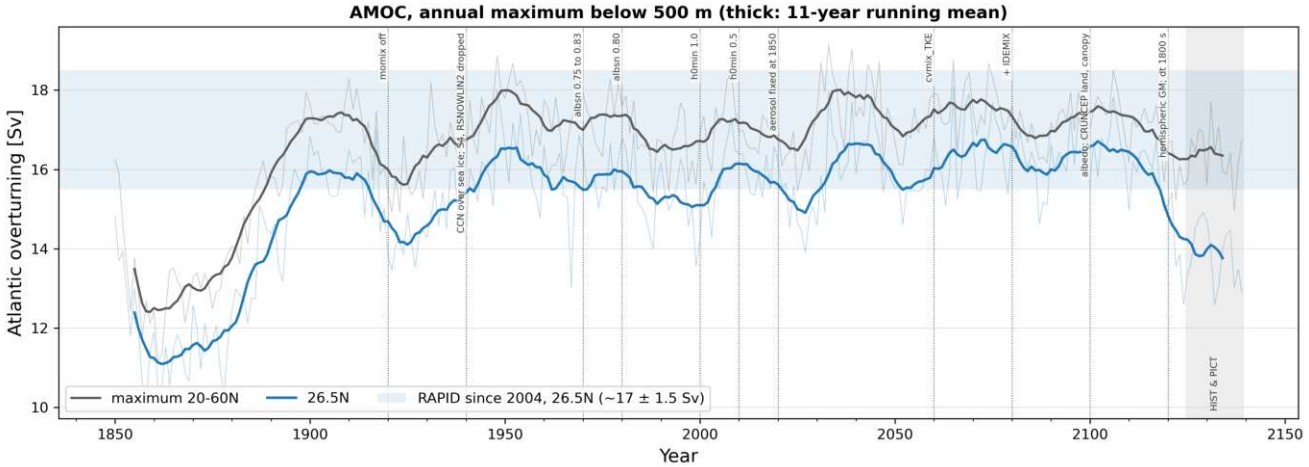


Figure 55: Atlantic overturning along the critical path: annual maximum below 500 m at 26.5 N and over 20–60 N. It spins up over the first 50 years, holds 15–16.5 Sv at 26.5 N for two centuries through every atmospheric and sea-ice change, and drops at the 2120 branch.

window	AMOC 26.5 N (Sv)	maximum 20–60 N (Sv)
1850–1869	11.8	13.1
1900–1919	15.8	17.1
1940–2059	15.8	17.2
2080–2099	16.3	17.1
2101–2119 (crunveg)	16.3	17.3
2121–2139, ob1800 (GM 2500)	15.0 ± 0.3	16.0
2121–2139, gmhemi1800	13.9 ± 0.2	16.3

The drop at 2120 has two parts. About half is common to the branch: ob1800, which keeps GM 2500 at the same 1800 s step, falls to 15.0 Sv and is still falling (14.4 over 2130–2139), in step with its Labrador collapses. The other 1.1 Sv is the hemispheric GM, which moves the cell north rather than weakening it: the 20–60 N maximum holds at 16.3 Sv near 36 N. RAPID is about 17 ± 1.5 Sv at 26.5 N, a present-day reference.

63.2 Heat by depth: IDEMIX is what warms the mid-depth ocean

Trends of the global-mean temperature per depth band, from the yearly profiles behind the Hovmöller (`cdo fldmean` per level, averaged by layer thickness; mK/decade, ±2 s.e.):

segment of the line	100–300 m	300–700 m	700–1500 m	1500–2500 m
KPP 2021–2059	+20	+7	$+8.5 \pm 0.9$	+2
`cvmix_TKE` 2061–2079	−142	−27	$+0.3 \pm 1.3$	+6
TKE+IDEMIX 2081–2099	−39	−19	$+26.9 \pm 1.0$	+10
crunveg 2101–2119	+50	+17	$+28.3 \pm 2.4$	+11
gmhemi1800 2121–2139	0	+42	$+42.0 \pm 1.0$	+7
ob1800 2121–2139	+1	+39	$+31.6 \pm 1.4$	+12

The 700–1500 m layer is flat under TKE alone and warms from the leg IDEMIX came in. The segments are confounded by everything else that changed, so the same quantity was computed for the four arms of Round 39 (§56), which share one restart and one period (`data/clim/mixing_arms_temp_trend_by_depth.txt`):

arm, 2021–2039	0–100 m	100–300 m	300–700 m	700–1500 m	1500–2500 m
KPP (control)	+28	+38	+25	$+11.3 \pm 1.7$	−2
`cvmix_KPP`	−91	−52	+4	$+15.0 \pm 0.9$	+3
`cvmix_TKE`	−214	−148	−18	$+8.9 \pm 1.8$	+4
`cvmix_TKE+IDEMIX`	−307	−128	+3	$+31.4 \pm 1.1$	+9

- **IDEMIX, not TKE, drives the mid-depth warming**: +31 mK/decade at 700–1500 m against +9 for TKE alone and +11 for KPP.
- **Both TKE arms drain the upper ocean**: by 2030–2039 the top 300 m is 0.26–0.34 K colder than under KPP. This is the upper-ocean heat drain of §57, seen at its source.
- **`cvmix_KPP` is a null for sea ice and not for heat**: its top 100 m is 0.17 K colder than native KPP by 2030–2039, its 100–300 m 0.15 K. Part of "TKE minus KPP" may therefore belong to the CVMix code path rather than to the scheme (§.3).
- **The 2120 branch adds to it**: 700–1500 m warms at +42 in gmhemi1800 against +32 in ob1800, so about +10 is the hemispheric GM, taken from the 1500–2500 m layer (+7 against +12). The doubling at 300–700 m (+17 → +40) is in both arms and is not the GM.

Round 39 adopted TKE+IDEMIX on Antarctic summer ice, cavity melt and the Southern Ocean cloud term, and stated a northern cost. It did not score heat by depth. On that measure KPP was the best of the four and IDEMIX the worst, while KPP held the heat at the surface (the line was at 14.4 °C in 2060–2064) and lost Antarctic summer ice.

In flight: PICAL__crunveg__gmhemi__tke, 2140–2159 from gmhemi1800's 2139 restart, one variable changed (`mix_scheme cvmix_TKE`, IDEMIX off), same FESOM library. It replaces the NH-1500 arm as the next run of the last round. There is no same-years control.

63.3 The carbon-cycle configuration: PICAL_cc

The gmhemi1800 tuning on `awiesm3-develop-cc`, 2140 onward: FESOM physics and LPJ-GUESS warm from gmhemi1800 2139–12–31, REcoM (2p3z2d) cold from climatology, OpenIFS cold. FESOM is `main` `82c67a4a` plus PR #1112 (`90074f13`), double precision.

- **Pre-industrial, concentration-driven, no emissions**: `with_co2_concdriven`. Radiation uses the CMIP 1850 concentration, OpenIFS sends REcoM the same uniform value (`ece_recom_set_state.F90`), and LPJ-GUESS uses its fixed 1850 CO_2; the coupled value it receives is only checksummed. The prognostic CO_2 tracer is diagnostic, started at 284 ppm.
- **REcoM cold under a physics restart**: with `recom_restart = .false.` the tracers are optional on the restart read and keep the climatology (GLODAP PI DIC and alkalinity, WOA nutrients and oxygen, PISCES iron).
- **Cost**: 0.674 s per model hour on 25 nodes, 6.7 times the physics-only line. A 5-year leg takes 8 h 20.
- **Stopped at 2145–01–01** with a complete restart including REcoM, pending performance work on the configuration.

Two traps, both silent, and one that was not a trap. The REcoM `bin_2p3z2d/namelist.io` is a different output list: it drops 37 outputs of the physics run, among them the ice budget tendencies. And with `cmip7_cmor_output` the FESOM XIOS templates (`xios_xml_cmip7`) defined no BGC field at all, so **the 2140–2144 leg has no REcoM diagnostics**. Both are fixed in `esm_tools` (`fdde4730e`: a `namelist.io.recom` that is the physics list plus the REcoM streams, and a gated BGC file group in the CMIP7 templates), checked by resolving a runscript without any override; no leg has yet written through the new file group. `recom_input_path` is now set per machine. The third suspicion, that `bin_2p3z2d/namelist.tra` brings surface salinity restoring back (`surf_relax_S` 1.929×10^{-6}), was wrong: `awiesm3.yaml` forces it to 0 for every v3.5 variant, and the resolved namelist has 0 without any runscript entry.

64 Round 47 (2026–10–03→05): TKE alone, snow albedo 0.82, and the IDEMIX control arm

Round 46 left one question in flight: does `cvmix_TKE` without IDEMIX keep the Antarctic ice and stop the mid-depth warming on today's configuration? Three runs answer it, each one change from its parent, all on hemispheric GM at 1800 s.

64.1 PICAL_crunveg_gmhemi_tke, 2140–2159: the mid-depth warming stops, the surface warms

Against its parent gmhemi1800 (last 15 years of each): net TOA $+0.53 \rightarrow +0.21\,\mathrm{W\,m^{-2}}$, the 700–1500 m trend $+42 \rightarrow +9$ mK/decade and 300–700 m $+43 \rightarrow 0$. The price is at the surface: global T2m rises $+0.19 \pm 0.06$ K/decade, April Arctic volume falls $-1.5 \times 10^3\,\mathrm{km^3}$/decade and February Antarctic extent goes 1.81 to 1.34 (`data/clim/tke_vs_idemix_branch_score.txt`).

64.2 PICAL_crunveg_tke_albsn082, 2160–2209: the surface warming stops

`albsn` $0.80 \rightarrow 0.82$, the bracket left open in Round 38 (0.83 ran away in the Arctic). Over 50 years global T2m is flat ($+0.008 \pm 0.020$ K/decade), the Arctic is stable (melt-season snow-free fraction 3.3–5.0 %, against 1.2 % in the runaway), and the pre-industrial bias is +0.55 K. The overturning is not collapsing: 13.5–14.0 Sv at 26.5 N in density space for all five decades, 18.6–20.1 at 55 N (`data/clim/dmoc_atlantic_index.txt`). What does not hold is the Antarctic summer pack: February extent 1.88 in the 2160s, 1.13 in the 2200s, most of it after 2190, with the clear-sky absorbed solar south of 60 S rising alongside.

64.3 PICAL_crunveg_idemix_albsn082, 2170–2209: IDEMIX is what holds the Antarctic

A control arm with one change, IDEMIX back on, over the same years.

trend 2180–2209, per decade	TKE only	TKE + IDEMIX
Antarctic extent, February (10^6 km^2)	-0.33 ± 0.20	-0.01 ± 0.24
Antarctic volume, September (10^3 km^3)	-0.75 ± 0.38	-0.10 ± 0.39
clear-sky absorbed solar, 60–90 S (W m^{-2})	$+0.67 \pm 0.53$	-0.11 ± 0.50
T2m global (K)	$+0.04 \pm 0.04$	-0.03 ± 0.05
ocean 700–1500 m (mK)	$+8 \pm 1$	$+29 \pm 1$
ocean 300–700 m (mK)	-1 ± 2	$+16 \pm 2$

In 2200–2209 the IDEMIX arm is 0.27 K colder globally (13.69 against 13.97 °C), has 0.69×10^6 km^2 more February ice in the south and 1.5×10^3 km^3 more April ice in the north, and 0.4 Sv less overturning at 26.5 N. Its net TOA is *not* lower (+0.36 against +0.31; +0.50 against +0.16 the decade before): it is colder at the surface because it buries the heat, not because it is closer to balance (`data/clim/idemix_arm_pairs.txt`).

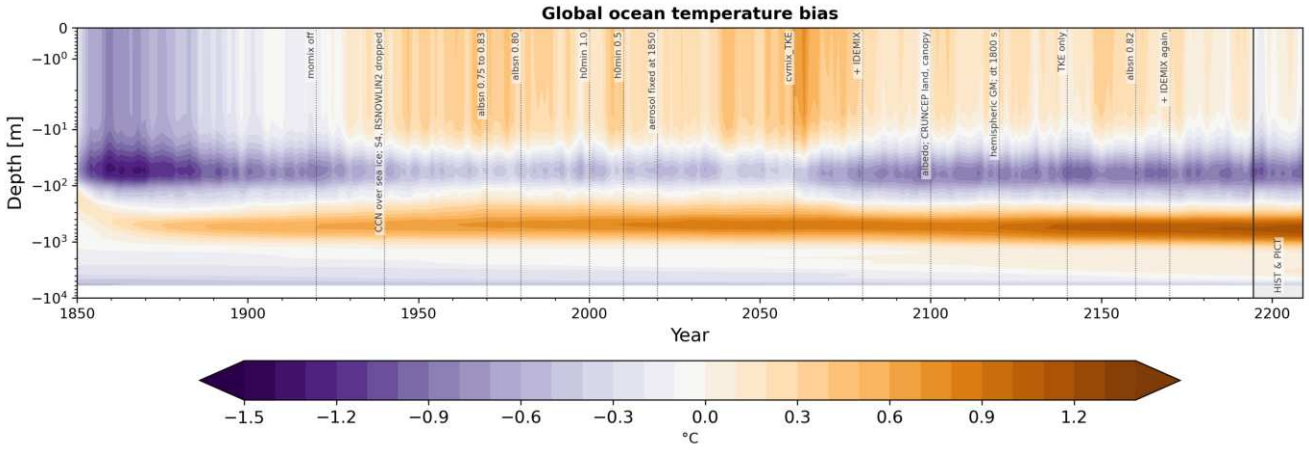


Figure 56: Global-mean ocean temperature against PHC3 along the critical path continued on the IDEMIX arm, 1850–2209. The mid-depth warm layer pauses over 2140–2169 (TKE only) and resumes when IDEMIX returns at 2170. The TKE-only continuation is `plots/reval_critical_path_tke_1850-2209/`.

So the two candidates trade the same heat: TKE alone leaves it at the surface and the Antarctic erodes; IDEMIX takes it down and the deep ocean drifts. Three critical-path evaluations now exist on albedo (to 2119, to 2139 and to 2209 on either line) and one on levante for the KPP era (1850–2059); all are under `plots/reval_critical_path_*`.

65 Round 48 (2026–10–05): AWI-ESM3 v3.5.0, every component on a tag

The release is pinned to tags only; `esm_tools` never points at a branch (`d4aafd571`). One set for the physics-only, carbon-cycle and ice-sheet variants.

component	tag	commit
FESOM	`2.8.1`	`44e4f9c0` (main, with #1112 hemispheric GM)
OpenIFS	`48r1v5`	`f6d1860d`
LPJ-GUESS	`4.1.13`	`179fb0d6`
OASIS3-MCT	`5.3`	`898c8b2e`
XIOS	`2.5.4`	`9c33506d`
runoff mapper	`1.4`	`36c888f7`
PISM, dEBM, ismm (ice-sheet variant)	`1.2.0-awi.1`, `1.0-awi.1`, `4.1.8-awi.1`	forks

Validation. A tree built from exactly these commits repeats 2190–2199 of the tuning line from the same restart (**PICAL_v350val**): every score is inside the noise of the pair except NH ice volume (+1.3 and $+1.6 \times 10^3$ km^3, within the parent's decadal range) and 300–700 m (+12 mK) (`data/clim/v350_validation_pairs.txt`). The carbon-cycle build (**PICAL_cc_v350val**) is identical to PICAL_cc in ocean and ice and writes its 70 BGC files. The ice-sheet variant builds on albedo and has not run there. The tuned settings (mixing, GM values, `albsn`, 1800 s) are still in runscripts, not in `awiesm3.yaml`, because the ocean configuration is not settled (§67). Details: `notes/awiesm3_v3.5.0_component_tags.md`.

66 Round 49 (2026–10–05→06): Bug XIII, REcoM exchanged gases against a vacuum

The first carbon-cycle leg with BGC output showed global-mean surface pCO_2 falling from 253 to 151 μatm in 24 months, surface DIC from 2006 to 1882 mmol m^{-3} with alkalinity unchanged, and a CO_2 flux out of the ocean at every node. The giveaway was that `dpCO2s` equalled `pCO2s` + 6.7 in every month: the flux saw an atmospheric pCO_2 of -6.7 μatm, which is $xCO_2\,(P_{\mathrm{atm}} - pH_2O)$ at $P_{\mathrm{atm}} = 0$ (recomputed from the model SST: -6.6, node by node $r = 0.996$).

Cause. `recom_main.F90` takes one atmosphere under `__oasis` and FESOM's `press_air` otherwise. FESOM proper is compiled with `__oasis`; the REcoM library has its own `CMakeLists.txt` and gets `__oifs __recom __usetp` only. `press_air` is filled on the standalone forcing path and stays zero in a coupled run. Every OpenIFS-coupled REcoM run since the CO_2 coupling went in is affected, including the tagged v3.5.0-cc; the physics is not, because nothing feeds back in concentration-driven mode.

Fix, in two steps (one-month tests from the same start, January 2140):

	tag 2.8.1	one atmosphere	OpenIFS pressure
atmospheric pCO_2 seen by the flux (μatm)	-6.7	277.6	276.8
surface ocean pCO_2 (μatm)	253.4	271.4	271.3
CO_2 flux into the ocean (mmol C m^{-2} d^{-1})	-32.2	$+1.6$	$+1.4$
O_2 flux into the ocean (mmol m^{-2} d^{-1})	-325	-5.7	-7.1

A constant atmosphere is 2 % too high over the Southern Ocean all year, so the second step couples the real field: OpenIFS sends `A_MSLP` to FESOM (`ECE_CPL_MSLP`), FESOM receives `mslp_oce` into `press_air` (`use_atm_mslp`), and REcoM uses it where it is positive. The pressure REcoM used, recovered from the output, follows the OpenIFS monthly mean (0.979 atm at 78–60 S, $r = 0.997$ node by node), giving 5.9 μatm less atmospheric pCO_2 south of 60 S. SST and SSS are identical in all three runs.

Status: REcoM #34 merged; FESOM #1115 open (the field and the submodule bump); OpenIFS on `awiesm3-develop` and as draft merge request !25 against the consortium's `ecearth`; `esm_tools` switch `with_mslp_oce_coupling` written, default on for the moving carbon-cycle versions, not committed until FESOM #1115 is in. The fix needs FESOM 2.8.2 and a v3.5.1 for the carbon-cycle variant. Existing REcoM states have lost carbon and oxygen and must not be used as BGC restarts. Write-up: `notes/recom_zero_sea_level_pressure.md`.

67 Round 50 (2026–10–06): back to KPP with uniform GM 1000, and the Southern Ocean opens

67.1 Why it was tried

Put on one axis by mixing era, the deep ocean drifted least under FESOM's own KPP, and the difference is in the abyss (`data/clim/mixing_eras_temp_drift_by_depth.txt`):

trend, mK/decade	300–700 m	700–1500 m	1500–2500 m	below 2500 m
KPP, 2020–2059	+8	+9	+2	+5.9
TKE + IDEMIX, 2100–2139	+34	+36	+9	+11.8
TKE, 2160–2209	−4	+8	+2	+12.6
TKE + IDEMIX arm, 2170–2209	+15	+29	+5	+13.4

Between 700 and 2500 m TKE alone is as quiet as KPP was; below 2500 m every CVMix scheme, `cvmix_KPP` included, warms about twice as fast. KPP's upper-ocean warming is acceptable because it can be countered in the atmosphere. The hemispheric GM was also to go: the 26.5 N overturning stepped down by about 1 Sv when it came in at 2120, and a uniform 2500 caps the Labrador Sea.

67.2 PICAL__crunveg__kpp__gm1000, 2170–2189

Two changes against the TKE-only line from the 2169 restart: `mix_scheme` KPP and `K_GM_max` 1000 everywhere.

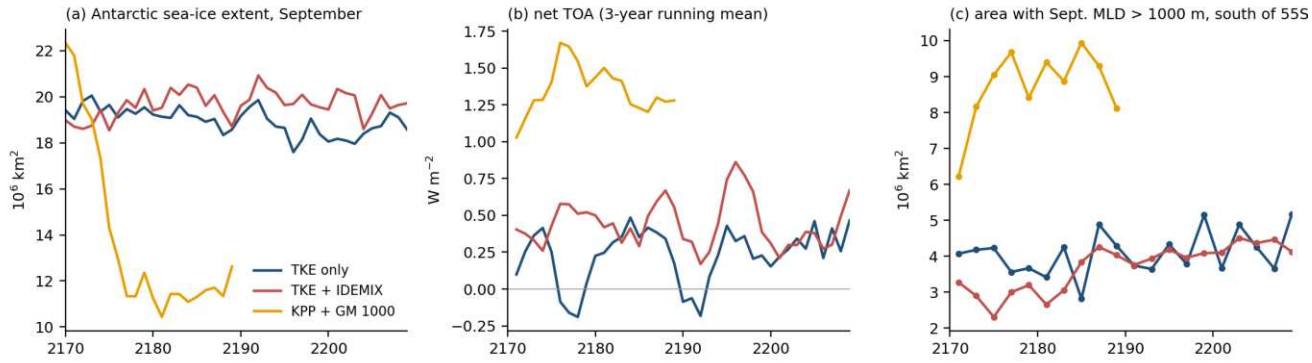


Figure 57: KPP with uniform GM 1000 against the TKE-only line and the IDEMIX arm from one restart. (a) Antarctic winter extent falls by 40 % in eight years and stays there. (b) Net TOA sits at +1.3 W m^{-2}. (c) The area south of 55 S with a September mixed layer deeper than 1000 m doubles within three years, before the ice responds: the convection leads.

2180–2189	TKE only	TKE + IDEMIX	KPP + GM 1000
net TOA (W m^{-2})	+0.33	+0.47	+1.33
T2m 60–90 S (°C)	−16.8	−17.2	−14.4
Antarctic extent, September / February (10^6 km^2)	19.0 / 1.77	19.8 / 1.67	11.4 / 0.21
AMOC 26.5 N (Sv)	13.3	12.4	14.3
Labrador March mixed layer (m)	1750	1690	1820
trend 300–700 m (mK/decade)	0	+5	+96
trend 700–1500 m	+5	+35	+34
trend below 2500 m	+13.0	+14.8	+5.2

Two of the aims are met: the abyss falls to the KPP-era +5 mK/decade, and the North Atlantic convects without the hemispheric setting, with 1 Sv more overturning. The Southern Ocean fails. The absorbed solar is 1.3 W m^{-2} above the TKE-only line, so the imbalance is now held up by lost ice. The second leg was not run (`data/clim/kpp_gm1000_2180-2189.txt`).

68 Round 51 (2026–10–06): the Southern Ocean cap, what holds it and what it lacks

68.1 Why the column destabilised

In the region that convects in the arm and not in the TKE-only line (5.9×10^6 km^2 of open ocean south of 55 S), winter water is cold and fresh over warm and salty: temperature destabilises and only salt holds the column. The margin is 0.25 kg m^{-3}.

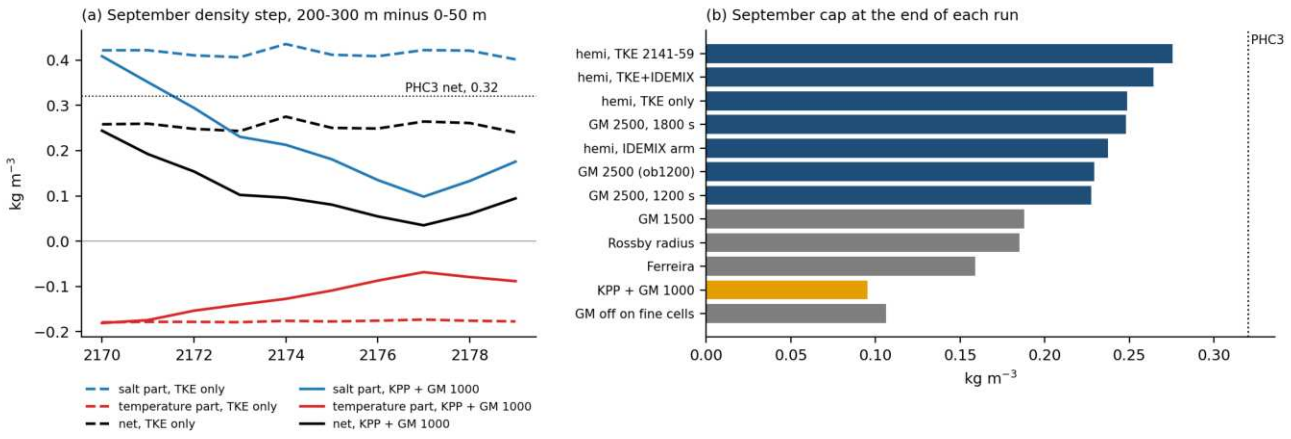


Figure 58: (a) September density step between 0–50 m and 200–300 m in the newly convecting region, split into its salt and temperature parts. In the arm the salt part collapses from 0.42 to 0.10 in seven years. (b) The same step at the end of every run with 3D output: it orders by the effective GM south of 55 S, and no run reaches the PHC3 value.

The salt comes from below, not from the surface (the arm receives slightly *more* surface freshwater): the upper 150 m gains 0.03–0.04 per year while 200–500 m freshens and cools by 0.57 K. Two measured mechanisms (`data/clim/so_destabilisation.txt`, `data/clim/so_residual_upwelling.txt`):

- **GM.** The westerlies drive +50 to +56 m yr^{-1} of upwelling at 200 m. The eddy-induced velocity opposes −28 with the southern GM 2500 and −15 with GM 1000, so net upwelling rises from 23–26 to 34–42 m yr^{-1}. A budget of that extra upwelling across the salinity contrast gives +0.02 to +0.04 per year for the top 150 m, the observed rate.
- **KPP.** The March mixed layer is 21–26 m against 38–46 m with TKE and K_v at 30 m is 10–90 times lower, so the summer meltwater stays in half the depth. The 50–150 m layer is already 0.07 saltier and 0.5 K warmer in the first winter. (2170 is also the arm's atmosphere spin-up year, so the size of this term is less certain.)

68.2 "2500" is 980: the coefficient the model applies

`K_GM_max` is the value for a 100 km cell; `scaling_resolution` reduces it with cell size. CORE3 is about 28 km in both the Southern Ocean and the Labrador Sea, where the Rossby radius is 5–9 km and nothing is resolved (`data/clim/gm_k_rossby.txt`).

effective surface K (m^2 s^{-1})	AWI-CM3 v3.3, CORE2	CORE3, `K_GM_max` 2500	CORE3, 1000
Southern Ocean 40–55 S	2560	1660	≈ 660
Southern Ocean 55–65 S	1960	980	≈ 390
Antarctic margin, south of 65 S	1170	870	≈ 350
Labrador Sea	100	980	≈ 390

The production line runs half the Southern Ocean GM of the released CORE2 model, and the arm ran a fifth. In effective terms the hemispheric setting is 980 in the south against 390 in the Labrador Sea.

68.3 The cap against observations

September, TKE-only line 2200–2209	PHC3	model	
density step, newly convecting region (kg m^{-3})	0.32	0.25	78 %
density step, Weddell region (convecting in every run)	0.24	0.075	31 %
surface salinity 0–50 m	34.04	34.20	+0.16
winter water 50–150 m: salinity / temperature (°C)	34.24 / −1.12	34.45 / +0.15	+0.21 / +1.3 K
deep water 200–500 m: salinity / temperature (°C)	34.62 / 0.97	34.72 / 1.45	+0.10 / +0.5 K

The cold, fresh winter-water layer is largely missing: warm deep water sits directly under the surface layer. March surface salinity matches (33.89 against 33.88); by September it has risen twice as much as observed. The Weddell region, at a third of the observed cap, overturns every winter in every run, which is the Weddell warm patch of the T2m maps (+3.6 K in the annual mean against ERA5, +7 K in autumn; +2.9 and +5 with IDEMIX). Winter observations under the ice are sparse, so the second decimal is not to be trusted (`data/clim/so_cap_vs_phc.txt`).

Across the thirteen runs with 3D output the cap gains about 0.05 kg m^{-3} per 400 m^2 s^{-1} of effective coefficient, so GM alone would need about 1600–1800 at 55–65 S, near the CORE2 value. The deep water was already 1.34 °C in 2101: a standing bias, with +0.1 K added in the IDEMIX years. The smallest deep-water drift of any run is the TKE-only line, −0.004 K/decade over 50 years; a *cooling* of 200–500 m marks the runs that convect (`data/clim/so_cap_drift_by_run.txt`).

68.4 Two more contributors: the wind and where the icebergs melt

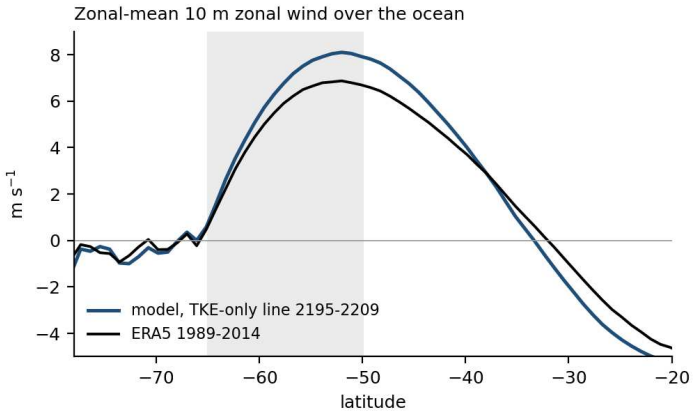


Figure 59: Zonal-mean 10 m zonal wind over the ocean. The jet is at the observed latitude (52 S) and 18 % too fast; over 50–65 S (shaded) the mean is 6.15 against 5.29 m s^{-1} in every season. The stress is only 10 % above ERA5 (see text). ERA5 is present day; the pre-industrial jet was weaker.

~~With ERA5-like winds the 50 m yr^{-1} of upwelling would be about 37: the wind excess is as large as the whole GM difference that broke the cap in the arm.~~ That used u^2 as a proxy for the stress and was wrong. Compared directly, the eastward stress over 50–65 S is 0.159 N m^{-2} against 0.145 in ERA5, +10 %: the model's bulk drag (τ/u_{10}^2) is 16 % lower, so the 10 m wind is faster for nearly the same stress. Aloft the excess shrinks (+0.9 m s^{-1} at 850 hPa, +0.6 at 500). An atmosphere-only run with observed SST over the ERA5 years shows the same (`amip_PD1_v35`: wind +11 %, stress +9 %), so the bias is the atmosphere's own and not a product of the coupled state. The wind is therefore a minor contributor to the upwelling, a few m yr^{-1} (`data/clim/so_wind_stress_amip_levante.txt`). (Round 45 found the winds over the pack close to NCEP-2; that was wind speed south of the ice edge, this is the zonal wind of the open-ocean jet.)

Levers for the wind (source survey, 2026–10–06; nothing run yet). Relative-wind stress is already on: FESOM surface currents reach OpenIFS (`LNEMOLIMCUR`, `LECURR`) and enter the momentum boundary condition, so that explanation is closed. The wave model does not run in its source default: `wam_namelist` has `LLGCBZ0` and `LLNORMAGAM` off where ECWAM's default has both on, which selects a different constant set (background Charnock 0.0065 against 0.0055, `TAUWSHELTER` 0.25 against 0). That is the only sea-drag lever reachable from a namelist and its sign at Southern Ocean wind speeds is not known from the source. `RSBLB` (2.0 against a default of 5.0, adopted in the 11-series) acts on momentum too and was never scored for wind. Orographic and non-orographic drag are poorly targeted. Since the stress is only 10 % high and the low drag is what makes the 10 m wind fast, a roughness lever would mainly correct the wind speed, not the ocean forcing; it is no longer a priority for the cap.

The land-ice freshwater is not too small but misplaced. OpenIFS removes all snow above 10 m water equivalent as "calving": 2475 Gt yr^{-1} from Antarctica, which closes against the surface mass balance on land south of 60 S to 1 %. The runoff mapper has a single Antarctic basin and delivers all of it, at a uniform 530 mm yr^{-1}, into three patches of together 4.7×10^6 km^2: the north-west Weddell Sea, the Scotia Sea at 50–60 S and a strip in the Ross Sea. Two thirds therefore enter north of 60 S, and the Indian and Pacific sectors receive nothing. The ice-shelf cavities melt another 1378 Gt yr^{-1} (2200–2209; `fw` at cavity nodes, area at the top wet level) on top, so the land-ice total exceeds the accumulation. Under `linfs` the global balance absorbs the excess as a uniform open-ocean correction, so this is an error of placement, not a salt drift (`data/clim/so_westerlies_and_freshwater.txt`).

Gt yr^{-1}	basal melt	calving delivered	calving by origin	origin − melt
Weddell 60 W–20 E	335	2212	310	−25
Indian 20–90 E	133	0	407	+274
Pacific 90–160 E	167	0	599	+432
Ross 160 E–130 W	166	213	241	+75
Amundsen–Bellingshausen	577	55	900	+323
total	1378	2480	2457	+1079

"By origin" is the OpenIFS surface mass balance on land per sector, an estimate: the model itself carries no sector information.

68.5 What does not help

- **A GEOMETRIC eddy-energy closure.** Its energy growth rate, computable offline from the density field, is 1.4–1.6 times the dissipation rate in the Labrador and Nordic Seas and 0.6–0.9 in the Southern Ocean; the ranking does not depend on its two parameters. At switch-on it would add restratification in the north and remove GM in the south (`data/clim/geometric_growth_rate.txt`). A first-tendency diagnostic, not a simulation of the scheme.
- **Scalings already run.** Rossby-radius scaling gives low K at high latitude in both hemispheres (cap 0.19); Ferreira scaling 0.16.
- **Dynamic backscatter** (`opt_visc` 8). Stable for two months at 1800 s, no change in the resolved flow, and a sub-grid energy between -0.44 and $+2.4\,\mathrm{m^2\,s^{-2}}$ with default parameters. CORE3 has a mean cell-size to Rossby-radius ratio of 3.7. Dropped.
- **`use_momix`.** A second wind-mixing term south of 50 S only, inherited from a regional Weddell model; no physical reason for the latitude limit. A diagnostic at most.
- **Counter-tuning the imbalance in the atmosphere.** In every comparison on this line a colder surface goes with a *larger* imbalance, at 1.0–1.3 $\mathrm{W\,m^{-2}\,K^{-1}}$; over 20–50 years the ocean uptake does not fall when the surface cools. The atmosphere sets the surface temperature and is a slow lever on the deep drift.

69 Round 52 (2026–10–06): the working hypothesis and the plan

Working hypothesis. The Southern Ocean surface cap is too weak in every run of the line (78 % of observed in the open ocean, 31 % in the Weddell region), and three measured errors push the same way: (i) too little eddy compensation, because the cell-size scaling halves the GM coefficient on a mesh that resolves no eddies there; (ii) a wind stress about 10 % too strong (a minor term); (iii) iceberg freshwater delivered mostly north of 60 S. The deep water under the cap is 0.5 K too warm, a standing bias that IDEMIX made worse. The hemispheric GM does nothing for the south: it keeps the effective 980 the line always had there, and lowers the north to 390 so the Labrador Sea convects. In the north it is very likely compensating for something else (the GEOMETRIC measure says the Labrador Sea should get *more* eddy restratification, not less).

Target. A September density step of about 0.3 $\mathrm{kg\,m^{-3}}$ in the open Southern Ocean and 0.24 in the Weddell region, with the 50–150 m layer near -1 °C. This can be read after five years of a test, long before the sea ice answers.

Plan, each from the 2169 restart on the TKE-only line unless stated:

1. **More effective GM in the south.** First with the existing hemispheric option (southern `K_GM_max` 4000, about 1570 at 55–65 S). Then a coefficient read from a map (about 50 lines in `oce_fer_gm.F90`, off by default), so the Southern Ocean can be raised without doubling the subtropics and the latitude step goes away. The map is static and is prescription, not physics; the rule behind it has to be physical and written down.
2. **The westerlies.** Checked and set aside: the stress is 10 % high, in AMIP as in the coupled runs. No tuning run planned; the low sea-surface drag is noted as an atmospheric bias.
3. **Icebergs.** Deliver (mapper calving − cavity basal melt) as icebergs and switch the patch calving off, so that accumulation = basal melt + icebergs and the freshwater lands where icebergs melt: $2475 - 1378 \approx 1100\,\mathrm{Gt\,yr^{-1}}$. No Fortran is needed for a first version: the runoff mapper 1.4 can drop the Antarctic basin (`IgnoreCalvingBasins: 66`, already an `esm_tools` switch), and FESOM's iceberg module is seeded from files. To build: a generator for the seed files and the per-leg bookkeeping that today exists only in the ice-sheet coupling. A cavity-and-iceberg piControl without an ice-sheet model exists on a colleague's HR branch; its seed files should be asked for first. The fully online version (a reservoir in FESOM) is a second step.
4. **Mixing.** KPP is kept as a candidate for its quiet abyss (+5 against +13 mK/decade below 2500 m), to be tested on a Southern Ocean that holds; not before. Why every CVMix scheme warms the abyss twice as fast is open.
5. **The release** is not at risk: TKE-only and TKE + IDEMIX on the hemispheric GM are both stable, evaluated configurations (§64).

70 Round 53 (2026–10–06→07): four one-change arms for the Southern Ocean cap

Round 52's first item, a southern `K_GM_max` of 4000, was withdrawn before it ran: more eddy compensation means less upwelled deep water, which is what closes the Atlantic overturning, and at 13–14 Sv there is no room (GM 2500 was 1 Sv weaker at 26.5 N than GM 1500). The cap was attacked through freshwater and sea ice instead. Four arms from the 2169 restart of **PICAL__crunveg__tke__albsn082** (TKE only, hemispheric GM, `albsn` 0.82, 1800 s), each one change against it; OpenIFS is cold-started in every arm, so 2170 is discarded.

run	the one change	years
PICAL__crunveg__icb	Antarctic calving through the mapper off; (calving − basal melt) as icebergs	2170–2209
PICAL__crunveg__wamz0	wave model on ECWAM default roughness (`LLGCBZ0`, `LLNORMAGAM` on)	2170–2189
PICAL__crunveg__cdoi10	ice–ocean drag `Cd_oce_ice` 0.0055 → 0.010	2170–2189
PICAL__crunveg__cpres15	ice strength `c_pressure` 20 → 15 (4× the strength at concentration 0.74)	2170–2189

70.1 The iceberg scheme

Item 3 of the plan, built without Fortran and without a change to the `esm_tools` Python core: two workflow jobs declared in `awiesm3.yaml` call a standalone script (`couplings/fesom/iceberg_budget.py`). At the end of a leg the job measures the OpenIFS calving and the cavity basal melt of that leg and keeps a ledger; before the next leg the seeding job writes FESOM's iceberg seed files for the owed mass, split between sectors by the observed present-day calving (Rignot et al. 2013), so that accumulation = basal melt + icebergs. The runoff mapper 1.4 drops the Antarctic basin (`IgnoreCalvingBasins`). The first leg starts from a rate of 1100 Gt yr^{-1}, or from a rate file in the pool.

It works: runoff south of 50 S falls from 2557 to 188 Gt yr^{-1}, the budget stays at 1105–1161 across four legs, seeded mass closes to 0.5 % (45 264 against 45 024 Gt owed), the restart carries the population (917, 1347, 1639 model bergs) and the run time is unchanged. The population spins up slowly: melt is 417, 692, 848 and 939 Gt yr^{-1} by decade against about 1120 seeded, so for the first two decades the Southern Ocean gets *less* land-ice freshwater than the control.

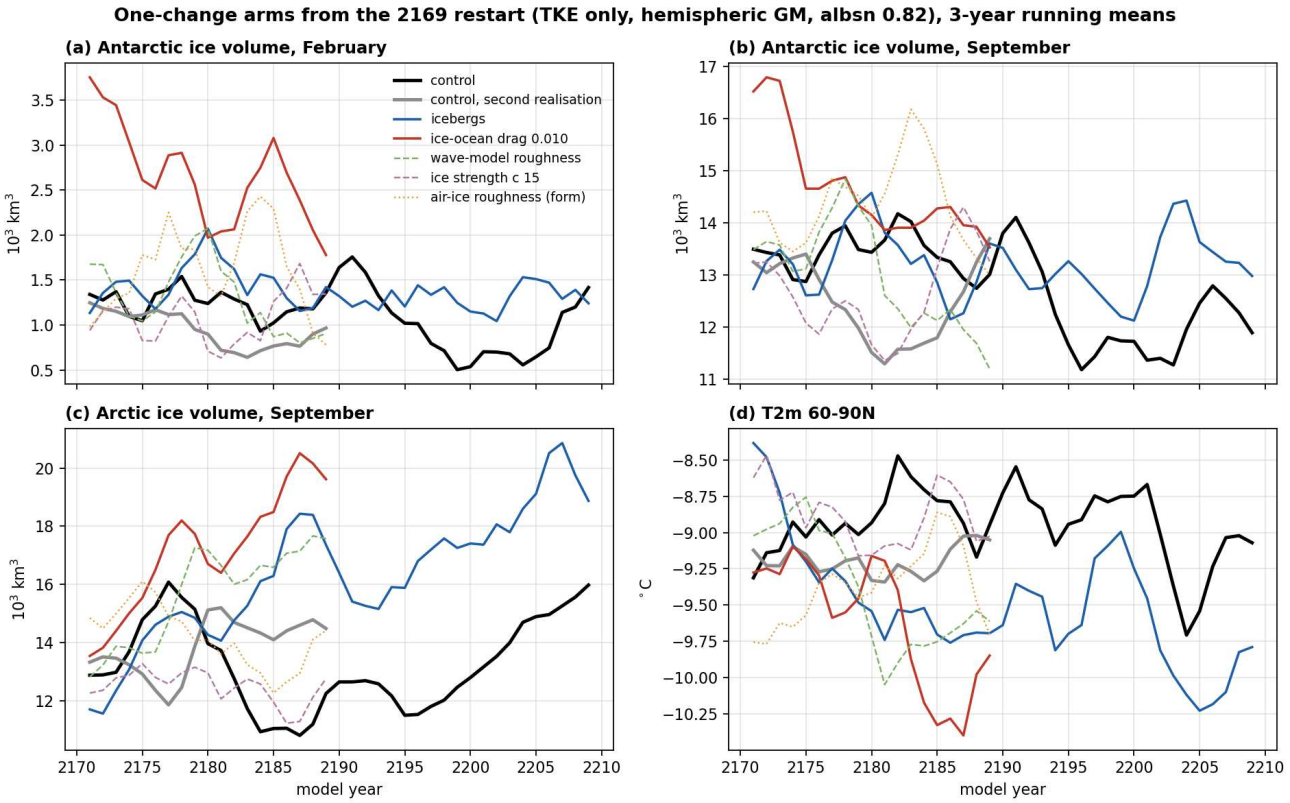


Figure 60: The arms against the control (black) and its second realisation (grey, §71), 3-year running means. (a, b) Only the ice–ocean drag (red) moves the Antarctic clearly; the iceberg run (blue) stays above a control that keeps losing summer ice after 2190. (c, d) The iceberg, wave and drag arms gain Arctic ice and cool north of 60 N in the 2180s; the second control moves half as far the same way with no change at all. The air–ice roughness run (orange, §72) is on the newer build.

70.2 Results

2180–2189, arm − control	icb	wamz0	cdoi10	cpres15	control
Antarctic volume, February (10^3 km^3)	+0.29	−0.03	+1.15*	−0.12	1.18
Antarctic volume, September	−0.15	−1.23*	+0.62	−0.70	13.4
Arctic volume, September	+4.41*	+5.01*	+6.50*	+0.25	11.9
T2m 60–90 N (K)	−0.84*	−0.92*	−0.99*	−0.13	−8.84
net TOA (W m^{-2})	−0.26*	+0.02	−0.24	−0.10	0.33
September cap, newly convecting region (kg m^{-3}; PHC3 0.32)	0.233	0.234	0.213	0.238	0.234

* beyond two standard errors of the interannual spread; §71 shows that this test is too generous. Tables: `data/clim/overnight_arms_volumes_drift.txt`, `data/clim/overnight_arms_2180-2189.txt`, `data/clim/icb_four_decades.txt`.

- **No arm strengthens the cap.** The best is the iceberg run in its last decade, 0.258 against 0.246 (Weddell 0.093 against 0.077), small against 0.32 and 0.24.
- **Ice–ocean drag** is the only robust Antarctic lever: February volume 1.3→3.0 and 1.2→2.3, drift 18.9→15.3 cm s^{-1} (−18 %). It is a compensation: the coefficient is already five times the value measured under winter Weddell pack (1.13×10^{-3}, Wamser and Martinson 1993), and the cap gets weaker, not stronger. Not adopted.
- **Wave-model roughness does nothing.** The first decade's 10 m wind over 50–65 S was 0.3 m s^{-1} below the control, the second's 0.4 above; τ/u_{10}^2 4.10 and 3.83 against the control's 4.01–4.12 and ERA5's 4.78.
- **Ice strength** loses a little Antarctic winter volume in the first decade and nothing after.
- **Icebergs, 40 years.** February Antarctic volume is above the control in all four decades (+0.15, +0.29, +0.15, +0.48) and September gains +1.40 by the 2200s, while the control's summer pack falls 1.29→0.81. AMOC within ±0.8 Sv with no trend, abyss +12.2 against +12.5 mK per decade, Southern Ocean convection area 4.2 against 4.2–4.4×10^6 km^2; 700–1500 m warms a third faster (+9.9 against +7.3). The Labrador Sea keeps convecting (March mixed layer 1720 against 1890–2020 m) with more ice (0.14–0.15 against 0.09), which goes with the larger Arctic ice.

71 Round 54 (2026–10–07): a second control, and the noise is twice the test

Three of the four arms grew Arctic ice through the 2180s, two of them by a change with no route to the Arctic. **PICAL_crunveg_ctl2** repeats the control from the same restart with no change, differing only in what every arm differs in: a cold-started atmosphere at 2170 and 10-year legs.

ctl2 − control	2171–2179	2180–2189
Arctic volume, September (10^3 km^3)	−1.45*	+2.74*
Antarctic volume, September	−0.49	−1.26*
Antarctic volume, February	−0.15	−0.40*
T2m 60–90 N (K)	−0.16	−0.34*
T2m global (K)	−0.02	+0.10*

A run that changes nothing passes the two-standard-error test on eleven of fifteen quantities in its second decade: the interannual spread does not contain the decadal spread between realisations. Read against ctl2 as the yardstick (about 2.7 in Arctic September volume, 1.3 and 0.4 in Antarctic September and February volume, 0.35 K at 60–90 N):

- **Stands:** the ice–ocean drag arm in the Antarctic (+1.73 and +1.15 in February).
- **At the edge:** the iceberg run's fourth-decade Antarctic gain; what speaks for it is that the sign holds in every decade.
- **Above the noise, not by enough to claim:** the Arctic growth of +4.4 to +6.5 in three arms. The iceberg run has held at about +4 for three decades, so it is not a runaway. The likely reading is an Arctic that is only marginally stable at `albsn` 0.82, so the outcome depends on the weather a realisation gets after the atmosphere restart.
- **Does not stand:** every first-decade difference of the wave and ice-strength arms.

`z0def`, the control on the newer build (§72), sits inside the ctl2 noise on every quantity, which validates that build (`data/clim/z0def_build_check.txt`). Earlier single-realisation Arctic comparisons, the IDEMIX arm's included (§64.3), carry the same caveat.

72 Round 55 (2026–10–07): the air–ice drag, and why the atmosphere gives it back

The drag arm worked by raising a coefficient that is already too high. The term that is high is on the air side: OpenIFS gives a neutral 10 m air–ice drag of 1.89×10^{-3} for compact ice, rising to 2.3 at concentration 0.74 and 3.0 at 0.5, against 1.44–1.68 measured. The roughness $z_{0m} = \max(z_{\mathrm{floor}},\, (1-c)\,C + A\,e^{-B(c-0.5)^2})$ was hard-coded; its constants are now `NAMSURFTUNE` entries (`ECE_TUNE_RZ0ICEM`, `ECE_TUNE_RAZ0ICE`; OpenIFS `pr12/seaice-roughness`, MR !26, follow-up to !23). With the namelist silent the released values apply. Three runs from 2169 on the tag tree with the newer OpenIFS and LPJ-GUESS: **PICAL_crunveg_z0def** (released values), **z0floor** (floor $10^{-3} \to 0.5 \times 10^{-3}$, compact-ice drag 1.63) and **z0form** (floor halved and A 6.05 $\to 3.0 \times 10^{-3}$).

Antarctic pack, JJA	z0def	z0floor	z0form	cdoi10
air–ice drag (10^{-3})	2.09	1.86	1.82	—
wind forcing $\langle \lvert U\rvert U\rangle$ (m^2 s^{-2})	35.5	38.0	39.5	—
air–ice stress, 2180–2189 (mN m^{-2})	93.7	92.9	92.6	93.1
drift speed, 2180–2189 (cm s^{-1})	18.9	18.6	18.4	15.3

Two reasons it delivers little (`data/clim/ice_drag_effective_z0*.txt`, `data/clim/z0_ice_speed_so_convection.txt`):

- **Size.** Free drift goes as the square root of the drag ratio. The ocean arm raised its coefficient by 82 % (−26 % drift expected, −18 measured); the observations allow the air side only 13–30 % (−7 % expected, −2 to −5 measured).

- **Compensation.** The momentum the atmosphere hands down is set by the large-scale pressure gradient. A smoother surface removes less momentum, so the 10 m wind over the pack speeds up by 7–11 % until nearly the same stress arrives (1–7 % less). The ocean has no such compensation. Buoy wind factors are quoted against the geostrophic wind for this reason.

z0form has more Antarctic ice in both decades (February +0.47, +0.53; September +1.9, +1.3), at the ctl2 noise limit, and is 0.56 and 0.23 K colder at 60–90 N on top of a cold Arctic winter. z0floor changes sign between decades. **Decision:** the code is on the OpenIFS tag `48r1v6` with the released values as default; the entries are not set in the spin-up.

73 Round 56 (2026–10–07): where the abyss warms, and the decision to restart from PHC3

No arm touched the abyssal drift (12.2–13.1 mK per decade below 2500 m in all of them). In the control, by basin (bias 2205–2209 against PHC3, trend 2171–2209; `data/clim/deep_warming_where.txt`):

K / mK per decade	200–500 m	500–1500 m	2500–4000 m	below 4000 m
Antarctic, south of 60 S	+0.90 / −2	+0.38 / −8	+0.72 / +10	+0.88 / +18
Southern Ocean 60–35 S	+1.51 / −2	+0.99 / −5	+0.50 / +18	+0.72 / +25
Atlantic	+0.40 / −11	+1.53 / +23	+1.20 / +28	+0.54 / +35
Pacific	−0.66 / −15	+0.35 / +9	−0.10 / +3	+0.05 / +5
Arctic and Nordic Seas	+0.66 / −6	+1.35 / +5	+2.02 / +9	+2.14 / +50
global	+0.19 / −11	+0.64 / +7	+0.37 / +12	+0.35 / +16

The Atlantic takes 40 % of the heat gained below 2500 m and the Southern Ocean north of 60 S about 30 %; the Pacific is almost still. The warm layer at 200–500 m around Antarctica is an established bias that is cooling slightly, not the drift. The Atlantic at 500–1500 m, the source of the deep Atlantic, is 1.5 K too warm and still warming (a reading of the pattern, not a budget). Below 2500 m the global rate shows no curvature over 360 years: +4 to +9 mK per decade under KPP (1850–2089), +11 to +13 since the CVMix schemes came in. No affordable spin-up length ends it.

Decision (Jan): a fresh spin-up from PHC3 with the final settings, and only as long as the carbon fluxes need to meet the drift thresholds, "because the longer the spinup runs, the worse the deep ocean gets". A restart does not lower the rate but starts the deep Atlantic 1.2 K and the deep Arctic 2 K closer to observations, and that gap takes centuries to rebuild. The assistant's advice went from restart to continue and back within the hour; the case for continuing (known circulation, time) was real but weaker than first presented.

74 Round 57 (2026–10–07→08): the spin-up configuration and the final v3.5.0 tags

Configuration of the CMIP7 spin-up (`notes/v3.5.0_final_tag_round.md`), on the carbon-cycle variant `v3.5.0-cc`:

- FESOM cold from PHC3, REcoM cold (GLODAP pre-industrial, WOA, PISCES iron) and concentration-driven at 1850 CO_2, receiving the OpenIFS sea-level pressure (Bug XIII fixed); LPJ-GUESS from the 2000-year offline spin-up with 1850 nitrogen deposition (state of year 3901).
- `cvmix_TKE` without IDEMIX; `K_GM_max` 2500 south / 1000 north; 1800 s; `albsn` 0.82; icebergs on (start rate 1105 Gt yr^{-1}, mapper calving off); everything else as in the control: ice–ocean drag 0.0055, `c_pressure` 20, wave and air–ice roughness released.
- Light output (monthly OpenIFS, FESOM and LPJ-GUESS), 5-year legs, 36 nodes.

Tags. FESOM `2.8.2` (adds the pressure coupling, PR #1115), OpenIFS `48r1v6` (`fc271eb`: the pressure coupling, the DMS namelist fix and the roughness entries); `esm_tools` `v3.5.0` updated in place, its tag set last. LPJ-GUESS 4.1.13, OASIS 5.3, XIOS 2.5.4 and mapper 1.4 unchanged.

Smoke test (**TEST_ccspin_1y**, 1850): icebergs with REcoM on 2688 ocean ranks from a cold start, all components through 365 days, restarts written. 37 minutes per year, 6.1 s per model day against 4.55 without icebergs. First-year basal melt is 903 Gt yr^{-1} (cold cavities, 1323 in the warm 2209 state), so the owed iceberg rate rises to 1464. A 50-year test spin-up (**SPIN_cc_test1**) ran on `develop-cc`; the production run, **AWI-ESM3-CC-LR-CMIP7-Spinup**, is run by a colleague on the tagged build from 2026–10–08.

75 Round 58 (2026–10–08→15): the CMIP7 spin-up, the first 160 years

75.1 A crash in 1902: an iceberg pinned to one cell

The leg 1900–1904 died in model year 1902: a cell at 102 W, 73.5 S was cooled to −5.6 °C. The iceberg on it (12.4 km long, 19.5 m thick) had a basal heat flux of 3.0×10^{10} W, about 200 W m^{-2} over its own footprint, but FESOM drew all of it from one cell of about 3 km^2, giving −9645 W m^{-2}. With grounding mode 2 a grounded berg stands still and keeps hitting the same cell; 1922 of 4498 bergs were grounded at the end of 1899 (1584 of 3926 in 1889).

Fix, from the 1900 restart: `l_allowgrounding` 1 (grounded bergs creep) and `l_cap_ibhf_n` (no iceberg may cool a cell below −1.85 °C). Both were set because the crashed berg could not be shown to be grounded. The cap is a numerical safeguard: it removes the heat flux and keeps the melt, so the latent heat comes from nowhere. It acts every step and discards 2.1×10^{14} J per step, 0.0003 W m^{-2} over the ocean or about 1 % of the iceberg melt heat. The physical fix, spreading a large berg's melt over the cells it covers, is open.

75.2 State

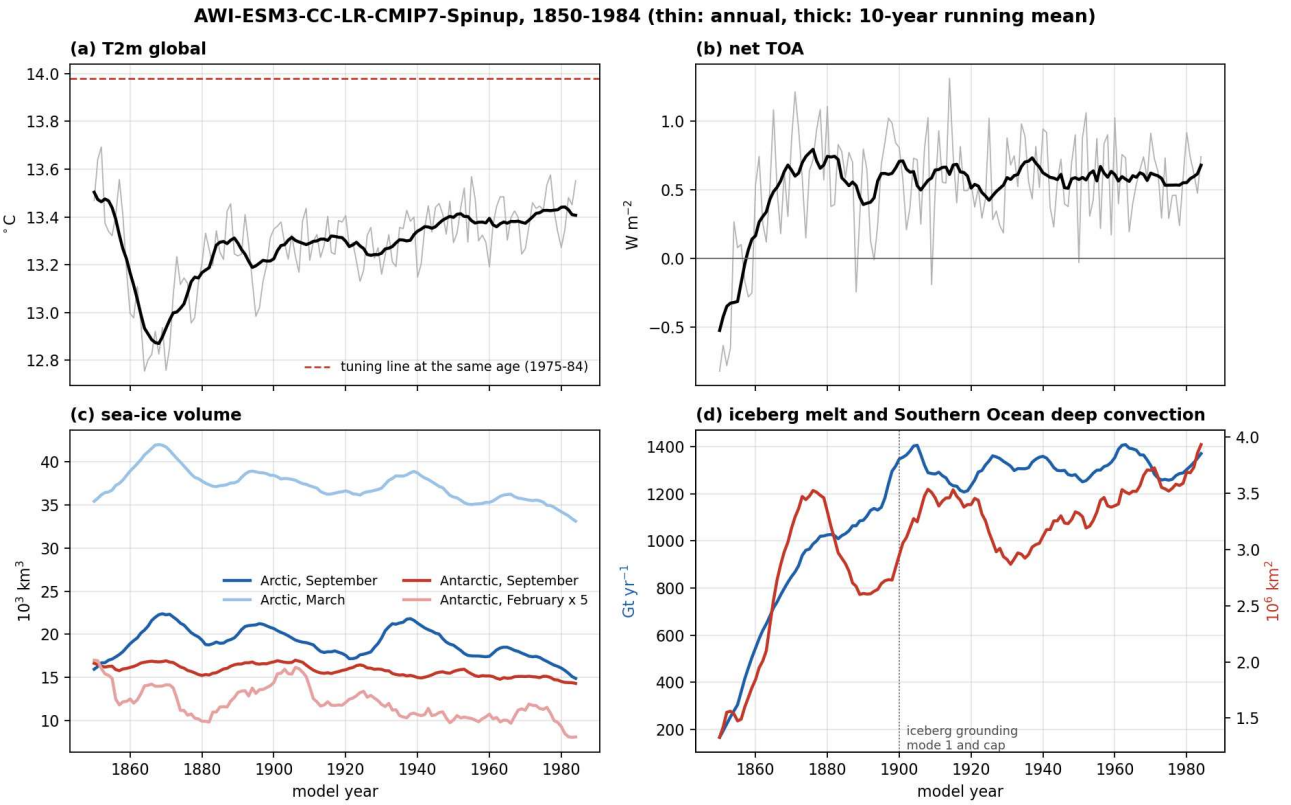


Figure 61: The CMIP7 spin-up, 1850–1984. (a) T2m dips by 0.6 K in the first 15 years and recovers only to its starting value, 0.5 K below the tuning line at the same age. (b) The imbalance settles at +0.5–0.7 W m^{-2}. (c) Arctic ice is within the control noise of the tuning line; the Antarctic summer pack declines. (d) Iceberg melt levels at about 1300 Gt yr^{-1} after the 1900 restart while Southern Ocean deep convection keeps spreading.

	1850–59	1890–99	1930–39	1975–84	tuning line, same age
T2m global (°C)	13.44	13.20	13.31	13.44	13.98
T2m 60–90 N (°C)	−10.9	−11.3	−10.9	−10.2	−9.1
net TOA (W m^{-2})	−0.31	+0.62	+0.70	+0.55	−0.04
AMOC 26.5 N (Sv)	13.6	14.8	14.4	15.0	—

- **Colder than the tuned state.** Against the tuning line's equilibrated 2195–2209, 0.42 K colder (land 0.58, ocean 0.35); the northern mid-latitudes absorb 0.85 W m^{-2} less sunlight and land albedo at 30–60 N is 0.272 against 0.261. Tree fraction settled by 1910 at 20 % for 30–60 N (from 26) and 6 % for 60–90 N (from 9).

The forest lost in the first decades is the obvious candidate, and on ccnice a lost forest did not regrow in 30 years (§58); a forest costs about 2×10^3 km^3 of NH ice and +0.24 K globally, about half of this gap.

- **The deep ocean drifts at half the tuning line's late rate**: whole ocean +10 mK per decade, below 2500 m +6 (tuning line 12–16), bias against PHC3 +0.12 K there by 1984. It sits in the deep Atlantic (+56 at 1500–2500 m, +41 at 2500–4000 m) and the Arctic and Nordic Seas; the Antarctic abyss warms at +24 below 4000 m. Mid-depth warming is slowing (630 m: 36→18).
- **Icebergs**: melt about 1300 Gt yr^{-1} against 1360–1440 supplied, ledger closed to 0.1 %.
- **Throughput**: 41–42 simulated years per day running, about 2700 core-hours per year.

75.3 Carbon: the criterion that ends the spin-up

The spin-up ends when the carbon drift of ocean and land together is below 0.1 GtC yr^{-1}. Pools from REcoM's DIC and organic carbon on the mesh and LPJ-GUESS's vegetation, litter and soil carbon (reval part 27, §75.4):

PgC	1850–59	1910–19	1950–59	1975–84	2000–09
ocean (DIC + organic)	36 200	36 175	36 165	36 158	36 152
land	2483	2314	2273	2260	2236
vegetation	645	523	503	500	494
soil and litter	1838	1792	1770	1760	1743

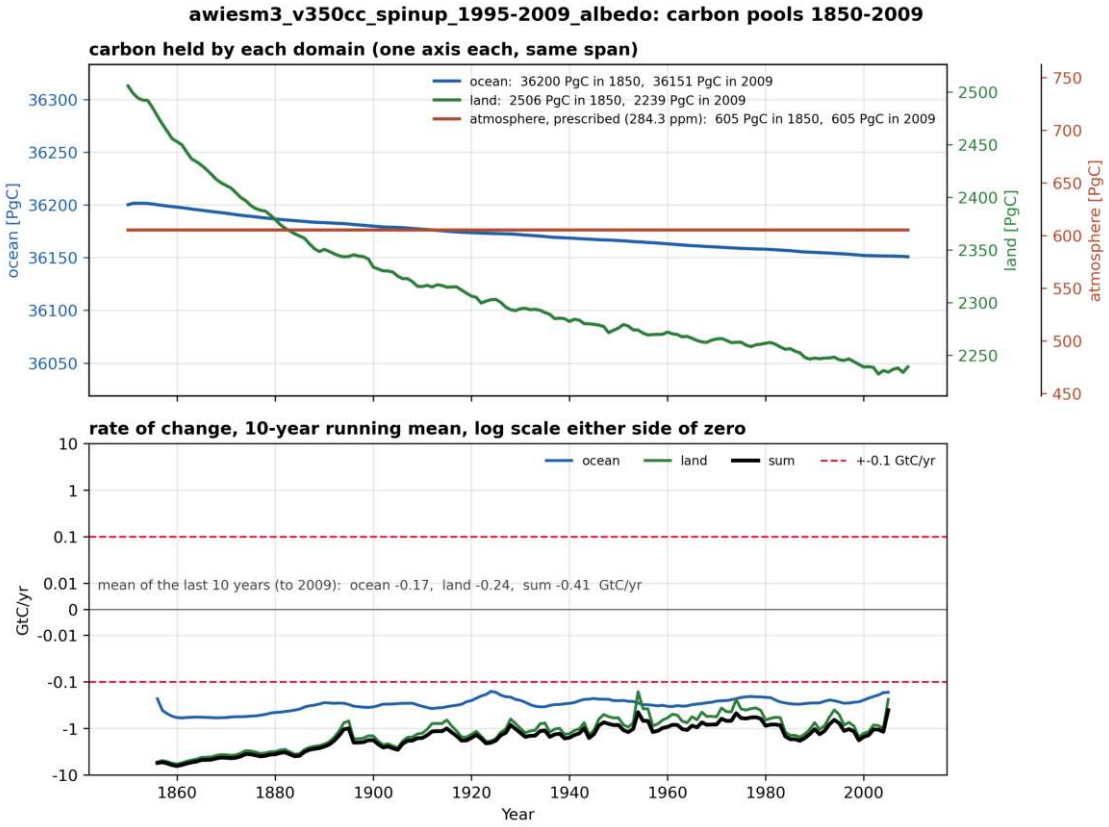


Figure 62: Carbon pools of the spin-up, 1850–2009. (a) Carbon held by each domain on axes of equal span; the atmosphere is prescribed. (b) Rate of change, 10-year running mean, logarithmic on either side of zero with the ±0.1 GtC yr^{-1} target. Over 2000–2009 the ocean loses 0.17 and the land 0.24, together 0.41 GtC yr^{-1}.

The ocean outgasses less each decade (air–sea flux −0.55 in the 1870s, −0.22 in 1975–84); the land loss has moved from vegetation to soil and litter, which respond slowly. The criterion is reachable, ocean first; on the 1975–2009 trend that is several decades out, not years.

75.4 Measuring land carbon from the remapped files

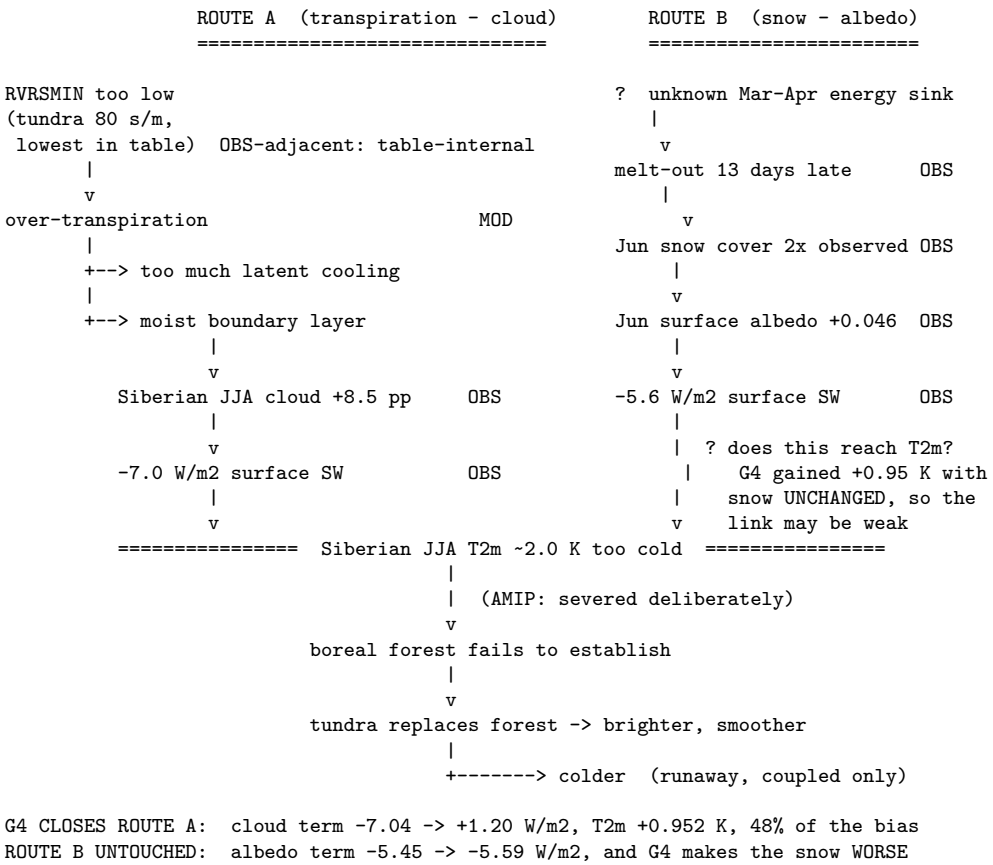
LPJ-GUESS output now arrives only on a 1° grid (lpjg2nc `-remap`). A conservative remap of land-only values gives a coast cell the mean over its land part and does not keep how large that part is, so the 1° files cover $1.457 \times 10^{14}\,\mathrm{m}^2$ against 1.304×10^{14} of model land and their totals are 12 % high. Scaling by the global area ratio left a 20 PgC step where two legs had kept the native table. The land fraction of each 1° cell is now rebuilt from the model's own land points (octahedral cell boxes intersected with the 1° boxes, $1.3038 \times 10^{14}\,\mathrm{m}^2$, equal to the native total); the step shrinks to 2–8 PgC, a residual of 0.3 %. The post-processing now keeps the native file as well (`-nc_keep_unstruc`, `esm_tools cabf48221`).

76 Synthesis: the causal web, and what is still dark

This section is the campaign's current best account of *why* the boreal cold bias exists, assembled from every round. It is written to be falsifiable: each link carries a status marker, and the unknowns are listed rather than glossed.

✓OBS	established against observation (CERES, Rutgers/IMS satellite)
●MOD	established only model-vs-model (ERA5 = HTESSEL sibling); suggestive, not authoritative
✗DEAD	tested and falsified
?	unknown — no evidence either way

76.1 The picture



76.2 Route A — transpiration to cloud: closed

Status: the strongest *measured* result of the campaign — but the mechanism is only partly established. The chain as proposed is: excess stomatal conductance cools the surface latently *and moistens the boundary layer*; the moisture becomes low cloud; the cloud removes shortwave; the surface cools.

The moisture step is contradicted by this report's own profiles. §21.1 finds the excess boreal cloud is *thermally* driven, not moisture-driven: Siberian RH is +5.6 to +8.2 % too high while q is *too low* (−0.2 to $-0.3\,\mathrm{g\,kg^{-1}}$). What was measured is the *correlation* F4 → less cloud → warmer; the moisture pathway is [**?**], not ✓OBS.

- ✓ **OBS** Siberian JJA cloud fraction was 78.1 % against CERES 69.6 %, and the cloud share of the SW deficit was $-7.04\,\mathrm{W\,m^{-2}}$ (§24.6).
- ✓ **OBS** G4 removes it: cloud term $-7.04 \rightarrow +1.20\,\mathrm{W\,m^{-2}}$, worth +0.952 K.
- The lever is defensible on *table-internal* grounds for tundra (`RVRSMIN(9)`= 80, the lowest of any vegetated type, §24.4) but **not** for needleleaf, where 250 → 1000 remains an unanchored 4× excursion that does not saturate (§24.6).
- **?** G4 slightly *overshoots* in August ($+3.19\,\mathrm{W\,m^{-2}}$ cloud term). A new bias in the making; watch it.

76.3 Route B — snow to albedo: open, and the interesting one

- ✓ **OBS** June surface albedo +0.046 too high (CERES, §25.1); June snow cover 0.380 vs satellite 0.203; melt-out 13 days late (§25.2).
- ● **MOD** Snow *accumulation* is right; the loss term is mistimed — too weak in Mar–Apr, too strong in May (§24.2).
- ✗ **DEAD** It is not the snow-*cover formulation*: round 13 changed `RQSNCR` and moved the albedo response into Sep–Nov instead (§23.2).
- ✗ **DEAD** It is not an energy *supply* problem: the model absorbs *more* net SW than CERES in Mar–Apr (+2.5, +1.2), because its spring albedo is too *low* (§25.1).
- ✗ **DEAD** It is not snow *brightness*: model snow is *darker* than both ERA5 and CERES. Brightening snow would worsen June.
- ● **MOD** It is probably not cold content: `tsn` is within a few tenths of a K of ERA5 — but this is model-vs-model (§25.3).
- ✓ **RESOLVED for the cover half** — see §26: HTESSEL has no sub-grid snow depletion, so cover stays pinned near 1 while the median cell still holds 27 cm in May. That explains the late melt-out and the June albedo, and explains why round 13 could not have worked.
- **? The March–April mass deficit remains open.** ERA5-only, The absorbed shortwave is present and the snow does not melt. Candidates: turbulent flux to the atmosphere (*unconstrained by any observation*); conduction/refreeze within the pack; or the canopy-shaded snow tile (7) behaving differently from exposed snow — the round-13 residual showed canopy masking already dominates in the melt months (§24.1).

76.4 The link that may not exist

The most consequential uncertainty in this report. Route B is a well-measured *radiative* bias, but its effect on *temperature* is unproven:

> G4 raised Siberian JJA T2m by +0.952 K while May–June snow mass did *not* change (§24.6), and while snow cover and melt date got *worse* (§25.2).

Temperature moved a full kelvin with the snow term stationary or degrading. Either the June albedo error is weakly coupled to T2m — a real radiative bias that the atmosphere largely exports — or the two routes offset each other in a way we have not separated. **Until this is resolved, the size of the prize in fixing Route B is unknown**, and it is possible that closing it buys far less than its $-5.6\,\mathrm{W\,m^{-2}}$ suggests. A clean test would be a run that changes the snow term alone with Route A held fixed; round 13 attempted exactly that and failed for an unrelated reason (it moved the wrong season).

76.5 The coupled amplifier, deliberately severed

The AMIP configuration exists to cut the vegetation–albedo feedback so the root cause can be hunted with the loop open. In coupled mode with LPJ-GUESS the chain closes: cold summers → boreal forest fails to establish → tundra replaces forest → higher albedo and lower roughness → colder. That runaway is why a 2 K AMIP bias becomes forest loss in the coupled model, and it is why 48 % closure may or may not be sufficient — **a question AMIP structurally cannot answer.**

A new risk sits precisely here. G4 delays melt-out from 24 to 28 May (§25.2). Later melt is later growing-season onset for LPJ-GUESS. G4 may therefore improve AMIP T2m while *harming* forest establishment through a pathway AMIP cannot see. This should be weighed before G4 is committed to the PI spin-up, and it is an argument for running coupled with G4 sooner rather than later.

76.6 The energy target, now decomposed PER RUN

Where the imbalance lives was settled long ago: §11.2.2 showed the global budget is right *by cancellation*, with the Southern Ocean carrying $+5.8\,\mathrm{W\,m^{-2}}$ and contributing $+0.57$ on its own, from a cloud deficit — the Hyder et al. signature. Nothing below revises that.

What was missing is that the decomposition existed **only for the control**. `amip_toa_decomp.py` never used the `RUNS` list, so across the tuning runs the energy target was tracked by a single global number with no idea which band any lever was moving. That is now part of the standard table and produced for every run automatically (§76.7). The new information is the **response**:

Δ net TOA vs control [$\mathrm{W\,m^{-2}}$]	60–90S	30–60S	tropics	30–60N	60–90N	global
control, absolute	−90.44	−22.24	+42.61	−21.78	−97.74	+0.64
CERES	−98.82	−21.63	+45.11	−24.29	−97.98	—
gap to close	**+8.38**	−0.61	−2.50	+2.51	+0.24	
A1a `ovl=0.10`	−3.33	−2.80	−1.06	−2.44	−2.75	−1.90
D2a `inpsea.2`	**−2.83**	−1.91	−0.38	−0.79	−0.81	−0.93
D2b `inp+p700`	−1.76	−0.83	+0.09	−0.19	−0.34	−0.28
B3 `clddiff`	−0.11	+0.89	**+2.02**	+0.86	+0.25	+1.34
B4 `entshalp3`	−0.56	−2.00	−1.55	−0.75	−0.24	−1.33
B5 `capdcycl0`	−0.47	+0.06	**−1.93**	−0.14	+0.18	−1.00
F4 `rsmin1000`	−0.40	+0.04	+0.07	+0.30	+0.38	+0.10
G1 = F4+D2b	−2.03	−0.83	+0.10	+0.26	−0.02	−0.20
G4 = G1+tundra	−1.98	−0.65	+0.12	+0.28	+0.13	−0.13

Three things this view gives that the global number could not:

(i) The 60–90S band is the most responsive in the model, moving −0.4 to −3.3 under levers that were designed for entirely different purposes. It is not inert; it has simply never been *targeted*.

(ii) The G-series buys about −2.0 of the +8.4 needed there, all of it from D2b, and D2a alone buys −2.8 — but D2a costs −0.93 globally and was rejected on other grounds. *Roughly a quarter of the Southern Ocean TOA gap is already closed as a by-product*, and the remainder is the same ∼ 6 pp cloud-area deficit that no lever has moved (§76.8).

(iii) The surface levers are confirmed regionally clean. F4 moves the tropics by +0.07 and G4 by +0.12, against B3's +2.02 and B5's −1.93. The tropical guardrail was being checked as a single number; it is now visible as a band, and it shows the F/G family does what it was designed to do and nothing else.

One trap the guardrail table invited. The PI target is **net TOA** ≈ 0, *not* the CERES column. CERES is present-day (2005–2015) and carries the real planetary warming imbalance, so it reads ∼ $+1.1\,\mathrm{W\,m^{-2}}$ globally: the control's +0.64 is *too positive* even though it sits *below* CERES. Per-band CERES differences are the right way to use it — for locating error, never as a global tuning target.

76.7 One command: `evaluate.sh`

The protocol had grown to five scripts run from memory, and skipping one has repeatedly produced wrong conclusions here — a lever promoted on 4-year noise (B8), a winter penalty unnoticed for eleven rounds (B5), and a TOA decomposition done only for the control. `scripts/analysis/evaluate.sh` now runs the whole protocol, with the TOA decomposition included in the main table rather than as a separate invocation. The way to make a guardrail effective is to make skipping it require effort.

76.8 The open list

1. **?** Where the March–April melt energy goes. Unconstrained; no satellite product measures turbulent flux.
2. **?** Why F4/G1 *increases* May–June snow mass (+5.58, +5.68 mm, both significant). Sublimation was the suspect and is *not* implicated in Mar–Apr; May snowfall rises significantly instead, for reasons unknown.
3. **?** Whether the June albedo bias reaches T2m at all (§76.4).
4. **?** Why G4 *repairs* G1's Nordic Seas SW RMSE damage (+0.300 → +0.089). Adding tundra resistance should not act on a Nordic ocean box. An unexplained improvement is as much a loose end as an unexplained degradation.
5. **?** The Southern Ocean cloud-*area* deficit (∼ 6 pp), untouched by every lever in all 34 evaluated runs.
6. **?** Tropics net TOA $2.4\,\mathrm{W\,m^{-2}}$ too low.

7. ✓ **but unaddressed** Global net TOA $+0.52\,\mathrm{W\,m^{-2}}$ against a ~ 0 target — *one of the two original problems*, barely moved by the whole campaign (control +0.64). For a PI spin-up this drift matters more than the last tenths of Siberian warming.

77 State of the campaign (2026–10–15)

A snapshot, not a story. Numbers are 44-year means over 1872–1915 unless stated.

77.1 Proven

Added 2026–10–15 (rounds 53–58, §70–§75).

- **Two realisations of one configuration differ by 2.7×10^3 km^3 of Arctic September ice and 0.35 K at 60–90 N over a decade**, and pass the two-standard-error test on eleven of fifteen quantities: decadal differences between single runs need a second realisation to be read (§71).
- **Icebergs from the model's own mass budget work** with no change to the `esm_tools` core: ledger closed to 0.5 %, Antarctic summer ice above the control in all four decades, nothing degraded; the cap barely moves (0.258 against 0.246) (§70.2).
- **No freshwater or sea-ice lever strengthens the Southern Ocean cap**: icebergs, wave roughness, ice–ocean drag, ice strength and air–ice roughness all leave it at 0.21–0.26 against 0.32 (§70, §72).
- **The ice–ocean drag is the one strong Antarctic lever and a compensation**: February volume doubles at 0.010, five times the measured value (§70.2).
- **The air–ice drag is not a lever on ice export in a coupled model**: the 10 m wind over the pack speeds up until nearly the same stress arrives (§72).
- **The abyssal drift sits in the deep Atlantic and Southern Ocean and has not slowed in 360 years**; only the mixing scheme ever changed its rate (§73).
- **The CMIP7 spin-up from PHC3 runs on tagged components** (FESOM 2.8.2, OpenIFS 48r1v6) with icebergs and REcoM at 41 simulated years per day; after 160 years it is 0.4–0.5 K colder than the tuning line at the same age, with a deep drift half the tuning line's late rate and a carbon drift of $0.41\,\mathrm{GtC\,yr^{-1}}$ (§75).

Added 2026–10–06 (rounds 47–52, §64–§69).

- **Dropping IDEMIX is what erodes the Antarctic summer pack**: February extent $-0.33 \pm 0.20 \times 10^6$ km^2/decade under TKE alone, -0.01 ± 0.24 with IDEMIX back on, same years, same restart. IDEMIX costs +29 against +8 mK/decade at 700–1500 m and does not lower the imbalance (§64.3).
- **`albsn` 0.82 stops the surface warming of the TKE-only line** without an Arctic runaway; pre-industrial bias +0.55 K (§64.2).
- **The overturning is not collapsing**: one step at 2120, then 13.5–14.0 Sv at 26.5 N for 90 years; the subpolar collapse belongs to uniform GM 2500 at 1800 s (§64.2).
- **v3.5.0 is tagged in every component** and the tagged build reproduces the tuning line within noise (§65).
- **REcoM coupled to OpenIFS exchanged gases against zero pressure** (Bug XIII); fixed, and sea-level pressure is now coupled (§66).
- **KPP quiets the abyss** (+5 against +13 mK/decade below 2500 m) **and uniform GM 1000 opens the Southern Ocean**: 40 % of the winter ice gone in eight years, net TOA $+1.3\,\mathrm{W\,m^{-2}}$ (§67.2).
- **The Southern Ocean column is held by salt alone, with a margin of 0.25 kg m^{-3}**, and the eddy-induced velocity compensates half of $50\,\mathrm{m\,yr^{-1}}$ of wind-driven upwelling with GM 2500 and a quarter with GM 1000 (§68.1).
- **`K_GM_max` 2500 is an effective 980 at 55–65 S**, half the CORE2 value (§68.2).
- **The cap is too weak in every run**: 78 % of PHC3 in the open ocean, 31 % in the Weddell region, with the winter-water layer 1.3 K too warm; it orders by effective GM across thirteen runs (§68.3).
- **The Southern Ocean 10 m westerlies are 16 % too fast but the stress only 10 % too strong**, in AMIP as in the coupled runs, at the right latitude; and **two thirds of the Antarctic calving freshwater enters north of 60 S** (§68.4).

Added 2026–10–03 (round 46, §63).

- **IDEMIX is what warms the mid-depth ocean**: +31 mK/decade at 700–1500 m against +9 for `cvmix_TKE` alone and +11 for KPP, four arms from one restart; on the production line that layer is flat under TKE alone and warms from the leg IDEMIX came in (§63.2).
- **Both TKE schemes drain the upper ocean**: the top 300 m is 0.26–0.34 K colder than under KPP after a decade (§63.2).
- **The AMOC held 15–16.5 Sv at 26.5 N for two centuries** through every atmospheric and sea-ice change, and dropped at the 2120 branch: about half common to the branch, 1.1 Sv a northward shift under hemispheric GM (§63.1).
- **The tuning runs on the carbon-cycle configuration** in concentration-driven PI mode, at 6.7 times the cost of the physics-only line (§63.3).

Added 2026–10–03 (rounds 41–45, §58–§62).

- **A boreal forest is affordable and survives on the merged LPJ-GUESS**: crunveg keeps 81 % of its BNS after 19 years against 34 and 20 % for the pre-merge runs, and costs about 2×10^3 km^3 of NH ice and +0.24 K globally (§58).
- **The Labrador ice-capped regime is a GM 2500 / Redi 1000 regime**, not a time-step or build effect: 61 runs, one clean pair (0.684 against 0.280), and the persistent capped state only under 2500/1000 (§59.3). The IDEMIX per-step smoothing is real and fixed (FESOM #1103) but does not restore convection (§59.1).
- **The GM heat-uptake benefit is Southern Ocean ventilation**: three pairs agree that the extra GM-1500 uptake enters at 34–55 S and is stored in the Indo-Pacific at 700–2000 m (§60.3).
- **Hemispheric GM is adopted**: Labrador mixed layer near 2050 m with no collapses, global T2m, TOA, Siberia and ice volume unchanged; it overshoots, March Labrador ice 0.10–0.13 against 0.16 (§60.6). FESOM PR #1112.
- **Against a PI reference the model is +0.46 K too warm**, the excess in NH summer land and SH summer ocean, with the Arctic winter cold (§61.3).
- **The Antarctic winter warm ring is a diffuse pack, not a missing edge**, and the winter area deficit opens in April–May; the drift over-responds to correct winds and turns wrong in the Ross and Amundsen–Bellingshausen sectors (§62).
- **`h0min` and `h0/h0_s` are not levers in AWI-ESM3**: hard-coded in `ice_thermo_cpl.F90`, inert in the namelist (§62.3).

Added 2026–09–25 (rounds 37–40, §54–§57).

- **The Southern Ocean winter bias resists every lever tried.** Salt plumes, Fox-Kemper MLE, a deeper lead-closing floor, all three together, and a calibrated ice-shelf basal melt all fail to move it in 20 years (§54.1). It is a freshwater regime, not a mixing deficiency.
- **Ice-shelf basal melt is calibrated.** `cavity_gamma_scale` 0.6 gives 1190 Gt yr^{-1} against 1893 uncalibrated and 1100–1325 observed; upstream as FESOM PR #1064.
- **The two snow albedos act in different hemispheres, by measured surface state.** $\partial\alpha/\partial$ `albsn` is 0.908 in SH DJF against 0.149 in NH JJA; `albsnm` is the reverse. Both stay global. The sensitivities are state-dependent and moved by a factor of three as the Arctic cooled (§55.1).
- **`cvmix_TKE+cvmix_IDEMIX` is the adopted mixing scheme**, on a four-way 20-year comparison from one restart. Both TKE variants grow the Antarctic summer pack where both KPP variants lose it; cavity melt falls 1255.6 → 1062.1 Gt yr^{-1} towards Adusumilli (2020). The cost is northern and was stated at adoption (§56).
- **Two real bugs in the CVMix forcing path**, an implicit netCDF fill value and an `MPI_DOUBLE_PRECISION` buffer on a `real(WP)` variable, both in FESOM PR #1087. The gate on any IDEMIX leg is `Etot_bot` 0.9135067 → 0.9179869 TW (§56.1).
- **The cold drift is an ocean redistribution, not an energy leak.** Total ocean heat content is flat at +0.04 ZJ/decade while the upper 700 m loses to below 700 m at 0.32 W m^{-2} equivalent, and the surface cools 0.21 K/decade with net TOA positive (§57.2).

- **The GM tuning cost 16.7 Sv of Drake Passage transport**, 110.0 → 93.3 Sv on identical years, monotonic in K_{GM}^{max} (§57.4).
- **The boreal forest is effectively gone** at 2099: BNS 0.001, TREEFPC 0.050, grass beating tree on 84.9 % of the Siberian box, stable across 2090–99. By competition with C3 grass, with every establishment gate open (§57.5).
- **The model skill degrades in the tropics, not the poles.** CMPI 0.844 → 0.903 across 1980–2099, driven by nino34 +0.155 and tropics +0.134 against arctic +0.003 (§57.3).

Added 2026–09–11 (round 36, §53).

- **The single-precision ocean heat leak is fixed.** 15F closes at $-0.069\,\mathrm{W\,m^{-2}}$ over 10 years and −0.035 over 20, like the double-precision 13A (−0.073); the unfixed 15C reads −0.544 (FESOM PR #1054).
- **Decade-4 TOA follows mid-depth heat storage, with a shallow slope.** Across 20 runs with 1850 forcing and at least 40 years (now including 16C, 16D, 16E), decade-4 TOA against the 700–2000 m heating rate averaged over years 1–40 gives $r = +0.72$, and +0.58 ($p = 0.015$) without 080, 11M and 11L. The slope without the extremes is about $1.3\,\mathrm{W\,m^{-2}}$ per $\mathrm{W\,m^{-2}}$, not the 4.1 of the first 17-run fit (§53.3).
- **Only the new-sea-ice chain lets mid-depth uptake decay**, to 0.21–0.44 $\mathrm{W\,m^{-2}}$ by years 25–45 (11G, 11I). 11X levels off at 0.50–0.62, and 15F tracks it (§53.2).
- **Stronger GM is the lever that reaches mid-depth.** 16C (`K_GM_max` 1500) cuts first-decade 700–2000 m uptake by about 30 % (0.67/0.62 against 0.92/0.96). Weaker Redi raises it and MLE leaves it unchanged (§53.4).
- **16E (`K_GM_max` 2500, Redi 1000) settles at the lowest TOA plateau the campaign has kept**: $+0.70\,\mathrm{W\,m^{-2}}$ in decade 4, -0.45 ± 0.08 against 15F and -0.24 ± 0.06 against 16C in clean pairs. Its deep uptake below 2000 m is nearly shut (0.04–0.09). The costs: NH March ice $+1.3\,\mathrm{M\,km^2}$ against 15F (17.2 in decade 4, obs 15.0), global T2m +0.21 K (§53.5).
- **The AMOC response to GM depends on the dose.** GM 1500 strengthens it ($+0.83 \pm 0.21$ Sv at 26.5N against 15F); GM 2500 is 1.0 ± 0.2 Sv weaker than GM 1500 at 26.5N and 1.7 ± 0.2 at 40–60N. MLE changes neither (-0.25 ± 0.18, §53.7).
- **`RSBLB = 2` adds late mid-depth uptake but not TOA**: +0.17/+0.19 $\mathrm{W\,m^{-2}}$ at 700–2000 m in decade 4 of both clean pairs, net TOA unchanged within ±0.07 (§53.6).

Added 2026–09–09 (rounds 34–35, §51, §52).

- **The ocean heat gain accounts for the imbalance, and it lands below 300 m.** 11X gains $+1.502\,\mathrm{W\,m^{-2}}$ per ocean area over 30 years; the top 100 m gains +0.013 and the 100–300 m layer *loses* 0.170, while 300–2000 m gains +1.388 (§52.3).
- **The tropical mixed layer is not too deep.** 43.6 against WOA18's 51.6 m annually and 81.0 against 80.8 at the seasonal maximum, both on WOA's criterion. The model is shallower than observed in all twelve months (§52.2).
- **The tropical deep warming is heave, not diapycnal flux.** At $\sigma_0 = 27.0$ (575 m) the water warmed +0.047 K while the surface sank +42.4 m, accounting for 0.42 of the +0.477 K seen at fixed depth (§52.4).
- **Heat crosses density surfaces at the mid- and high-latitude winter outcrops**, where the mixed layer is +48.9 to +155.7 m too deep: 30–45N +0.58 K at 500 m, 45–30S +0.57 K at 950 m (§52.4).
- **The Southern Ocean loses heat rather than burying it.** $-0.076\,\mathrm{W\,m^{-2}}$ of the global budget, cooling at every depth from 70 to 2400 m, with 18.7 % of month-nodes mixing to the sea floor (§52.3, §52.2).
- **The coupled ice skin was diverging because `oasis_put` returns `OASIS_Waitgroup` for every field in a group but the last.** The anchor was 0.0 in all 63,923 logged events; the skin reached −36,470 K. After four fixes the minimum is 224.39 K with zero points below 200 K (§51).
- **OASIS `GSMART` on a sign-changing field is unsafe.** 9105 excursions above $1\,\mathrm{kW\,m^{-2}}$ onto real ice in one coupled year, maximum 22,550; `GSSMAR` and `GLOBAL` deliver none (§51.3).

Added 2026–08–23 (round 33, §50).

- **The Southern Ocean generates the TOA imbalance rather than responding to it.** The 90–60S and 60–45S bands contribute +0.383 and $+0.353\,\mathrm{W\,m^{-2}}$ to the global mean, the two largest positive terms, against a tropical −0.872 (§50.2).

- **Every arm drifts, including the 1850 controls**: +0.8 K and -2×10^6 km^2 of September ice per 40 years. The air was NOT warm at the branch point (−0.06 to −0.65 K), so the ocean was not warmed by the atmosphere — both warm together (§50.1).
- **The polar cloud residual is an OPACITY problem, the sub-Antarctic one an AMOUNT problem.** At 90–60S cover is right to 2 pp but each cloud is 8 % too dim; at 60–45S there are 5 pp too few clouds each 9 % too bright. Every campaign lever raises opacity, which is why 60–45S is now overcorrected at −2.97 while 90–60S still reads +6.98 (§50.4).
- **Cloud phase flips sign at 60S**: +0.015 too liquid at 45–65S, −0.028 too icy poleward. The INP family was retired on a measurement taken in the one band where the model is too liquid (§50.4).
- **The ice melts from above in summer and fails to refreeze from below in autumn.** +6.98 W m^{-2} of DJF cloud shortwave, and a +0.9 K anomaly at exactly 100 m that autumn mixed-layer deepening entrains — which is why the MAM anomaly is longwave with no shortwave component (§50.6, Fig. 37).
- **The sea ice is not too dark** (compact-ice albedo 0.760–0.778); it is too diffuse. No cell exceeds 90 % concentration in January–March, so the area-mean albedo falls to 0.49–0.51 (§50.5).

Added 2026–08–23 (round 32, §49).

- **Stable-BL mixing is the winter land lever, and it works coupled.** 11Q/11R = 11N/11P + `RSBLB 2.0` give Arctic DJF screen +0.614 */+0.674 * against AMIP's non-significant +0.285, with T925/T850 null — coupled, the inversion weakens by warming the screen, not by chilling the air above (§49.3, Fig. 36b).
- **It is free in energy terms coupled.** Global net TOA −0.001/−0.027, both null, against AMIP's +0.175 *: the tropical shortwave gain is absorbed as ocean warming and cancelled by the longwave response (§49.3).
- **Winter and summer move together for the first time.** Siberian JJA improves 1.000 K over base, 0.767 of it from `RSBLB`, against a campaign history of levers that buy one by spending the other (§49.5).
- **11R is the best-scoring run in the campaign**, CMPI 0.7577 against 11E's 0.8237, and it reverses 11G's `tas`/`zg` regression (§49.4).
- **The stable-mixing trade is linear in dose and cannot be dodged.** Tropical SW CRE per K of Arctic inversion is 0.88/0.87 for `RSBLB` b=3/b=2 and 3.43 for the mixing-length route, so the shortwave cost is a property of the scheme rather than of the knob (§49.2, Fig. 36a).
- **Best coupled land state so far:** annual all-land bias −2.509 → −1.854 K from base at 1850, a 26 % reduction, at unchanged net TOA (§49.5).

Added 2026–08–21 (rounds 30–31, §46–§48).

- **The Southern Ocean bias is an AMIP problem, not a coupling problem.** 84 % of the coupled SO SW CRE error is already present in AMIP over a prescribed, correct ocean; cloud cover matches to 0.001 and CRE to 0.91 W m^{-2} (§46.1).
- **Tuning net TOA to zero does stop the ocean drift.** The apparent "−1.0 W m^{-2} structural leak" was the omitted snow-enthalpy term; with it the column conserves to 0.02–0.04 W m^{-2} and 11M (TOA −0.016) drifts −0.02 K/century (§46.3).
- **The winter land bias sits between the screen and 925 hPa.** In 11P at 60–90N DJF the air-side inversion bias is +2.81 K while the skin-side bias is +0.11 K, and the profile decays to −0.42 K by 850 hPa (§48.3, Fig. 35).
- **The surface-side levers are retired as a class, by measurement.** Skin conductivity (+0.125 K against a 0.41 threshold, §48.1) and Raupach z_0 (DJF land–ocean gap −0.017 and −0.003 in two windows, §48.2) both score null, for the reason the decomposition gives: they adjust a part of the column that carries no error.
- **Only about half the residual land bias is reachable by any mixing lever.** Of the −2.23 K all-land DJF bias, ∼1.0 K is excess inversion and ∼1.2 K is a deep cold bias still present at 850 hPa (§48.3).
- **The LX4 stack is the best coupled state so far**, all-land annual −2.79 → −2.11 K at fixed forcing basis and 60–90N DJF −6.35 → −4.49 K (§47.3). Note 11N→11P is a change of comparison basis, not a further gain.

Added 2026–08–16 (round 29, §45).

- **Raupach canopy roughness is dead, structurally rather than statistically.** Its winter mechanism needs a canopy; the model's Siberian trees are 2.89 m tall with 2 cm stems, giving $\lambda_{\text{stem}} = 0.0092$ and $z_0 = 0.0132$ m against IFS's 0.058 m. It makes the winter surface 4.4× smoother. The lever cannot bootstrap the forest it requires in order to work.
- **The land-surface repair is a trade.** Paired, it warms Siberian DJF soil +0.308 K ($t = 2.49$, still growing) and recovers TREEFPC +0.0094 of what S4 cost — but it moves the Siberian JJA bias against ERA5 from −0.673 to −1.123, worse than the 11E baseline.
- **This design cannot resolve anything below ≈0.35 K of Siberian JJA soil at 50 coupled years.** The interannual noise floor on the paired difference is 1.2 K; the measured +0.312 K would need 57 years. Future land levers must be screened on a lower-noise diagnostic — the surface momentum budget resolved `skt−2t` at $p = 3 \times 10^{-5}$ where soil temperature saw nothing.
- **Branched coupled arms must be scored paired.** All arms share the 1350 state, so the difference series starts at zero and grows; the unpaired $1.96\,\text{sd}\sqrt{2/n}$ threshold discards the pairing and hid a resolved result.

Added 2026–08–11 (round 28, §44).

- **S4 (`RCL_INPPMIN` 70000 → 50000) is the best configuration the campaign has.** Southern Ocean surface bias +3.66 → −0.96, global net TOA +0.64 → **+0.25** — nearest the piControl target of anything run — Siberia JJA +1.03 and DJF +0.05, tropics +0.26. One namelist number, and it had sat unevaluated since 2026–08–09.
- **The round-26/27 levers are global dimming knobs, not Southern Ocean levers.** LX3 moves the tropics −4.05 against the Southern Ocean's −5.61, near-uniformly across every basin. DMS's one-year seasonal selectivity does not become spatial selectivity at 44 years.
- **`RSNOWLIN2` is confirmed at full length on both counts**: it delivers the tropical longwave gain (+2.06 global LW CRE) and warms the Southern Ocean (+0.61 net) while worsening global energy (+0.42). Not adoptable; still the only lever that moves the tropical longwave at all.
- **A1a's boreal damage reproduces exactly** — −0.67 K Siberian JJA on the new base against −0.749 on the bare base, both ≈3× threshold.
- **The campaign's SO "band mean" and "SW RMSE" are different fields**: TOA cloud forcing and surface net shortwave. They disagree about which arm is best, and the surface error exceeds the TOA error everywhere — an atmospheric-absorption signature.
- **Krakatoa is visible in the AMIP runs** (1883–86, all arms), independently confirming that these "pre-industrial" runs read transient forcing.

Added 2026–08–10 (round 27, vegetation, §43.12).

- **Warming restores the boreal forest, and the transfer function is now measured.** Offline 2000-year LPJ-GUESS equilibria at TCO95: AMIP forcing gives TREE 0.202 and BNS 0.034; +1 K gives 0.261/0.058; +2 K gives 0.327/0.099 against the observed-forcing 0.415/0.227. Monotonic, not saturating, and +2 K recovers 59 % of the gap. **With the AMIP cold bias at ≈2 K period-clean, closing it should recover most of the forest** — the campaign's founding premise, previously assumed and now calibrated.
- **The coupled model dismantles an inherited forest in 50 years.** 11E starts at BNS 0.213 / TREE 0.394 (the CRUNCEP restart) and ends at 0.017/0.083 — a 92 % loss of larch — with grass beating trees on 70 % of cells against 28 % at the start. It is a transient toward a much poorer equilibrium, not an endpoint.
- **Warming does not dissolve the competition.** `grass>tree` sits at 43–45 % across AMIP, +1 K and +2 K against CRUNCEP's 29 %. Forcing is the driver; grass-versus-larch remains the mechanism (§31.2).
- **Siberia is NOT solved.** 11E fixes the *atmospheric* half — summer warmth and the winter penalty every previous snow lever carried — and leaves the vegetation untouched.

Added 2026–08–10 (round 27, §43).

- **`RSNOWLIN2` 0.030 → 0.04 closes the tropical longwave deficit.** +2.581 W m^{-2} alone and +2.740 on the adopted stack, against a ±0.279 threshold — 120–127 % of the −2.16 deficit and five times the largest response among the other fifty arms. It is the temperature exponent of ice→snow autoconversion, so it builds cold thin cirrus rather than thickening anvils, and thin cirrus is longwave-dominant.

- **It also warms the Southern Ocean.** On the adopted stack its SO longwave +2.626 swamps its shortwave −0.507, giving net +2.294, resolved. On the bare base it was neutral: the sign reversal is produced by the base. **The campaign's two remaining regional targets are in tension through this one parameter.**
- **Southern Ocean levers must be scored on NET, not SW CRE.** Longwave cancels a very consistent 25–29 % across the whole 44-year archive, so SO SW × 0.72 is the rule for ordinary cloud levers — and the new levers break it. Best net lever in the archive is A1a at −4.567, ahead of D2a's −3.602.
- **DMS is the cleanest of the new Southern Ocean levers on net forcing** (−1.976, only 17 % longwave cancellation) and the only one that reaches DJF, where the deficit is +10.86 and austral-summer insolation is 460 W m^{-2} against 87 in JJA. Its annual mean coincides with `RCL_INPPMIN`'s; its seasons do not.
- **A jointly-tuned eight-parameter cloud stack for TCO95L91 existed unused** in the v3.4.2 DARS2 runscripts, with no `fort.4` on disk setting any of it.
- **The AMIP runs are not at 1850 forcing** — `NCMIPFIXYR` sits in the dead `&NAMECECMIP6` block, runtime is −1, GHG runs 1869→1911. Deltas are unaffected; the absolute energy baseline is offset ≈0.15 W m^{-2}. The coupled runs *are* correctly pinned. Z1 verified a one-line fix.
- **S4 − P5**: SO SW CRE −1.98, global net TOA −0.41, tropics −0.10, Siberia within noise — but SO SW RMSE +0.255 against ±0.019. Best mean, worse pattern.

Added 2026–08–10 (round 26, §42).

- **Marine biogenic CCN is a real Southern Ocean lever, and the best one found.** At $S = 166$ it moves SO SW CRE −7.25 W m^{-2} in DJF against a ±3.82 threshold — 67 % of the +10.86 DJF deficit, in the season the deficit is worst, from the bloom rather than from a tuned constant.
- **It cannot be adopted as a tuning knob at any published coefficient.** The tropical cost is 0.0129 W m^{-2} per unit S, so the ±0.5 tolerance binds at $S \approx 39$, below Woodhouse et al.'s floor of 43; and Twomey's $\Delta \ln N_d$ scaling makes the bound conservative. Scaling S cannot separate the two bands because tropical background CCN is *lower* than the SO's.
- **Detection thresholds for a 1-year pair, per band**: SO SW CRE annual ±1.97, SO DJF ±3.82, tropics annual ±0.99, tropics DJF ±1.22, global ±1.11. Two of the three bracket arms fall below their own thresholds and carry no information.
- **The tropical longwave deficit gates the Southern Ocean fix.** The tropical cost of the DMS lever is shortwave; the tropical bias is longwave. They are independent terms, so closing the LW deficit makes the SO lever affordable rather than competing with it — and the two are complementary in the global budget (−1.72 against $\approx +1$).

Added 2026–08–09 (round 25, §41).

- **The tropopause is not displaced.** Model and ERA5 put the WMO lapse-rate tropopause on the same level in all five bands. The 200/100 hPa sign reversal is a *sharper* transition, not a moved one — the model cools +1.8 K km^{-1} faster than ERA5 through 300–250 at the poles and −1.5 to −1.9 slower above 200.
- **Cold and dry share one entry point.** A −2.07 K tropical cold point predicts −28.5 % of entry-air drying by Clausius–Clapeyron against a measured −46.9 % at 100 hPa, i.e. **61 %**. They are not the independent compensating pair §38.4 recorded, so a lever that warms the tropical cold point should move both the right way (§41.3).

Added 2026–08–09 (round 24, §40).

- **Southern Ocean droplet number is correct in the mean** — $N_d \approx 73$ cm^{-3} against an observed 60–120, evaluated on 6-hourly wind. The marine CCN path is mapped end to end and `RCCNSEA` is inert because `LCCNO=.TRUE.`
- **But the seasonal phase is inverted**: model N_d is lowest in DJF (68.5) and highest in JJA (76.1), because it is wind-driven, where observations peak in DJF from biogenic sulphate. The measured SO CRE error is worst in exactly those months.
- **The cloud-lifetime CCN route would divide by zero** at `NAERCLD`= 0, so the amount two-thirds is unreachable from the CCN path (§40.3).
- **A DMS emission climatology already ships at TCO95 and nothing reads it** — `emdms`, paramId 210043, on the model's own N96 grid, with `ICMCL` having headroom for an eighth field (§40.4).

Added 2026–08–09 (round 23, §39).

- **The Southern Ocean amount deficit is confirmed against the right reference.** CERES and MODIS put SO cloud area at 89.5 and 89.3 % against the model's 83.1; ERA5 reads 81.6 but reflects 6.31 W m^{-2} less than CERES against the model's 7.45, i.e. it has low area *and* low reflection in proportion. **ERA5 reproduces the model's bias because ERA5 is IFS**, so it is not an independent cloud reference (§39.3).
- **The Southern Ocean and the tropics fail through different cloud types.** SO loses energy via SW cloud (+7.45); the tropics via LW cloud (−2.16). No global microphysics knob can serve both, which is why `RVICE` trades one for the other (§39.2).
- **Nothing in 50 runs warms the tropics through the longwave**, the term that is actually deficient there. The only tropics-warming levers on record act in the shortwave.
- **CERES cloud area is valid over dark ocean, not over ice** — CERES and MODIS disagree by 7.7 pp at 90S–65S.

Added 2026–08–09 (round 22, §38).

- **The period-clean radiative error is +1.23 W m^{-2}, about twice what the PI framing reports.** `amip_presentday` reads net TOA +2.20 against CERES's +0.97 (§38.2).
- **The tropical TOA deficit is mostly the reference period** — −2.51 PI-vs-CERES against −0.67 period-clean. The Southern Ocean is the opposite: robust to the epoch and *larger* period-clean, +7.46 W m^{-2} and +0.735 of the global mean, now unambiguously the largest single regional term.
- **The SO deficit is cloud AMOUNT,** 65.5 % **of it**, zonally uniform to 1.42 pp and worst in austral spring and summer — and **no lever in the campaign moves cloud area**: D2a +0.63 pp and D2b +0.43 pp of a 6.43 pp gap, against +0.48 and +0.45 pp from two boreal-*land* levers that cannot physically act there (§38.3).
- **The global cold bias is extratropical and lives at the tropopause**, peaking at 200 hPa over both poles (−4.20, −4.45) and reversing sign at 100 hPa, with the tropics the *least*-biased band. The convective-heating hypothesis is excluded (§38.4).
- **The atmosphere is cold AND dry, and the two cancel in OLR.** −1.62 K and −10.1 % over 700–300 hPa; a Planck response to the cold alone predicts −6.10 W m^{-2} of OLR against −2.53 measured. Fixing either alone moves net TOA several W m^{-2} the wrong way.
- **The `tanh` falsification reproduces coupled on a controlled pair**: 11B minus 11A is −18.99 K of Siberian DJF soil (§38.1).
- **The coupled branch already carries D2b**, not just G4 — 11A/11B/11D all set `RCL_INPSEA` with the 700 hPa gate. K1 and P5 are what remain missing.

Added 2026–08–07.

- **The `tanh` depletion is falsified on observations, not on preference.** Its range is open at 1, so complete snow cover is unrepresentable at any parameters, while Russian snow courses report exactly 10/10 in 99.74 % of 14 369 Siberian DJF surveys. Cost: −20.2 K against 174 676 station soil observations where the scheme *off* is right to +0.8 K (§33.3).
- **The damage was distributional, not a matter of mean cover.** DJF cover is identical to 0.002 across N1/N2/O1/O2 while the soil differs by 23 K; the clipping ramp leaves 93.5 % of the box perfectly insulated, the `tanh` smears the same deficit into every cell (§33.2).
- **P3 is adopted** — both falsifiers passed, and it beats the scheme-off baseline on bias, RMSE and JJA against 105 RIHMI stations (§34).
- **Aerosol is ∼2 % of the Southern Ocean clear-sky excess, not 13 %**, measured rather than inferred, and the two reflection columns are not separable (§35).

- **K1 is the carried-forward configuration (adopted 2026–08–05)** = G4 + the snow-free land albedo correction. Siberian JJA +1.036 K (t=8.29), global T2m RMSE 1.579 → **1.524** (the campaign's best bar K2), every season inside its own threshold, and no Arctic energy penalty — 60–90N net TOA −97.463 against a control −97.737 and CERES −97.98 (§30.2). Namelist-only on top of G4.
- **Land surface albedo is CLOSED as an explanation for the universal bias.** Removing 74 % of a directly measured bias bought 0.089 K of ∼0.7 K; a perfect fix buys ∼0.12 K, one sixth of the target (§30.3).

- **G4 is the base K1 is built on**: Siberian JJA T2m +0.952 K (t=7.68). *Closure depends on a denominator this report does not pin down*: against the ≈ 2.0 K working figure it is 48 %, but §17.2 gives an honest range of −1.3 to −2.0 K and §27.3 measures −2.58 K against ERA5, so the defensible span is **37–73 %**, SO SW RMSE 6.88 → 4.80, subpolar N Atl 5.01 → 4.74, tropics untouched (+0.124), every season inside its own threshold. Namelist-only (§24.7).
- **The boreal budget is nearly spent.** Siberia's JJA −2.58 K is ∼0.7 universal (present over prescribed-SST ocean too), ∼0.3 orographic and 1.3–1.6 boreal-specific; G4 has taken ∼0.95 of the last (§21.3). The remainder is not reachable by a boreal-indexed lever, and this — not a parameter limit — is why F5 = F4.
- **The boreal forest fails by TWO routes, not one** (§31.1). BNE/BINE are *gate-excluded* — 0.7 % of NE-Siberian cells clear `tcmin_est`= −30 and their LAI is exactly 0.000. BNS (larch) has *no* cold gate, clears its GDD gate on 84 % of cells, and is *outcompeted*: grass carries 5.8× its LAI. Since NE Siberia is ∼80 % larch, the winter-gate mechanism does not explain the region that matters most.
- **The near-surface cold is global, and the soil is not the cause.** All 20 surface types are cold in every season, and over low-lying land the soil and skin are the *least*-biased parts of the column (§21.2, §21.3).
- **The boreal cold bias had two routes.** Route A (over-transpiration → low cloud → SW loss) is **closed**: the Siberian JJA cloud term went −7.04 → +1.20 W m^{-2}. Route B (late melt → June albedo) is **diagnosed and fixable** but carries almost no temperature leverage (§27.2).
- **The snow field can be made correct, but not at this price**: melt-out 12 May against a satellite 11 May (§27.1) — and the scheme that does it costs −16.2 K of coupled DJF soil for +1.2 K of summer (§32). **Rejected**; round 19 tests a re-parameterisation (§32.2).
- **The winter damage is seeded in AUTUMN, not winter.** Midwinter snow cover is identical between the schemes (Δ`ZCVS` = −0.0004 in January); the deficit peaks in October (−0.075) exactly as the soil begins to diverge, then vanishes while the soil runs to −24 K (§32.1).
- **The control soil is right, verified against direct observations.** RIHMI-WDC station soil temperature under natural cover puts the control within ∼1 K at every depth the Russian network reports in winter, and its thermal offset within 0.7–2.3 K of the observed +17.8 to +24.2 K (§27.8). The campaign no longer depends on ERA5 — an HTESSEL sibling — for any soil claim.
- **I3 is the strongest summer lever built**: Siberian JJA +1.322 K (t=10.61), and it closes the Siberian surface SW deficit to 166.58 against CERES 166.26 — the deficit §22 opened the albedo thread with. It is **not adopted**: it buys that with +1.72 W m^{-2} at 60–90N, the largest NH−SH albedo shift of any run, and the campaign's only significant Nordic Seas RMSE degradation (§27.6).
- **The Siberian winter penalty is not radiative.** DJF snow albedo, SWE and absorbed SW are identical across control/G4/I1/I2/I3 to three decimals, on an 8 W m^{-2} midwinter budget that cannot build −2.4 K (§27.7). The route is thermal.
- **Half the land albedo excess is the snow tile.** Of +0.0154 against CERES, +0.0074 is snow and +0.0080 the snow-free surface; every forest type is inside +0.006 once snow is masked per cell per month (§29.1).
- `RVRSMIN` **does not saturate**; 1000 is a cap set by winter damage, not a knee (§24.6).
- **The energy target is Antarctic/Southern Ocean**, and the two campaign targets share that root cause (§11.2.2, §76.6).
- **Tuning is namelist-driven and mergeable with EC-Earth** — one added line in an ECMWF file (§24.7).

77.2 Open, in priority order

Added 2026–10–15 (rounds 53–58).

1. **When does the spin-up meet the carbon criterion?** 0.41 GtC yr^{-1} over 2000–2009 (ocean 0.17, land 0.24) against 0.1; the land loss is now in soil and litter (§75.3).
2. **Is a spin-up 0.4–0.5 K colder than the tuned state acceptable?** The lost forest and its higher land albedo are the candidate; continuing will not close it (§75.2).
3. **The Southern Ocean cap is still at 70–80 % of observed**, and the levers that do not weaken the AMOC are exhausted. GM 4000 was withdrawn for the AMOC (§70).
4. **Icebergs**: a large berg's melt is drawn from one cell; the heat-flux cap discards about 1 % of the melt heat and is not physical. Changing leg lengths are untested (§75.1).

5. **Is the Arctic at `albsn` 0.82 only marginally stable?** Three of four arms gained 4–6.5×10^3 km^3 (§71).

Added 2026–10–06 (rounds 47–52). The plan is §69.

1. **How much of the missing cap is GM and how much freshwater placement?** Two measured contributors, neither yet tested in a run; the wind is a minor third.
2. **Why does the Labrador Sea need an effective 390 when a physical measure asks for more?** Probably a too fresh or too ice-prone Labrador Sea; not tested.
3. **Why does every CVMix scheme warm the abyss twice as fast as FESOM's KPP?** `cvmix_KPP` does it too, so it is the code path, not TKE.
4. **TKE-only or TKE + IDEMIX for the release**, if the Southern Ocean work does not land first.
5. **Carbon cycle**: FESOM #1115 and OpenIFS !25 open; a control with the corrected gas exchange has not started.
6. **Housekeeping**: the tuned settings are not in `awiesm3.yaml`; the ice-sheet variant has not run on albedo; the AMIP inputs are not in the albedo pool.

Added 2026–10–03 (round 46).

1. **Does TKE without IDEMIX keep the Antarctic ice and stop the mid-depth warming on today's configuration? PICAL_crunveg_gmhemi_tke**, 2140–2159, in flight. Its known cost on the old line was a system losing energy (net TOA −0.47 W m^{-2} over 2060–2079).
2. **`cvmix_KPP` against native KPP**: 0.17 K in the top 100 m between two implementations of one scheme is unexplained.
3. **PICAL_cc is paused at 2145** for performance work; its BGC output definition is fixed in `esm_tools` (`fdde4730e`) and not yet exercised by a leg.
4. **The AMOC of ob1800 is still falling** (14.4 Sv over 2130–2139); gmhemi1800's is steady but 2.5 Sv below the pre-2120 line at 26.5 N.

Added 2026–10–03 (rounds 41–45).

1. **The Labrador overshoot under hemispheric GM**: March ice 0.10–0.13 against 0.16 and NH mid-latitude MLD the largest CMPI loss. Next: NH `K_GM_max` 1500.
2. **The line still cools**: −0.059 ± 0.025 K/decade over 2101–2139 with ocean uptake 0.69 W m^{-2}, mostly below 700 m (§61.3).
3. **NH summer land is warm and dry**, central Asia +2.9 K with half the observed rain and LPJ-GUESS cover 0.31 against IFS's 0.57; cause and effect not separable (§61.4).
4. **The diffuse Antarctic pack and its over-responsive drift**, documented as a known bias, not tuned (§62.5).
5. **Structural biases of v3.5 to document**: Southern Ocean cloud, the Antarctic pack, the stratocumulus coasts, NH summer land.

Added 2026–09–25 (rounds 37–40).

1. **The upper-ocean heat drain.** 0.32 W m^{-2} equivalent out of the top 700 m is the mechanism behind the cold drift and nobody has tried to stop it. Whether it is the mixing scheme, the Southern Ocean freshwater regime, or both, is open.
2. **NH April volume** decides whether `cvmix_TKE+cvmix_IDEMIX` keeps its place: +1.145 ± 0.968 per decade over 2080–2099, sitting 3.37 above the early GIOMAS record.
3. **The Antarctic at 62 % of observed winter volume**, unchanged in character by any Round 37 lever and not a vertical-mixing problem.
4. **The ACC is 30 % weak** and slowly weakening, and the GM setting that improved the CDW made it worse. The two constraints have not been reconciled.
5. **Does the canopy work do anything without trees? PICAL_crunveg** was built to answer it and cancelled before starting.

6. **Will the CRUNCEP state load on `TCO95-land`?** It was written on an 11538-cell `L096` gridlist; the old respin handoff recorded a missing-cell case as fatal with peatland on, and the code has changed substantially since. Untested.
7. **The tropical CMPI degradation** has no mechanism attached to it yet.

Re-ranked 2026–09–11 after round 36 (§53). The goal is now to use the relation between mid-depth heat storage and TOA to bring TOA close to zero without degrading the rest of the climate; then retune the melt-pond albedo for NH sea ice.

1. **Bring TOA to about zero.** The three 40-year GM arms are done (§53.5). 16E is the best, at +0.70 in decade 4, and GM alone will not close the rest: the pairs give 1.2–1.6 $\mathrm{W\,m^{-2}}$ of TOA per $\mathrm{W\,m^{-2}}$ of mid-depth uptake, 16E's mid-depth rate is already 0.28, and the step from GM 1500 to 2500 cost 1.7 Sv of AMOC at 40–60N. Open: which second lever takes the remaining 0.7.
2. **Then retune the melt-pond albedo for NH sea ice.** 16E has NH March extent 17.2 $\mathrm{M\,km^2}$ in decade 4, against 16C's 15.7 and the observed 15.0. This comes after TOA, because the ice responds to the ocean state.
3. **SH sea ice should improve by itself** once the ocean holds less heat and warms the ice less from below. Verify this rather than tune it. 16E: September extent $-0.4\,\mathrm{M\,km^2}$ against 15F over 30 years; the direction holds, the size is small.
4. ~~Check the AMOC for the GM arms.~~ Done 2026–09–11: GM strengthens it (§53.7). Open instead: why 15F has the weakest AMOC of the older configurations, and why GM 2500 gives back the gain GM 1500 made.
5. **16E_1990** (constant 1990 forcing as 11P/11R/11V, with 1850 land use; running, 38 years at 1387) gives the present-day pair for period-clean scoring against observations. 16C_1990 failed: LPJ-GUESS refused the 1990 land use on the 1850 state.

Previous ranking, 2026–09–09:

Re-ranked 2026–09–09 after round 35 (§52). The ocean side now has a located mechanism, and it is not where the plan assumed.

1. **Price a winter mixed-layer lever at the subtropical and subpolar outcrops.** That is where heat crosses density surfaces, and where the model over-mixes by +48.9 (30–45N, 45–30S) to +155.7 m (45–60N) at the seasonal maximum. The coincidence of location is established; the causal link is not, so the first arm is a measurement, not an adoption.
2. **The Southern Ocean convection, still unfixed and now quantified.** 18.7 % of month-nodes reach the sea floor and the band cools at every depth. The corrected CORE3 mesh did not fix it. This is a stratification and salinity problem, not a diffusivity one, and it is the largest single ocean defect in the model.
3. **Re-baseline the energy target** against 11X now that the ocean heat gain (+1.502 $\mathrm{W\,m^{-2}}$ of ocean area) can be set against a TOA number directly.
4. **Is the tropical surface uptake itself too large?** +13.9 $\mathrm{W\,m^{-2}}$ net into the ocean at 30S–30N is the source of the heat that the mid-latitudes then bury (§52.6). ERA5 and NCEP2 are on disk and this has never been scored.
5. **Implement Fox–Kemper mixed-layer-eddy restratification.** FESOM 2.7 has none, and its absence is the standard reason coarse models over-deepen winter mixed layers at the outcrops. Implementation work rather than a knob, but it acts exactly where the measurement points.
6. **Write `opottempdiff` and `opottemprmadvect` in 3D** in the next arm's `io_list`. The column-integrated versions cannot separate interior diapycnal mixing from the surface flux.
7. **Dropped: tropical interior ocean mixing**, and **dropped: the `linfs` virtual salt flux** — both falsified in round 35.
8. **Excluded by instruction: sea surface salinity restoring.** `surf_relax_s` stays 0 in every arm, including as a suppressant for the Weddell convection. Restoring hides the model's error rather than fixing it and makes a coupled PI spin-up non-conservative.
9. **A PHC3 climatology on the corrected CORE3.** Only `core3_beta` (211567 nodes) exists, which forces every obs comparison onto a regular grid or onto drift rather than bias.

Campaign plan, set 2026–08–10 (evening): finish the Southern Ocean and the tropics, then one final land round. The vegetation result (§43.12) shows Siberian tuning is not finished, but it also shows what finishing it requires — closing the ≈2 K cold bias, for which the offline $+1/+2$ K series is now the calibration. That is a separate campaign from the radiative one, and mixing them is what made round 27 sprawl.

1. ~~**Finish SO and tropics.** LX1/LX3/LY2 completed 48 years on 2026–08–10 and await `evaluate.sh`.~~ **Done 2026–08–11** (`data/evaluate_L.out`); scored in §44.3. `RSNOWLIN2`'s tropical gain does *not* survive: LX3 and LY2 reach the Southern Ocean by dimming the planet (−0.88 and −0.58 global TOA), and LX1 pays 0.83 W m^{-2} of imbalance for 0.4 of SO surface. **S4 wins on all three constraints at once.** The one decision still genuinely open from this item is **A1b**, credited here with solving the energy target while appearing in no run of the adopted stack.
2. **Then the final land round**, with three things it did not have before: the $+1/+2$ K transfer function, the three unread `LPJG_COMP_EXP*` competition experiments, and the knowledge that 50 coupled years shows a transient, not an equilibrium, so vegetation must be judged offline or over centuries.
3. **Re-baseline the energy target** with one fixed-forcing control (§43.9); Z1 proved the one-line fix works.
4. **11E** is complete at 50 coupled years and is the best coupled configuration on Siberian T2m, but net TOA is still +0.765 and SO CRE 5.65 short of CERES.

Re-ranked 2026–08–10 after round 27 (§43). The tropical longwave deficit now has a lever that closes it, and the lever costs the Southern Ocean. Everything below follows from that.

1. **Price `RSNOWLIN2` at 44 years.** LX1/LX3/LY2 are in flight against P5. The tropical gain and the Southern Ocean penalty are both established at one year; what is completely unmeasured is **Siberia**, and there is no headroom — the land stack recovers 1.15 K of a ≈2 K period-clean bias, leaving ≈0.9 K cold. LY2 carries A1a's value, rejected in round 3 *for boreal damage*.
2. **Resolve the tropical-LW / Southern-Ocean tension.** They are coupled through one parameter. If `RSNOWLIN2` survives Siberia, the question becomes whether DMS or A1a can pay back the SO longwave it adds.
3. **Decide A1b.** `RCL_OVERLAPLIQICE` 0.65 → 0.35 is credited in this report with solving the energy target and appears in *no* run of the adopted stack. Y1 shows it is essentially energy-neutral on the new base (global −0.005) and reaches only −1.96 of SO SW. The inconsistency should be closed explicitly.
4. **Re-baseline the energy target** with one fixed-forcing control run (§43.9). Do *not* mix fixed and transient arms in one comparison.
5. **11E coupled**, resumed at year 1370 with 20 of 50 banked; still the gate on the PI spin-up.
6. **The tropical cold point**, −2.07 K (§41), untouched.

Re-ranked 2026–08–10 after round 26 (§42). The Southern Ocean lever now exists and is priced; what blocks it is the tropics. The list collapses accordingly:

1. **The tropical longwave cloud deficit**, −2.16 W m^{-2}. Promoted to first because it is no longer a parallel item: it *gates* the Southern Ocean fix (§42.5). The lever must be keyed on **deep convection**, not on cloud ice in general — `RVICE` is the counter-example. *Do the free check first*: no arm of ∼52 has ever been scored on this term, and `tropical_lw_scan.py` is written to answer it from data already on disk.
2. **11E coupled**, running since 2026–08–10. Decides whether the coupled configuration is level with AMIP, and gates the PI spin-up. **11F** is running alongside it but has been **reclassified as a mechanism probe rather than a tuning arm** (§42.6): its $S = 166$ is not adoptable, and S4 delivers the same Southern Ocean forcing at twenty times the tropical selectivity. It is kept only for the generic question — what ≈2 W m^{-2} of SO brightening does with free SST and sea ice — which transfers to S4 and which AMIP cannot answer at any length. First to cut if compute tightens.
3. **The tropical cold point**, −2.07 K, the shared entry point for the cold bias and 61 % of the stratospheric dry bias (§41).
4. **S4** (`RCL_INPPMIN` 70000 → 50000) completed 48 years on 2026–08–09 and is **unevaluated**.
5. **The SO cloud-*amount* two-thirds** stays open with no identified lever; the CCN path reaches the opacity third only.
6. **Dropped:** the Blomquist et al. (2017) transfer velocity. It is less steep in wind than the Nightingale quadratic, so it moves flux away from the windy Southern Ocean toward the calm tropics — worse selectivity, not better (§42.5).

Re-ranked after round 24 (§40), which answers round 23's question. Round 23 asked for a *regionally selective* Southern Ocean reflection lever. Round 24 found one and priced it: marine CCN is selective by construction, but it reaches only the **opacity third** — 2.57 W m^{-2} in-band, ≈0.25 global — and the amount two-thirds is unreachable from the CCN path because the lifetime route divides by zero at `NAERCLD`= 0. The list is therefore:

1. **11E coupled** (11D at SWEMIN = 15 plus K1). A namelist edit on a lever already validated over 44 AMIP years, and the only route to three questions AMIP cannot price. Cheapest decisive run available.
2. **The tropical longwave cloud deficit**, −2.16 W m^{-2}, untouched by any lever in 50 runs and the one term that is deficient in the direction the tropics need.
3. **The tropical cold point**, −2.07 K, which §41 shows is the shared entry point for the cold bias and 61 % of the stratospheric dry bias. The tropopause *height* is correct, so this is a cold-point temperature problem, not a displacement one, and one lever should move both errors.
4. **Marine biogenic CCN via `emdms`** — real missing physics with the input already on disk, but ≈0.25 W m^{-2} global and it needs a published flux→CCN relation, so it follows the cheaper items.
5. **The SO cloud-*amount* two-thirds** stays open with no identified lever.

Re-ranked again after round 23 (§39). The Southern Ocean stays first but the requirement has sharpened: it needs a *regime-selective* reflection lever, because every global low-cloud knob overshoots a tropical band that is only −0.67 W m^{-2} from CERES. Second is the tropical *longwave* cloud deficit, untouched by any lever. Third is the extratropical tropopause. Fourth is 11E coupled.

Re-ranked 2026–08–09 after round 22 (§38). The top of this list is now: (i) **the Southern Ocean cloud-*amount* deficit**, 65.5 % of a +7.46 W m^{-2} band error, untouched by every lever and *not* reachable by the INP branch (§38.3); (ii) **the extratropical tropopause cold-and-dry pair**, which is the dominant land-T2m term and simultaneously the reason the global energy budget cancels (§38.4); (iii) **11D re-run at** SWEMIN = 15 **plus K1**, the first coupled configuration level with the AMIP one.

Two entries below are **superseded** and struck rather than deleted: ~~The snow-free land albedo residual +0.0080 is untouched by all 41 runs.~~ K1 is exactly that lever and removed 74 % of it for +0.089 K; the entry predates K1's adoption and survived unedited (§30.3, which also retracted K3 before it ran). ~~Tropics net TOA 2.4 W m^{-2} too low.~~ Period-clean it is 0.67 (§38.2).

Added 2026–08–07. (i) **The autumn cover excess P3 introduced** — +0.259 in September against Rutgers, where as-released was +0.107. P5/P6 test `SWEMIN` = 15 and 30; the offline optimum is 30 (§34.1). (ii) **The Southern Ocean unexplained residual has grown to** ∼1.9 **W m^{-2}** now that the aerosol term is measured (§35). (iii) **Scale awareness is a two-anchor interpolation, not a law** — a new resolution needs validation (§34.2). (iv) The M1 clear-sky sign anomaly is significant and untraced.

1. **The blocker: a −4.7 K Siberian winter bias** accompanying the snow fix (§27.3). Not a radiation-supply problem (§27.4) and now positively excluded as radiative at all (§27.7), so the route is thermal. Round 16 tests skin conductivity; if that fails, boundary-layer mixing. *J1/J2 are built on I1, not I3* — diagnostic either way, but the production pair needs re-running on the better base (§28).
2. **Decide the coupled base: I3 or I1.** I3 gives +0.072 K more JJA and a better global T2m RMSE, but carries the Arctic energy bill of §27.6, which **AMIP structurally cannot price** — prescribed SST and ice absorb 1.7 W m^{-2} without responding, and the coupled spin-up will not. Open question: whether a recalibrated mode 2 keeps the +1.3 K without the 60–90N surplus, since the SDOR scaling does most of its work outside the box it was tuned in (§29.2).
3. **The snow-free land albedo residual** +0.0080 — crops +0.0204, semidesert +0.0169, bare soil +0.0145, tundra +0.0105 — is untouched by all 41 runs and is the one lever aimed at the ∼0.7 K *universal* bias rather than the boreal one. It **must not simply be stacked on I3** (§29.3): the mid-latitude entries are safe, the tundra entry competes for the same Arctic energy budget.
4. **Global net TOA** +0.52 **W m^{-2}** against a PI target of ∼ 0 — one of the two original faults, barely moved by the whole campaign. Note this sits oddly beside a cold troposphere: in AMIP the prescribed SST absorbs the imbalance without warming, so the two are not contradictory, but the pairing is unexplained.
5. **The global tropospheric cold bias** of §21.1 (−0.9 K at 700–850 hPa, −2.8 to −3.0 K at 200 hPa) is now the dominant remaining term for land 2 m temperature and has never been worked on. It is out of reach of every lever this campaign has built.
6. **SO cloud-area deficit** ∼6 **pp**, untouched by all 34 evaluated levers, and now known to drive the energy target too.

7. $-12\,\mathbf{W\,m^{-2}}$ **downward-LW deficit in Oct–Nov**, unexplained.
8. Tropics net TOA $2.4\,\mathrm{W\,m^{-2}}$ too low; Nordic Seas SW RMSE moved by G1 and partly repaired by G4, both unexplained.
9. Whether 48 % of the boreal bias suffices for forest survival — **AMIP structurally cannot answer this**; it needs the coupled tests.

78 Falsified claims, kept

Every mechanism this campaign has killed, with the measurement that killed it. Retained because the history has to stay honest, and because three of them were killed only after being built on.

claim	what killed it	where
~~Decade-4 TOA falls about 4 W m^{-2} per W m^{-2} of 40-year mid-depth uptake removed ($r = +0.86$, 17 runs), so a GM arm that cuts the uptake by 0.2–0.3 reaches TOA balance~~	16E cut the 40-year 700–2000 m uptake to 0.34, the lowest of any run; the fit predicted decade-4 TOA −0.03, and it measured +0.70 (4.4 residual sd). The slope was set by 080 and 11M; without them it is about 1.3.	§53.3
~~Stronger GM strengthens the AMOC, so it is no obstacle to a larger GM dose~~	True for GM 1500 only. GM 2500 (16E) is 1.0 ± 0.2 Sv weaker than GM 1500 at 26.5N and 1.7 ± 0.2 at 40–60N over 40 years, and below 15F at 40–60N.	§53.7
~~Fox-Kemper MLE takes back the AMOC gain of GM 1500~~	Read off 16D's second decade (11.4 Sv). By decade 4 16D is at 15.1, and 16D minus 16C is -0.25 ± 0.18 Sv over 40 years.	§53.7
~~The `linfs` virtual salt flux is accumulating salt at the mid- and high-latitude outcrops, which deepens the winter mixed layer there and is the real cause of the heat burial~~	Global mean salinity drifts -2.5×10^{-5} psu in 30 years, so salt is conserved. And at 45–60N the virtual salt flux is *freshening* while the layer becomes 0.30 psu saltier — the salinification is advective, a consequence of the deep mixing rather than its cause.	§52.6
~~The ocean heat uptake is spin-up from the initial condition and will equilibrate, so it should not be tuned against~~	The three decade-to-decade steps of `thetaoga` are +1.461, +1.433 and +1.472 W m^{-2} — flat to 3 % over forty years, with no deceleration after the first five.	§52.6
~~The model buries surface heat by mixing it down in the tropics, which is why the surface stays cold while TOA stays positive — so tropical ocean mixing is the lever~~	The tropical mixed layer is 8.1 m too *shallow* annually and level at the seasonal maximum (81.0 against 80.8), and 90 % of the tropical deep warming is heave ($\sigma_0 = 27.0$ sinks +42.4 m while the water on it warms +0.047 K). Heat crosses density surfaces at the mid-latitude outcrops instead. The burial itself is real: the ocean gains +1.502 W m^{-2} and none of it stays in the top 100 m.	§52.2, §52.4
~~The Southern Ocean convection is a route by which surface heat reaches the deep ocean~~	90–60S is the only band with a net heat *loss*, −0.076 W m^{-2} of the global budget, cooling at every depth from 70 to 2400 m and peaking −0.26 K near 1000 m. Convection ventilates: it connects the abyss to a cold surface.	§52.3
~~`force_flux_consv` in FESOM's coupling receive is amplifying `A_Q_ice` into the kilowatt excursions that reach the ice skin solve~~	A probe comparing the field immediately before and after the two calls found `raw_max` = `corrected_max` in all 20,760 samples of a coupled year. The path is inert in this configuration; the excursions came from OASIS's `GSMART` conservation instead.	§51.3
~~The Southern Ocean surface is too cold, so the ice-albedo feedback works in the model's favour as it warms toward equilibrium~~	Measured with SST, which under sea ice is pinned near freezing and cannot show the error. T2m is the free variable and reads +1.5 to +1.9 K at 70–80S, peaking at +4.5 K in MAM. The feedback works against us.	§50.8
~~The residual Southern Ocean cloud deficit is in a band the campaign has never worked on~~	It has: 90–60S DJF SW CRE has gone +13.63 → +6.98 across 11E→11R. The real structure is a meridional gradient in lever effectiveness, not an untouched region.	§50.3
~~`RSBLLMIN` spares the tropics: the 30 m floor binds only where the stable layer is shallower than 300 m, so a shallow Arctic winter SBL feels it and a deep subtropical Sc boundary layer does not~~	It also sets the length in free-shear layers, so it enhances entrainment across the Sc capping inversion. Same Arctic gain as SB2 at four times the tropical cost (+2.106 against +0.541 of SW CRE) and +1.673 W m^{-2} of net TOA. The risk was recorded in the runscript before the run.	§49.2
~~The tropical low-cloud loss will degrade `nino34`, where 11G already regressed~~	`cmpitool` on 11R: `nino34` is the MOST improved region, −0.030 (`tas` −0.258), and the tropics are neutral at −0.0002. The cost is real but lands on `thetao` (+0.018 global), the ocean absorbing the shortwave.	§49.4
~~Snow skin conductivity is the winter lever: λ_{sk} over exposed snow is too small, letting the skin radiatively decouple in polar night, so raising `ECE_LAMSK_SN` from 7 to 15–25 should warm DJF by +1 to +3 K~~	Measured null at 48 yr, paired: +0.125 K at 60–90N DJF against a 0.41 threshold (J1) and −0.039 against 0.45 (J2). Verified live first — the symbol is in `libsurf.SP.so`, `fort.4` carries the values, and the arms differ locally by up to 21 K. The decomposition then gave the reason: the skin-side bias is already only +0.11 K.	§48.1
~~Raupach roughness will move the coupled winter screen temperature~~	Scored on the 11J−11I run-time-gated pair: DJF land–ocean gap −0.017 (1380–89) and −0.003 (1360–79), sign flipping at 60–90N between windows. Confirms by measurement the structural argument already on this list.	§48.2
~~Raupach (1994) canopy roughness is a WINTER lever: a leafless boreal canopy is the roughest state, so DJF grid-box z_0 rises 0.058 → 0.140 m and warms Siberian winter soil~~	The winter mechanism requires a canopy the model does not have. Measured in 11J: $h = 2.89$ m, stem diameter 0.0198 m, so $\lambda_{\text{stem}} = 0.0092$ — an order of magnitude below the peak of the Raupach z_0/h curve. Its own constants then give $z_0 = 0.0132$ m, the bare-soil value, against IFS's 0.058 m: 4.4× *smoother*, which is the sign the momentum budget measures (DJF $\lvert\tau\rvert$ −4.1 %, 10 m wind +3.0 %), DJF soil +0.054 K, $p = 0.670$.	§45.5
~~The land-surface repair is free — nothing resolves against its threshold, so adopt~~	Scored with the wrong statistic and against no target. Paired (the arms branch from one 1350 state) Siberian	§45.4

Two further items are *provisional* rather than dead: the March–April mass deficit rests on ERA5 SWE (forest-biased), and mode 2's scale-awareness carries only ∼16 % of the length scale over this box (§26.4).

.1 Round 37–40 (2026–09–13→25)

- ~~The Southern Ocean winter over-ice bias is a vertical mixing deficiency, fixable with a salt-plume scheme, mixed-layer eddies, or a deeper lead-closing floor.~~
  Killed by **PI200_spp**, **PI200_mle**, **PI200_h0** and **PI200_all3**, four 20-year branches from one restart: none moves it. It is a freshwater regime, salty-and-venting or fresh-with-warm-CDW, and these levers do not cross between them (§54.1).

- ~~Reducing ice-shelf basal melt breaks the loop warm CDW → melt → fresh shelf → no dense water → warmer CDW.~~
  Killed by **PI200_cavg06**: the calibration is right on its own terms (1190 against 1893 Gt yr^{-1}, observed 1100–1325) and the Southern Ocean does not respond in 20 years.

- ~~Raising the lead-closing floor `h0min` 0.5 → 1.0 thickens the Antarctic pack.~~
  The flux response was exactly as designed, area-to-volume of freezing 1.71 → 1.10, and the state did not follow: thickness 0.61 → 0.62 m and the 60–90 S bias +0.82 → +1.45 K. Reverted (§55.2).

- ~~The ice is too dark because the melt ponds are too bright; darken `albpnd`.~~
  Wrong direction and double-counting: the four `ice_therm` albedos are pond-free and `albpnd` is applied over them afterwards. Ponds stayed dark and the ice and snow were brightened instead (§55.1).

- ~~`albsn` 0.75 and `albsnm` 0.65 are FESOM's own defaults.~~
  They are not: FESOM ships 0.81 and 0.77, so those values were deliberate darkenings, not inherited. Corrected before the numbers were committed.

- ~~The albedo sensitivities are fixed properties of the configuration.~~
  They are state-dependent: NH $\partial\alpha/\partial$ `albsn` moved 0.149 → 0.504 as the Arctic cooled. Re-measure rather than reuse.

- ~~IDEMIX fails because the bottom forcing file is wrong, or unnormalised, or carries NaNs that the 3-D interpolator mishandles, or NaNs that the 2-D reader mishandles.~~
  Four wrong hypotheses in a row. The file has *zero* NaNs, proven by a 30-line offline Fortran harness. The real defects were an implicit netCDF fill value and an MPI type mismatch (§56.1, FESOM PR #1087).

- ~~A matching `fesom` executable checksum proves the right FESOM is running.~~
  It proves nothing: the executable is a driver stub, byte-identical across these builds. Only `libfesom.so` distinguishes them, and it can be a symlink that `md5sum` silently follows (§57.1).

- ~~Net TOA is positive, so the run is warming.~~
  Net TOA is +0.167 W m^{-2} over 2080–2099 and the surface cools at 0.21 K/decade. Total ocean heat content is flat; the upper 700 m is losing heat to below 700 m at 0.32 W m^{-2} equivalent (§57.2).

- ~~GM tuning is free at the Southern Ocean.~~
  It cost 16.7 Sv of Drake Passage transport, 110.0 → 93.3 Sv on identical years, monotonic in K_{GM}^{max} (§57.4).

- ~~The coupled run reads `namelists/lpj_guess/global.ins`; `ecearth.ins.j2` is unused by this setup.~~
  Exactly backwards. `guess.ins` is hard-wired in `framework.cpp:1063`, rendered per rank, and imports `ecearth.ins`; `global.ins` is staged and never reached. The two had drifted apart, `iftreefracca` 1 against 0 (§57.6).

- ~~tripyview is not installed on this system.~~
  It is, in `~/.conda/envs/esm-tools_auto_tripyview`; its editable-install `egg-link` points at a path that no longer exists, so the import fails misleadingly. `PYTHONPATH=/work/ab0246/a270092/software/tripyview` fixes it.

.2 Round 53–58 (2026–10–06→15)

- ~~More effective GM in the south (southern `K_GM_max` 4000) is the first lever for the cap.~~
  Withdrawn before it ran: less upwelled deep water weakens the AMOC, which is at 13–14 Sv; GM 2500 was already 1 Sv weaker at 26.5 N than GM 1500 (§70).

- ~~Icebergs, wave roughness and ice–ocean drag grow Arctic ice: $+4.4$ to $+6.5 \times 10^3$ km^3 in September, all significant.~~
  A second realisation of the control with no change moves +2.7 the same way and passes the same test on eleven of fifteen quantities (§71).

- ~~The wave model's default roughness slows the Southern Ocean 10 m wind (0.3 m s^{-1} in the first decade).~~
  The second decade is 0.4 faster than the control; the iceberg run, which changes nothing in the atmosphere, moved as much in the first (§70.2).

- ~~The iceberg run's fourth-decade Antarctic gain (+0.48 February, +1.40 September) is outside the noise.~~
  It is at the edge of the ctl2 yardstick (0.4 and 1.3); only the constant sign speaks for it (§71).

- ~~Lowering the air–ice roughness slows the Antarctic drift the physical way, as the higher ocean drag did by compensation.~~
  A 13 % lower coefficient sped the wind over the pack up by 7–11 %; the stress fell 1–7 % and the drift 2–5 % (§72).

- ~~A fresh start from PHC3 only resets the clock on the abyssal drift, so continuing the 350-year state is the better spin-up.~~
  The rate is the same either way, but the restart begins the deep Atlantic 1.2 K and the deep Arctic 2 K closer to observations, a gap that takes centuries to rebuild (§73). Reversed by decision the same evening.

- ~~Spin-up land carbon is 2553 PgC in 1975–84 and the 1890s legs show a 20 PgC dip.~~
  The 1° LPJ-GUESS files count whole coast cells as land, 12 % too much area; with the land fraction rebuilt from the native points it is 2260 PgC and the dip is 2–8 PgC (§75.4).

.3 Round 46 (2026–10–03)

- ~~`cvmix_KPP` reproduces the KPP control closely enough to serve as the null.~~
  For sea ice it does. For heat it does not: its top 100 m is 0.17 K and its 100–300 m 0.15 K colder than native KPP by 2030–2039 (§63.2).

- ~~The ocean starts gaining heat near 1000 m in the last run, under hemispheric GM.~~
  That layer has warmed since IDEMIX came in at 2080 (+27 mK/decade); gmhemi1800 is the fastest (+42), not the first, and ob1800 without the GM change warms at +32 (§63.2). The eye was caught by a contour level being crossed.

- ~~The 2120 drop of the AMOC at 26.5 N is the hemispheric GM.~~
  Half of it is in ob1800 as well; the GM part is a northward shift of 1.1 Sv with the 20–60 N maximum unchanged (§63.1).

.4 Round 41–45 (2026–09–26→10–03)

- ~~The Labrador Sea stops convecting at 1800 s because IDEMIX mixes less at the longer step; fixing the smoothing brings convection back.~~
  The deficit is real and the fix removes about 85 % of it, but **PICAL_crunveg_nx1800** still sat at 414 m against ihf0's 370 (§59.1).

- ~~The 2120s Labrador weakening is caused by the new FESOM build and the 1800 s step.~~
  Every such difference was taken against ob1200, a single realisation at the 28th percentile of the null; every other arm, and the unbranched ccnice continuation, sit at the bottom of it. The weakening begins in ccnice in the IDEMIX era (§59.2).

- ~~The GM 2500 heat-uptake benefit is separable from the Labrador capping: switch GM off on the fine mesh and keep the rest.~~
  gmramp (v3.3 ramp, off below 30 km) opened the Labrador Sea and gave back the whole benefit, TOA +0.442*, while losing 40 % of SH September volume. The benefit is Southern Ocean ventilation at 34–55 S, and the Southern Ocean margin sits on the same mesh spacing as the Labrador Sea (§60.2, §60.3).

- ~~A Rossby-radius GM separates the Labrador Sea from the Southern Ocean.~~
  R_d is 8.0 km in the Labrador Sea and 9.3 km at 55–65 S. The online arm gmrd1800 lost 1.67×10^3 km^3 of SH September volume (§60.4, §60.6).

- ~~The coupled surface imbalance of gmhemi1800 is 1.51 W m^{-2}~~ (and ~~gm2500's net surface flux is 1.47~~).
  Both omitted the snow-enthalpy term through L_f in J kg^{-1} applied to snowfall in metres; corrected 0.67 and 0.641 (§61.2).

- ~~OpenIFS double-counts the LPJ-GUESS vegetation cover by applying `RVCOV` on top of the Lambert–Beer FPC.~~
  `RVCOV` is applied only when LPJ-GUESS is not coupled (`surfbc_ctl_mod.F90` ∼285–320). Raised and falsified the same day (§61.4).

- ~~The central Asian steppe LAI of 4.9 is too high.~~
  Per-tile LAI; the cover-weighted value is 0.82 (§61.4).

- ~~The Antarctic winter warm ring is a missing ice edge.~~
  37.7 % of the area is pack in both model and HadISST and is +1.77 K at concentration 0.74 against 0.84; the missing-ice class is 2.4 % (§62.1).

- ~~A McPhee St^*u^* ocean-to-ice heat flux thickens the Antarctic pack.~~
  ihf1 − ihf0: SH September volume −1.25*, NH September +1.04* × 10^3 km^3 — the wrong direction in both hemispheres (§62.2).

- ~~The namelist `h0`/`h0_s` set lead closing in AWI-ESM3.~~
  They are read only by `ice_thermo_oce.F90`, compiled out when coupled; `ice_thermo_cpl.F90` hard-codes `h0min` 0.5, `h0max` 1.5 (§62.3).

- ~~The Antarctic drift and deformation errors come from the winds.~~
  Wind speed, the Amundsen Sea Low's relative depth and the wind divergence are close to or weaker than NCEP-2; the ice over-responds (wind factor 2.4 against 1.4 %) and turns +63° in the Ross against +24° observed (§62.5).

- ~~Going back to KPP with a uniform GM of 1000 gives a quiet deep ocean, a healthy overturning and a surface the atmosphere can correct; expected net TOA +0.6 to +0.8.~~
  The abyss and the overturning behaved as expected, but the Southern Ocean convected over twice the area, 40 % of the winter ice went in eight years and net TOA was +1.3 (§67.2).

- ~~The hemispheric GM is where the AMOC started to collapse.~~
  The 2120 branch changed GM and the time step together. 26.5 N stepped down about 1.5 Sv and has been flat for 90 years; the subpolar collapse is in ob1800, which kept the uniform GM (§64.2).

- ~~A GEOMETRIC eddy-energy closure would produce the north–south GM contrast by itself.~~
  Its energy growth rate is 1.4–1.6 times the dissipation in the Labrador and Nordic Seas and 0.6–0.9 in the Southern Ocean, independent of its parameters (§68.5).

- ~~A −0.7 W m^{-2} retune of the atmosphere removes about a third of the imbalance, and −2 all of it.~~
  That assumed a CMIP-typical ocean uptake efficiency. On this line a colder surface goes with a larger imbalance in every comparison (§68.5).

- ~~The convection in the KPP arm taps heat left in the upper Southern Ocean by 320 years of GM 2500.~~
  Not needed: deep-water temperature and salinity were identical in both runs at the branch, and the salt budget closes with the change in upwelling (§68.1).

- ~~The Southern Ocean wind stress is 35 % too strong, as large a contributor to the upwelling as the GM change.~~
  That was u_{10}^2. The stress itself is +10 % against ERA5 (0.159 against 0.145 N m^{-2} over 50–65 S), in AMIP too; the bulk drag is 16 % lower (§68.4).

- ~~Dynamic backscatter dramatically lowers the stable time step.~~
  Not at two months: it ran at 1800 s. Not tested longer, and the sub-grid energy was not well behaved (§68.5).

A Parameter catalogue (distinguishing change per run)

Run	Change (file/namelist)	Baseline → New	Process
06A	`albpnd` (`namelist.ice`)	0.2 → 0.28	melt-pond albedo
06B	+ `rfracmax`	1.0 → 0.75	pond water retention
06C	+ `pndaspect`	0.8 → 1.3	pond depth:area (=common pond)
06D *HRlike*	(only common pond shift)	–	no other diff found
06G	`albpnd/rfracmax/pndaspect`	0.35/0.6/1.6	stronger meltpond
06L	`albpnd/rfracmax/pndaspect`	0.4/0.55/1.8	extra-strong meltpond
06H	strong pond + `RVICE` (`&NAMCLDP`)	0.16 → 0.18	meltpond + cloud-ice fall speed
06Q	`albw/albocn`	0.1 → 0.045	open-water/ocean albedo
06R	`albw/albocn`	0.1 → 0.075	open-water/ocean albedo
06S	`whichevp` (`&ice_dyn`)	1 → 0	sea-ice rheology mEVP→EVP
06E	`RVICE`+`GGAUSSB` (`&NAMGWWMS`)	0.18; −0.5 → −0.6	cloud-ice + gravity-wave
06F	`LRDALB` (`NALBEDOSCHEME=3`)	F → T	diagnosed surface albedo
06P	`ENTSTPC3` (`&NAMCUMF`)	absent → 1	deep-convection entrainment
06N	`use_momix` (`namelist.tra`)	F → T	near-surface ocean mixing (mospp)
06M	OASIS coupling period + LAG	7200 → 3600 s	atmos↔ocean 2h→1h (1hcpl)
06O	1hcpl + mospp	–	coupling + mixing
06O4	O + `momix_kv`	0.01 → 0.012	+ larger M-O mixing
06O5	O + `momix_kv`	0.01 → 0.02	+ even larger mixing
06T	O + KPP-low (`kpp_av0/kv0` 0.005 → 0.003, bckg ↓ 10×)	–	+ weaker KPP mixing
06U	T + open-water albedo 0.075	–	combo
06V	T + `ENTSTPC3=1`	–	combo (most aggressive)

B Implementation & lab-notebook detail

Operational detail moved here to keep the work-plan science legible. Retained for reproducibility.

A.1 esm-tools fixes for the AMIP–noLPJG generator (§10.2). Model tree `oifsamip-cy48` git-aligned to `awiesm3-develop` (oifs `local_combined_fixes` @ 1303a90, oasis `afac6a3`, xios `2.5.3`) + bundle/field patches. Three bugs fixed, committed & pushed (commit `9921d95b`, branch `development_interactiv_mesh_awiesm3`, and a follow-up for the WAM recipe):

- `xios.yaml` dereferenced the `fesom` namespace via `choose_fesom.*` and crashed *any* FESOM-less setup — now gated on `"$(( 'fesom' in config ))"`.
- `oifsamip.yaml` never overrode the OIFS `comp_command`, so the binary compiled *without* XIOS or coupling despite `LXIOS:TRUE` — now adds `-with-cplng2 -with-oifs-xios -build-type=release` (like awicm3/awiesm3).
- WAM aborted (`CDATEA`≠`CDATEF`) at any start year ≠1990 because the OIFS `preprocess` step (`change_icm_date.sh`, which re-dates ICMGG/ICMSH/wave init files) never ran: `oifsamip.yaml` lacked `prepcompute_recipe: ${oifs.prepco` and fell back to the default recipe with no preprocess step. Added (matches `awicm3.yaml`).

Runscript `oifsamip-cy48-levante-TCO95L91_lpjgforcing.yaml` overrides XIOS output to `file_def_lpjg_spinup` + `field_def_lpjg_safe`, emitting daily `tas,tasmin,tasmax,pr,rsns,rlns,sfcWind` + `tdps,ps,psl,clt,prsn,evspsbl,p` *esm_master gotcha:* the single-component target is `recomp-<setup>/oifs-48r1` (the *component* name); the alias `/oifs` silently falls back to the setup and doubles the path (`oifsamip-cy48/oifsamip-cy48`).

A.2 Flux-accumulation fix (§10.2), at root. The de-accumulation divisor in `field_def_lpjg_safe.xml` was the hard-coded `/21600` (6 h). It is now templated: `field_def_lpjg_safe.xml.j2` uses `/{{oifs.flux_accumulation_period}` with `oifs.flux_accumulation_period` defaulting to `${oifs.time_step}` (= NFRHIS×TSTEP; with NFRHIS=1 this is the timestep). A validated ×6 scalar patch (`AMIP_noLPJG_1d_1870-1879_TCO95_PI_fluxfix.nc`) corrected the existing forcing without a rerun.

Controlled temperature test now ready. Because the AMIP–CRUNCEP comparison shows meaningful forcing sensitivity but the CERES replacement rules out net radiation as its dominant source, two additional AMIP forcing cycles were created for the next attribution step. They add exactly +1 or +2 K to `tas`, `tasmin` and `tasmax` at all TCO95 cells north of or at 45°N; every field south of 45°N and every non-temperature variable is unchanged. Their NH-land forcing-cycle means are −3.57 and −2.57 °C, compared with −4.57 °C in the source AMIP file. These are *prepared inputs, not spin-up results.* Running them with unchanged LPJG parameters will quantify whether a uniform boreal temperature increment crosses a vegetation threshold or merely produces a modest continuous response.

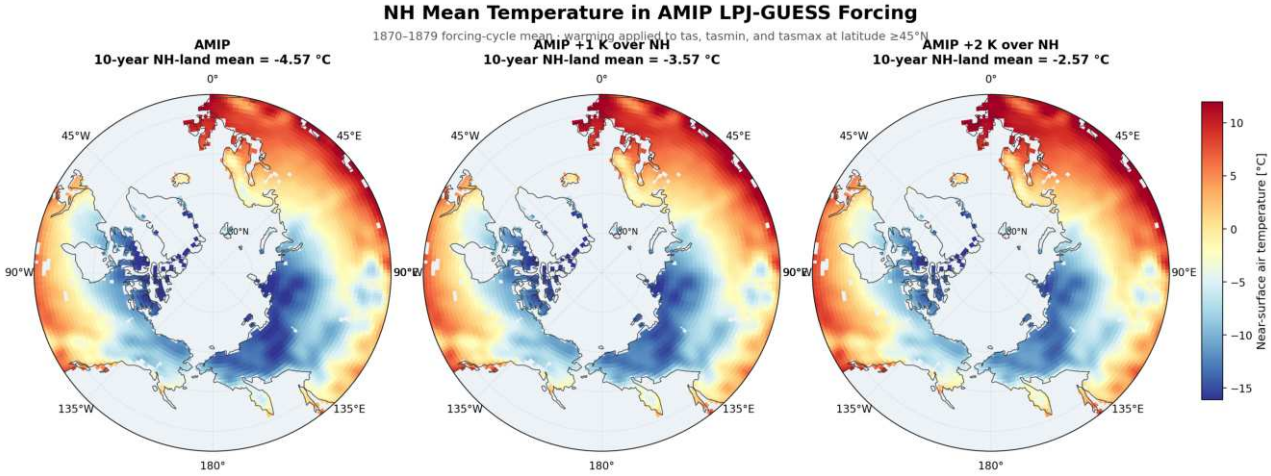


Figure 63: Input check for the planned AMIP temperature sensitivity. Ten-year NH-land mean near-surface temperature in the original, +1 K and +2 K forcing cycles. The increments were applied consistently to `tas`, `tasmin` and `tasmax` at latitude ≥ 45°N; no vegetation response is shown because these two spin-ups have not yet been run.

A.3 CERES forcing construction and validation (§7). The direct source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_200003-202106.nc`; the climatological source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. The output files are `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERES_rsns_rlns.nc` and `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERESclim_daily_variability.nc`. Both contain 3652 daily timesteps and 40 320 TCO95 cells. Coordinates and all non-radiation fields are unchanged from CRUNCEP v3. The direct file has `rsns` range 0–358.23 and `rlns` range −197.30–34.76 W m^{-2}. The daily-variability file has ranges 0–455.61 and −206.92–93.85 W m^{-2}; every mapped CERES cell-month target is reproduced within 8×10^{-6} W m^{-2}. Construction and preanalysis scripts, cached intermediates and the detailed provenance document are retained under `/work/ab0995/a270270/analysis/Offline_LPJ_GUESS_analysis/`.

A.4 Phase-1B OpenIFS source edits (§10.3). Paths under `oifs-48r1/ifs-source/arpifs/phys_ec/`:

knob	file:line	current	proposed	effect
`ZLMIN`	`vdfexcu.F90`:201	30 m	40–50 m	min stable mixing length
`ZCB`	`vdfexcu.F90`:190	5.0	3.0–4.0	LTG b: lower ⇒ longer tail ⇒ more K

Secondary companions if needed: the 0.1 / 300 m stable mixing-length limits (`vdfexcu.F90`:374,376) and the surface-layer constants `RCHETA`/`RCHETB`/`RCHBHDL` (`surf/module/suvexcs_mod.F90`, mirror in `suvdfs.F90`).

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